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Inventor(s)	Viswanathan; Karthik et al.

Antibody molecules to C5AR1 and uses thereof

Abstract

Antibody molecules that specifically bind to C5aR1 are disclosed. The antibody molecules can be used to treat, prevent, and/or diagnose disorders, such as ANCA-vasculitis.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application is a divisional application of U.S. application Ser. No. 17/147,842, filed Jan. 13, 2021, which claims priority to U.S. Provisional Application Ser. No. 62/960,544, filed Jan. 13, 2020, the disclosures of which are hereby incorporated by reference in their entireties.

INCORPORATION-BY-REFERENCE-OF SEQUENCE LISTING

(1) The contents of the file named “20230929_SVI-001US3_ST26.txt”, which was created on Aug. 9, 2023, and is 1,840 KB in size, are hereby incorporated by reference in their entirety.

BACKGROUND

(2) Anti-neutrophil cytoplasmic autoantibody (ANCA)-associated vasculitis (AAV) is an autoimmune disease characterized by inflammation and destruction of small- and medium-sized blood vessels and the presence of circulating antibodies against myeloperoxidase (MPO-ANCA) or proteinase 3 (PR3-ANCA). AAV affects various organs, but most commonly the kidney, with >75% of patients characterized by rapidly progressive glomerulonephritis. The potent anaphylatoxin, complement component 5a (C5a), promotes chemotaxis and activation of neutrophils, a key driver in inflammatory diseases driven by type III hypersensitivities such as ANCA-vasculitis.

(3) There is a need for developing new approaches for treating, preventing and diagnosing various disorders associated with C5 and its receptors, e.g., complement 5 receptor 1 (C5aR1), including ANCA-vasculitis and other disorders that share similar disease mechanisms.

SUMMARY

(4) The present invention provides, among other things, anti-C5aR1 antibodies with increased specificity to C5aR1 and therapeutic uses of such antibodies in effectively treating diseases or disorders associated with C5 and its receptors, such as, ANCA-vasculitis, COVID-19, Acute respiratory distress syndrome (ARDS), Influenza A, typical hemolytic uremic syndrome, age-related macular degeneration, rheumatoid arthritis, sepsis, severe burn, antiphospho lipid syndrome, asthma, lupus nephritis, Goodpasture's syndrome, and chronic obstructive pulmonary disease. As described herein, the present invention is, in part, based on identification of a new class of anti-C5aR1 specific antibodies that bind to certain regions on Site I and/or Site II of C5aR1 and have significantly reduced cross reactivity to C5aR2 or any other G protein-coupled receptors. In particular, anti-C5aR1 antibodies of the present invention are characterized with high binding affinity to C5aR1 (e.g., with K_{sub}.D less than 50 nM) and minimal cross-reactivity with C5aR2. This is significant because C5aR1-antibodies of the present invention allow potent inhibition of C5aR1 signaling in the presence of high C5a concentrations. As a result, C5aR1-antibodies of the present invention can be used at a lower dose to achieve therapeutic effect relative to the other anti-C5aR1 antibodies or C5a-antibodies. This is demonstrated by the surprisingly high potency observed in functional assays, relative to prior-art antibodies, as described herein. Moreover, highly potent Site I C5aR1 antibodies of the present invention compete with each other for Site I, and highly potent Site II C5aR1 antibodies of the present invention compete with each other on Site II. Additionally, the present invention provides methods and compositions for inhibiting C5aR1 and/or C5a signaling by targeting both Site I and Site II of C5aR1. Simultaneous targeting of Site I and Site II significantly enhances inhibitory activity. For example, combination of Site I and Site II antibodies or bispecific antibodies (e.g., biparatopic), but not two Site I or two Site II antibodies, significantly enhance activity. Inventive anti-C5aR1 antibodies of the present invention promise a more potent treatment of complement mediated diseases and disorders, particularly ANCA-vasculitis.

(5) The C5a-C5aR1 axis is of particular interest for therapeutic intervention in order to block attraction of neutrophils to local sites, inhibit neutrophil activation as well as vascular destruction.

The compositions and methods disclosed herein, may include a step of administering a C5aR1 antagonist, as well as methods of treating a subject in need of such a treatment

(6) In one aspect, described herein is an antibody molecule capable of binding to complement component of 5a receptor 1 (C5aR1), comprising a heavy chain variable region (VH) having a sequence disclosed herein and/or a light chain variable region (VL) having a sequence disclosed herein.

(7) In one aspect, described herein is an antibody molecule capable of binding to complement component of 5a receptor 1 (C5aR1), comprising a heavy chain variable region (VH) and/or a light chain variable region (VL), wherein the VH comprises an HCDR1, an HCDR2, and an HCDR3, each of which differs by no more than 1, 2, 3 amino acid residues from, or has at least 85, 90, 95, 99 or 100% identity with the HCDR1, HCDR2, and HCDR3, respectively, of a VH described in Table 2A, and/or wherein the VL comprises an LCDR1, an LCDR2, and an LCDR3, each of which differs by no more than 1, 2, or 3 amino acid residues from, or has at least 85, 90, 95, 99 or 100% identity with, the LCDR1, LCDR2, and LCDR3, respectively, of a VL described in Table 2B.

(8) In some embodiments, a VH comprises an HCDR1, an HCDR2, and an HCDR3, each of which differs by no more than 1, 2, 3 amino acid residues from, or has at least 85, 90, 95, 99 or 100% identity with the HCDR1, HCDR2, and HCDR3, respectively, of a VH described in Table 1A.

(9) In some embodiments, a VH comprises an HCDR1, an HCDR2, and an HCDR3, each of which differs by no more than 1, 2, 3 amino acid residues from, or has at least 85, 90, 95, 99 or 100% identity with the HCDR1, HCDR2, and HCDR3, respectively, of a VH described in Table 3A.

(10) In some embodiments, a VL comprises an LCDR1, an LCDR2, and an LCDR3, each of which differs by no more than 1, 2, or 3 amino acid residues from, or has at least 85, 90, 95, 99 or 100% identity with, the LCDR1, LCDR2, and LCDR3, respectively, of a VL described in Table 1B.

(11) In some embodiments, a VL comprises an LCDR1, an LCDR2, and an LCDR3, each of which differs by no more than 1, 2, or 3 amino acid residues from, or has at least 85, 90, 95, 99 or 100% identity with, the LCDR1, LCDR2, and LCDR3, respectively, of a VL described in Table 3B

(12) In one aspect, described herein is an antibody molecule capable of binding to C5aR1 (e.g., human C5aR1), comprising a VH and/or a VL, wherein the VH comprises (i) an HCDR1 comprises an amino acid sequence that differs by no more than 1, 2, or 3 amino acid residues from, or has at least 85, 90, 95, 99 or 100% identity with, the amino acid sequence of any of SEQ ID NOs: 601-661; (ii) an HCDR2 comprises an amino acid sequence that differs by no more than 1, 2, or 3 amino acid residues from, or has at least 85, 90, 95, 99 or 100% identity with the amino acid sequence of any of SEQ ID NOs: 721-781; and/or (iii) an HCDR3 comprises an amino acid sequence that differs by no more than 1, 2, or 3 amino acid residues from, or has at least 85, 90, 95, 99 or 100% identity with the amino acid sequence of any of SEQ ID NOs: 841-901; and/or wherein the VL comprises: (i) an LCDR1 comprising an amino acid sequence that differs by no more than 1, 2, or 3 amino acid residues from, or has at least 85, 90, 95, 99 or 100% identity with the amino acid sequence of any of SEQ ID NOs: 662-720, an LCDR2 comprising an amino acid sequence that differs by no more than 1, 2, or 3 amino acid residues from, or has at least 85, 90, 95, 99 or 100% identity with the amino acid sequence of any of SEQ ID NOs: 782-840, and/or (iii) an LCDR3 comprising an amino acid sequence that differs by no more than 1, 2, or 3 amino acid residues from, or has at least 85, 90, 95, 99 or 100% identity with the amino acid sequence of any of SEQ ID NOs: 902-960.

(13) In one aspect, described herein is an antibody molecule capable of binding to complement component 5a receptor 1 (C5aR1), wherein the antibody molecule competes with a C5aR1 antibody molecule comprising a HCDR1 comprising an amino acid sequence of SEQ ID NO: 656, a HCDR2 comprising an amino acid sequence of SEQ ID NO: 776, and a HCDR3 comprising an amino acid sequence of SEQ ID NO: 896, a LCDR1 comprising an amino acid sequence of SEQ ID NO: 715, a LCDR2 comprising an amino acid sequence of SEQ ID NO: 835, and a LCDR3 comprising an amino acid sequence of SEQ ID NO: 955.

- (14) In one aspect, the present invention provides, among other things, an antibody molecule capable of binding to complement component 5a receptor 1 (C5aR1), wherein the antibody molecule comprises a VH and/or a VH, wherein the VH comprises an HCDR1 comprising an amino acid sequence of SEQ ID NO: 1456, a HCDR2 comprising an amino acid sequence of SEQ ID NO: 1457, and a HCDR3 comprising and amino acid sequence of SEQ ID NO: 1458, a LCDR1 comprising an amino acid sequence of SEQ ID NO: 1459, a LCDR2 comprising an amino acid sequence of SEQ ID NO: 1460, and a LCDR3 comprising an amino acid sequence of SEQ ID NO: 1461.
- (15) In some embodiments, an antibody molecule capable of binding to complement component 5a receptor 1 (C5aR1) competes with a C5aR1 antibody molecule comprising a HCDR1 comprising an amino acid sequence of SEQ ID NO: 656, a HCDR2 comprising an amino acid sequence of SEQ ID NO: 776, and a HCDR3 comprising and amino acid sequence of SEQ ID NO: 896.
- (16) In some embodiments, an antibody molecule capable of binding to complement component 5a receptor 1 (C5aR1) competes with a C5aR1 antibody molecule comprising a LCDR1 comprising an amino acid sequence of SEQ ID NO: 715, a LCDR2 comprising an amino acid sequence of SEQ ID NO: 835, and a LCDR3 comprising an amino acid sequence of SEQ ID NO: 955.
- (17) In one aspect, described herein is an antibody molecule capable of binding to complement component 5a receptor 1 (C5aR1), wherein the antibody molecule binds to an epitope on Site I of C5aR1, wherein the epitope on Site I comprises amino acid residues of SEQ ID NO: 1449 or SEQ ID NO: 1452.
- (18) In some embodiments, the antibody molecule binds to amino acid residues T8-D18 in SEQ ID NO: 1449 or SEQ ID NO: 1452.
- (19) In some embodiments, the antibody molecule binds to amino acid residues T8-G12 in SEQ ID NO: 1449 or SEQ ID NO: 1452.
- (20) In some embodiments, the antibody molecule to amino acid residues T8, D10, Y11, Y14, and/or D15 in SEQ ID NO: 1449 or SEQ ID NO: 1452.
- (21) In one aspect, described herein is an antibody molecule capable of binding to complement component 5a receptor 1 (C5aR1), comprising a heavy chain variable region (VH) and a light chain variable region (VL), wherein the VH has at least 80% identity with the amino acid sequence of any of SEQ ID NOs: 481-541; and wherein the VL has at least 70% identity with the amino acid sequence of any of SEQ ID NOs: 542-600.
- (22) In some embodiments, an antibody molecule comprises a heavy chain variable region (VH) has at least 75% identity with the amino acid sequence of any of SEQ ID NO: 481-541. In some embodiments, an antibody molecule comprises a heavy chain variable region (VH) has at least 78% identity with the amino acid sequence of any of SEQ ID NO: 481-541. In some embodiments, an antibody molecule comprises a heavy chain variable region (VH) has at least 80% identity with the amino acid sequence of any of SEQ ID NO: 481-541. In some embodiments, an antibody molecule comprises a heavy chain variable region (VH) has at least 82% identity with the amino acid sequence of any of SEQ ID NO: 481-541. In some embodiments, an antibody molecule comprises a heavy chain variable region (VH) has at least 85% identity with the amino acid sequence of any of SEQ ID NO: 481-541. In some embodiments, an antibody molecule comprises a heavy chain variable region (VH) has at least 87% identity with the amino acid sequence of any of SEQ ID NO: 481-541. In some embodiments, an antibody molecule comprises a heavy chain variable region (VH) has at least 90% identity with the amino acid sequence of any of SEQ ID NO: 481-541. In some embodiments, an antibody molecule comprises a heavy chain variable region (VH) has at least 92% identity with the amino acid sequence of any of SEQ ID NO: 481-541. In some embodiments, an antibody molecule comprises a heavy chain variable region (VH) has at least 95% identity with the amino acid sequence of any of SEQ ID NO: 481-541. In some embodiments, an antibody molecule comprises a heavy chain variable region (VH) has at least 97% identity with the amino acid sequence of any of SEQ ID NO: 481-541. In some embodiments, an antibody molecule

comprises a heavy chain variable region (VH) has at least 98% identity with the amino acid sequence of any of SEQ ID NO: 481-541. In some embodiments, an antibody molecule comprises a heavy chain variable region (VH) has at least 99% identity with the amino acid sequence of any of SEQ ID NO: 481-541. In some embodiments, an antibody molecule comprises a heavy chain variable region (VH) has 100% identity with the amino acid sequence of any of SEQ ID NO: 481-541.

(23) In some embodiments, an antibody molecule comprises a light chain variable region (VH) has at least 68% identity with the amino acid sequence of any of SEQ ID NO: 542-600. In some embodiments, an antibody molecule comprises a light chain variable region (VH) has at least 70% identity with the amino acid sequence of any of SEQ ID NO: 542-600. In some embodiments, an antibody molecule comprises a light chain variable region (VH) has at least 72% identity with the amino acid sequence of any of SEQ ID NO: 542-600. In some embodiments, an antibody molecule comprises a light chain variable region (VH) has at least 75% identity with the amino acid sequence of any of SEQ ID NO: 542-600. In some embodiments, an antibody molecule comprises a light chain variable region (VL) has at least 78% identity with the amino acid sequence of any of SEQ ID NO: 542-600. In some embodiments, an antibody molecule comprises a light chain variable region (VL) has at least 80% identity with the amino acid sequence of any of SEQ ID NO: 542-600. In some embodiments, an antibody molecule comprises a light chain variable region (VL) has at least 82% identity with the amino acid sequence of any of SEQ ID NO: 542-600. In some embodiments, an antibody molecule comprises a light chain variable region (VL) has at least 85% identity with the amino acid sequence of any of SEQ ID NO: 542-600. In some embodiments, an antibody molecule comprises a light chain variable region (VL) has at least 87% identity with the amino acid sequence of any of SEQ ID NO: 542-600. In some embodiments, an antibody molecule comprises a light chain variable region (VL) has at least 90% identity with the amino acid sequence of any of SEQ ID NO: 542-600. In some embodiments, an antibody molecule comprises a light chain variable region (VL) has at least 92% identity with the amino acid sequence of any of SEQ ID NO: 542-600. In some embodiments, an antibody molecule comprises a light chain variable region (VL) has at least 95% identity with the amino acid sequence of any of SEQ ID NO: 542-600. In some embodiments, an antibody molecule comprises a light chain variable region (VL) has at least 97% identity with the amino acid sequence of any of SEQ ID NO: 542-600. In some embodiments, an antibody molecule comprises a light chain variable region (VL) has at least 98% identity with the amino acid sequence of any of SEQ ID NO: 542-600. In some embodiments, an antibody molecule comprises a light chain variable region (VL) has at least 99% identity with the amino acid sequence of any of SEQ ID NO: 542-600. In some embodiments, an antibody molecule comprises a light chain variable region (VL) has 100% identity with the amino acid sequence of any of SEQ ID NO: 542-600.

(24) In one aspect, described herein is an antibody molecule capable of binding to complement component 5a receptor 1 (C5aR1), comprising a VH and/or a VL, wherein the VH comprises: an HCDR1 comprising an amino acid sequence of SEQ ID NO: 656; an HCDR2 comprising an amino acid sequence of SEQ ID NO: 776; and an HCDR3 comprising an amino acid sequence of SEQ ID NO: 896; and wherein the VL comprises: an LCDR1 comprising an amino acid sequence of SEQ ID NO: 715, an LCDR2 comprising an amino acid sequence of SEQ ID NO: 835, and an LCDR3 comprising an amino acid sequence of SEQ ID NO: 955.

(25) In one aspect, described herein is an antibody molecule capable of to complement component 5a receptor 1 (C5aR1), comprising a VH and/or a VL, wherein the VH comprises an amino acid sequence with at least 80% identity with amino acid sequence of SEQ ID NO: 536; and wherein the VL comprises an amino acid sequence with at least 70% identity with amino acid sequence of SEQ ID NO: 595.

(26) In one embodiment, the antibody molecule comprises the VH comprising the amino acid sequence of SEQ ID NO: 536 and the VL comprises the amino acid sequence of SEQ ID NO: 595.

- (27) In one aspect, the present invention provides, among other things, an antibody molecule capable of binding to complement component 5a receptor 1 (C5aR1), wherein the antibody molecule comprises a VH and/or a VL, wherein the VH comprises an HCDR1 comprising an amino acid sequence of SEQ ID NO: 1462, a HCDR2 comprising an amino acid sequence of SEQ ID NO: 1463, and a HCDR3 comprising an amino acid sequence of SEQ ID NO: 1464, a LCDR1 comprising an amino acid sequence of SEQ ID NO: 1465, a LCDR2 comprising an amino acid sequence of SEQ ID NO: 1466, and a LCDR3 comprising an amino acid sequence of SEQ ID NO: 1467.
- (28) In one aspect, described herein is an antibody molecule capable of binding to complement component 5a receptor 1 (C5aR1), comprising a VH and/or a VL, wherein the VH comprises: an HCDR1 comprising an amino acid sequence of SEQ ID NO: 603; an HCDR2 comprising an amino acid sequence of SEQ ID NO: 723; and an HCDR3 comprising an amino acid sequence of SEQ ID NO: 843; and wherein the VL comprises: an LCDR1 comprising an amino acid sequence of SEQ ID NO: 663, an LCDR2 comprising an amino acid sequence of SEQ ID NO: 783, and an LCDR3 comprising an amino acid sequence of SEQ ID NO: 903.
- (29) In one embodiment, the antibody molecule comprises the VH comprising the amino acid sequence of SEQ ID NO: 483 and the VL comprises the amino acid sequence of SEQ ID NO: 543.
- (30) In one aspect, described herein is an antibody molecule capable of binding to complement component 5a receptor 1 (C5aR1), comprising a VH and/or a VL, wherein the VH comprises: an HCDR1 comprising an amino acid sequence of SEQ ID NO: 611; an HCDR2 comprising an amino acid sequence of SEQ ID NO: 731; and an HCDR3 comprising an amino acid sequence of SEQ ID NO: 851; and wherein the VL comprises: an LCDR1 comprising an amino acid sequence of SEQ ID NO: 671, an LCDR2 comprising an amino acid sequence of SEQ ID NO: 792, and an LCDR3 comprising an amino acid sequence of SEQ ID NO: 911.
- (31) In some embodiments, the antibody molecule comprises the VH comprising the amino acid sequence of SEQ ID NO: 491 and the VL comprises the amino acid sequence of SEQ ID NO: 551.
- (32) In one aspect, described herein is an antibody molecule capable of binding to complement component 5a receptor 1 (C5aR1), wherein the antibody molecule competes with a C5aR1 antibody molecule comprising: an HCDR1 comprising an amino acid sequence of SEQ ID NO: 612, an HCDR2 comprising an amino acid sequence of SEQ ID NO: 732, and an HCDR3 comprising an amino acid sequence of SEQ ID NO: 852, an LCDR1 comprising an amino acid sequence of SEQ ID NO: 672, an LCDR2 comprising an amino acid sequence of SEQ ID NO: 792, and an LCDR3 comprising an amino acid sequence of SEQ ID NO: 912.
- (33) In one aspect, described herein is an antibody molecule capable of binding to complement component 5a receptor 1 (C5aR1), wherein the antibody molecule binds to an epitope on Site II of C5aR1, wherein the epitope on Site II comprises amino residues of SEQ ID NO: 1450.
- (34) In some embodiments, the antibody molecule binds to amino acid residues R175-G189 in SEQ ID NO: 1448.
- (35) In some embodiments, the antibody molecule binds to amino acid residues E180-P183 in SEQ ID NO: 1448.
- (36) In some embodiments, the antibody molecule binds to amino acid residues E180-P184 in SEQ ID NO: 1448.
- (37) In some embodiments, wherein the antibody molecule binds to amino acid residues E178-P183 in SEQ ID NO: 1448.
- (38) In one aspect, described herein is an antibody molecule capable of binding to complement component 5a receptor 1 (C5aR1), comprising a VH and/or a VL, wherein the VH comprises: an HCDR1 comprising an amino acid sequence of SEQ ID NO: 612; an HCDR2 comprising an amino acid sequence of SEQ ID NO: 732; and an HCDR3 comprising an amino acid sequence of SEQ ID NO: 852; and wherein the VL comprises: an LCDR1 comprising an amino acid sequence of SEQ ID NO: 672, an LCDR2 comprising an amino acid sequence of SEQ ID NO: 792, and an LCDR3

comprising an amino acid sequence of SEQ ID NO: 912.

(39) In one aspect, described herein is an antibody molecule capable of binding to complement component 5a receptor 1 (C5aR1), comprising a VH and/or a VL, wherein the VH comprises: an HCDR1 comprising amino acid sequence that differs no more than 1 or 2 amino acid residues from SEQ ID NO: 612; an HCDR2 comprising amino acid sequence that differs no more than 1 or 2 amino acid residues from SEQ ID NO: 732; and an HCDR3 comprising amino acid sequence that differs no more than 1 or 2 amino acid residues from SEQ ID NO: 852; and wherein the VL comprises: an LCDR1 comprising amino acid sequence that differs no more than 1 or 2 amino acid residues from SEQ ID NO: 672, an LCDR2 comprising amino acid sequence that differs no more than 1 or 2 amino acid residues from SEQ ID NO: 792, and an LCDR3 comprising amino acid sequence that differs no more than 1 or 2 amino acid residues from of SEQ ID NO: 912.

(40) In one aspect, described herein is an antibody molecule capable of to complement component 5a receptor 1 (C5aR1), comprising a VH and/or a VL, wherein the VH comprises an amino acid sequence with at least 80% identity with amino acid sequence of SEQ ID NO: 492; and wherein the VL comprises an amino acid sequence with at least 70% identity with amino acid sequence of SEQ ID NO: 552.

(41) In some embodiments, the antibody molecule comprises the VH comprising the amino acid sequence of SEQ ID NO: 492 and the VL comprises the amino acid sequence of SEQ ID NO: 552.

(42) In one aspect, described herein is an antibody molecule capable of binding to complement component 5a receptor 1 (C5aR1), comprising a VH and/or a VL, wherein the VH comprises: an HCDR1 comprising an amino acid sequence of SEQ ID NO: 602; an HCDR2 comprising an amino acid sequence of SEQ ID NO: 722; and an HCDR3 comprising an amino acid sequence of SEQ ID NO: 842; and wherein the VL comprises: an LCDR1 comprising an amino acid sequence of SEQ ID NO: 662, an LCDR2 comprising an amino acid sequence of SEQ ID NO: 782, and an LCDR3 comprising an amino acid sequence of SEQ ID NO: 902.

(43) In some embodiments, the antibody molecule comprises the VH comprising the amino acid sequence of SEQ ID NO: 482 and the VL comprises the amino acid sequence of SEQ ID NO: 542.

(44) In one aspect, described herein is an antibody molecule capable of binding to complement component 5a receptor 1 (C5aR1), comprising a VH and/or a VL, wherein the VH comprises: an HCDR1 comprising an amino acid sequence of SEQ ID NO: 617; an HCDR2 comprising an amino acid sequence of SEQ ID NO: 737; and an HCDR3 comprising an amino acid sequence of SEQ ID NO: 857; and wherein the VL comprises: an LCDR1 comprising an amino acid sequence of SEQ ID NO: 677, an LCDR2 comprising an amino acid sequence of SEQ ID NO: 797, and an LCDR3 comprising an amino acid sequence of SEQ ID NO: 917.

(45) In some embodiments, the antibody molecule comprises the VH comprising the amino acid sequence of SEQ ID NO: 497 and the VL comprises the amino acid sequence of SEQ ID NO: 557.

(46) In some embodiments, the antibody molecule reduces (e.g., inhibits or blocks) localized enrichment of C5a to C5aR1.

(47) In some embodiments, the antibody molecule binds to C5aR1 with a dissociation constant ($K_{sub.D}$) of less than about 100 nM. In some embodiments, the antibody molecule binds to C5aR1 with a dissociation constant ($K_{sub.D}$) of less than about 90 nM. In some embodiments, the antibody molecule binds to C5aR1 with a dissociation constant ($K_{sub.D}$) of less than about 80 nM. In some embodiments, the antibody molecule binds to C5aR1 with a dissociation constant ($K_{sub.D}$) of less than about 75 nM. In some embodiments, the antibody molecule binds to C5aR1 with a dissociation constant ($K_{sub.D}$) of less than about 70 nM. In some embodiments, the antibody molecule binds to C5aR1 with a dissociation constant ($K_{sub.D}$) of less than about 65 nM. In some embodiments, the antibody molecule binds to C5aR1 with a dissociation constant ($K_{sub.D}$) of less than about 60 nM. In some embodiments, the antibody molecule binds to C5aR1 with a dissociation constant ($K_{sub.D}$) of less than about 60 nM. In some embodiments, the antibody molecule binds to C5aR1 with a dissociation constant ($K_{sub.D}$) of less than about 55 nM.

In some embodiments, the antibody molecule binds to C5aR1 with a dissociation constant (K.sub.D) of less than about 50 nM. In some embodiments, the antibody molecule binds to C5aR1 with a dissociation constant (K.sub.D) of less than about 45 nM. In some embodiments, the antibody molecule binds to C5aR1 with a dissociation constant (K.sub.D) of less than about 40 nM. In some embodiments, the antibody molecule binds to C5aR1 with a dissociation constant (K.sub.D) of less than about 35 nM. In some embodiments, the antibody molecule binds to C5aR1 with a dissociation constant (K.sub.D) of less than about 30 nM. In some embodiments, the antibody molecule binds to C5aR1 with a dissociation constant (K.sub.D) of less than about 25 nM. In some embodiments, the antibody molecule binds to C5aR1 with a dissociation constant (K.sub.D) of less than about 20 nM. In some embodiments, the antibody molecule binds to C5aR1 with a dissociation constant (K.sub.D) of less than about 15 nM. In some embodiments, the antibody molecule binds to C5aR1 with a dissociation constant (K.sub.D) of less than about 10 nM. In some embodiments, the antibody molecule binds to C5aR1 with a dissociation constant (K.sub.D) of less than about 8 nM. In some embodiments, the antibody molecule binds to C5aR1 with a dissociation constant (K.sub.D) of less than about 5 nM. In some embodiments, the antibody molecule binds to C5aR1 with a dissociation constant (K.sub.D) of less than about 3 nM. In some embodiments, the antibody molecule binds to C5aR1 with a dissociation constant (K.sub.D) of less than about 1 nM. In some embodiments, the antibody molecule binds to C5aR1 with a dissociation constant (K.sub.D) of less than about 0.5 nM. In some embodiments, the antibody molecule binds to C5aR1 with a dissociation constant (K.sub.D) of less than about 0.1 nM. In some embodiments, the antibody molecule binds to C5aR1 with a dissociation constant (K.sub.D) of less than about 100 pM. In some embodiments, the antibody molecule binds to C5aR1 with a dissociation constant (K.sub.D) of less than about 80 pM. In some embodiments, the antibody molecule binds to C5aR1 with a dissociation constant (K.sub.D) of less than about 50 pM. In some embodiments, the antibody molecule binds to C5aR1 with a dissociation constant (K.sub.D) of less than about 25 pM. In some embodiments, the antibody molecule binds to C5aR1 with a dissociation constant (K.sub.D) of less than about 10 pM.

(48) In some embodiments, the antibody molecule binds to C5aR1 with a dissociation constant (K.sub.D) of between 1 pM and 500 nM. In some embodiments, the antibody molecule binds to C5aR1 with a dissociation constant (K.sub.D) of between 5 pM and 100 nM. In some embodiments, the antibody molecule binds to C5aR1 with a dissociation constant (K.sub.D) of between 10 pM and 50 nM. In some embodiments, the antibody molecule binds to C5aR1 with a dissociation constant (K.sub.D) of between 500 pM and 10 nM. In some embodiments, the antibody molecule binds to C5aR1 with a dissociation constant (K.sub.D) of between 1 nM and 10 nM. In some embodiments, the antibody molecule binds to C5aR1 with a dissociation constant (K.sub.D) of between 1 nM and 50 nM. In some embodiments, the antibody molecule binds to C5aR1 with a dissociation constant (K.sub.D) of between 10 nM and 50 nM.

(49) In some embodiments, the antibody molecule reduces C5a-induced chemotaxis.

(50) In some embodiments, the antibody molecule is capable of binding to neutrophils.

(51) In some embodiments, the antibody molecule does not, or does not substantially bind to C5aR2, or other GPCRs.

(52) In one aspect, described herein is an antibody molecule capable of binding to complement component 5a receptor 1 (C5aR1), wherein the antibody molecule comprises: a first immunoglobulin variable region that binds to a first epitope, and a second immunoglobulin variable region that binds to a second epitope, wherein the first epitope is on Site II of C5aR1, comprising amino acid residues of SEQ ID NO: 1450; and wherein the second epitope is on Site I of C5aR1 comprising amino acid residues of SEQ ID NO: 1449 or SEQ ID NO: 1452.

(53) In some embodiments, the multispecific antibody molecule comprises the first immunoglobulin variable region that binds to amino acid residues R175-G189 in SEQ ID NO: 1450.

(54) In some embodiments, the multispecific antibody molecule comprises the second immunoglobulin variable region that binds to amino acid residues T8-D18 in SEQ ID NO: 1449 or SEQ ID NO: 1452.

(55) In some embodiments, the multispecific antibody molecule comprises the first immunoglobulin variable region that competes with a C5aR1 antibody molecule comprising: a HCDR1 comprising an amino acid sequence of SEQ ID NO: 612, a HCDR2 comprising an amino acid sequence of SEQ ID NO: 732, and a HCDR3 comprising an amino acid sequence of SEQ ID NO: 852, a LCDR1 comprising an amino acid sequence of SEQ ID NO: 672, a LCDR2 comprising an amino acid sequence of SEQ ID NO: 792, and a LCDR3 comprising an amino acid sequence of SEQ ID NO: 912; and wherein the second immunoglobulin variable region competes with a C5aR1 antibody molecule comprising: a HCDR1 comprising an amino acid sequence of SEQ ID NO: 656, a HCDR2 comprising an amino acid sequence of SEQ ID NO: 776, and a HCDR3 comprising an amino acid sequence of SEQ ID NO: 896, a LCDR1 comprising an amino acid sequence of SEQ ID NO: 715, a LCDR2 comprising an amino acid sequence of SEQ ID NO: 835 and a LCDR3 comprising an amino acid sequence of SEQ ID NO: 955.

(56) In one aspect, the present invention provides, among other things a multispecific antibody molecule capable of binding to complement component 5a receptor 1 (C5aR1), wherein the antibody molecule comprises: a first immunoglobulin variable region that binds to a first epitope, and a second immunoglobulin variable region that binds to a second epitope, wherein the first epitope is on Site II of C5aR1; wherein the first immunoglobulin variable region comprises: a HCDR1 comprising an amino acid sequence of SEQ ID NO: 612, a HCDR2 comprising an amino acid sequence of SEQ ID NO: 732, and a HCDR3 comprising an amino acid sequence of SEQ ID NO: 852, a LCDR1 comprising an amino acid sequence of SEQ ID NO: 672, a LCDR2 comprising an amino acid sequence of SEQ ID NO: 792, and a LCDR3 comprising an amino acid sequence of SEQ ID NO: 912; wherein the second epitope is on Site I of C5aR; and wherein the second immunoglobulin variable region comprises: HCDR1 comprising an amino acid sequence of SEQ ID NO: 656, a HCDR2 comprising an amino acid sequence of SEQ ID NO: 776, and a HCDR3 comprising an amino acid sequence of SEQ ID NO: 896, a LCDR1 comprising an amino acid sequence of SEQ ID NO: 715, a LCDR2 comprising an amino acid sequence of SEQ ID NO: 835 and a LCDR3 comprising an amino acid sequence of SEQ ID NO: 955.

(57) A multispecific antibody molecule capable of binding to complement component 5a receptor 1 (C5aR1), wherein the antibody molecule comprises: a first immunoglobulin variable region that binds to a first epitope, and a second immunoglobulin variable region that binds to a second epitope, wherein the first epitope is on Site II of C5aR1; wherein the first immunoglobulin variable region comprises: a HCDR1 comprising an amino acid sequence of SEQ ID NO: 1462, a HCDR2 comprising an amino acid sequence of SEQ ID NO: 1463, and a HCDR3 comprising an amino acid sequence of SEQ ID NO: 1464, a LCDR1 comprising an amino acid sequence of SEQ ID NO: 1465, a LCDR2 comprising an amino acid sequence of SEQ ID NO: 1466, and a LCDR3 comprising an amino acid sequence of SEQ ID NO: 1467; wherein the second epitope is on Site I of C5aR; and wherein the second immunoglobulin variable region comprises: HCDR1 comprising an amino acid sequence of SEQ ID NO: 1456, a HCDR2 comprising an amino acid sequence of SEQ ID NO: 1457, and a HCDR3 comprising an amino acid sequence of SEQ ID NO: 1458, a LCDR1 comprising an amino acid sequence of SEQ ID NO: 1459, a LCDR2 comprising an amino acid sequence of SEQ ID NO: 1460, and a LCDR3 comprising an amino acid sequence of SEQ ID NO: 1461.

(58) In some embodiments, the multispecific antibody molecule is a bispecific or a biparatopic antibody.

(59) In some embodiments, the multispecific antibody is a bispecific or biparatopic antibody molecule comprises a half antibody, or fragment thereof, having binding specificity for the first epitope and a half antibody, or fragment thereof, having binding specificity for the second epitope.

(60) In some embodiments, the multispecific antibody is a bispecific or biparatopic antibody molecule comprises a scFv, or fragment thereof, having binding specificity for the first epitope.

(61) In some embodiments, the multispecific antibody is a bispecific or biparatopic antibody molecule comprises a scFv, or fragment thereof, having binding specificity for the second epitope.

(62) In some embodiments, the multispecific antibody molecule reduces C5a-induced chemotaxis.

(63) In some embodiments, the multispecific antibody molecule is capable of binding to neutrophils.

(64) In some embodiments, the multispecific antibody molecule of does not, or does not substantially bind to C5aR2, or other GPCRs.

(65) In one aspect, described herein is a pharmaceutical composition comprising the antibody molecule disclosed herein and optionally further comprising a pharmaceutically acceptable carrier or excipient.

(66) In one aspect, described herein is a combination comprising an antibody molecule of the present invention and a second therapeutic agent.

(67) In some embodiments, a combination comprises an antibody molecule capable of binding to Site I on C5aR1 and an antibody molecule capable of binding to Site II on C5aR1.

(68) In some embodiments, a combination comprises an antibody Ab 329 and an antibody Ab583.

(69) In some embodiments, the second therapeutic agent comprises a small molecule (e.g., avacopan).

(70) In some embodiments, the combination comprises a second antibody molecule.

(71) In some embodiments, the combination comprises the second antibody molecule is an antibody molecule described herein.

(72) In some embodiments, the second antibody molecule is selected from antibodies 3C5, 7F3, or 7h3.

(73) In one aspect, described herein is a nucleic acid molecule comprising a nucleotide sequence encoding the antibody molecule described herein.

(74) In one aspect, described herein is a vector comprising the nucleic acid antibody molecules described herein.

(75) In one aspect, described herein is a host cell comprising the nucleic acid molecule described herein.

(76) In one aspect, described herein is a method of producing an anti-C5aR1 antibody molecule, the method comprising culturing the host cell under conditions that allow production of an antibody molecule, thereby producing the antibody molecule.

(77) In one embodiment, the method further comprises isolating the antibody molecule.

(78) In one aspect, the method of treating a disease or disorder, the method comprising administering to a subject in need thereof an effective amount of the antibody molecule described herein, the pharmaceutical composition described herein, or the combination described herein, thereby treating the disorder.

(79) In some embodiments, the disorder is a C5aR1-associated (e.g., associated with C5aR1-activation) disorder.

(80) In some embodiments, the disorder is an autoimmune disorder.

(81) In some embodiments, the disorder is rheumatoid arthritis.

(82) In some embodiments, the disorder is a kidney disorder.

(83) In some embodiments, the disorder is ANCA-vasculitis or lupus.

(84) In some embodiments, the disorder is a cancer.

(85) In some embodiments, the method further comprises administering to the subject a second therapeutic agent.

(86) In one aspect, described herein, is a method of modulating (e.g., decreasing) a C5aR1 activity, the method comprising contacting a cell or a subject in need thereof an effective amount of the antibody molecule described herein, the pharmaceutical composition described herein, or the

combination of described, thereby modulating (e.g., decreasing) the C5aR1 activity.

(87) In some embodiments, the C5aR1 activity is modulated in vitro, ex vivo, or in vivo.

(88) In one aspect, described herein is an antibody molecule capable of binding to complement component 5a receptor 1 (C5aR1), wherein the antibody molecule binds to an epitope on Site I of C5aR1, wherein the epitope on Site I comprises amino acid residues of SEQ ID NO: 1449 or SEQ ID NO: 1452, and wherein the antibody molecule reduces C5a-induced chemotaxis.

(89) In one aspect, described herein is an antibody molecule capable of binding to complement component 5a receptor 1 (C5aR1), wherein the antibody molecule binds to an epitope on Site I of C5aR1, wherein the epitope on Site I comprises amino acid residues of SEQ ID NO: 1449 or SEQ ID NO: 1452, and wherein the antibody molecule reduces C5a-induced calcium release.

(90) In one aspect, described herein is an antibody molecule capable of binding to complement component 5a receptor 1 (C5aR1), wherein the antibody molecule binds to an epitope on Site I of C5aR1, wherein the epitope on Site I comprises amino acid residues of SEQ ID NO: 1449 or SEQ ID NO: 1452, and wherein the antibody molecule reduces C5a-induced CD11b expression.

(91) In one aspect, described herein is an antibody molecule capable of binding to complement component 5a receptor 1 (C5aR1), wherein the antibody molecule binds to an epitope on Site I of C5aR1, wherein the epitope on Site I comprises amino acid residues of SEQ ID NO: 1449 or SEQ ID NO: 1452, and wherein the antibody molecule binds C5aR1 on human neutrophils.

(92) In one aspect, described herein is an antibody molecule capable of binding to complement component 5a receptor 1 (C5aR1), wherein the antibody molecule binds to an epitope on Site I of C5aR1, wherein the epitope on Site I comprises amino acid residues of SEQ ID NO: 1449 or SEQ ID NO: 1452, and wherein the antibody molecule does not substantially bind C5aR2 or other GPCRs.

(93) In one aspect, described herein is an antibody molecule capable of binding to complement component 5a receptor 1 (C5aR1), wherein the antibody molecule binds to an epitope on Site II of C5aR1, wherein the epitope on Site II comprises amino acid residues of SEQ ID NO: 1450, and wherein the antibody molecule reduces C5a-induced chemotaxis.

(94) In one aspect, described herein is an antibody molecule capable of binding to complement component 5a receptor 1 (C5aR1), wherein the antibody molecule binds to an epitope on Site II of C5aR1, wherein the epitope on Site II comprises amino acid residues of SEQ ID NO: 1450, and wherein the antibody molecule reduces C5a-induced calcium release.

(95) In one aspect, described herein is an antibody molecule capable of binding to complement component 5a receptor 1 (C5aR1), wherein the antibody molecule binds to an epitope on Site II of C5aR1, wherein the epitope on Site II comprises amino acid residues of SEQ ID NO: 1450, and wherein the antibody molecule reduces C5a-induced CD11b expression.

(96) In one aspect, described herein is an antibody molecule capable of binding to complement component 5a receptor 1 (C5aR1), wherein the antibody molecule binds to an epitope on Site II of C5aR1, wherein the epitope on Site II comprises amino acid residues of SEQ ID NO: 1450, and wherein the antibody molecule binds C5aR1 on human neutrophils.

(97) In one aspect, described herein is an antibody molecule capable of binding to complement component 5a receptor 1 (C5aR1), wherein the antibody molecule binds to an epitope on Site II of C5aR1, wherein the epitope on Site II comprises amino acid residues of SEQ ID NO: 1450, and wherein the antibody molecule does not substantially bind C5aR2 or other GPCRs.

(98) In one aspect, described herein is an antibody molecule capable of binding to complement component 5a receptor 1 (C5aR1), wherein the antibody molecule binds to an epitope on Site I and Site II of C5aR1, wherein the epitope on Site I and Site II comprises amino acid residues of SEQ ID NO: 1449, 1452 and 1450, and wherein the antibody molecule reduces C5a-induced chemotaxis.

(99) In one aspect, described herein is an antibody molecule capable of binding to complement component 5a receptor 1 (C5aR1), wherein the antibody molecule binds to an epitope on Site I and

Site II of C5aR1, wherein the epitope on Site I and Site II comprises amino acid residues of SEQ ID NO: 1449, 1452 and SEQ ID NO: 1450, and wherein the antibody molecule reduces C5a-induced calcium release.

(100) In one aspect, described herein is an antibody molecule capable of binding to complement component 5a receptor 1 (C5aR1), wherein the antibody molecule binds to an epitope on Site I and Site II of C5aR1, wherein the epitope on Site I and Site II comprises amino acid residues of SEQ ID NO: 1449, 1452 and 1450, and wherein the antibody molecule reduces C5a-induced CD11b expression.

(101) In one aspect, described herein is an antibody molecule capable of binding to complement component 5a receptor 1 (C5aR1), wherein the antibody molecule binds to an epitope on Site I and Site II of C5aR1, wherein the epitope on Site I or Site II comprises amino acid residues of SEQ ID NO: 1449, 1452 or 1450, and wherein the antibody molecule binds C5aR1 on human neutrophils.

(102) In one aspect, described herein is an antibody molecule capable of binding to complement component 5a receptor 1 (C5aR1), wherein the antibody molecule binds to an epitope on Site I and Site II of C5aR1, wherein the epitope on Site I and Site II comprises amino acid residues of SEQ ID NO: 1449, 1452 and 1450, and wherein the antibody molecule does not substantially bind C5aR2 or other GPCRs.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a diagram showing that the combination of Site I and Site II antibodies or biparatopic antibodies that bind to Site I and Site II on C5aR1 can result in formation of large clusters, which are not formed by comparable monospecific antibodies.

(2) FIGS. 2A-2D are a series of graphs showing the functionality of exemplary anti-C5aR1 antibodies quantified in the GeneBLAzer assay (clone 66 (FIG. 2A), clone 11-v2 (FIG. 2B), clone 583 (FIG. 2C), and clone 329 (FIG. 2D)). C5a concentration was plotted on the x-axis and the percent activation of C5aR1 was plotted on the y-axis. The C5a dose response in the absence of antagonist was plotted in black and the dose response in the presence of varying concentrations of antibody was plotted in shades of gray.

(3) FIGS. 3A-3C are a series of graphs showing that exemplary Site I and Site II antibodies (clones 66 and 79, respectively) in combination (FIG. 3C) were more effective than either antibody alone (clone 66 (FIG. 3A), and clone 79 (FIG. 3B)) in the GeneBLAzer assay.

(4) FIG. 4A is a superposed model of the mAb 329 interacting with ECL2 of C5aR1.

(5) FIG. 4B, shows the residues of the LCDR1 & 3 interacting with the ECL2 of C5aR1.

(6) FIG. 4C is an exemplary cartoon diagram of crystal structure of 329-Fab-peptide complex (left panel). The VH and VL are shown in ribbon representations, and ECL2 cyclic peptide is shown in spheres. The central panel shows the primary structure of the cyclic peptide where the highlighted residues (REEYFPPKVL corresponding to residues 178-187 of SEQ ID NO: 1448) were derived from ECL2 of human C5aR1 and its 3D structure found in Fab-peptide complex. The peptide is found in a hairpin loop confirmation similar to that of found in the C5aR1 receptor molecule. The right panel shows the binding and molecular interactions of the cyclic peptide with Fab domain. The hydrogen bonds are shown in dotted line and the relevant residues are labeled. The Fab-peptide interactions are mostly with the VL domain CDR residues and E180 residue seems to make most interactions with the antibody.

(7) FIG. 5A is a diagram showing the results of alanine scanning mutations of site I interaction with mab 583. The N-terminal amino acid sequence of C5aR1 (SEQ ID NO: 1452) is listed across the top of the figure. Glycosylation sites are indicated by rectangle boxes. The only natural glycosylation site is at position N5. The other sites were introduced by mutagenesis. Mutations that

abolished clone 583 binding are circled. The sulfated tyrosines at positions 11 and 14 are marked with a star sign. FIG. 5B is a diagram showing results of alanine scanning mutations of site I interaction with mab 66. The N-terminal amino acid sequence of C5aR1 (SEQ ID NO: 1452) is listed across the top of the figure. Mutations that abolished clone 66 binding are circled. The sulfated tyrosines at positions 11 and 14 are marked with a star sign. FIG. 5C shows the results of Ala scanning mutations in ECL1 (residues 4-14 of SEQ ID NO: 1453), FIG. 5D shows the results of Ala scanning mutations in ECL3 (residues 1-9 of SEQ ID NO: 1454). FIG. 5E shows structural position of ECL1 and ECL3, with respect to ECL2. FIG. 5F shows the ala scanning mutations of C5aR1 ECL2 (residues 173-197 of SEQ ID: 1448) for clones 11, 336, and 329. Mutations that abolished binding are circled.

(8) FIG. 6 is a graph showing binding of exemplary anti-C5aR1 antibodies to human neutrophils.
(9) FIG. 7 is a graph showing antagonism of C5aR1 in the GeneBLAzer assay by exemplary anti-C5aR1 antibodies, compared to an isotype control (HYHEL-10), a no-antagonist control, several reference anti-C5aR1 antibodies (i.e., antibodies 3C5, 7F3, and 7h3), and the small molecule inhibitor avacopan.

(10) FIGS. 8A-8F are a series of graphs showing cell chemotaxis induced by various concentrations of exemplary anti-C5aR1 antibodies 11v2 (FIG. 8A), 66 (FIG. 8B), 322 (FIG. 8C), 329 (FIG. 8D), 336v2-1 (FIG. 8E), and 583 (FIG. 8F).

(11) FIGS. 9A-9D are a series of graphs showing neutrophil chemotaxis induced by various concentrations of exemplary anti-C5aR1 antibodies 11v2 (FIG. 9A), 66 (FIG. 9B), 329 (FIG. 9C), and 583 (FIG. 9D).

(12) FIGS. 10A-10I are a series of graphs showing C5aR1 inhibition of chemotaxis with different combination of clones of anti-C5aR1 antibodies—Clone 11+Clone 66 (FIG. 10A), Clone 66+Clone 329 (FIG. 10B), Clone 66+Clone 336v2 (FIG. 10C), Clone 11+Clone 583 (FIG. 10D), Clone 583+Clone 329 (FIG. 10E), Clone 583+Clone 336v2 (FIG. 10F), Clone 11+Clone 322 (FIG. 10G), Clone 322+Clone 329 (FIG. 10H), Clone 322+Clone 336v2 (FIG. 10I).

(13) FIGS. 11A-11C are series of graphs showing C5aR1 inhibition of the C5a induced calcium efflux by C5aR1 antibodies in U937 cells—Clone 11v2 (FIG. 11A), Clone 329 (FIG. 11B), and Clone 583 (FIG. 11C).

(14) FIGS. 12A-12C are series of graphs showing C5aR1 inhibition of the C5a induced calcium efflux by C5aR1 antibodies in human neutrophil cells—Clone 11v2 (FIG. 12A), Clone 329 (FIG. 12B), and Clone 583 (FIG. 12C).

(15) FIGS. 13A-13B are a series of graphs showing C5aR1 signaling inhibition in the GeneBLAzer assay by an exemplary anti-C5aR1 antibody (FIG. 13A), as well as inhibition of C5a-induced calcium flux in C5aR1-U937 cells by the same exemplary anti-C5aR1 antibody (FIG. 13B). -9.5, -9, -8.5 and -8 corresponds to 10.sup.-9.5 M, 10.sup.-9.0 M, 10.sup.-8.5 M and 10.sup.-8.0 M concentration of antibody 329.

(16) FIGS. 14A-14C are a series of graphs showing inhibition of C5aR1 signaling in the GeneBLAzer assay by antibody 329 (FIG. 14A), avacopan (FIG. 14B), or a combination of antibody 329 and avacopan (FIG. 14C).

(17) FIGS. 15A-15C are a series of graphs showing inhibition of C5aR1 signaling in the GeneBLAzer assay by antibody 329 (FIG. 15A), antibody 66 (FIG. 15B), or a combination of antibody 329 and antibody 66 (FIG. 15C).

(18) FIGS. 16A-16B are a series of graphs showing binding to cells expressing C5aR1 on their surfaces by the Site II antibody 329 either alone or in combination with a Site I antibody (either clone 66 (FIG. 16A); or clone 583 (FIG. 16B)). A HYHEL10 isotype control is included for comparison.

(19) FIGS. 17A-17C are a series of graphs showing inhibition of calcium flux in C5aR1-U937 cells by antibody 66 (FIG. 17A), antibody 329 (FIG. 17B), or a combination of antibody 66 and antibody 329 (FIG. 17C).

(20) FIGS. **18A-18C** is a series of graphs showing inhibition of chemotaxis by antibody 583 (FIG. **18A**), antibody 329 (FIG. **18B**), or a combination of antibody 583 and antibody 329 (FIG. **18C**).

(21) FIGS. **19A-19E** is a series of graphs showing of C5aR1 mediated inhibition of Ga signaling as a biparatopic antibody—Clone 329 IgG4, with 583 scFv linked to light chain (FIG. **19A**), Clone 336 IgG4, with 583 scFv linked to light chain (FIG. **19B**), Clone 11 IgG4, with 583 scFv linked to heavy chain (FIG. **19C**), Clone 329 IgG4, with 583 scFv linked to heavy chain (FIG. **19D**), Clone 336 IgG4, with 583 scFv linked to heavy chain (FIG. **19E**),

(22) FIGS. **20A-20G** is a series of graphs showing of C5aR1 mediated inhibition of C5aR1 mediated G α signaling as a biparatopic antibody, targeting both Site I and Site II, in the presence of C5-alpha—Clone 329 IgG4 with Clone 583 scFv linked to heavy chain (FIG. **20A**), Clone 329 IgG4 with Clone 583 scFv, linked to light chain (FIG. **20B**), Clone 336 IgG4 with Clone 583 scFv linked to heavy chain (FIG. **20C**), Clone 11 IgG4 with Clone 583 scFv linked to heavy chain (FIG. **20D**) and two IgG4 antibodies—Clone 329 and Clone 583 (FIG. **20E**), only IgG-clone 329 (FIG. **20F**) and only Clone 583 (FIG. **20G**).

(23) FIGS. **21A-21C** are a series of graphs showing of C5aR1 mediated inhibition of CD11b expression in C5aR1 mice (Jackson Laboratories). Bp=biparatopic antibody. FIG. **21A** shows treatment with Avacopan 5 hrs before injection of Clone 329 or Clone 583 or biparatopic antibody, blood draw at 10 min before injection of antibody, 1 min post injection, 5 min post injection and 2 hours post injection. FIG. **21B** shows treatment with Avacopan 5 hrs before injection of Clone 329 or Clone 583 or biparatopic antibody, blood draw at 5 min before injection of antibody, 1 min post injection, 5 min post injection and 2 hours post injection. FIG. **21C** shows treatment with Avacopan 1 hr before injection of Clone 329 or Clone 583 or biparatopic antibody, blood draw at 5 min before injection of antibody, 1 min post injection, 5 min post injection and 2 hours post injection.

(24) FIGS. **22A-22D** are a series of graphs showing inhibition of C5aR1 signaling in the GeneBLazer assay by the Site I-targeting antibody 3c5 (FIG. **22A**), the Site II-targeting antibody shc7 (FIG. **22B**), a combination of the two antibodies (FIG. **22C**), or a biparatopic antibody comprising one antigen-binding domain from 3c5 and one antigen-binding domain from shc7 (FIG. **22D**).

DETAILED DESCRIPTION

(25) The present invention provides, among other things, anti-C5aR1 antibodies that can effectively inhibit C5a-C5aR inhibition and block C5a signaling in the presence of high C5a concentrations. Additionally, the present invention provides methods and compositions for inhibiting C5aR1 and/or C5a signaling by targeting both Site I and Site II of C5aR1. Simultaneous targeting of Site I and Site II significantly enhances inhibitory activity.

(26) The disclosure herein is base, at least in part, on the identification of antagonistic antibody molecules that inhibit the binding and/or signaling of soluble C5a to C5aR1. Without wishing to be bound by theory, it is believed that in some embodiments, the antibody molecules described herein have one or more of the following characteristics: targeting an epitope on C5aR1 that sterically blocks C5a binding and allosterically prevents C5aR1 from adopting the signaling conformation; strong binding affinity/avidity and a slow off-rate to prevent competition and displacement by C5a; no substantial cross reactivity with the structurally homologous C5aR2 or other GPCRs; lack of Fc effector function to prevent C5aR1-expressing cells from cytotoxic effects; or favorable biophysical, pharmacokinetic, and/or biodistribution properties to allow for prolonged therapeutic effect compared to small molecule antagonists.

(27) C5a/C5aR1 Interaction

(28) The binding of C5a to C5aR1 is a terminal event in the complement pathway that can have a pleiotropic effect, with one major consequence being the migration, trafficking, and/or activation of leukocytes, including, but not limited to, neutrophils. By targeting a terminal step of C5aR1 signaling, the rest of the complement system can remain fully functional. C5aR1 blockade by a small molecule allosteric antagonist, CCX168 (avacopan), has been clinically validated and in a

Phase III clinical trial for patients with ANCA vasculitis. The antibody molecules described herein can, at least in part, reduce aberrant inflammation and/or tissue damage by inhibiting (e.g., blocking) C5aR1 signaling and subsequent chemotaxis and activation of leukocytes to the site of inflammation. An antibody therapeutic targeting C5aR1 provides an approach for specific blockade of C5a/C5aR1 signaling (e.g., blocking C5a signaling in the presence of high C5a concentrations). Additionally, the biophysical properties, biodistribution, and/or PK properties of an antibody molecule are often superior to small molecule drugs—for example, prolonged serum half-life and reduced administration frequency, high specificity and/or affinity, and/or a favorable developability profile. Furthermore, many small molecules targeting GPCRs often have promiscuity across multiple GPCRs due to the conserved canonical structure and the small number of contact residues. Antibodies can provide improved specificity by engaging with multiple contacts across a conformational landscape.

(29) Due to the biparatopic nature of C5a, which binds independently to two distinct regions of C5aR1, antagonistic mimicry by engagement of C5aR1 with a biparatopic antibody targeting binding Site I and Site II can provide a potent strategy for inhibiting C5a signaling. For example, engagement of Site I and Site II can provide the most potent antagonism by orthosterically blocking both C5a binding sites and possibly increase the avidity and/or cross-link receptors. Without wishing to be bound by theory, it is believed that in some embodiments, intramolecular or intermolecular engagement of Site I and Site II by a single biparatopic antibody can mask multiple (e.g., all) C5a binding sites, while also increasing the off-rate and residence time of the antibody molecule. The life span of a neutrophil is estimated to be less than 24 hours; therefore, it is believed that in some embodiments, an antibody molecule with a long residence time can occupy C5aR1 for the lifespan of the cell. Without wishing to be bound by theory, it is believed that in some embodiments, the antibody molecules described herein can have improved properties (e.g., improved target engagement, pharmacokinetics (PK), and efficiency) due to improved residence time compared to small molecule antagonists for improved pharmacodynamics (PD).

(30) Disclosed herein are antibody molecules that bind to C5aR1, e.g., human C5aR1 with high affinity and specificity. Advantageously, several of the antibody molecules described herein have improved ability to reduce (e.g., inhibit, block, or neutralize) one or more biological activities of C5aR1. Nucleic acid molecules encoding the antibody molecules, expression vectors, host cells, compositions (e.g., pharmaceutical compositions), kits, and methods for making the antibody molecules, are also provided. The antibody molecules and pharmaceutical compositions disclosed herein can be used (alone or in combination with other agents or therapeutic modalities) to treat, prevent and/or diagnose disorders and conditions, e.g., disorders and conditions associated with C5aR1, e.g., ANCA-vasculitis.

Definitions

(31) As used herein, the articles “a” and “an” refer to one or to more than one (e.g., to at least one) of the grammatical object of the article.

(32) The term “or” is used herein to mean, and is used interchangeably with, the term “and/or”, unless context clearly indicates otherwise.

(33) “About” and “approximately” shall generally mean an acceptable degree of error for the quantity measured given the nature or precision of the measurements. Exemplary degrees of error are within 20 percent (%), typically, within 10%, and more typically, within 5% of a given value or range of values.

(34) The compositions and methods disclosed herein encompass polypeptides and nucleic acids having the sequences specified, or sequences substantially identical or similar thereto, e.g., sequences at least 85%, 90%, 95% identical or higher to the sequence specified.

(35) In the context of an amino acid sequence, the term “substantially identical” is used herein to refer to a first amino acid that contains a sufficient or minimum number of amino acid residues that are i) identical to, or ii) conservative substitutions of aligned amino acid residues in a second amino

acid sequence such that the first and second amino acid sequences can have a common structural domain and/or common functional activity. For example, amino acid sequences that contain a common structural domain having at least about 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identity to a reference sequence, e.g., a sequence provided herein.

(36) In the context of nucleotide sequence, the term “substantially identical” is used herein to refer to a first nucleic acid sequence that contains a sufficient or minimum number of nucleotides that are identical to aligned nucleotides in a second nucleic acid sequence such that the first and second nucleotide sequences encode a polypeptide having common functional activity, or encode a common structural polypeptide domain or a common functional polypeptide activity. For example, nucleotide sequences having at least about 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identity to a reference sequence, e.g., a sequence provided herein.

(37) The term “functional variant” refers polypeptides that have a substantially identical amino acid sequence to the naturally occurring sequence, or are encoded by a substantially identical nucleotide sequence, and are capable of having one or more activities of the naturally-occurring sequence.

(38) Calculations of homology or sequence identity between sequences (the terms are used interchangeably herein) are performed as follows.

(39) To determine the percent identity of two amino acid sequences, or of two nucleic acid sequences, the sequences are aligned for optimal comparison purposes (e.g., gaps can be introduced in one or both of a first and a second amino acid or nucleic acid sequence for optimal alignment and non-homologous sequences can be disregarded for comparison purposes). In a typical embodiment, the length of a reference sequence aligned for comparison purposes is at least 30%, e.g., at least 40%, 50%, 60%, e.g., at least 70%, 80%, 90%, 100% of the length of the reference sequence. The amino acid residues or nucleotides at corresponding amino acid positions or nucleotide positions are then compared. When a position in the first sequence is occupied by the same amino acid residue or nucleotide as the corresponding position in the second sequence, then the molecules are identical at that position.

(40) The percent identity between the two sequences is a function of the number of identical positions shared by the sequences, taking into account the number of gaps, and the length of each gap, which need to be introduced for optimal alignment of the two sequences.

(41) The comparison of sequences and determination of percent identity between two sequences can be accomplished using a mathematical algorithm. In an embodiment, the percent identity between two amino acid sequences is determined using the Needleman and Wunsch ((1970) *J. Mol. Biol.* 48:444-453) algorithm which has been incorporated into the GAP program in the GCG software package, using either a Blossum 62 matrix or a PAM250 matrix, and a gap weight of 16, 14, 12, 10, 8, 6, or 4 and a length weight of 1, 2, 3, 4, 5, or 6. In certain embodiments, the percent identity between two nucleotide sequences is determined using the GAP program in the GCG software package, using a NWSgapdna.CMP matrix and a gap weight of 40, 50, 60, 70, or 80 and a length weight of 1, 2, 3, 4, 5, or 6. One suitable set of parameters (and the one that should be used unless otherwise specified) are a Blossum 62 scoring matrix with a gap penalty of 12, a gap extend penalty of 4, and a frameshift gap penalty of 5.

(42) The percent identity between two amino acid or nucleotide sequences can be determined using the algorithm of E. Meyers and W. Miller ((1989) CABIOS, 4:11-17) which has been incorporated into the ALIGN program (version 2.0), using a PAM120 weight residue table, a gap length penalty of 12 and a gap penalty of 4.

(43) The nucleic acid and protein sequences described herein can be used as a “query sequence” to perform a search against public databases to, for example, identify other family members or related sequences. Such searches can be performed using the NBLAST and XBLAST programs (version 2.0) of Altschul, et al. (1990) *J. Mol. Biol.* 215:403-10. BLAST nucleotide searches can be performed with the NBLAST program, score=100, wordlength=12 to obtain nucleotide sequences homologous to a nucleic acid as described herein. BLAST protein searches can be performed with

the XBLAST program, score=50, wordlength=3 to obtain amino acid sequences homologous to protein molecules described herein. To obtain gapped alignments for comparison purposes, Gapped BLAST can be utilized as described in Altschul et al., (1997) *Nucleic Acids Res.* 25:3389-3402. When utilizing BLAST and gapped BLAST programs, the default parameters of the respective programs (e.g., XBLAST and NBLAST) can be used. See NCBI.

(44) As used herein, the term “hybridizes under low stringency, medium stringency, high stringency, or very high stringency conditions” describes conditions for hybridization and washing. Guidance for performing hybridization reactions can be found in *Current Protocols in Molecular Biology*, John Wiley & Sons, N.Y. (1989), 6.3.1-6.3.6, which is incorporated by reference. Aqueous and nonaqueous methods are described in that reference and either can be used. Specific hybridization conditions referred to herein are as follows: 1) low stringency hybridization conditions in 6× sodium chloride/sodium citrate (SSC) at about 45° C., followed by two washes in 0.2×SSC, 0.1% SDS at least at 50° C. (the temperature of the washes can be increased to 55° C. for low stringency conditions); 2) medium stringency hybridization conditions in 6×SSC at about 45° C., followed by one or more washes in 0.2×SSC, 0.1% SDS at 60° C.; 3) high stringency hybridization conditions in 6×SSC at about 45° C., followed by one or more washes in 0.2×SSC, 0.1% SDS at 65° C.; and preferably 4) very high stringency hybridization conditions are 0.5M sodium phosphate, 7% SDS at 65° C., followed by one or more washes at 0.2×SSC, 1% SDS at 65° C. Very high stringency conditions 4) are suitable conditions and the ones that should be used unless otherwise specified.

(45) It is understood that the molecules described herein may have additional conservative or non-essential amino acid substitutions, which do not have a substantial effect on their functions.

(46) The term “amino acid” is intended to embrace all molecules, whether natural or synthetic, which include both an amino functionality and an acid functionality and capable of being included in a polymer of naturally-occurring amino acids. Exemplary amino acids include naturally-occurring amino acids; analogs, derivatives and congeners thereof; amino acid analogs having variant side chains; and all stereoisomers of any of any of the foregoing. As used herein the term “amino acid” includes both the D- or L-optical isomers and peptidomimetics.

(47) A “conservative amino acid substitution” is one in which the amino acid residue is replaced with an amino acid residue having a similar side chain. Families of amino acid residues having similar side chains have been defined in the art. These families include amino acids with basic side chains (e.g., lysine, arginine, histidine), acidic side chains (e.g., aspartic acid, glutamic acid), uncharged polar side chains (e.g., glycine, asparagine, glutamine, serine, threonine, tyrosine, cysteine), nonpolar side chains (e.g., alanine, valine, leucine, isoleucine, proline, phenylalanine, methionine, tryptophan), beta-branched side chains (e.g., threonine, valine, isoleucine) and aromatic side chains (e.g., tyrosine, phenylalanine, tryptophan, histidine).

(48) The terms “polypeptide,” “peptide” and “protein” (if single chain) are used interchangeably herein to refer to polymers of amino acids of any length. The polymer may be linear or branched, it may comprise modified amino acids, and it may be interrupted by non-amino acids. The terms also encompass an amino acid polymer that has been modified; for example, disulfide bond formation, glycosylation, lipidation, acetylation, phosphorylation, or any other manipulation, such as conjugation with a labeling component. The polypeptide can be isolated from natural sources, can be produced by recombinant techniques from a eukaryotic or prokaryotic host, or can be a product of synthetic procedures.

(49) The terms “nucleic acid,” “nucleic acid sequence,” “nucleotide sequence,” or “polynucleotide sequence,” and “polynucleotide” are used interchangeably. They refer to a polymeric form of nucleotides of any length, either deoxyribonucleotides or ribonucleotides, or analogs thereof. The polynucleotide may be either single-stranded or double-stranded, and if single-stranded may be the coding strand or non-coding (antisense) strand. A polynucleotide may comprise modified nucleotides, such as methylated nucleotides and nucleotide analogs. The sequence of nucleotides

may be interrupted by non-nucleotide components. A polynucleotide may be further modified after polymerization, such as by conjugation with a labeling component. The nucleic acid may be a recombinant polynucleotide, or a polynucleotide of genomic, cDNA, semisynthetic, or synthetic origin which either does not occur in nature or is linked to another polynucleotide in a non-natural arrangement.

(50) The term “isolated,” as used herein, refers to material that is removed from its original or native environment (e.g., the natural environment if it is naturally occurring). For example, a naturally-occurring polynucleotide or polypeptide present in a living animal is not isolated, but the same polynucleotide or polypeptide, separated by human intervention from some or all of the co-existing materials in the natural system, is isolated. Such polynucleotides could be part of a vector and/or such polynucleotides or polypeptides could be part of a composition, and still be isolated in that such vector or composition is not part of the environment in which it is found in nature.

(51) As used herein, the term “treat,” e.g., a disorder (e.g., a disorder described herein), means that a subject (e.g., a human) who has a disorder, e.g., a disorder described herein, and/or experiences a symptom of a disorder, e.g., a disorder described herein, will, in an embodiment, suffer less a severe symptom and/or recover faster when an antibody molecule is administered than if the antibody molecule were never administered. Other assays, e.g., urine tests, blood tests, iothalamate clearance tests, or imaging (e.g., ultrasound, X-rays, or cystoscopy), can also be used to monitor treatment in a patient, or to detect the presence, e.g., decreased presence (or absence), of a symptom of the disorder, after treatment of the disorder in the subject. Treatment can, e.g., partially or completely, alleviate, ameliorate, relieve, inhibit, or reduce the severity of, and/or reduce incidence, and optionally, delay onset of, one or more manifestations of the effects or symptoms, features, and/or causes of a disorder, e.g., a disorder described herein. In an embodiment, treatment is of a subject who does not exhibit certain signs of a disorder, e.g., a disorder described herein, and/or of a subject who exhibits only early signs of a disorder, e.g., a disorder described herein. In an embodiment, treatment is of a subject who exhibits one or more established signs of a disorder, e.g., a disorder described herein. In an embodiment, treatment is of a subject diagnosed as suffering from a disorder, e.g., a disorder described herein.

(52) As used herein, the term “prevent,” a disorder, e.g., a disorder described herein, means that a subject (e.g., a human) is less likely to have the disorder, e.g., a disorder described herein, if the subject receives the antibody molecule.

(53) Various aspects of the compositions and methods herein are described in further detail below. Additional definitions are set out throughout the specification.

(54) C5aR1

(55) C5aR1 (complement component 5a receptor 1), also known as complement 5a receptor 1, C5a receptor, cluster of differentiation 88 (CD88), or C5a anaphylatoxin chemotactic receptor 1, is a G protein-coupled receptor encoded by the Complement C5a Receptor 1 (C5AR1) gene. Most GPCR molecules are randomly distributed on the cell surface and most class A GPCRs are monomers. C5aR1 belongs to class A of GPCRs but is likely to be present as oligomers. The oligomeric GPCR molecules may become monomers upon their respective antagonist binding suggesting that the oligomerization is likely due to nonspecific interactions. C5aR1 is the receptor for Complement Component 5a (C5a) and plays a role in a number of biological processes, such as modulation of inflammatory responses, obesity, development, and cancer. C5aR1 is broadly expressed, for example, in granulocytes (e.g., neutrophils, eosinophils, basophils, and mast cells), and immature dendritic cells, as well as in the brain, lung, heart, kidney, liver, ovary, and testis. C5aR1 is also found on the plasma membrane. The number of C5 receptors per cell is significantly high, up to about 200,000 sites per leukocytes.

(56) C5a has a high affinity for C5aR1, with measurements ranging from low nanomolar to low picomolar. Due to low molecular weight, on rate is fast (about 4.8×10^8 M⁻¹ s⁻¹). The K_d for C5aR1 on neutrophil is typically in the low nanomolar to high picomolar range (depending

on assay format and presence of G-proteins). Engagement for only 1-2 seconds is typically needed for G-protein signaling, which supports the importance of having high residence time of a C5aR1 inhibitor (e.g., an antibody molecule described herein). GPCRs can have multiple conformational states that will vary in their level of activity (e.g., not binary). Without wishing to be bound by theory, it is believed that in some embodiments, the antibody molecules described herein can shift the energetics to favor an inactive state upon engagement.

(57) C5aR1 is a class 1 G-coupled protein receptor (GPCR), which possesses the canonical GPCR structure, including 7 transmembrane helices, 4 extracellular domains, and 4 intracellular domains. C5aR1 comprises two regions exposed to the extracellular milieu that are involved in its interaction with the C5a ligand. One region, referred to herein as “Site I,” typically comprises the N-terminal residues (e.g., N-terminal 37 residues) of C5aR1. Site I generally forms a flexible random coil structure. In an embodiment, Site I comprises or consists of the amino acid sequence of MNSFNYTTPDYGHYDDKDTLNLTPVDKTSNTRLRVPD (SEQ ID NO: 1449). In an embodiment, an antibody molecule that binds to Site I binds to an amino acid sequence comprising or consisting of SEQ ID NO: 1449, or a sequence having at least 85%, 90%, 95%, 96%, 97%, 98%, or 99% identity thereto, or a sequence comprising no more than 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10 amino acid differences therefrom. In an embodiment, Site I comprises or consists of the amino acid sequence of MDSFNYTTPDYGHYDDKDTLNLTPVDKTSNTRLRVPD (SEQ ID NO: 1452). In an embodiment, an antibody molecule that binds to Site I binds to an amino acid sequence comprising or consisting of SEQ ID NO: 1452, or a sequence having at least 85%, 90%, 95%, 96%, 97%, 98%, or 99% identity thereto, or a sequence comprising no more than 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10 amino acid differences therefrom. An antibody molecule that binds to Site I can typically bind to all residues in Site I or a subset thereof. For example, an antibody molecule that binds to Site I makes contact with one or more residues in Site I. In an embodiment, Site I generally comprises a number of sulfated residues (e.g., sulfated tyrosine residues) and a number of Asp residues. Without wishing to be bound by theory, it is believed that in some embodiments, Site I recruits C5a to the receptor by binding to C5a as a flexible extension from the rest of the C5aR1 protein, thereby enriching the local concentration of C5a around a C5aR1-expressing cell.

(58) C5aR1 comprises three extracellular loops (ECLs), referred to here as extracellular loop 1 (ECL1), extracellular loop 2 (ECL2) and extracellular loop 3 (ECL3). ECL1 comprises residues 94-110 of C5aR1 and is defined by amino acid sequence of SEQ ID NO: 1453. ECL2 comprises residues 175-200 of C5aR1 and is defined by amino acid sequence of SEQ ID NO: 1450. ECL3 comprises residues 266-282 of C5aR1 and is defined by amino acid sequence of SEQ ID NO: 1454. Certain amino acids of C5aR1 have natural variants such as N2D and N279K.

(59) A second region, referred to herein as “Site II,” may comprise the three extracellular loops (e.g., ECL1, ECL2, and ECL3) and the transmembrane residues forming vestibule of C5aR1. Site II typically comprises ECL 2 and optionally, the transmembrane residues forming vestibule of C5aR1 comprising amino acid sequences of SEQ ID NO: 1450. An antibody molecule that binds to Site II can typically bind to all residues in Site II or a subset thereof. For example, an antibody molecule that binds to Site II makes contact with one or more residues in Site II. Similarly, an antibody molecule that binds to ECL1, ECL2, and/or ECL3 makes contact with one or more residues in ECL1, ECL2, and/or ECL3. In an embodiment, an antibody molecule that binds to Site II binds to ECL2. In an embodiment, an antibody molecule that binds to Site II binds to ECL1. In an embodiment, an antibody molecule that binds to Site II binds to ECL3. In an embodiment, an antibody molecule that binds to Site II binds to ECL1 and ECL2. In an embodiment, an antibody molecule that binds to Site II binds to ECL2 and ECL3. In an embodiment, an antibody molecule that binds to Site II binds to ECL1 and ECL3. In an embodiment, an antibody molecule that binds to Site II binds to ECL1, ECL2, and ECL3. In an embodiment, an antibody molecule that binds to Site II binds to ECL2, but does not bind, or does not substantially bind, to ECL1 and/or ECL3. In an embodiment, an antibody molecule that binds to Site II binds to ECL1, but does not bind, or

does not substantially bind, to ECL2 and/or ECL3. In an embodiment, an antibody molecule that binds to Site II binds to ECL2, but does not bind, or does not substantially bind, to ECL1 and/or ECL3. In an embodiment, an antibody molecule that binds to Site II binds to ECL2 and to one or more residues in ECL1 and/or ECL3. In an embodiment, an antibody molecule that binds to Site II binds to ECL1 and ECL2, but does not bind, or does not substantially bind, to ECL3. In an embodiment, an antibody molecule that binds to Site II binds to ECL1 and ECL3, but does not bind, or does not substantially bind, to ECL2. In an embodiment, an antibody molecule that binds to Site II binds to ECL2 and ECL3, but does not bind, or does not substantially bind, to ECL1. In an embodiment, an antibody molecule that binds to Site II further binds one or more transmembrane residues (e.g., the transmembrane residues forming the vestibule region). In an embodiment, an antibody molecule that binds to Site II does not bind, or does not substantially bind, to one or more transmembrane residues (e.g., the transmembrane residues forming the vestibule region). The ECL2 generally comprises solvent-exposed Phe and Tyr residues. Without wishing to be bound by theory, it is believed that in some embodiments, C5a binding to Site II is important for activation of downstream signaling by C5aR1.

(60) The energy of agonist binding to the extracellular domain transmits an allosteric conformational change to the transmembrane and intracellular domains, allowing for G-protein binding and signaling. C5aR1 has two known agonists: C5a and C5a.sup.desArg. C5a has a short half-life in serum as the C-terminal arginine is quickly cleaved by carboxypeptidase N to form C5a.sup.desArg, which binds to C5aR1 with reduced affinity and displays biased signaling. C5a.sup.desArg, unlike C5a, does not signal the G α pathway and does not stimulate granulocyte release. However, C5a.sup.desArg does stimulate neutrophil chemotaxis, so it is of interest to also block C5a.sup.desArg binding to prevent neutrophil migration to the site of inflammation. C5a.sup.desArg signaling favors chemotaxis and is not prone to desensitization (e.g., neutrophils continue to migrate until reach high concentrations of C5a, not C5a.sup.desArg). In an embodiment, an orthosteric antagonist that blocks C5a binding (e.g., an antibody molecule as described herein) also inhibits (e.g., blocks) C5a.sup.desArg binding. Inhibition of C5a binding is, in some embodiments, needed at the inflammation site while inhibition of C5a.sup.desArg is at periphery and can prevent the migration of neutrophils to the inflammation site. Targeting C5aR1 typically leave the membrane attack complex pathway (C5b) untouched.

(61) C5a/C5aR1 signaling generally involves the following steps: receptor recognition, activation, and signaling. C5a engagement with C5aR1 involves binding to two sites (sometimes referred to herein as Site I and Site II) in a step-wise manner. C5a first engages with Site I on the N-terminal domain of C5aR1, an interaction that is largely driven by the sulfated tyrosines and acidic residues on the N-terminus with basic residues on C5a. Additional contacts are made with basic residues on Site II, e.g., located on the second extracellular loop (ECL2) of C5aR1, and acidic residues on C5a. Once fully engaged, the C-terminus of C5a promotes activation, primarily driven by the terminal arginine on the C-terminus of C5a with aspartic acid at position 282 of C5aR1. C5a binding Site II is essential for C5aR1 signaling, while Site I may not be required. Once engaged, activation is initiated by a rearrangement of the contacts made between transmembrane helices 3, 6 and 7, which drives the allosteric conformational change to the intracellular side of C5aR1 required for G-protein binding. Functional validation of the importance of Site I and Site II for C5a engagement and signaling are evident by the existence of orthosteric antagonists that target either Site I or Site II.

(62) An exemplary amino acid sequence of human C5aR1 (SEQ ID NO: 1448) is provided as follows.

(63) TABLE-US-00001 >Human NP_001727.1 C5a anaphylatoxin chemotactic receptor 1 [*Homo sapiens*] (SEQ ID NO: 1448)

MNSFNYTTPDYGHYDDKDTLNLTPVDKTSNTLRVPDILALVIFAVVF
LVGVLGNALVVWVTAFEAKRTINAIWFLNLAVADFLSCLALPILFTSI

VQHHHWPFGGAAACSIPLSLILLNMYASILLLATISADRFLLVFKPIWC
 QNFRGAGLAWIACAWWGLALLLTIPSFLYRVVREEYFPPKVLGVDY
 SHDKRRERAVAIVRLVLGFLWPLLTLTICYTFILLRTWSRRATRSTKT
 LKVVVAVVASFFIFWLPYQVTGIMMSFLEPSSPTFLLLNLKLDLSCVSF
 AYINCCINPIIYVVAGQGFQGRRLKSLPSLLRNVLTEESVVRESKSFT RSTVDTMAQKTQAV
 (64) The domains and key regions of the exemplary human C5aR1 sequence of SEQ ID NO: 1448 are depicted in the annotated sequence provided below. The N-terminal peptide (residues 1-37) is shown in bold and italic; the transmembrane helix residues (residues 38-64, 70-93, 111-132, 154-174, 201-226, 243-265 and 283-303) are underlined; ECL1 (residues 94-110), ECL2 (residues 175-200) and ECL3 (residues 266-282) are shown in bold, without italics. Certain amino acids have natural variants (N2D and N279K), shown as lowercase letters below.

(65) TABLE-US-00002

	10	20	
30	40 <i>MnSFNYTTPD</i>	<i>YGHYDDKDTL</i>	<i>DLNTPVDKTS</i>
<i>NTLRVPDILA</i>	50	60	
70	80 <u>LVIFAVVFLV</u>	<u>GVLGNALVVW</u>	<u>VTAFEAKRTI</u> <u>NAIWFLNLAV</u>
	90	100	110 120
<u>ADFLSCLALP</u>	<u>ILFTSIVQHH</u>	HWPFGGAACS	<u>ILPSLILLNM</u>
130	140	150	160 <u>YASILLLATI</u>
<u>SADRFLLVFK</u>	<u>PIWCQNFRGA</u>	<u>GLAWIACAVA</u>	170
180	190	200 <u>WGLALLLTIP</u>	<u>SFLYRVVREE</u>
YFPPKVLGCV	DYSHDKRRER	210	
220	230	240 <u>AVAIVRLVLG</u>	<u>FLWPLLTLTI</u>
<u>CYTFILLRTW</u>	<u>SRRATRSTKT</u>	250	260
270	280 <u>LKVVVAVVAS</u>	<u>FFIFWLPYQV</u>	<u>TGIMMSFLEP</u>
SSPTFLLLnK	290	300	
310	320 <u>LDSLSCVSFAY</u>	<u>INCCINPIIY</u>	<u>VVAGQGFQGR</u> <u>LRKSLPSLLR</u>
	330	340	350 NVLTEESVVR
ESKSFTRSTV	DTMAQKTQAV		

(66) As used herein, when an anti-C5aR1 antibody molecule binds, or substantially binds, to human C5aR1, it binds, or substantially binds, to one or more isoforms of human C5aR1, e.g., one or more isoforms of human C5aR1 described herein. In an embodiment, the antibody molecule binds or substantially binds to human C5aR1 having the amino acid sequence of SEQ ID NO: 1448.

(67) In an embodiment, the antibody molecule binds to Site I of C5aR1 (e.g., human C5aR1). An exemplary human C5aR1 Site I comprises or consists of the amino acid sequence of: MNSFNYYTTPDYGHYDDKDTLNLNTPVDKTSNTLRVPD (SEQ ID NO: 1449). In an embodiment, the antibody molecule binds to the amino acid sequence of SEQ ID NO: 1449, or a sequence having at least 85%, 90%, 95%, 96%, 97%, 98%, or 99% identity thereto, or a sequence comprising no more than 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10 amino acid differences therefrom. In an embodiment, the antibody molecule binds to a region comprising at least 5 (e.g., at least 5, 10, 15, 20, 25, 30, 31, 32, 33, 34, 35, 36, or 37) amino acids (e.g., consecutive amino acids) of SEQ ID NO: 1449. Another exemplary human C5aR1 Site I comprises or consists of the amino acid sequence of:

(68) TABLE-US-00003 (SEQ ID NO: 1452)
 MDSFNYYTTPDYGHYDDKDTLNLNTPVDKTSNTLRVPD.

In an embodiment, the antibody molecule binds to the amino acid sequence of SEQ ID NO: 1452, or a sequence having at least 85%, 90%, 95%, 96%, 97%, 98%, or 99% identity thereto, or a sequence comprising no more than 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10 amino acid differences therefrom. In an embodiment, the antibody molecule binds to a region comprising at least 5 (e.g., at least 5, 10, 15, 20, 25, 30, 31, 32, 33, 34, 35, 36, or 37) amino acids (e.g., consecutive amino acids) of SEQ ID NO: 1452.

(69) In an embodiment, the antibody molecule binds to ECL2 of Site II of C5aR1 (e.g., human C5aR1). An exemplary human C5aR1 ECL2 comprises or consists of: RVVREEYFPPKVLCGVDYSHDKRRER (SEQ ID NO: 1450) (core epitope region underlined). In an embodiment, the antibody molecule binds to the amino acid sequence of SEQ ID NO: 1450, or a sequence having at least 85%, 90%, 95%, 96%, 97%, 98%, or 99% identity thereto, or a sequence comprising no more than 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10 amino acid differences therefrom. In an embodiment, the antibody molecule binds to a region comprising at least 5 (e.g., at least 5, 10, 15, 20, 25, 26, or 27) amino acids (e.g., consecutive amino acids) of SEQ ID NO: 1450. In an embodiment, the antibody molecule binds to a region comprising at least 5 (e.g., at least 5, 10, 15, 16, 17, 18, or 19) amino acids (e.g., consecutive amino acids) of the underlined portion of SEQ ID NO: 1450.

(70) In an embodiment, the antibody molecule binds to ECL1 of Site II of C5aR1 (e.g., human C5aR1). An exemplary human C5aR1 ECL1 comprises or consists of: TSIVQHHHWPFGGAACS (SEQ ID NO: 1453). In an embodiment, the antibody molecule binds to the amino acid sequence of SEQ ID NO: 1453, or a sequence having at least 85%, 90%, 95%, 96%, 97%, 98%, or 99% identity thereto, or a sequence comprising no more than 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10 amino acid differences therefrom. In an embodiment, the antibody molecule binds to a region comprising at least 5 (e.g., at least 5, 10, 15, 16, or 17) amino acids (e.g., consecutive amino acids) of SEQ ID NO: 1453.

(71) In an embodiment, the antibody molecule binds to ECL3 of Site II of C5aR1 (e.g., human C5aR1). An exemplary human C5aR1 ECL3 comprises: SFLEPSSPTTELLLNKLD (SEQ ID NO: 1454). In an embodiment, the antibody molecule binds to the amino acid sequence of SEQ ID NO: 1454, or a sequence having at least 85%, 90%, 95%, 96%, 97%, 98%, or 99% identity thereto, or a sequence comprising no more than 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10 amino acid differences therefrom. In an embodiment, the antibody molecule binds to a region comprising at least 5 (e.g., at least 5, 10, 15, 16, or 17) amino acids (e.g., consecutive amino acids) of SEQ ID NO: 1454. Another exemplary human C5aR1 ECL3 comprises or consists of: SFLEPSSPTFLLKLD (SEQ ID NO: 1455). In an embodiment, the antibody molecule binds to the amino acid sequence of SEQ ID NO: 1455, or a sequence having at least 85%, 90%, 95%, 96%, 97%, 98%, or 99% identity thereto, or a sequence comprising no more than 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10 amino acid differences therefrom. In an embodiment, the antibody molecule binds to a region comprising at least 5 (e.g., at least 5, 10, 15, 16, or 17) amino acids (e.g., consecutive amino acids) of SEQ ID NO: 1455.

(72) An exemplary amino acid sequence of mouse C5aR1 (SEQ ID NO: 1451) is provided as follows.

(73) TABLE-US-00004 mouse C5aR1 (SEQ ID NO: 1451)
MDPIDNSSFEINYDHYGTMDPNIPADGIHLPKRQPGDVAALIIYSVVF
LVGVPGNALVWVTAFEARRAVNAIWFLNLAVADLLSCLALPVLFTTV
LNHNYWYFDATAACIVLPSLILLNMYASILLLATISADRFLLVFKPIWC
QKVRGTGLAWMACGVAWVLALLLTIPSFVYREAYKDFYSEHTVCGINY
GGGSFPKEKAVAILRLMVG FVLPLLTNICYTFLLLR TWSRKATRSTK
TLKVVMMAVICFFIFWLPYQVTGVMIAWLPPSSPTLKRVEKLNSLCVS
LAYINCCVNPIIYVMAGQGFGHRLRLPSIIRNALSEDSVGRDSKTF TPSTTDTSTRKSQAV

(74) As used herein, when an anti-C5aR1 antibody molecule binds, or substantially binds, to C5aR1, it binds, or substantially binds, to one or more isoforms of C5aR1, e.g., one or more isoforms of C5aR1 described herein. In an embodiment, the antibody molecule binds or substantially binds to C5aR1 having the amino acid sequence of SEQ ID NO: 1451.

(75) As used herein, when an anti-C5aR1 antibody molecule does not bind, or does not substantially bind, to C5aR1, it does not bind, or does not substantially bind, to one or more isoforms of C5aR1, e.g., one or more isoforms of C5aR1 described herein. In an embodiment, the antibody molecule does not bind, or does not substantially bind, to C5aR1 having the amino acid sequence of SEQ ID NO: 1451.

(76) Epitope

(77) The antibody molecule described herein can bind to an epitope on C5aR1 (e.g., human C5aR1). For example, an epitope bound by an antibody molecule described herein can include one or more epitope contact points.

(78) The antibody molecules described herein can bind to one or more residues on C5aR1. The amino acids bound by the antibody molecules described herein are defined as “epitope” or “epitope contact points.”

(79) In some embodiments, the antibody molecules described herein are designed to target sulfated N-terminal peptide or Site I of C5aR1, defined by SEQ ID NO: 1449 or SEQ ID NO: 1452. In some embodiments, the Site I residues targeted by the antibody molecules described herein, are sulfated. In some embodiments one or more of amino acid residues T8 (threonine 8), D10 (aspartate 10), Y11 (tyrosine 11), Y14 (tyrosine 14) and/or D15 (aspartate 15) are critical Site I epitope contact points. In some embodiments all of amino acid residues T8 (threonine 8), D10 (aspartate 10), Y11 (tyrosine 11), Y14 (tyrosine 14) and/or D15 (aspartate 15) are critical Site I epitope contact points. In some embodiments, the sulfation at Y11 and/or Y14 are critical for binding of Site I antibody molecules described herein. In some embodiments, core epitope spans 12 amino acids from T7 to D18 of SEQ ID NO: 1448, for binding of Site I antibody molecules described herein. In some embodiments, core epitope spans amino acids from T8 to D18 of SEQ ID NO: 1448, for binding of Site I antibody molecules described herein.

(80) The initial site of engagement for C5a binding; the *S. aureus* CHIPS protein binds the N-terminal peptide, which makes contact with both sulfated tyrosines, potentially blocks C5aR1 signaling by C5a; and mutation of the tyrosines at positions 11 and 14 to phenylalanine significantly reduces the functionality of C5aR1 signaling by C5a; and sulfate groups can provide strong binding energy to produce a high affinity antibody.

(81) In some embodiments, the antibody molecules described herein are designed to target Site II, defined by amino acids of SEQ ID NO: 1450. In some embodiments amino acid encompassing R175 to G189 of SEQ ID NO: 1448 are core epitopes for binding of Site II antibody molecules described herein. In some embodiments, amino acid encompassing E180-P183 of SEQ ID NO: 1448 are core epitopes for binding of Site II antibody molecules described herein. In some embodiments, amino acid encompassing E180-P184 of SEQ ID NO: 1448 are core epitopes for binding of Site II antibody molecules described herein. In some embodiments, amino acid encompassing E178-P183 of SEQ ID NO: 1448 are core epitopes for binding of Site II antibody molecules described herein. In some embodiments, one or more of the residues R35, H101, V176, V177, R178, E179, E180, Y181, F182, P183 P184, K185, L187, D191, S193, H194, E266, P267, S268, F272, L273 and/or K276 of SEQ ID NO: 1448 are important for binding of Site II antibodies described herein. In some embodiments one or more of the residues E180, Y181, F182, and/or P183 of SEQ ID NO: 1448 are critical epitopes for binding of Site II antibodies described herein. In one embodiment, the amino acid residue W102 of SEQ ID NO: 1448 is critical for binding of Site II antibodies described herein. Without wishing to be bound by theory, it is believed that in some embodiments, the antibody molecules described herein are designed to target Site II of C5aR1, at least in part, because it is an orthosteric inhibitor to prevent docking of the C5a C-terminus into the binding pocket, which is responsible for C5aR1 signaling.

(82) In an embodiment, the antibody molecules described herein are designed to target both Site I and Site II of C5aR1. In an embodiment, the antibody molecules described herein are designed to target both sulfated N-terminal peptide within Site I of C5aR1 and the ECL2 within Site II of C5aR1.

(83) In an embodiment, the antibody molecule contacts (e.g., binds, or substantially binds, to) one or more residues, or one or more regions, of Site I and/or Site II of C5aR1 (e.g., human C5aR1). In an embodiment, the antibody molecule binds to an epitope comprising ECL2 of C5aR1 or a portion thereof (e.g., a Site II epitope). In an embodiment, the antibody molecule binds to an epitope

comprising ECL1 of C5aR1 or a portion thereof (e.g., a Site II epitope). In an embodiment, the antibody molecule binds to an epitope comprising ECL3 of C5aR1 or a portion thereof (e.g., a Site II epitope). In an embodiment, the antibody molecule binds to an epitope comprising a sulfated N-terminal region (e.g., a sulfated Site I epitope). In an embodiment, the antibody molecule binds to an epitope comprising a non-sulfated N-terminal region (e.g., a non-sulfated Site I epitope).

(84) In an embodiment, the antibody molecule binds to one or more epitopes described herein. In an embodiment, the antibody molecule binds, or substantially binds, to one or more, e.g., 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, or all, residues (e.g., consecutive residues) within a region of human C5aR1 Site I sequence, e.g., comprising SEQ ID NO: 1449 or 1452. In an embodiment, the antibody molecule binds, or substantially binds, to one or more, e.g., 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, or all, residues (e.g., consecutive residues) within a region of human C5aR1 Site II sequence, e.g., comprising any of SEQ ID NOs: 1450, 1453, 1454, or 1455. In an embodiment, the antibody molecule binds, or substantially binds, to one or more, e.g., 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, or more, residues (e.g., consecutive residues) within a core epitope region of human C5aR1 Site II sequence, e.g., comprising amino acids 1-20 of SEQ ID NO: 1450. In an embodiment, the antibody molecule binds, or substantially binds, to one or more, e.g., 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, or all, residues (e.g., consecutive residues) within a region of human C5aR1 Site I sequence, e.g., comprising SEQ ID NO: 1449 or 1452; and one or more, e.g., 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, or all, residues (e.g., consecutive residues) within a region of human C5aR1 Site II sequence, e.g., comprising any of SEQ ID NOs: 1450, 1453, 1454, or 1455. In an embodiment, the antibody molecule binds, or substantially binds, to one or more, e.g., 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, or all, residues (e.g., consecutive residues) within a region of human C5aR1 Site I sequence, e.g., comprising SEQ ID NO: 1449 or 1452; and one or more, e.g., 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, or more, residues (e.g., consecutive residues) within a core epitope region of human C5aR1 Site II sequence, e.g., comprising amino acids 1-20 of SEQ ID NO: 1450.

(85) Antibody Molecules

(86) Disclosed herein are antibody molecules or antigen binding fragments that bind to C5aR1. As described above the antibodies in this disclosure inhibit binding of C5a to C5aR1, leading to inhibition of diseases associated with dysfunctional C5a/C5aR1 axis pathway, such as ANCA-associated vasculitis.

(87) As used herein, the term “antibody molecule” refers to a protein, e.g., an immunoglobulin chain or a fragment thereof, comprising at least one immunoglobulin variable domain sequence. The term “antibody molecule” includes, for example, full-length, mature antibodies and antigen-binding fragments of an antibody. For example, an antibody molecule can include a heavy (H) chain variable domain sequence (abbreviated herein as VH), and a light (L) chain variable domain sequence (abbreviated herein as VL). In another example, an antibody molecule includes two heavy (H) chain variable domain sequences and two light (L) chain variable domain sequence, thereby forming two antigen binding sites, such as Fab, Fab', F(ab')₂, Fc, Fd, Fd', Fv, single chain antibodies (scFv for example), single variable domain antibodies, diabodies (Dab) (bivalent and bispecific), and chimeric (e.g., humanized) antibodies, which may be produced by the modification of whole antibodies or those synthesized de novo using recombinant DNA technologies. These functional antibody fragments retain the ability to selectively bind with their respective antigen or receptor. Antibodies and antibody fragments can be from any class of antibodies including, but not limited to, IgG, IgA, IgM, IgD, and IgE, and from any subclass (e.g., IgG1, IgG2, IgG3, and IgG4) of antibodies. The antibody molecules can be monoclonal or polyclonal. The antibody molecule can also be a human, humanized, CDR-grafted, or in vitro generated antibody. The antibody

molecule can have a heavy chain constant region chosen from, e.g., IgG1, IgG2 (for e.g., SEQ ID NO: 1444, SEQ ID NO: 1445, SEQ ID NO: 1446, SEQ ID NO: 1447), IgG3, or IgG4. The antibody molecule can also have a light chain chosen from, e.g., kappa or lambda. The term “immunoglobulin” (Ig) is used interchangeably with the term “antibody” herein.

(88) Examples of antigen-binding fragments include: (i) a Fab fragment, a monovalent fragment consisting of the VL, VH, CL and CHI domains; (ii) a F(ab')₂ fragment, a bivalent fragment comprising two Fab fragments linked by a disulfide bridge at the hinge region; (iii) a Fd fragment consisting of the VH and CHI domains; (iv) a Fv fragment consisting of the VL and VH domains of a single arm of an antibody, (v) a diabody (dAb) fragment, which consists of a VH domain; (vi) a camelid or camelized variable domain; (vii) a single chain Fv (scFv), see e.g., Bird et al. (1988) *Science* 242:423-426; and Huston et al. (1988) *Proc. Natl. Acad. Sci. USA* 85:5879-5883; (viii) a single domain antibody. These antibody fragments may be obtained using any suitable method, including several conventional techniques known to those with skill in the art, and the fragments can be screened for utility in the same manner as are intact antibodies.

(89) The term “antibody” includes intact molecules as well as functional fragments thereof. Constant regions of the antibodies can be altered, e.g., mutated, to modify the properties of the antibody (e.g., to increase or decrease one or more of: Fc receptor binding, antibody glycosylation, the number of cysteine residues, effector cell function, or complement function).

(90) The antibody molecule can be a single chain antibody. A single-chain antibody (scFv) may be engineered (see, for example, Colcher, D. et al. (1999) *Ann N Y Acad Sci* 880:263-80; and Reiter, Y. (1996) *Clin Cancer Res* 2:245-52). The single chain antibody can be dimerized or multimerized to generate multivalent antibodies having specificities for different epitopes of the same target protein.

(91) The antibody molecules disclosed herein can also be single domain antibodies. Single domain antibodies can include antibodies whose complementary determining regions are part of a single domain polypeptide. Examples include, but are not limited to, heavy chain antibodies, antibodies naturally devoid of light chains, single domain antibodies derived from conventional 4-chain antibodies, engineered antibodies and single domain scaffolds other than those derived from antibodies. Single domain antibodies may be any of the art, or any future single domain antibodies. Single domain antibodies may be derived from any species including, but not limited to mouse, human, camel, llama, fish, shark, goat, rabbit, and bovine. According to some aspects, a single domain antibody is a naturally occurring single domain antibody known as heavy chain antibody devoid of light chains. Such single domain antibodies are disclosed in WO 94/04678, for example. For clarity reasons, this variable domain derived from a heavy chain antibody naturally devoid of light chain is known herein as a VHH or nanobody to distinguish it from the conventional VH of four chain immunoglobulins. Such a VHH molecule can be derived from antibodies raised in Camelidae species, for example in camel, llama, dromedary, alpaca and guanaco. Other species besides Camelidae may produce heavy chain antibodies naturally devoid of light chain; such VHHs are also contemplated.

(92) The VH and VL regions can be subdivided into regions of hypervariability, termed “complementarity determining regions” (CDR), interspersed with regions that are more conserved, termed “framework regions” (FR or FW). The terms “complementarity determining region,” and “CDR,” as used herein refer to the sequences of amino acids within antibody variable regions which confer antigen specificity and binding affinity. As used herein, the terms “framework,” “FW” and “FR” are used interchangeably.

(93) The extent of the framework region and CDRs has been precisely defined by a number of methods (see, Kabat, E. A., et al. (1991) *Sequences of Proteins of Immunological Interest*, Fifth Edition, U.S. Department of Health and Human Services, NIH Publication No. 91-3242; Chothia, C. et al. (1987) *J. Mol. Biol.* 196:901-917; and the AbM definition used by Oxford Molecular's AbM antibody modeling software. See, generally, e.g., *Protein Sequence and Structure Analysis of Antibody Variable Domains*. In: *Antibody Engineering Lab Manual* (Ed.: Duebel, S. and

Kontermann, R., Springer-Verlag, Heidelberg). In an embodiment, the following definitions are used: AbM definition of CDR1 of the heavy chain variable domain and Kabat definitions for the other CDRs. In an embodiment, Kabat definitions are used for all CDRs. In addition, embodiments described with respect to Kabat or AbM CDRs may also be implemented using Chothia hypervariable loops. Each VH and VL typically includes three CDRs and four FRs, arranged from amino-terminus to carboxy-terminus in the following order: FR1, CDR1, FR2, CDR2, FR3, CDR3, and FR4.

(94) As used herein, an “immunoglobulin variable domain sequence” refers to an amino acid sequence which can form the structure of an immunoglobulin variable domain. For example, the sequence may include all or part of the amino acid sequence of a naturally-occurring variable domain. For example, the sequence may or may not include one, two, or more N- or C-terminal amino acids, or may include other alterations that are compatible with formation of the protein structure.

(95) The term “antigen-binding region” refers to the part of an antibody molecule that comprises determinants that form an interface that binds to an antigen, e.g., C5aR1, or an epitope thereof. With respect to proteins (or protein mimetics), the antigen-binding region typically includes one or more loops (of at least, e.g., four amino acids or amino acid mimics) that form an interface that binds to the antigen, e.g., C5aR1. Typically, the antigen-binding region of an antibody molecule includes at least one or two CDRs and/or hypervariable loops, or more typically at least three, four, five or six CDRs and/or hypervariable loops.

(96) The terms “compete” or “cross-compete” are used interchangeably herein to refer to the ability of an antibody molecule to interfere with binding of an anti-C5aR1 antibody molecule, e.g., an anti-C5aR1 antibody molecule provided herein, to a target, e.g., C5aR1. The interference with binding can be direct or indirect (e.g., through an allosteric modulation of the antibody molecule or the target). The extent to which an antibody molecule is able to interfere with the binding of another antibody molecule to the target, and therefore whether it can be said to compete, can be determined using a competition binding assay, for example, a FACS assay, an ELISA or BIACORE assay. In an embodiment, a competition binding assay is a quantitative competition assay. In an embodiment, a first anti-C5aR1 antibody molecule is said to compete for binding to the target with a second anti-C5aR1 antibody molecule when the binding of the first antibody molecule to the target is reduced by 10% or more, e.g., 20% or more, 30% or more, 40% or more, 50% or more, 55% or more, 60% or more, 65% or more, 70% or more, 75% or more, 80% or more, 85% or more, 90% or more, 95% or more, 98% or more, 99% or more in a competition binding assay (e.g., a competition assay described herein).

(97) The terms “monoclonal antibody” or “monoclonal antibody composition” as used herein refer to a preparation of antibody molecules of single molecular composition. A monoclonal antibody composition displays a single binding specificity and affinity for a particular epitope. A monoclonal antibody can be made by hybridoma technology or by methods that do not use hybridoma technology (e.g., recombinant methods).

(98) An “effectively human” protein is a protein that does not evoke a neutralizing antibody response, e.g., the human anti-murine antibody (HAMA) response. HAMA can be problematic in a number of circumstances, e.g., if the antibody molecule is administered repeatedly, e.g., in treatment of a chronic or recurrent disease condition. A HAMA response can make repeated antibody administration potentially ineffective because of an increased antibody clearance from the serum (see, e.g., Saleh et al., *Cancer Immunol. Immunother.*, 32:180-190 (1990)) and also because of potential allergic reactions (see, e.g., LoBuglio et al., *Hybridoma*, 5:5117-5123 (1986)).

(99) The antibody molecule can be a polyclonal or a monoclonal antibody. In an embodiment, the antibody can be recombinantly produced, e.g., produced by any suitable phage display or combinatorial methods.

(100) Various phage display and combinatorial methods for generating antibodies are known in the

art (as described in, e.g., Ladner et al. U.S. Pat. No. 5,223,409; Kang et al. International Publication No. WO 92/18619; Dower et al. International Publication No. WO 91/17271; Winter et al. International Publication WO 92/20791; Markland et al. International Publication No. WO 92/15679; Breitling et al. International Publication WO 93/01288; McCafferty et al. International Publication No. WO 92/01047; Garrard et al. International Publication No. WO 92/09690; Ladner et al. International Publication No. WO 90/02809; Fuchs et al. (1991) *Bio/Technology* 9:1370-1372; Hay et al. (1992) *Hum Antibod Hybridomas* 3:81-85; Huse et al. (1989) *Science* 246:1275-1281; Griffiths et al. (1993) *EMBO J* 12:725-734; Hawkins et al. (1992) *J Mol Biol* 226:889-896; Clackson et al. (1991) *Nature* 352:624-628; Gram et al. (1992) *PNAS* 89:3576-3580; Garrad et al. (1991) *Bio/Technology* 9:1373-1377; Hoogenboom et al. (1991) *Nuc Acid Res* 19:4133-4137; and Barbas et al. (1991) *PNAS* 88:7978-7982, the contents of all of which are incorporated by reference herein).

(101) In an embodiment, the antibody molecule is a fully human antibody (e.g., an antibody made in a mouse which has been genetically engineered to produce an antibody from a human immunoglobulin sequence), or a non-human antibody, e.g., a rodent (mouse or rat), goat, primate (e.g., monkey), camel antibody. In an embodiment, the non-human antibody is a rodent (mouse or rat antibody). Methods of producing rodent antibodies are known in the art.

(102) Human monoclonal antibodies can be generated using transgenic mice carrying the human immunoglobulin genes rather than the mouse system. Splenocytes from these transgenic mice immunized with the antigen of interest are used to produce hybridomas that secrete human mAbs with specific affinities for epitopes from a human protein (see e.g., Wood et al. International Application WO 91/00906, Kucherlapati et al. PCT publication WO 91/10741; Lonberg et al. International Application WO 92/03918; Kay et al. International Application 92/03917; Lonberg, N. et al. 1994 *Nature* 368:856-859; Green, L. L. et al. 1994 *Nature Genet.* 7:13-21; Morrison, S. L. et al. 1994 *Proc. Natl. Acad. Sci. USA* 81:6851-6855; Bruggeman et al. 1993 *Year Immunol* 7:33-40; Tuailon et al. 1993 *PNAS* 90:3720-3724; Bruggeman et al. 1991 *Eur J Immunol* 21:1323-1326).

(103) An antibody can be one in which the variable region, or a portion thereof, e.g., the CDRs, are generated in a non-human organism, e.g., a rat or mouse. Chimeric, CDR-grafted, and humanized antibodies are within the invention. Antibodies generated in a non-human organism, e.g., a rat or mouse, and then modified, e.g., in the variable framework or constant region, to decrease antigenicity in a human are within the invention.

(104) Chimeric antibodies can be produced by any suitable recombinant DNA technique. Several are known in the art (see Robinson et al., International Patent Application Publication No. WO1987/002671; Akira, et al., European Patent Application Publication No. 184,187; Taniguchi, M., European Patent Application Publication No. 171,496; Morrison et al., European Patent Application Publication No. 173,494; Neuberger et al., International Patent Application Publication No. WO 86/01533; Cabilly et al. U.S. Pat. No. 4,816,567; Cabilly et al., European Patent Application Publication No. 125,023; Better et al. (1988 *Science* 240:1041-1043); Liu et al. (1987) *PNAS* 84:3439-3443; Liu et al., 1987, *J. Immunol.* 139:3521-3526; Sun et al. (1987) *PNAS* 84:214-218; Nishimura et al., 1987, *Canc. Res.* 47:999-1005; Wood et al. (1985) *Nature* 314:446-449; and Shaw et al., 1988, *J. Natl Cancer Inst.* 80:1553-1559).

(105) A humanized or CDR-grafted antibody will have at least one or two but generally all three recipient CDRs (of heavy and or light immunoglobulin chains) replaced with a donor CDR. The antibody may be replaced with at least a portion of a non-human CDR or only some of the CDRs may be replaced with non-human CDRs. It is only necessary to replace the number of CDRs required for binding of the humanized antibody to lipopolysaccharide. In an embodiment, the donor will be a rodent antibody, e.g., a rat or mouse antibody, and the recipient will be a human framework or a human consensus framework. Typically, the immunoglobulin providing the CDRs is called the "donor" and the immunoglobulin providing the framework is called the "acceptor." In

an embodiment, the donor immunoglobulin is a non-human (e.g., rodent). The acceptor framework is typically a naturally-occurring (e.g., a human) framework or a consensus framework, or a sequence about 85% or higher, e.g., 90%, 95%, 99% or higher identical thereto.

(106) As used herein, the term “consensus sequence” refers to the sequence formed from the most frequently occurring amino acids (or nucleotides) in a family of related sequences (See e.g., Winnaker, From Genes to Clones (Verlagsgesellschaft, Weinheim, Germany 1987). In a family of proteins, each position in the consensus sequence is occupied by the amino acid occurring most frequently at that position in the family. If two amino acids occur equally frequently, either can be included in the consensus sequence. A “consensus framework” refers to the framework region in the consensus immunoglobulin sequence.

(107) As described above, the antibody molecules or antigen binding fragments thereof, can bind C5aR1 at specific residues. For example, in some embodiments, the antibody molecules or antigen binding fragments thereof can bind C5aR1 at Site I (SEQ ID NO: 1449 or SEQ ID NO: 1452) and/or Site II (SEQ ID NO: 1448).

(108) In some embodiments, the C5aR1 antibodies disclosed herein show antagonistic activity against C5a-mediated C5aR1 signaling, e.g., resulting in inhibited calcium mobilization and chemotaxis of neutrophils in the presence of C5a. Antibodies against Site I and antibodies against Site II were both identified and selected for their ability to inhibit C5a binding to C5aR1. Approximately 61 C5aR1 antagonistic antibodies were evaluated and were sequenced. The sequences are summarized in Tables 1, 2 and 3.

(109) In some embodiments, the antibodies presented herein, comprise AYAMS (SEQ ID NO: 1456), SISTGGNTY (SEQ ID NO: 1457), and GYQRFSGFAY (SEQ ID NO: 1458) or a variant thereof. In some cases, SEQ ID NOs: 1456-1458 are referred to as CDRs. In some embodiments the antibody variant, comprises a sequence that differs by no more than 5, 4, 3, 2 or 1 amino acids from amino acids of SEQ ID NOs: 1456, 1457 and/or 1458. In some embodiments, the antibody comprising SEQ ID NOs: 1456, 1457 and/or 1458, bind Site I defined by SEQ ID NO: 1449 or SEQ ID NO: 1452. In some embodiments, SEQ ID NOs: 1456, 1457 and/or 1458 are part of HCDR.

(110) In some embodiments, the antibodies presented herein, comprise RSSQSLVHSNGNTYLN (SEQ ID NO: 1459), KVSNRLS (SEQ ID NO: 1460), and SQSTHVPYT (SEQ ID NO: 1461). In some cases, SEQ ID NOs: 1459-1461 are referred to as CDRs. In some embodiments the antibody variant, comprises a sequence that differs by no more than 5, 4, 3, 2 or 1 amino acids from amino acids of SEQ ID NOs: 1459, 1460 and/or 1461. In some embodiments, the antibody comprising SEQ ID NOs: 1459, 1460 and/or 1461, bind Site II defined by SEQ ID NO: 1449 or SEQ ID NO: 1452. In some embodiments, SEQ ID NOs: 1459, 1460 and/or 1461 are part of LCDR.

(111) In some embodiments, the antibodies presented herein, comprise NYWMH (SEQ ID NO: 1462), YLNPSSGYTKY (SEQ ID NO: 1463), and SGGDNYGNPYYFDR (SEQ ID NO: 1464) or a variant thereof. In some cases, SEQ ID NOs: 1462-1464 are referred to as CDRs. In some embodiments the antibody variant, comprises a sequence that differs by no more than 5, 4, 3, 2 or 1 amino acids from amino acids of SEQ ID NOs: 1462, 1463 and/or 1464. In some embodiments, the antibody comprising SEQ ID NOs: 1462, 1463 and/or 1464, bind Site II defined by SEQ ID NO: 1450 or SEQ ID NO: 1452. In some embodiments, SEQ ID NOs: 1462, 1463 and/or 1464 are part of HCDR.

(112) In some embodiments, the antibodies presented herein, comprise VHSNGNTYLH (SEQ ID NO: 1465), YLNPSSGYTKY (SEQ ID NO: 1466), and SGGDNYGNPYYFDR (SEQ ID NO: 1467) or a variant thereof. In some cases, SEQ ID NOs: 1466-1467 are referred to as CDRs. In some embodiments the antibody variant, comprises a sequence that differs by no more than 5, 4, 3, 2 or 1 amino acids from amino acids of SEQ ID NOs: 1465, 1466 and/or 1467. In some embodiments, the antibody comprising SEQ ID NOs: 1465, 1466 and/or 1467, bind Site II defined by SEQ ID NO: 1450 or SEQ ID NO: 1452. In some embodiments, SEQ ID NOs: 1465, 1466

and/or 1467 are part of LCDR.

(113) In some embodiments, the anti-C5aR1 antibody comprises a Fab, linked to a Fc domain, comprising two heavy chains and two light chains. Each heavy chain and light chain comprises a heavy chain variable region (HCVR or VH) and a light chain variable region (LCVR or VL). The heavy chain and light chain also comprise three heavy chain complementarity determining regions (HCDR1, HCDR2 and HCDR3) and three light chain complementarity determining regions (LCDR1, LCDR2 and LCDR3).

(114) In some embodiments, the antibody molecule capable of binding to complement component 5a receptor 1 (C5aR1), is Ab583, comprising a HCDR1 comprising an amino acid sequence of SEQ ID NO: 656, a HCDR2 comprising an amino acid sequence of SEQ ID NO: 776, and a HCDR3 comprising an amino acid sequence of SEQ ID NO: 896; and a LCDR1 comprising an amino acid sequence of SEQ ID NO: 715, a LCDR2 comprising an amino acid sequence of SEQ ID NO: 835, and a LCDR3 comprising an amino acid sequence of SEQ ID NO: 955 or comprises a sequence that differs by no more than 5, 4, 3, 2 or 1 amino acids from amino acids of SEQ ID Nos: 656, 776, 896, 715, 955 and/or 835. In some embodiments, the antibody molecule capable of binding to complement component 5a receptor 1 (C5aR1), is Ab583, comprising a variable heavy chain comprises an amino acid sequence of SEQ ID NO: 536; or has at least 70, 80, 90, 95, 96, 97, 98, 99, or 100% identity with, the amino acid sequence of SEQ ID NO: 536. In some embodiments, the antibody molecule capable of binding to complement component 5a receptor 1 (C5aR1), is Ab583, comprising a variable light chain comprises amino acid sequence of SEQ ID NO: 595 or has at least 70, 80, 90, 95, 96, 97, 98, 99, or 100% identity with, the amino acid sequence of SEQ ID NO: 595. In some embodiments, Ab583 can bind C5aR1 at Site I (SEQ ID NO: 1449 or SEQ ID NO: 1452).

(115) In some embodiments, the antibody molecule capable of binding to complement component 5a receptor 1 (C5aR1), is the antibody Ab66, comprising a HCDR1 comprising an amino acid sequence of SEQ ID NO: 603, a HCDR2 comprising an amino acid sequence of SEQ ID NO: 723; and a HCDR3 comprising an amino acid sequence of SEQ ID NO: 843; and a LCDR1 comprising an amino acid sequence of SEQ ID NOs: 663, a LCDR2 comprising an amino acid sequence of SEQ ID NOs: 783, and a LCDR3 comprising an amino acid sequence of SEQ ID NOs: 903; or comprises a sequence that differs by no more than 5, 4, 3, 2 or 1 amino acids from amino acids of SEQ ID Nos: 603, 723, 843, 663, 783 and/or 903. In some embodiments, the antibody molecule capable of binding to complement component 5a receptor 1 (C5aR1), is Ab66, comprising a variable heavy chain comprises amino acid sequence of SEQ ID NO: 483, or has at least 70, 80, 90, 95, 96, 97, 98, 99, or 100% identity with, the amino acid sequence of SEQ ID NO: 483. In some embodiments, the antibody molecule capable of binding to complement component 5a receptor 1 (C5aR1), is Ab66, comprising a variable light chain comprises amino acid sequence of SEQ ID NO: 543, or has at least 70, 80, 90, 95, 96, 97, 98, 99, or 100% identity with, the amino acid sequence of SEQ ID NO: 543. In some embodiments, Ab66 can bind C5aR1 at Site I (SEQ ID NO: 1449 or SEQ ID NO: 1452).

(116) In some embodiments, the antibody molecule capable of binding to complement component 5a receptor 1 (C5aR1), is the antibody Ab322, comprising a HCDR1 comprising an amino acid sequence of SEQ ID NO: 611, a HCDR2 comprising an amino acid sequence of SEQ ID NO: 731; and a HCDR3 comprising an amino acid sequence of SEQ ID NO: 851; and a LCDR1 comprising an amino acid sequence of SEQ ID NOs: 671, a LCDR2 comprising an amino acid sequence of SEQ ID NOs: 791, and a LCDR3 comprising an amino acid sequence of SEQ ID NOs: 911; or comprises a sequence that differs by no more than 5, 4, 3, 2 or 1 amino acids from amino acids of SEQ ID Nos: 611, 731, 851, 671, 791 and/or 911. In some embodiments, the antibody molecule capable of binding to complement component 5a receptor 1 (C5aR1), is Ab322, comprising a variable heavy chain comprises amino acid sequence of SEQ ID NO: 491, or has at least 70, 80, 90, 95, 96, 97, 98, 99, or 100% identity with, the amino acid sequence of SEQ ID NO: 491. In some

embodiments, the antibody molecule capable of binding to complement component 5a receptor 1 (C5aR1), is Ab322, comprising a variable light chain comprises amino acid sequence of SEQ ID NO: 551, or has at least 70, 80, 90, 95, 96, 97, 98, 99, or 100% identity with, the amino acid sequence of SEQ ID NO: 551. In some embodiments, Ab322 can bind C5aR1 at Site I (SEQ ID NO: 1449 or SEQ ID NO: 1452).

(117) In some embodiments, the antibody molecule capable of binding to complement component 5a receptor 1 (C5aR1), is the antibody Ab329, comprising a HCDR1 comprising an amino acid sequence of SEQ ID NO: 612, an HCDR2 comprising an amino acid sequence of SEQ ID NO: 732, and an HCDR3 comprising an amino acid sequence of SEQ ID NO: 852, a LCDR1 comprising an amino acid sequence of SEQ ID NO: 672, a LCDR2 comprising an amino acid sequence of SEQ ID NO: 792, and a LCDR3 comprising an amino acid sequence of SEQ ID NO: 912; or comprises a sequence that differs by no more than 5, 4, 3, 2 or 1 amino acids from amino acids of SEQ ID Nos: 612, 732, 852, 672, 792 and/or 912. In some embodiments, the antibody molecule capable of binding to complement component 5a receptor 1 (C5aR1), is Ab329, comprising a variable heavy chain comprises amino acid sequence of SEQ ID NO: 492, or has at least 70, 80, 90, 95, 96, 97, 98, 99, or 100% identity with, the amino acid sequence of SEQ ID NO: 492. In some embodiments, the antibody molecule capable of binding to complement component 5a receptor 1 (C5aR1), is Ab329, comprising a variable light chain comprises amino acid sequence of SEQ ID NO: 552, or has at least 70, 80, 90, 95, 96, 97, 98, 99, or 100% identity with, the amino acid sequence of SEQ ID NO: 552. In some embodiments, Ab329 can bind C5aR1 at Site II (SEQ ID NO: 1450).

(118) In some embodiments, the antibody molecule capable of binding to complement component 5a receptor 1 (C5aR1), is the antibody Ab336v2, comprises a HCDR1 comprising an amino acid sequence of SEQ ID NO: 617, a HCDR2 comprising an amino acid sequence of SEQ ID NO: 737 and a HCDR3 comprising an amino acid sequence of SEQ ID NO: 857; and a LCDR1 comprising an amino acid sequence of SEQ ID NO: 677, a LCDR2 comprising an amino acid sequence of SEQ ID NO: 797, and a LCDR3 comprising an amino acid sequence of SEQ ID NO: 917; or comprises a sequence that differs by no more than 5, 4, 3, 2 or 1 amino acids from amino acids of SEQ ID Nos: 617, 737, 857, 677, 797 and/or 917. In some embodiments, the antibody molecule capable of binding to complement component 5a receptor 1 (C5aR1), is Ab336v2, comprising a variable heavy chain comprises amino acid sequence of SEQ ID NO: 497, or has at least 70, 80, 90, 95, 96, 97, 98, 99, or 100% identity with, the amino acid sequence of SEQ ID NO: 497. In some embodiments, the antibody molecule capable of binding to complement component 5a receptor 1 (C5aR1), is Ab336v2, comprising a variable light chain comprises amino acid sequence of SEQ ID NO: 557, or has at least 70, 80, 90, 95, 96, 97, 98, 99, or 100% identity with, the amino acid sequence of SEQ ID NO: 557. In some embodiments, Ab336v2 can bind C5aR1 at Site II) (SEQ ID NO: 1450).

(119) In some embodiments, the antibody molecule capable of binding to complement component 5a receptor 1 (C5aR1), is the antibody Ab11v2, comprises a HCDR1 comprising an amino acid sequence of SEQ ID NO: 602, a HCDR2 comprising an amino acid sequence of SEQ ID NO: 722; and a HCDR3 comprising an amino acid sequence of SEQ ID NO: 842; and a LCDR1 comprising an amino acid sequence of SEQ ID NO: 662, a LCDR2 comprising an amino acid sequence of SEQ ID NO: 782, and a LCDR3 comprising an amino acid sequence of SEQ ID NO: 902 or comprises a sequence that differs by no more than 5, 4, 3, 2 or 1 amino acids from amino acids of SEQ ID Nos: 602, 722, 842, 662, 782 and/or 902. In some embodiments, the antibody molecule capable of binding to complement component 5a receptor 1 (C5aR1), is Ab11v2, comprising a variable heavy chain with amino acid sequence of SEQ ID NO: 482, or has at least 70, 80, 90, 95, 96, 97, 98, 99, or 100% identity with, the amino acid sequence of SEQ ID NO: 482. In some embodiments, the antibody molecule capable of binding to complement component 5a receptor 1 (C5aR1), is Ab11v2, comprising a variable light chain with amino acid sequence of SEQ ID NO: 542, or has at least 70, 80, 90, 95, 96, 97, 98, 99, or 100%

identity with, the amino acid sequence of SEQ ID NO: 542. In some embodiments, Ab11v2 can bind C5aR1 at Site II (SEQ ID NO: 1450).

(120) In some embodiments, the antibody molecule capable of binding to complement component 5a receptor 1 (C5aR1), is Ab583, comprising a HCDR1 comprising an amino acid sequence of SEQ ID NO: 176, a HCDR2 comprising an amino acid sequence of SEQ ID NO: 296, and a HCDR3 comprising an amino acid sequence of SEQ ID NO: 416; and a LCDR1 comprising an amino acid sequence of SEQ ID NO: 235, a LCDR2 comprising an amino acid sequence of SEQ ID NO: 355, and a LCDR3 comprising an amino acid sequence of SEQ ID NO: 475 or comprises a sequence that differs by no more than 5, 4, 3, 2 or 1 amino acids from amino acids of SEQ ID Nos: 176, 296, 416, 235, 355 and/or 475. In some embodiments, the antibody molecule capable of binding to complement component 5a receptor 1 (C5aR1), is Ab583, comprising a variable heavy chain comprises an amino acid sequence of SEQ ID NO: 56; or has at least 70, 80, 90, 95, 96, 97, 98, 99, or 100% identity with, the amino acid sequence of SEQ ID NO: 56. In some embodiments, the antibody molecule capable of binding to complement component 5a receptor 1 (C5aR1), is Ab583, comprising a variable light chain comprises amino acid sequence of SEQ ID NO: 115 or has at least 70, 80, 90, 95, 96, 97, 98, 99, or 100% identity with, the amino acid sequence of SEQ ID NO: 115. In some embodiments, Ab583 can bind C5aR1 at Site I (SEQ ID NO: 1449 or SEQ ID NO: 1452).

(121) In some embodiments, the antibody molecule capable of binding to complement component 5a receptor 1 (C5aR1), is the antibody Ab66, comprising a HCDR1 comprising an amino acid sequence of SEQ ID NO: 123, a HCDR2 comprising an amino acid sequence of SEQ ID NO: 243; and a HCDR3 comprising an amino acid sequence of SEQ ID NO: 363; and a LCDR1 comprising an amino acid sequence of SEQ ID NOs: 183, a LCDR2 comprising an amino acid sequence of SEQ ID NOs: 303, and a LCDR3 comprising an amino acid sequence of SEQ ID NOs: 423; or comprises a sequence that differs by no more than 5, 4, 3, 2 or 1 amino acids from amino acids of SEQ ID Nos: 123, 243, 363, 183, 303 and/or 423. In some embodiments, the antibody molecule capable of binding to complement component 5a receptor 1 (C5aR1), is Ab66, comprising a variable heavy chain comprises amino acid sequence of SEQ ID NO: 3, or has at least 70, 80, 90, 95, 96, 97, 98, 99, or 100% identity with, the amino acid sequence of SEQ ID NO: 3. In some embodiments, the antibody molecule capable of binding to complement component 5a receptor 1 (C5aR1), is Ab66, comprising a variable light chain comprises amino acid sequence of SEQ ID NO: 63, or has at least 70, 80, 90, 95, 96, 97, 98, 99, or 100% identity with, the amino acid sequence of SEQ ID NO: 63. In some embodiments, Ab66 can bind C5aR1 at Site I (SEQ ID NO: 1449 or SEQ ID NO: 1452).

(122) In some embodiments, the antibody molecule capable of binding to complement component 5a receptor 1 (C5aR1), is the antibody Ab322, comprising a HCDR1 comprising an amino acid sequence of SEQ ID NO: 131, a HCDR2 comprising an amino acid sequence of SEQ ID NO: 251; and a HCDR3 comprising an amino acid sequence of SEQ ID NO: 371; and a LCDR1 comprising an amino acid sequence of SEQ ID NOs: 191, a LCDR2 comprising an amino acid sequence of SEQ ID NOs: 311, and a LCDR3 comprising an amino acid sequence of SEQ ID NOS: 431; or comprises a sequence that differs by no more than 5, 4, 3, 2 or 1 amino acids from amino acids of SEQ ID Nos: 131, 251, 371, 191, 311 and/or 431. In some embodiments, the antibody molecule capable of binding to complement component 5a receptor 1 (C5aR1), is Ab322, comprising a variable heavy chain comprises amino acid sequence of SEQ ID NO: 11, or has at least 70, 80, 90, 95, 96, 97, 98, 99, or 100% identity with, the amino acid sequence of SEQ ID NO: 11. In some embodiments, the antibody molecule capable of binding to complement component 5a receptor 1 (C5aR1), is Ab322, comprising a variable light chain comprises amino acid sequence of SEQ ID NO: 71, or has at least 70, 80, 90, 95, 96, 97, 98, 99, or 100% identity with, the amino acid sequence of SEQ ID NO: 71. In some embodiments, Ab322 can bind C5aR1 at Site I (SEQ ID NO: 1449 or SEQ ID NO: 1452).

(123) In some embodiments, the antibody molecule capable of binding to complement component 5a receptor 1 (C5aR1), is the antibody Ab329, comprising a HCDR1 comprising an amino acid sequence of SEQ ID NO: 132, an HCDR2 comprising an amino acid sequence of SEQ ID NO: 252, and an HCDR3 comprising an amino acid sequence of SEQ ID NO: 372, a LCDR1 comprising an amino acid sequence of SEQ ID NO: 192, a LCDR2 comprising an amino acid sequence of SEQ ID NO: 312, and a LCDR3 comprising an amino acid sequence of SEQ ID NO: 432; or comprises a sequence that differs by no more than 5, 4, 3, 2 or 1 amino acids from amino acids of SEQ ID Nos: 132, 252, 372, 192, 312 and/or 432. In some embodiments, the antibody molecule capable of binding to complement component 5a receptor 1 (C5aR1), is Ab329, comprising a variable heavy chain comprises amino acid sequence of SEQ ID NO: 12, or has at least 70, 80, 90, 95, 96, 97, 98, 99, or 100% identity with, the amino acid sequence of SEQ ID NO: 12. In some embodiments, the antibody molecule capable of binding to complement component 5a receptor 1 (C5aR1), is Ab329, comprising a variable light chain comprises amino acid sequence of SEQ ID NO: 72, or has at least 70, 80, 90, 95, 96, 97, 98, 99, or 100% identity with, the amino acid sequence of SEQ ID NO: 72. In some embodiments, Ab329 can bind C5aR1 at Site II (SEQ ID NO: 1450).

(124) In some embodiments, the antibody molecule capable of binding to complement component 5a receptor 1 (C5aR1), is the antibody Ab336v2, comprises a HCDR1 comprising an amino acid sequence of SEQ ID NOs: 137, a HCDR2 comprising an amino acid sequence of SEQ ID NOs: 257 and a HCDR3 comprising an amino acid sequence of SEQ ID NOs: 377; and a LCDR1 comprising an amino acid sequence of SEQ ID NOs: 197, a LCDR2 comprising an amino acid sequence of SEQ ID NOs: 317, and a LCDR3 comprising an amino acid sequence of SEQ ID NOs: 437; or comprises a sequence that differs by no more than 5, 4, 3, 2 or 1 amino acids from amino acids of SEQ ID Nos: 137, 257, 377, 197, 317 and/or 437. In some embodiments, the antibody molecule capable of binding to complement component 5a receptor 1 (C5aR1), is Ab336v2, comprising a variable heavy chain comprises amino acid sequence of SEQ ID NO: 17, or has at least 70, 80, 90, 95, 96, 97, 98, 99, or 100% identity with, the amino acid sequence of SEQ ID NO: 17. In some embodiments, the antibody molecule capable of binding to complement component 5a receptor 1 (C5aR1), is Ab336v2, comprising a variable light chain comprises amino acid sequence of SEQ ID NO: 77, or has at least 70, 80, 90, 95, 96, 97, 98, 99, or 100% identity with, the amino acid sequence of SEQ ID NO: 77. In some embodiments, Ab336v2 can bind C5aR1 at Site II (SEQ ID NO: 1450).

(125) In some embodiments, the antibody molecule capable of binding to complement component 5a receptor 1 (C5aR1), is the antibody Ab11v2, comprises a HCDR1 comprising an amino acid sequence of SEQ ID NOs: 122, a HCDR2 comprising an amino acid sequence of SEQ ID NOs: 242; and a HCDR3 comprising an amino acid sequence of SEQ ID NOs: 362; and a LCDR1 comprising an amino acid sequence of SEQ ID NOs: 182, a LCDR2 comprising an amino acid sequence of SEQ ID NOs: 302, and a LCDR3 comprising an amino acid sequence of SEQ ID NOs: 422 or comprises a sequence that differs by no more than 5, 4, 3, 2 or 1 amino acids from amino acids of SEQ ID Nos: 122, 242, 362, 182, 302 and/or 422. In some embodiments, the antibody molecule capable of binding to complement component 5a receptor 1 (C5aR1), is Ab11v2, comprising a variable heavy chain with amino acid sequence of SEQ ID NO: 2, or has at least 70, 80, 90, 95, 96, 97, 98, 99, or 100% identity with, the amino acid sequence of SEQ ID NO: 2. In some embodiments, the antibody molecule capable of binding to complement component 5a receptor 1 (C5aR1), is Ab11v2, comprising a variable light chain with amino acid sequence of SEQ ID NO: 62, or has at least 70, 80, 90, 95, 96, 97, 98, 99, or 100% identity with, the amino acid sequence of SEQ ID NO: 62. In some embodiments, Ab11v2 can bind C5aR1 at Site II (SEQ ID NO: 1450).

(126) In some embodiments, the antibody molecule capable of binding to complement component 5a receptor 1 (C5aR1), is Ab583, comprising a HCDR1 comprising an amino acid sequence of SEQ ID NO: 1136, a HCDR2 comprising an amino acid sequence of SEQ ID NO: 1256, and a

HCDR3 comprising an amino acid sequence of SEQ ID NO: 1376; and a LCDR1 comprising an amino acid sequence of SEQ ID NO: 1195, a LCDR2 comprising an amino acid sequence of SEQ ID NO: 1315, and a LCDR3 comprising an amino acid sequence of SEQ ID NO: 1435 or comprises a sequence that differs by no more than 5, 4, 3, 2 or 1 amino acids from amino acids of SEQ ID Nos: 1136, 1256, 1376, 1195, 1315 and/or 1435. In some embodiments, the antibody molecule capable of binding to complement component 5a receptor 1 (C5aR1), is Ab583, comprising a variable heavy chain comprises an amino acid sequence of SEQ ID NO: 1016; or has at least 70, 80, 90, 95, 96, 97, 98, 99, or 100% identity with, the amino acid sequence of SEQ ID NO: 1016. In some embodiments, the antibody molecule capable of binding to complement component 5a receptor 1 (C5aR1), is Ab583, comprising a variable light chain comprises amino acid sequence of SEQ ID NO: 1075 or has at least 70, 80, 90, 95, 96, 97, 98, 99, or 100% identity with, the amino acid sequence of SEQ ID NO: 1075. In some embodiments, Ab583 can bind C5aR1 at Site I (SEQ ID NO: 1449 or SEQ ID NO: 1452).

(127) In some embodiments, the antibody molecule capable of binding to complement component 5a receptor 1 (C5aR1), is the antibody Ab66, comprising a HCDR1 comprising an amino acid sequence of SEQ ID NO: 1082, a HCDR2 comprising an amino acid sequence of SEQ ID NO: 1202; and a HCDR3 comprising an amino acid sequence of SEQ ID NO: 1322; and a LCDR1 comprising an amino acid sequence of SEQ ID NOs: 1143, a LCDR2 comprising an amino acid sequence of SEQ ID NOs: 1263, and a LCDR3 comprising an amino acid sequence of SEQ ID NOs: 1383; or comprises a sequence that differs by no more than 5, 4, 3, 2 or 1 amino acids from amino acids of SEQ ID Nos: 1082, 1202, 1322, 1143, 1263, and/or 1383. In some embodiments, the antibody molecule capable of binding to complement component 5a receptor 1 (C5aR1), is Ab66, comprising a variable heavy chain comprises amino acid sequence of SEQ ID NO: 962, or has at least 70, 80, 90, 95, 96, 97, 98, 99, or 100% identity with, the amino acid sequence of SEQ ID NO: 962. In some embodiments, the antibody molecule capable of binding to complement component 5a receptor 1 (C5aR1), is Ab66, comprising a variable light chain comprises amino acid sequence of SEQ ID NO: 1023, or has at least 70, 80, 90, 95, 96, 97, 98, 99, or 100% identity with, the amino acid sequence of SEQ ID NO: 1023. In some embodiments, Ab66 can bind C5aR1 at Site I (SEQ ID NO: 1449 or SEQ ID NO: 1452).

(128) In some embodiments, the antibody molecule capable of binding to complement component 5a receptor 1 (C5aR1), is the antibody Ab322, comprising a HCDR1 comprising an amino acid sequence of SEQ ID NO: 1091, a HCDR2 comprising an amino acid sequence of SEQ ID NO: 1211; and a HCDR3 comprising an amino acid sequence of SEQ ID NO: 1331; and a LCDR1 comprising an amino acid sequence of SEQ ID NOs: 1151, a LCDR2 comprising an amino acid sequence of SEQ ID NOs: 1271, and a LCDR3 comprising an amino acid sequence of SEQ ID NOs: 1391; or comprises a sequence that differs by no more than 5, 4, 3, 2 or 1 amino acids from amino acids of SEQ ID Nos: 1091, 1211, 1331, 1151, 1271 and/or 1391. In some embodiments, the antibody molecule capable of binding to complement component 5a receptor 1 (C5aR1), is Ab322, comprising a variable heavy chain comprises amino acid sequence of SEQ ID NO: 971, or has at least 70, 80, 90, 95, 96, 97, 98, 99, or 100% identity with, the amino acid sequence of SEQ ID NO: 971. In some embodiments, the antibody molecule capable of binding to complement component 5a receptor 1 (C5aR1), is Ab322, comprising a variable light chain comprises amino acid sequence of SEQ ID NO: 1031, or has at least 70, 80, 90, 95, 96, 97, 98, 99, or 100% identity with, the amino acid sequence of SEQ ID NO: 1031. In some embodiments, Ab322 can bind C5aR1 at Site I (SEQ ID NO: 1449 or SEQ ID NO: 1452).

(129) In some embodiments, the antibody molecule capable of binding to complement component 5a receptor 1 (C5aR1), is the antibody Ab329, comprising a HCDR1 comprising an amino acid sequence of SEQ ID NO: 1092, an HCDR2 comprising an amino acid sequence of SEQ ID NO: 1212, and an HCDR3 comprising an amino acid sequence of SEQ ID NO: 1332, a LCDR1 comprising an amino acid sequence of SEQ ID NO: 1152, a LCDR2 comprising an amino acid

sequence of SEQ ID NO: 1272, and a LCDR3 comprising an amino acid sequence of SEQ ID NO: 1392; or comprises a sequence that differs by no more than 5, 4, 3, 2 or 1 amino acids from amino acids of SEQ ID Nos: 1092, 1212, 1332, 1152, 1272 and/or 1392. In some embodiments, the antibody molecule capable of binding to complement component 5a receptor 1 (C5aR1), is Ab329, comprising a variable heavy chain comprises amino acid sequence of SEQ ID NO: 972, or has at least 70, 80, 90, 95, 96, 97, 98, 99, or 100% identity with, the amino acid sequence of SEQ ID NO: 972. In some embodiments, the antibody molecule capable of binding to complement component 5a receptor 1 (C5aR1), is Ab329, comprising a variable light chain comprises amino acid sequence of SEQ ID NO: 1032, or has at least 70, 80, 90, 95, 96, 97, 98, 99, or 100% identity with, the amino acid sequence of SEQ ID NO: 1032. In some embodiments, Ab329 can bind C5aR1 at Site II (SEQ ID NO: 1450).

(130) In some embodiments, the antibody molecule capable of binding to complement component 5a receptor 1 (C5aR1), is the antibody Ab336v2, comprises a HCDR1 comprising an amino acid sequence of SEQ ID NOs: 1097, a HCDR2 comprising an amino acid sequence of SEQ ID NOs: 1217 and a HCDR3 comprising an amino acid sequence of SEQ ID NOs: 1337; and a LCDR1 comprising an amino acid sequence of SEQ ID NOs: 1157, a LCDR2 comprising an amino acid sequence of SEQ ID NOs: 1277, and a LCDR3 comprising an amino acid sequence of SEQ ID NOs: 1397; or comprises a sequence that differs by no more than 5, 4, 3, 2 or 1 amino acids from amino acids of SEQ ID Nos: 1097, 1217, 1337, 1157, 1277 and/or 1397. In some embodiments, the antibody molecule capable of binding to complement component 5a receptor 1 (C5aR1), is Ab336v2, comprising a variable heavy chain comprises amino acid sequence of SEQ ID NO: 977, or has at least 70, 80, 90, 95, 96, 97, 98, 99, or 100% identity with, the amino acid sequence of SEQ ID NO: 977. In some embodiments, the antibody molecule capable of binding to complement component 5a receptor 1 (C5aR1), is Ab336v2, comprising a variable light chain comprises amino acid sequence of SEQ ID NO: 1037, or has at least 70, 80, 90, 95, 96, 97, 98, 99, or 100% identity with, the amino acid sequence of SEQ ID NO: 1037. In some embodiments, Ab336v2 can bind C5aR1 at Site II (SEQ ID NO: 1450).

(131) In some embodiments, the antibody molecule capable of binding to complement component 5a receptor 1 (C5aR1), is the antibody Ab11v2, comprises a HCDR1 comprising an amino acid sequence of SEQ ID NOs: SEQ ID NOs: 1082, a HCDR2 comprising an amino acid sequence of SEQ ID NOs: 1202; and a HCDR3 comprising an amino acid sequence of SEQ ID NOs: 1322; and a LCDR1 comprising an amino acid sequence of SEQ ID NOs: 1142, a LCDR2 comprising an amino acid sequence of SEQ ID NOs: 1262, and a LCDR3 comprising an amino acid sequence of SEQ ID NOs: 1382 or comprises a sequence that differs by no more than 5, 4, 3, 2 or 1 amino acids from amino acids of SEQ ID Nos: 1082, 1202, 1322, 1142, 1262 and/or 1382. In some embodiments, the antibody molecule capable of binding to complement component 5a receptor 1 (C5aR1), is Ab11v2, comprising a variable heavy chain with amino acid sequence of SEQ ID NO: 962, or has at least 70, 80, 90, 95, 96, 97, 98, 99, or 100% identity with, the amino acid sequence of SEQ ID NO: 962. In some embodiments, the antibody molecule capable of binding to complement component 5a receptor 1 (C5aR1), is Ab11v2, comprising a variable light chain with comprises amino acid sequence of SEQ ID NO: 1022, or has at least 70, 80, 90, 95, 96, 97, 98, 99, or 100% identity with, the amino acid sequence of SEQ ID NO: 1022. In some embodiments, Ab11v2 can bind C5aR1 at Site II (SEQ ID NO: 1450).

(132) In an embodiment, the antibody molecule comprises one, two, or three CDRs of the VH described herein, e.g., in Table 1A, 2A, or 3A (e.g., any of VH-11, VH-11v2, VH-66, VH-79, VH-184, VH-216, VH-272, VH-308, VH-317, VH-317v2, VH-322, VH-329, VH-330, VH-332, VH-335, VH-336v1, VH-336v2, VH-336v3, VH-338, VH-341v1, VH-341v2, VH-343, VH-399, VH-402, VH-416, VH-429, VH-430, VH-440, VH-453, VH-454, VH-465, VH-475, VH-481, VH-497, VH-502, VH-503, VH-504, VH-507, VH-508, VH-510, VH-511, VH-521, VH-530, VH-536, VH-541, VH-547, VH-549, VH-550, VH-553, VH-556, VH-557, VH-567, VH-568, VH-570, VH-573,

VH-583, VH-584, VH-585, VH-586, VH-588, or VH-592), using the IMGT, Kabat, or Chothia definitions of CDRs.

(133) In an embodiment, the antibody molecule comprises one, two, or three CDRs of the VL described herein, e.g., in Table 1B, 2B, or 3B (e.g., any of VK-11, VK-66, VK-79, VK-184, VK-184_C_Y, VK-216_272, VK-308, VK-317, VK-317v2, VK-322, VK-329, VK-330, VK-332, VK-335, VK-336v1, VK-336v2, VK-338v1, VK-338v2, VK-341v1, VK-341v2, VK-343, VK-399, VK-402, VK-416, VK-429, VK-430, VK-440, VK-453, VK-454, VK-465, VK-475, VK-481, VK-497, VK-502, VK-503, VK-504, VK-507, VK-508, VK-510, VK-511, VK-518, VK-528, VK-521, VK-530, VK-541, VK-547, VK-549, VK-550, VK-553, VK-556, VK-557, VK-567, VK-568, VK-583, VK-584, VK-585, VK-586, VK-588, VK-592), using the IMGT, Kabat, or Chothia definitions of CDRs.

(134) In an embodiment, the antibody molecule comprises one, two, or three CDRs of the VH of an antibody molecule described herein, e.g., in Table 1A, 2A, or 3A (e.g., any of antibodies 11v2, 322, 329, 330, 336v2-1, 402, 429, 430, 440, 453, 454, 465, 475, 507, 510, 511, 518, 528, 541, 547, 549, 550, 553, 556, 567, 568, 570, 573, 583, 584, 585, 586, 588, or 592, e.g., as listed in Table 1C), using the IMGT, Kabat, or Chothia definitions of CDRs.

(135) In an embodiment, the antibody molecule comprises one, two, or three CDRs of the VL region of an antibody molecule described herein, e.g., in Table 1B, 2B, or 3B (e.g., any of antibodies 11v2, 322, 329, 330, 336v2-1, 402, 429, 430, 440, 453, 454, 465, 475, 507, 510, 511, 518, 528, 541, 547, 549, 550, 553, 556, 567, 568, 570, 573, 583, 584, 585, 586, 588, or 592 (e.g., as listed in Table 1C))), using the IMGT, Kabat, or Chothia definitions of CDRs.

(136) In an embodiment, the antibody molecule is capable of binding, or substantially binding, to human C5aR1. In an embodiment, the antibody molecule binds to C5aR1 with high affinity, e.g., with a dissociation constant ($K_{sub.D}$) of less than about 100 nM, typically about 10 nM, and more typically, about $10^{sup.-0.001}$ nM, about $10^{sup.-0.01}$ nM, about $10^{sup.-0.01}$ nM, about 5-0.01 nM, about 3-0.05 nM, about 1-0.1 nM, or stronger, e.g., less than about 80, 70, 60, 50, 40, 30, 20, 10, 8, 6, 4, 3, 2, 1, 0.5, 0.2, 0.1, 0.05, 0.01, 0.005, or 0.001 nM. In an embodiment, the antibody molecule binds to C5aR1 with a $K_{sub.off}$ slower than $1 \times 10^{sup.-4}$, $5 \times 10^{sup.-5}$, or $1 \times 10^{sup.-5}$ s.^{sup.-1}. In an embodiment, the antibody molecule binds to C5aR1 with a $K_{sub.on}$ faster than $1 \times 10^{sup.4}$, $5 \times 10^{sup.4}$, $1 \times 10^{sup.5}$, or $5 \times 10^{sup.5}$ M.^{sup.-1} s.^{sup.-1}.

(137) In an embodiment, the antibody molecule is capable of binding, or substantially binding, to human C5aR1. In an embodiment, the antibody molecule binds to C5aR1 with high affinity, e.g., with a dissociation constant ($K_{sub.D}$) of less than about 100 nM, typically about 10 nM, and more typically, about $10^{sup.-0.001}$ nM, about $10^{sup.-0.01}$ nM, about $10^{sup.-0.01}$ nM, about 5-0.01 nM, about 3-0.05 nM, about 1-0.1 nM, or stronger, e.g., less than about 80, 70, 60, 50, 40, 30, 20, 10, 8, 6, 4, 3, 2, 1, 0.5, 0.2, 0.1, 0.05, 0.01, 0.005, or 0.001 nM. In an embodiment, the antibody molecule binds to C5aR1 with a $K_{sub.off}$ slower than $1 \times 10^{sup.4}$, $5 \times 10^{sup.5}$, or $1 \times 10^{sup.5}$ s.^{sup.-1}. In an embodiment, the antibody molecule binds to C5aR1 with a $K_{sub.on}$ faster than $1 \times 10^{sup.4}$, $5 \times 10^{sup.4}$, $1 \times 10^{sup.5}$, or $5 \times 10^{sup.5}$ M.^{sup.-1} s.^{sup.-1}.

(138) In an embodiment, the antibody molecule inhibits binding of human C5aR1 to human C5a by 50% or more, e.g., 60% or more, 70% or more, 80% or more, 85% or more, 90% or more, 95% or more, 96% or more, 97% or more, 98% or more, 99% or more, or 100%, as determined by a method described herein (e.g., normalized to the no antibody control).

(139) An antibody can be humanized by any suitable method, and several such methods known in the art (see e.g., Morrison, S. L., 1985, *Science* 229:1202-1207, by Oi et al., 1986, *BioTechniques* 4:214, and by Queen et al. U.S. Pat. Nos. 5,585,089, 5,693,761 and 5,693,762, the contents of all of which are hereby incorporated by reference).

(140) Humanized or CDR-grafted antibodies can be produced by CDR-grafting or CDR substitution, wherein one, two, or all CDRs of an immunoglobulin chain can be replaced. See e.g., U.S. Pat. No. 5,225,539; Jones et al. 1986 *Nature* 321:552-525; Verhoeyan et al. 1988 *Science*

239:1534; Beidler et al. 1988 *J. Immunol.* 141:4053-4060; Winter U.S. Pat. No. 5,225,539, the contents of all of which are hereby expressly incorporated by reference. Winter describes a CDR-grafting method which may be used to prepare humanized antibodies (UK Patent Application GB 2188638A, filed on Mar. 26, 1987; Winter U.S. Pat. No. 5,225,539), the contents of which is expressly incorporated by reference.

(141) Also provided are humanized antibodies in which specific amino acids have been substituted, deleted or added. Criteria for selecting amino acids from the donor are described in, e.g., U.S. Pat. No. 5,585,089, e.g., columns 12-16 of U.S. Pat. No. 5,585,089, the contents of which are hereby incorporated by reference. Other techniques for humanizing antibodies are described in Padlan et al. EP 519596 A1, published on Dec. 23, 1992.

(142) In an embodiment, the antibody molecule has a heavy chain constant region chosen from, e.g., the heavy chain constant regions of IgG1, IgG2 (e.g., IgG2a), IgG3, IgG4, IgM, IgA1, IgA2, IgD, and IgE; particularly, chosen from, e.g., the (e.g., human) heavy chain constant regions of IgG1, IgG2, IgG3, and IgG4. In another embodiment, the antibody molecule has a light chain constant region chosen from, e.g., the (e.g., human) light chain constant regions of kappa or lambda. The constant region can be altered, e.g., mutated, to modify the properties of the antibody molecule (e.g., to increase or decrease one or more of: Fc receptor binding, antibody glycosylation, the number of cysteine residues, effector cell function, and/or complement function). In an embodiment, the antibody molecule has effector function and can fix complement. In another embodiment, the antibody molecule does not recruit effector cells or fix complement. In certain embodiments, the antibody molecule has reduced or no ability to bind an Fc receptor. For example, it may be an isotype or subtype, fragment or other mutant, which does not support binding to an Fc receptor, e.g., it has a mutagenized or deleted Fc receptor binding region.

(143) In an embodiment, a constant region of the antibody molecule is altered. Methods for altering an antibody constant region are known in the art. Antibody molecules with altered function, e.g. altered affinity for an effector ligand, such as FcR on a cell, or the C1 component of complement can be produced by replacing at least one amino acid residue in the constant portion of the antibody with a different residue (see e.g., EP 388,151 A1, U.S. Pat. Nos. 5,624,821 and 5,648,260, the contents of all of which are hereby incorporated by reference). Amino acid mutations which stabilize antibody structure, such as S228P (EU nomenclature, S241P in Kabat nomenclature) in human IgG4 are also contemplated. Similar type of alterations could be described which if applied to the murine, or other species immunoglobulin would reduce or eliminate these functions.

(144) In an embodiment, the only amino acids in the antibody molecule are canonical amino acids. In an embodiment, the antibody molecule comprises naturally-occurring amino acids; analogs, derivatives and congeners thereof; amino acid analogs having variant side chains; and/or all stereoisomers of any of any of the foregoing. The antibody molecule may comprise the D- or L-optical isomers of amino acids and peptidomimetics.

(145) A polypeptide of an antibody molecule described herein may be linear or branched, it may comprise modified amino acids, and it may be interrupted by non-amino acids. The antibody molecule may also be modified; for example, by disulfide bond formation, glycosylation, lipidation, acetylation, phosphorylation, or any other manipulation, such as conjugation with a labeling component. The polypeptide can be isolated from natural sources, can be produced by recombinant techniques from a eukaryotic or prokaryotic host, or can be a product of synthetic procedures.

(146) The antibody molecule described herein can be used alone in unconjugated form, or can be bound to a substance, e.g., a toxin or moiety (e.g., a therapeutic drug; a compound emitting radiation; molecules of plant, fungal, or bacterial origin; or a biological protein (e.g., a protein toxin) or particle (e.g., a recombinant viral particle, e.g., via a viral coat protein). For example, the anti-C5aR1 antibody can be coupled to a radioactive isotope such as an α -, β -, or γ -emitter, or a β - and γ -emitter.

(147) An antibody molecule can be derivatized or linked to another functional molecule (e.g., another peptide or protein). As used herein, a “derivatized” antibody molecule is one that has been modified. Methods of derivatization include but are not limited to the addition of a fluorescent moiety, a radionucleotide, a toxin, an enzyme or an affinity ligand such as biotin. Accordingly, the antibody molecules are intended to include derivatized and otherwise modified forms of the antibodies described herein, including immunoadhesion molecules. For example, an antibody molecule can be functionally linked (by chemical coupling, genetic fusion, noncovalent association or otherwise) to one or more other molecular entities, such as another antibody (e.g., a bispecific antibody or a diabody), a detectable agent, a toxin, a pharmaceutical agent, and/or a protein or peptide that can mediate association of the antibody or antibody portion with another molecule (such as a streptavidin core region or a polyhistidine tag).

(148) Some types of derivatized antibody molecule are produced by crosslinking two or more antibodies (of the same type or of different types, e.g., to create bispecific antibodies). Suitable crosslinkers include those that are heterobifunctional, having two distinctly reactive groups separated by an appropriate spacer (e.g., m-maleimidobenzoyl-N-hydroxysuccinimide ester) or homobifunctional (e.g., disuccinimidyl suberate). Such linkers are available from Pierce Chemical Company, Rockford, Ill.

(149) Useful detectable agents with which an anti-dengue antibody molecule may be derivatized (or labeled) to include fluorescent compounds, various enzymes, prosthetic groups, luminescent materials, bioluminescent materials, fluorescent emitting metal atoms, e.g., europium (Eu), and other anthanides, and radioactive materials (described below). Exemplary fluorescent detectable agents include fluorescein, fluorescein isothiocyanate, rhodamine, 5dimethylamine-1-naphthalenesulfonyl chloride, phycoerythrin and the like. An antibody may also be derivatized with detectable enzymes, such as alkaline phosphatase, horseradish peroxidase, β -galactosidase, acetylcholinesterase, glucose oxidase and the like. When an antibody is derivatized with a detectable enzyme, it is detected by adding additional reagents that the enzyme uses to produce a detectable reaction product. For example, when the detectable agent horseradish peroxidase is present, the addition of hydrogen peroxide and diaminobenzidine leads to a colored reaction product, which is detectable. An antibody molecule may also be derivatized with a prosthetic group (e.g., streptavidin/biotin and avidin/biotin). For example, an antibody may be derivatized with biotin, and detected through indirect measurement of avidin or streptavidin binding. Examples of suitable fluorescent materials include umbelliferone, fluorescein, fluorescein isothiocyanate, rhodamine, dichlorotriazinylamine fluorescein, dansyl chloride or phycoerythrin; an example of a luminescent material includes luminol; and examples of bioluminescent materials include luciferase, luciferin, and aequorin.

(150) Labeled antibody molecules can be used, for example, diagnostically and/or experimentally in a number of contexts, including (i) to isolate a predetermined antigen by standard techniques, such as affinity chromatography or immunoprecipitation; (ii) to detect a predetermined antigen (e.g., in a cellular lysate or cell supernatant) in order to evaluate the abundance and pattern of expression of the protein; (iii) to monitor protein levels in tissue as part of a clinical testing procedure, e.g., to determine the efficacy of a given treatment regimen.

(151) An antibody molecule may be conjugated to another molecular entity, typically a label or a therapeutic (e.g., antimicrobial (e.g., antibacterial or bactericidal), immunomodulatory, immunostimulatory, cytotoxic, or cytostatic) agent or moiety. Radioactive isotopes can be used in diagnostic or therapeutic applications. Radioactive isotopes that can be coupled to the antibody molecules include, but are not limited to α -, β -, or γ -emitters, or β - and γ -emitters. Such radioactive isotopes include, but are not limited to iodine (^{131}I or ^{125}I), yttrium (^{90}Y), lutetium (^{177}Lu), actinium (^{225}Ac), praseodymium, astatine (^{211}At), rhenium (^{186}Re), bismuth (^{212}Bi or ^{213}Bi), indium (^{111}In), technetium ($^{99\text{m}}\text{Tc}$), phosphorus (^{32}P), rhodium (^{188}Rh), sulfur (^{35}S), carbon (^{14}C), tritium (^3H),

chromium (.sup.51Cr), chlorine (.sup.36Cl), cobalt (.sup.57Co or .sup.58Co), iron (.sup.59Fe), selenium (.sup.75Se), or gallium (.sup.67Ga). Radioisotopes useful as therapeutic agents include yttrium (.sup.90Y), lutetium (.sup.177Lu), actinium (.sup.225Ac), praseodymium, astatine (.sup.211At), rhenium (.sup.186Re), bismuth (.sup.212Bi or .sup.213Bi), and rhodium (.sup.188Rh). Radioisotopes useful as labels, e.g., for use in diagnostics, include iodine (.sup.131I or .sup.125I), indium (.sup.111In), technetium (.sup.99mTc), phosphorus (.sup.32P), carbon (.sup.14C), and tritium (.sup.3H), or one or more of the therapeutic isotopes listed above.

(152) The present disclosure provides radiolabeled antibody molecules and methods of labeling the same. In an embodiment, a method of labeling an antibody molecule is disclosed. The method includes contacting an antibody molecule, with a chelating agent, to thereby produce a conjugated antibody. The conjugated antibody is radiolabeled with a radioisotope, e.g., .sup.111Indium, .sup.90Yttrium and .sup.177Lutetium, to thereby produce a labeled antibody molecule.

(153) In another aspect, this disclosure provides a method of making an antibody molecule disclosed herein. The method includes, for example, providing an antigen, e.g., C5aR1 or a fragment thereof; obtaining an antibody molecule that specifically binds to the antigen; evaluating efficacy of the antibody molecule in modulating activity of the antigen and/or organism expressing the antigen, e.g., C5aR1. The method can further include administering the antibody molecule, including a derivative thereof (e.g., a humanized antibody molecule) to a subject, e.g., a human.

(154) This disclosure provides an isolated nucleic acid molecule encoding the above antibody molecule, vectors and host cells thereof. The nucleic acid molecule includes, but is not limited to, RNA, genomic DNA and cDNA.

(155) Amino acid sequences of exemplary antibody molecules are described in Tables 1A, 1B, 2A, 2B, 3A, and 3B. Tables 1A, 2A, and 3A list exemplary heavy chain variable region sequences and complementarity-determining regions (CDRs) thereof. Tables 1B, 2B, and 3B list exemplary light chain variable region sequences and complementarity-determining regions (CDRs) thereof. Table 1C lists matched pairs of heavy and light chain variable regions from exemplary antibody molecules described herein.

(156) TABLE-US-00005

TABLE	1A	The amino acid sequences of the heavy chain variable region (VH) and IMGT HCDRs of the exemplary anti-C5aR1 antibodies	SEQ	SEQ	SEQ	SEQ	ID	ID	ID	ID	Name	VH	sequence	NO:	HCDR1	NO:	HCDR2	NO:	HCDR3	NO:																																																																														
VH-11	QVQLQQSGPELVKPGASVK	1	GYSFSSSW	121	ISPGDGD	241	VRRFLITSTRYVMDYWG	361	ISCKASGYSFSSSWMNWVK	QRPGKGLEWIGRISPGDGD	TRYSGKFKGKATLTADKSS	STAYMQVTSLTSEDSAIYF	CVRRFLITSTRYVMDYWGQ	GTTVTVSS	VH-11v2	QVQLQQSGPELVKPGASVK	2	GYSFSSSW	122	ISPGDGD	242	VRFLITSTRYVMDYWG	362	ISCKASGYSFSSSWMNWVK	QRPGKGLEWIGRISPGDGD	TRYSGKFKGKATLTADKSS	STAYMQVTSLTSEDSAIYF	CVRFLITSTRYVMDYWGQG	TTVTVSS	VH-66	QVQLQQSDAELVKPGASVK	3	GYTFIDHA	123	ISPGNGE	243	KRALFYITGKYQPMYDYG	363	ISCKASGYTFIDHAIHWVK	QRPEQGLEWIGYISPGNGE	IKYNEKFKAKATLTADKSS	SAAYMQLNSLTADSAYYF	CKRALFYITGKYQPMYDYG	QGTTVTDSS	VH-79	QVQLKESGPGGLVAPSQSL	4	GFSLTSA	124	IWTGGG	244	ARDGDYVYYAMAYWG	364	ITCTVSGFSLTSAISWVR	QPPGKGLEWLGVITWTGGGT	KYNSALKSRLSISKDNSKS	HVFLKMNSLQSDDTARYYC	ARDGDYVYYAMAYWGQGTT	VTVSS	VH-184	QVQLQQSGAELVKPGASVK	5	GYAFSRYW	125	IYPGDGD	245	TRSLGVWG	365	ISCKASGYAFSRYWMNWVK	QRPGKGLEWIGQIYPGDGD	TKYNGKFKGKATLTADKSS	STAYMQLNSLTSEDSAYYF	CTRSLGVWGTTGTTVTVSS	VH-216	QIQLVQSGPELKKPGETVK	6	GYSFTTFG	126	INTYSGV	246	ARGLGRLLAYWG	366	ISCKASGYSTTFGMSWVK	QAPGKVLKWMGWINTYSGV	PTYADDFKGRFAFSLETS	STAYLQINNLIKNEDTATYF	CARGLGRLLAYWGQGT	LT	VSA	VH-272	QVQLQQSGAELVKPGASVK	7	GYAFSSYW	127	IYPGDGD	247	TRSLGVWG	367

ISCKASGYTFTSYWMHWWK QRPQGKLEWIGYIPGDDG TKYNGKFKGKATLTADKSS
STAYMQVSSLTSEDSAVYF CTRSLGVWGTGTTVTVSS VH-308
QVQLQQSGAELAKPGASVK 8 GYTFTSYW 128 INPSSGY 248 ARGMFAMDYWG 368
MSCKASGYTFTSYWMHWWK QRPQGQLEWIGYINPSSGY TEYNQKFKDKATLTADKSS
STAYMQLSSLTSEDSAVYY CARGMFAMDYWGQGTCTVTV SS VH-317
QVQLQQPGAELVMPGASVK 9 GYTFTSYW 129 IDPSDGY 249 ATEGFVG 369
LSCKASGYTFTSYWLHWVR QRPQGQLEWIGEIDPSDGY SNHNQKFKGKATLTVDKSS
STAYMQLSSLTSEDSAVYY CATEGFVGQGTCTVTVSS VH-317v2
QVQLQQPGAELVMPGASVK 10 GYTFTSYW 130 IDPSDSY 250 ARVAYYSNFGGFAYWG
370 LSCKASGYTFTSYWMHWWK QRPQGQLEWIGEIDPSDSY
TNYNQKFKGKATLTVDKSS STAYMQLSSLTSEDSAVYY CARVAYYSNFGGFAYWGQG
TTVTVSS VH-322 QVQLQQSDAALVKPGASVK 11 GHTFTDHA 131 ISPGNGD 251
KGPLFVRGQYYITMDYWG 371 ISCKASGHTFTDHAHWK QRPEQGLEWIGYISPGNGD
IKYNDKFKGKATLTADKSS STAYMQLNSLTPEDSAVYF CKGPLFVRGQYYITMDYWG
QGTCTVTVSS VH-329 QVQLQQSGAELTKPGASVK 12 GYTFTNYW 132 LNPSSGY 252
TRSGGDNYGNPYYFDRWG 372 LSCKASGYTFTNYWMHWWK
QRPQGQLEWIGYLNPSGY TKYNQKFKDKATLTADKSS STAYMQLNSLTYEDSAVYY
CTRSGGDNYGNPYYFDRWG QGTCTVTVSS VH-330 QVQLKQSGPGLVQPSQSL 13
GFSLSYG 133 IWRGGS 253 AKNSQLGNAMDYWG 373 IACTVSGFSLTSYGVHWVR
QSPGKGLEWLGVWIRGGST DYNAAFKSRLSITKDNSKS QVFFTMNRLHADDTAIYYC
AKNSQLGNAMDYWGQGTCTV TVSS VH-332 QVQLQQSDAELVKPGASVK 14 GYTFTDHS
134 ISPGNGD 254 KGPLLLRWRYFYPVDYWG 374 ISCKASGYTFTDHSIHWK
QRPEQGLEWIGYISPGNGD IKYDEKFKGKATLTADTSS STAYMQLNSLTSSEDSAVYF
CKGPLLLRWRYFYPVDYWG QGTCTVTVSS VH-335 QVQLQQSDAALVKPGASVK 15
GYTFTDHA 135 ISPGNGD 255 KGPLLRWRYITMDYWG 375
ISCKASGYTFTDHAHWK QRPEQGLEWIGYISPGNGD IKYNEKFKGKATLTADKSS
STAYMQLNSLTSSEDSAVYF CKGPLLRWRYITMDYWG QGTCTVTVSS VH-336v1
EVKLEESGGGLVQPGGSMK 16 GFTFSDAW 136 IRNKANN 256 TRGGYYVFAYWG 376
LSCAASGFTFSDAWMDWVR HA QSPEKGLEWVAEIRNKANN
HATYYAESVKGRFTISRDD SKSSVYLQMNSLRAEDTGI YYCTRGGYYVFAYWGQGT
TVTVSS VH-336v2 QVQLKQSGPGLVQSSQSL 17 GFSLSYG 137 IWSGGS 257
AKNSQLGNAMDYWG 377 ITCTVSGFSLISYGVHWVR QSPGKGLEWLGVWISGGST
DYNAAFKSRLSITKDNSKS QVFFKMNSLQADDTAIYYC AKNSQLGNAMDYWGQGTCTV
TVSS VH-336v3 QVQLQQPGAELVKPGASVK 18 GYTFTSYW 138 IHPNSNS 258
ARSLTGTKTYWG 378 LSCKASGYTFTSYWMHWWK QRPQGQLEWIGMIHPNSNS
TNYNEKFKSKATLTVDKSS STAYMQLSSLTSEDSAVYY CARSLTGTKTYWGQGTCTV VSS
VH-338 EVQLQQSGPELVKPGASVK 19 GYTFTDYN 139 INPNNGG 259 AHGEGDYAYWG
379 MSCKASGYTFTDYNMHWWK QSHGKSLEWIGYINPNNGG
TSYNQKFKGKATLTVNKSS STAYMELRSLTSEDSAVYY CAHGEEDYAYWGQGTCTVTV SS
VH-341v1 EFQLQQSGPELVKPGASVK 20 GYTFTKYV 140 INPYNDG 260 ATARATSYWG
380 MSCKASGYTFTKYVIHWK QKPGQGLEWIGYINPYNDG
TKYNEKFKGKARLTSDKSS NTVYMDLSSLTSEDSAVYY CATARATSYWGQGTCTVTVS S
VH-341v2 QVQLQQPGAELVKPGASVK 21 GYSFTSYW 141 IYPGRGT 261
ARWGTTGRSYWG 381 MSCKASGYSTSYWITWLK QRPQGQLEWIGDIYPGRGT
TDYNEKLKSRATLTVDTS TTAYMQLSSLTSEDSAVYY CARWGTTGRSYWGQGTCTV
VSS VH-343 QVQLQQSGAELVKPGASVK 22 GYTFTEYT 142 FYPGSGS 262
ARHGNYDGSWFAYWG 382 LSCKASGYTFTEYTIHWVN QRSGQGLEWIGWFYPGSGS
IKYNEKFKDKATLTADKSS HTVYMELSRSLTSEDSAVYF CARHGNYDGSWFAYWGQG
TLVTVSA VH-399 EVQLVESGGDLVKPGGSLK 23 GFTFSNYG 143 ITSGGTH 263
TRHGAYYSNPWFAYWG 383 LSCAASGFTFSNYGMSWVR QTPDKRLEWVATITSGGTH

TYYPDSVKGRFTISRDNAL NTLYLQMSLKSSEDAMYY CTRHGAYYNNPWFAYWGQG
 TLVTVS VH-402 QIQLVQSGPELKKPGETVK 24 GYTFTTTFG 144 INTNSGM 264
 ARKSLFYWG 384 ISCKASGYTFTTFGMSWVK QAPGKGLKWMGWINTNSGM
 PTYTDDFRGRFAFSLETS STAYLQISSLKNEEDATYF CARKSLFYWGQGTTLTVSS VH-
 416 QVQLQQSGAELAKPGASVK 25 GYTFISYW 145 INPRSDY 265
 ARVTGTEGPYYFDYWG 385 LSCASGYTFISYWMHWVK QRPQGQLEWIGYINPRSDY
 AKYNQKFKDKATLTNNKSS STAYMQLSSLTYEDYAVYY CARVTGTEGPYYFDYWGQG
 TTLTVSS VH-429 QVQLQQSGAELAKPGASVK 26 GYTFSSYW 146 INPRGDY 266
 VRVTGSEGPYYFDYWG 386 LSCASGYTFSSYWIHWVK QRPQGQLEWIGYINPRGDY
 TKYNQKFKDKATLTADKSS STAFMQLSSLTYEDSAVYY CVRVTGSEGPYYFDYWGQG
 TTLTVSS VH-430 QVQLQQSGADLAKPGASVK 27 GYTFTSYW 147 INPRGDY 267
 ARVTGTEGPYYFDYWG 387 LSCASGYTFTSYWIHWVK QRPQGQLEWIGYINPRGDY
 TKYNQKFKDKATLTADRSS STAYMQLSSLTYEDYAVYY CARVTGTEGPYYFDYWGQG
 TTLTVSS VH-440 QIQLVQSGPELKKPGETVK 28 GYTFTAYG 148 INTYSGV 268
 ARSRYDGYFDYWG 388 ISCKASGYTFTAYGMSWVK QTPGKGLKWMGWINTYSGV
 PANADDFKGRFAFSLETS STAYLQINNLLKNEEDATYF CARSRYDGYFDYWGQGTTL
 TVSS VH-453 QVQLKQSGPGQVAPSQSL 29 GFSLNSA 149 IWSDGS 269
 ARNGRLGNAMDYWG 389 ITCTVSGFSLNSAVHWVR QSPGKGLEWLGVWSDGST
 DYNATFISRLSISRDNSKS QVFFKMRSLQVDDTAVYYC ARNGRLGNAMDYWGQGTSTV
 TVSS VH-454 QVQLKQSGPGLVAPSQSL 30 GFSLTSYG 150 IWGVGS 270
 ASPYYSHYVPFAYWG 390 ITCTVSGFSLTSYGVDWIR QSPGKGLEWLGVWGVGST
 NYNSALKSRLSISKDNSRS QVFLKLNSLQTDDTAMYYC ASPYYSHYVPFAYWGQGTTL
 VTVSA VH-465 QVQLQQSGAELAKPGASVK 31 GYIFTSYW 151 INPSTTS 271
 ARPDNSGYVGFAYWG 391 LSCASGYIFTSYWMNWVK QRPQGQLEWIGYINPSTTS
 TKYNQKFKDKATLTADKSS TTAYMQLTSLTYEDSAVYY CARPDNSGYVGFAYWGQGT
 LVTVSA VH-475 QVQLKQSGPGLVAPSQSL 32 GFSLTSYG 152 IWGVGG 272
 ASPYYSHYVPFAYWG 392 ITCTVSGFSLTSYGVDWIR QSPGKGLEWLGVWGVGGT
 NYNSALKSRLSISKDNSRS QVFLKLNSLQTDDTAMYYC ASPYYSHYVPFAYWGQGTTL
 VTVSA VH-481 QVQLQQSGAELAKPGASVQ 33 GYSFTRYW 153 INPSTDY 273
 AGGLPHFDYWG 393 VSCASGYSFTRYWMHWIK QRPQGQLEWIGYINPSTDY
 SAYNQKFKDKATLTADKSS STAYLQLTSLTSEDSAVYY CAGGLPHFDYWGQGTTLTV SS
 VH-497 QVQLKQSGPGLVAPSQSL 34 GFSLTYYG 154 IWSDGS 274 ARNSRYGNSFAYWG
 394 ITCTVSGFSLTTYGVHWVR QPPGKGLEWLWVWSDGST TYNALKSRLSISKDNSKS
 QVFLKMNSLQPDDETAMYYC ARNSRYGNSFAYWGQGTTLV TVSA VH-502
 DVQLQESGPDLVKPSQSL 35 GYSITSGY 155 IHYSGS 275 VFWLPFDYWG 395
 LTCTVTGYSITSGYSWHWI S RQFPGNKLEWMGYIHYSGS TNYNPSLKSRLSITRDTSK
 NQFFLQLKSVTTEDTATYY CVFWLPFDYWGQGTTLTVS S VH-503
 DVQLQESGPGLVKPFQSL 36 GYSITSDY 156 INYSGS 276 ARMGYRYPWFAYWG 396
 LTCTVTGYSITSDYAWNWI A RQFPGNKLEWMGYINYSGS TSYNPSLRSRLSITRDTSK
 NQFFLHLNSVTTEDTATYY CARMGYRYPWFAYWGQGTTL VTVSA VH-504
 EVQLKQSGGGLVKPGGSLK 37 GFTFSSYT 157 ISSGGSY 277 TRGGGNYDAMDYWG 397
 LSCAASGFTFSSYTMSWVR QTPEKRLEWVATISSGGSY TYYPDSVKGRFTISRDNAL
 NTLYLQMSLKSSEDAMYY CTRGGGNYDAMDYWGQGTST VTVSS VH-507
 QVQLQQSGAELAKPGASVK 38 GYTFTNYW 158 INPSSAS 278
 ARVPLPYGSSYGPYFFDFWG 398 LSCASGYTFTNYWMHWVK
 QRPQGQLEWIGYINPSSAS SKYNQKFKDRATLTDDKSS STAFMHLSSLTYEDSAVYY
 CARVPLPYGSSYGPYFFDF WGQGTTLTVSS VH-508 QVQLQQSGAELARPGASVK 39
 GYTFTSYT 159 INPSSGY 279 ARSGDYDGFAYWG 399 MSCASGYTFTSYTMHWVK
 QRPQGQLEWIGYINPSSGY TKYNQKFKDKATLTADKSS STAYMQLSSLTSEDSAVYY
 CARSGDYDGFAYWGQGTTLV TVSA VH-510 QVQLQQSGAELAKPGASVK 40 GYTFTSYW

160 INPSGDY 280 ARVPLSYGSSYGPYFFDYWG 400 LSCKASGYTFTSYWMHWVK
QRPQGQLEWIGYINPSSGY TKYNQKFKDRATLTADKSS STAYMQLSSLTYEDSAVYY
CARVPLSYGSSYGPYFFDY WGQGTTLTVSS VH-511 QVQLQQSGAELAKPGASVK 41
GYTFTNYW 161 INPSGDY 281 ARVPLSYGSGNGPYFFDYWG 401
LSCKTSGYTFTNYWMHWIK QRPGLGLEWIGYINPSGDY TKHNQKFKDKATLTADRSS
STAYMQLSSLTYEDSAVYY CARVPLSYGSGNGPYFFDY WGQGTTLTVSS VH-521
QVQLQQSRAALAKPGASVK 42 GYTFTNHW 162 INPINGF 282
ARVPLSYGGSYGPYFFDFWG 402 LSCKASGYTFTNHWMHWVK
QRPQGQLEWIGYINPINGF NRYNQNFKDRATLTDDKAS STAFIHLNGLTYEDEFAVYY
CARVPLSYGGSYGPYFFDF WGQGTTLTVSS VH-530 QVQLQQSRAALAKPGASVK 43
GYTFTNHW 163 INPINGF 283 ARVPLSYGGSYGPYFFDFWG 403
LSCKASGYTFTNHWLHWVK QRPQGQLEWIGYINPINGF SKYNQNFKNRATLTDDSSS
STAFIHLNGLTYEDEFAVYY CARVPLSYGGSYGPYFFDF WGQGTTLTVSS VH-536
QVQLQQSGAALAKPGASVK 44 GYSFTNYW 164 INPINGY 284
ARVPLSYGGSYGPYFFDFWG 404 LSCKASGYTFTNHWMHWVK
QRPQGQLEWIGYINPINGY GKYNQNFKDRATLTDDKSS STAFIHLNGLTYEDSAVYY
CARVPLSYGGSYGPYFFDF WGQGTTLTVSS VH-541 QVQLQQSGPELVKPGESVK 45
GYTFTDYY 165 FYPNNGG 285 TRGSGPFAYWG 405 MSCKASGYTFTDYYMDWVK
QSHGKSLEWIGYFYPNNGG VKYSQKFKDKAALTVDKSS TTAYMELHSLTFEDSAVYY
CTRSGSGPFAYWGQGTTLTV SA VH-547 QVQLKQSGPGLVAPSQSLS 46 GFSLTNYG 166
IWGDGI 286 ASALDYSNYGFAYWG 406 ITCTVSGFSLTNYGVDWVR
QSPGKGLEWLGVWGDGIT KYNSALKSRLSISKDNSKS QVFLKMNSLQTDDTAMYYC
ASALDYSNYGFAYWGQGTTL VTVSA VH-549 QVQLKQSGPELVKPGDSVK 47 GYTFTDYY
167 FYPNNGG 287 TRGSGPFAYWG 407 MSCKVSGYTFTDYYIDWVK
QSHGKSLEWIGYFYPNNGG AKYNQKFKSKAALTVDKSS TTAYMELHSLTFEDSAVYY
CTRSGSGPFAYWGQGTTLTV SA VH-550 QVQLKQSGPGLVAPSQSLS 48 GFSLTNCG 168
IWGDGL 288 ASALDYSNYGFAYWG 408 ITCTVSGFSLTNCGVDWVR
QSPGKSLEWLGVWGDGLT KYNSALKSRLSISKDNSKS QVFLKVNLSLQTDDTAVYYC
ASALDYSNYGFAYWGQGTTL VTVSA VH-553 QVQLKQSGPELVKPGDSVK 49 GYTFTDYY
169 FYPNNGG 289 TRGSGPFAYWG 409 MSCKASGYTFTDYYMDWVK
QSHGKSLEWIGYFYPNNGG AKYNQKFKGKAALTVDKSS TTAYMELHSLTFEDSAVYY
CTRSGSGPFAYWGQGTTLTV SA VH-556 QVQLKQSGPGLVAPSQSLS 50 GFSLTNCG 170
IWGDGL 290 ASALDYSNFGFAYWG 410 ITCTVSGFSLTNCGVDWVR
QSPGKSLEWLGVWGDGLT KYNSALKSRLSISKDNSKS QVFLKVNRLQTDDTAMYYC
ASALDYSNFGFAYWGQGTTL VTVSA VH-557 QVQLKQSGPELVKPGDSVK 51 GYTFSDDY
171 FYPNNDG 291 ARGSGPFVYWG 411 MSCKASGYTFSDYYMDWVK
QSHGKSLEWIGYFYPNNDG IRYNQRFKGRASLTVDKSS NTAYMELHSLTSEDSAVYY
CARGSGPFVYWGQGTTLTV SA VH-567 DVQLQESGPGLVKPSQSLS 52 GYSITSDS 172
ISYSGR 292 ARASIGFDYWVG 412 LTCTVTGYSITSDSAWNWI A
RQFPGNKLEWMGYISYSGR ISYNPSLKSRLSITRDTSK NQFFLQFNSVTTEDTAKYY
CARASIGFDYWVGQGTTLTV SS VH-568 EVQLQQSGADLVKPGASVK 53 GFSITSDY 173
IDPTNVN 293 ARRLRQTYAMDYWG 413 LSCTVSGFNIKDSYIHWLK
QRPQGQLEWIGRIDPTNVN TKYDPKFQGKASITTDTS NTAYLQLSSLTSENTAVYY
CARRLRQTYAMDYWGQGTTS VTVSS VH-570 DVQLQESGPGLVKPSQSLS 54 GYSITSDS
174 ISYSGR 294 ARASIGFDYWVG 414 LTCTVTGYSITSDSAWNWI A
RQFPGNKLEWMGYISYSGR ISYNPSLKSRLSITRDTSK NQIFLQFNSVTTEDTAKYY
CARASIGFDYWVGQGTTLTV SS VH-573 DVQLQESGPGLVKPSQSLS 55 GYSITSDS 175
ISYSGR 295 ARASIGFDYWVG 415 LTCTVTGYSITSDSAWNWI A
RQFPGNKLEWMGYISYSGR ISYNPSLKSRLSITRDTSK NQFFLQFYSVTTEDTARYY
CARASIGFDYWVGQGTTLTV SS VH-583 EVQLVESGGGLVKPGGSLK 56 GFTFNAYA 176

ISTGGN 296 TRGYQRFSTGTFNAYAMSWVR
QTPEKRLEWVASISTGGNT YCPDSVKDRFTVSRDNVRN ILYLQMSSLRSEDAMYYC
TRGYQRFSGFAYWGQGTTLV TVS VH-584 DVQLQESGPDLVKPSQSLS 57 GYSITSGY 177
IHNSGS 297 ARSIGDYWG 417 LTCTVTGYSITSGYSWHWI S RQSPGNKLEWLAYIHNSGS
TNYNPSLKSISRITRDTSK NQFFLKLNSVTTEDTATYY CARSIGDYWGQGTTLAVSS VH-
585 DVQLQESGPGLVKPSQSLS 58 GYSITSDY 178 ISYSGS 298 ARYGGNYPTYAMDYWG
418 LTCTVTGYSITSDYAWNWI A RQFPGNKLEWMGYISYSGS TSYNPSLKSISRITRDTSK
NQFFLQLSSVTTEDTATYY CARYGGNYPTYAMDYWGQG TSVTVSS VH-586
DVQLQESGPDLVKPSQSLS 59 GFSLTDSY 179 IHYSGR 299 ARYDFAYWG 419
LTCIVAGFSLTDSYSWHWI S RQFPGNKLEWMGYIHYSGR TNYNPSLKTQFSITRNTSK
NQFFLQLISVPTEDTATYY CARYDFAYWGRGTSVTVSS VH-588
DVQLQESGPGLVKPSQSLS 60 GYSITSDY 180 ISYSGS 300 AITTGGYFDYWG 420
LTCTVTGYSITSDYAWNWI A RQFPGNKLEWMGYISYSGS IRYNPSLKSISRITRDTSK
NQFFLQLNSVTTEDTATYY CAITTGGYFDYWGQGTTLT VSS VH-592
QVQLKQSGAELVRPGSSVG 61 GYAFTNFW 181 LYPGDDD 301 AVTEVKRRRSFAYWG 421
ISCKASGYAFTNFWMNWVR QRPQGQLEWIGQLYPGDDD THYNGKFKGKVTLTADRSS
GTAYMQLSRLTSEDSAVYF CAVTEVKRRRSFAYWGQGT LVTVSA
(157) TABLE-US-00006 TABLE 1B The amino acid sequences of the light chain
variable region (VL) and IMGT LCDRs of the exemplary anti-C5aR1 antibodies
SEQ ID SEQ ID SEQ ID Name VL sequence NO: LCDR1 NO: LCDR2 NO:
LCDR3 NO: VK-11 DVVMTQTPLSLPVSLGDQA 62 QSLVHSNG 182 KV 302
SQSTLVPPFTFG 422 SISCRSSQSLVHSNGNTYL NTY HWYLQKPGQSPKLLIYKVS
NRFSGVPDRFSGSGSGTDF TLKISRVEAEDLGVYFCSQ STLVPPFTFGGGTKLEIK VK-66
DIVMTQSQKFMSTTAGDRV 63 QNVGSA 183 SS 303 QQYNSFPLTFTG 423
SITCKASQNVGSAVVWYQQ KPGRSPKLLIYSSSIRYTG VPDRFTGSGSGTDFTLTIN
SVQSEDLADYFCQQYNSFP LTFGAGTKLEIK VK-79 DVVMTQTPLSLPVSLGDQA 64
QSLVHSNG 184 KV 304 SQSTHVPPTFG 424 SISCRSSQSLVHSNGNTYL NTY
HWYLQKPGQSPKLLIYKVS NRFSGVPDRFSGSGSGTDF KLKISRVEAEDLGVYFCSQ
STHVPPTFGGGTKLEIK VK-184 DVVMTQTPLTSLVTIGQPA 65 QSLVHSNG 185 LV 305
VQGTHVPYTFG 425 SISCKSSQSLVHSNGKTYL KTY NWLLQRPQGQSPKRLIYLV
KLDSGVPDRFTGSGSGTDF TLKISRVEAEDLGVYCCVQ GTHVPYTFGGGTLEIK VK-
184_C_Y DVVMTQTPLTSLVTIGQPA 66 QSLVHSNG 186 LV 306 VQGTHVPYTFG 426
SISCKSSQSLVHSNGKTYL KTY NWLLQRPQGQSPKRLIYLV KLDSGVPDRFTGSGSGTDF
TLKISRVEAEDLGVYYCVQ GTHVPYTFGGGTLEIK VK-216_272
DVVMTQTPLTSLVTIGQPA 67 QSLVDSG 187 LV 307 LQATHFPWTFG 427
SISCKSSQSLVDSGKTYL KTY NWLLQRPQGQSPKRLIYLV KLDSGVPDRFTGSGSGTDF
TLKISRVEAEDLGVYYCLQ ATHFPWTFGGGTLEIK VK-308 DIVMTQSQNFMTSVGDRV
68 QYVGTY 188 SA 308 QQYSSSPYTFG 428 SVTCKASQYVGTYVAWYQQ
KPGQSPKALIYSASYRHTG VPDRFTGSGSGTDFTLTIS NVQSEDLADYFCQQYSSSP
YTFGGGTLEIK VK-317 DIVMTQSPSSLSVSAGEKV 69 QSLVSGN 189 GA 309
QNDHSHPYTFG 429 TMSCKSSQSLVSGNQKNY QKNY LAWHQQKPGQPPKLLIYGA
STRESGVPDRFTGSGSGTD FTLTISSVQAEDLAVYYCQ NDHSHPYTFGGGTLEIK VK-
317v2 DIVMSQSPSSLAVSAGEKV 70 QSLVNSRT 190 WA 310 KQSYNLYTFG 430
TMSCKSSQSLVNSRTRKNY RKNY LAWYQQKPGQSPKLLIYWA
STRESGVPDRFTGSGSGTD FTLTISSVQAEDLAVYYCK QSYNLYTFGGGTLEIK VK-322
DIVMTQSQKFMSTTVGDRV 71 QNVGAA 191 SA 311 QQYNSFPLTFTG 431
SITCKASQNVGAADVWYQQ KPGQSPKLLIYSASYRYS VPDRFTGSGSGTDFTLTIS
NMQSEDLADYFCQQYNSFP LTFGGGTLEIK VK-329 DVVMTQTPLSLTVSLGDQA 72
QSLVHSNG 192 KV 312 SQSTLVPLTFTG 432 SISCRSSQSLVHSNGNTYL NTY
HWYLQKPGQSPKFLIYKVS NRFSGVPDRFSGSGSGTDF RLKISRVEAEDLGVYFCSQ

STLVPLTFGAGTKLEIK VK-330 DIVMSQSPSSLAHSVGEKV 73 QSLFNSRT 193 WA 313
NQSYDLLTFG 433 TMSCKSSQSLFNSRTRKNY RKNY LAWYQQKPGQSPKLLIYWA
STRESGVPDRFTGSGSGTD FTLTINSVQAEDLALYYCN QSYDLLTFGAGTKLEIK VK-332
DIVMTQSQKFMSSAVGDRV 74 QNVGAA 194 SA 314 QQYNSFPLTFG 434
TITCKASQNVGA AVAWYQQ KPGQSPKLLLYSASIRYTG VPDRFTGSGSGTDFTLTIS
NIQSEDLAHFFCQQYNSFP LTFGGGGTKLEIK VK-335 DIVMTQSQKFMSTTVGDRV 75
QNVGAA 195 SA 315 QQYNSFPLTFG 435 SITCKASQNVGA AVVWYQQ
KPGQSPKLLIYSASYRYTG VPDRFTGSGSGTDFTLTIS NMQSEDVADYFCQQYNSFP
LTFGGGGTKLEIK VK-336v1 DIVMSQSPSSLAVSAGEKV 76 QSLSSRT 196 WA 316
NQSYDLLTFG 436 TMSCKSSQSLSSRTRKNY RKNY LAWYQQKPGQSPKLLIYWA
STRESGVPDRFTGSGSGTD FTLTISSVQAEDLALYFCN QSYDLLTFGAGTKLEIK VK-336v2
DIVMTQSPATLSVTPGDRV 77 QSIDY 197 YA 317 QNGHSFPPTFG 437
SLSCRASQSISDYLHWYQQ KSHESPRLLIKYASQSISG IPSRFGSGSGSGSDFTLSIN
SVEPEDVGVYYCQNGHSFP PTFGAGTKLEIK VK-338v1 DIVMTQSQKFMSTTLGDRV 78
QSVGAA 198 SA 318 QQYNSFPLTFG 438 SIPCKASQSVGA AVAWYQQ
KPGQSPKLLIYSASIRYAG VPDRFTGSGSGTDFTLTIS NMRSEDLADYFCQQYNSFP
LTFGGGGTKLEIK VK-338v2 DIQMTQSSSYLSVSLGGRV 79 DHINNW 199 GA 319
QQYWSTPYTFG 439 TITCKASDHINNW LAWYQQ KPGNAPRLLISGATSLETG
VPSRFGSGSGGKDYTLSIT SLQTEDVATYYCQQYWSTP YTFGGGGTKLEIK VK-341v1
DVVMTQTPLTSLVTIGQP 80 QSLLESDG 200 LV 320 LQATHFPHTFG 440
SISCKSSQSLLESDGKTYL KTY NWLLQRPGESPKLLIYLVS KLD SGVPDRFTGSGSGTDF
TLKISRVEAEDLG VFYCLQ ATHFPHTFGGGGTKLEIK VK-341v2 DIQMTQSPSSLSASLGERV
81 QEIKTY 201 AA 321 LQYASYPWTFG 441 SLTCRASQEIKTYLSWLQQ
KPDGTIKRLIYAATTLESV VPKRFGSWSGSEYSLTIS SLESEDFADYYCLQYASYP
WTFGGGGTKLEIK VK-343 DVQITQSPSYLAASPGETI 82 KSISKY 202 SG 322
QQHDEYPWTFG 442 TINCRAKSSISKYLAWYQE KPGKTNKLLIYSGSTLQSG
IPSRFGSGSGSGTDFTLTIS SLEPEDFAMYCYCQQHDEYP WTFGGGGTKLEIK VK-399
DIVMTQSQKFMSTSVGDRV 83 QNVGEN 203 SA 323 QQYNSSPWTFG 443
SVTCKASQNVGENVSWYQQ KPGQFPKALIYSASYRYSG VPDRFTGSGSGTDFSLTIS
NVQSEDLAEYFCQQYNSSP WTFGGGGTKLEIK VK-402 DVVMTQSPLTSLVTIGQPA 84
RSLDSDG 204 LV 324 WQGTHFPWTFG 444 SISCRSSRSLDSDGKTKL KTK
HWLLQRPQGSPKSLIYLVS KLD SGV PNRFTGGGSGTDF TLKINRVEAEDLG VYYCWQ
GTHFPWTFGVGKLEIK VK-416 DVVMTQTPLSLPVSLGQA 85 QSLVHSNG 205 KV
325 SQSTLVPLTFG 445 SISCRSSQSLVHSNGNTYL NTY HWYLQKPGQSPKFLIYKVS
NRISGV PDRFGSGSGTDF TLKISRVEAEDLG VFYFCSQ STLVPLTFGAGTKLELR VK-429
DVVMTQTPLSLPVSLGDQA 86 QSLVHSNG 206 KV 326 SQSTLVPLTFG 446
SISCRSSQSLVHSNGNTYL NTY QWYLQKPGQSPKLLIYKVS NRGSGVPDRFGSGSGGTEF
TLKIRRVEAEDLG VFYFCSQ STLVPLTFGAGTKLELK VK-430 DVVMTQTPLSLPVSLGDQV
87 QSLVHSNG 207 KV 327 SQSTLVPLTFG 447 SISCRSSQSLVHSNGNTYL NTY
QWYLQKPGQSPKLLIYKVS NRGSGVPDRFGSGSGTDF TLKISRVEAEDLG VFYFCSQ
STLVPLTFGAGTKLELR VK-440 ENVLTQSPAIMSASPGEKV 88 SSVSY 208 ST 328
QQRSSYPPTFG 448 TITCSASSSVSYMHWFQK PGTSPKLWIYSTSNLASGV
PTRFGSGSGTSYSLTISR MEAEDAATYYCQQRSSYP TFGGGAKEIK VK-453
DIVMTQSPSSLAVSVGEKV 89 QSLFSSRT 209 WA 329 KQSYNLLTFG 449
TMSCKSSQSLFSSRTRKNY RKNY LAWYQQKPGQSPKLLIYWA
STRESGVPYRFTGSGSGTD FTLTISSVQTEDLAVYYCK QSYNLLTFGAGTKLEL VK-454
DVVMTQTPLSLPVSLGDQA 90 QSLVHSNG 210 RV 330 SQSTHVPPTFG 450
SISCRSSQSLVHSNGNTYL NTY HWYLQKPGQSPKLLIYRVS NRGSGVPDRFGSGSGTDF
TLKISRVEAEDLG VFYFCSQ STHVPPTFGGGGTKLEIK VK-465 DVVMTQTPLSLPVSLGDQA
91 QSLVHSTG 211 KV 331 SQSTLVPLTFG 451 SVSCRSSQSLVHSTGNTFL NTF

HWYLQKPGQSPKLLIYKVS NRFSQVADRFSGSGTDF TLKISRVEAEDLGVYFCSQ
STLVPLTFGAGTKLELK VK-475 DVVMTQTPLSLPVSLGDQA 92 QSLVHSNG 212 KV
332 SQSTLVPPTEFG 452 SISCRSSQSLVHSNGNTYL NTY HWYLQKPGQSPKLLIYKVS
NRFSGLPDRFSGSGSGTDF TLKISRVEAEDLGVYFCSQ STLVPPTEFGGGTKLEIK VK-481
DVVMTQTPLSLPVSLGDQA 93 QSLVHNNG 213 KV 333 SQSTHVPITFG 453
SISCRSSQSLVHNNGVTYL VTY HWYLQKPGQSPKLLIYKVS NRFSQVPDRFSGSGSGTDF
TLKISRVEAEDLGVYFCSQ STHVPITFGAGTKLELK VK-497 DVVMTQTPLSLPVSLGDQA
94 QSLVHSNG 214 KV 334 SQSTHVPTEFG 454 SISCRSSQSLVHSNGNTYL NTY
HWYLQKPGQSPKLLIYKVS NRFSQVPDRFSGSGSGTDF TLKISRVEAEDLGVYFCSQ
STHVPTEFGSGTKLEIK VK-502 DVVMTQTPLSLPVSLGDQA 95 QSLVHSNG 215 KV 335
SQSTHVPYTFG 455 SISCRSSQSLVHSNGNTYL NTY HWYLQKSGQSPKLLIYKVS
NRFSQVPDRFSGSGSGTDF TLKISRVEAEDLGVYFCSQ STHVPYTFGGGTKEIK VK-503
DIVLTQSPASLAVSLGQRA 96 KSVSTSGY 216 LA 336 QHNRELPPTEFG 456
TISCRASKSVSTSGYSYLH SY WYQQKPGQPPKLLIYASN LESQVPARFSGSGSGTDF
LNIHPVEEEDAATYYCQHN RELPPTEFGGGTKLEIK VK-504 DVVMTQSPLTSLVTIGQPA
97 QSLLYSNG 217 LV 337 VQGTHFPHTFG 457 SISCKSSQSLLYSNGKTYL KTY
NWLLQRPQSPKRLIYLV KLDQVPDRFTGSGSGTDF TLKISRVEAEDLGVYYCVQ
GTHFPHTFGGGTKLEIK VK-507 DVVMTQTPLSLPVSLGDHA 98 QSLIHSNG 218 TV 338
SQSTLVPLTFG 458 SISCRSSQSLIHSNGNNYL NNY HWYLQKPGQSPKLLIYTVS
NRFSQVPDRFSGSGSGTDF TLKISRVEAEDLGVYFCSQ STLVPLTFGSGTKLEIK VK-508
DIVMTQSPSSLAMSVGQKV 99 QSLNLS 219 FA 339 QHYSTPTEFG 459
TMSCKSSQSLNLSQKNY QKNY LAWYQQKPGQSPKLLIYFA
STRESQVPDRFIGSGSGTD FTLTISNVQAEDLADYFCQ QHYSTPTEFGGGTKLEIK VK-510
DVVMTQTPLSLPVSLGDHA 100 QSLIHSNG 220 KV 340 SQSTLVPLTFG 460
SISCRSSQSLIHSNGNNYL NNY HWYLQKPGQSPQLLIYKVS NRFSQVPDRFSGSGSGTDF
TLKISRVEAEDLGVYFCSQ STLVPLTFGSGTKLELR VK-511 DVVMTQTPLSLPVSLGDHA
101 QSLVHSNG 221 KV 341 SQSTLVPLTFG 461 SISCRSSQSLVHSNGNIYL NIY
HWYLQRPQSPKLLIHKVS NRFSQVPDRFSGSGSGTDF TLKISRVEAEDVGIYFCSQ
STLVPLTFGSGTKLELK VK-518 DVVMTQTPLSLPVSLGDHA 102 QSLVHSNG 222 KV 342
SQSTLVPLTFG 462 SISCRSSQSLVHSNGNTYL NTY HWYLQRPQSPKLLIHKVS
NRFSQVPDRFSGSGSGTDF TLKISRVEAEDVGIYFCSQ STLVPLTFGSGTKLELK VK-528
DVVMTQTPLSLPVSLGDHA 103 QSLIHSNG 223 KV 343 SQSTLVPLTFG 463
SISCRSSQSLIHSNGNNYL NNY HWYLQKPGQSPKLLIYKVS NRFSQVPDRFSGSGSGTDF
TLKISRVEAEDLGVYFCSQ STLVPLTFGSGTKLELK VK-521 DVVMTQSPLSLPVSLGDHA
104 QSLIHSNG 224 TV 344 SQSTLVPLTFG 464 SISCRSSQSLIHSNGNNYL NNY
HWYLQKPGQSPKLLIYTVS NRFSQVPDRFSGSGSGTDF TLKISRVEAEDLGVYFCSQ
STLVPLTFGSGTKLEV VK-530 DVVMTQTPLSLPVSLGDHA 105 QSLIHSNG 225 TV 345
SQSTLVPLTFG 465 SISCRSSQSLIHSNGNNYL NNY HWYLQKPGQSPKLLIYTVS
NRFSQVPDRFSGSGSGTDF TLKISRVEAEDLGVYFCSQ STLVPLTFGSGTKLEV VK-541
DIVMTQTPLSLPVSLGDQA 106 QSLVHSSG 226 KV 346 SQSTLVPTFG 466
SISCRSSQSLVHSSGNTYL NTY HWYLQKPGQSPKLLIYKVS NRFSQVPDRFSGSGSGTDF
TLRISVAAEDLGVYFCSQ STLVPTFGAGTELELK VK-547 DVVMTQTPLSLPVSLGNQA
107 QRLVHSNG 227 KV 347 AQSTLVPPTEFG 467 SISCRSSQRLVHSNGNTYL NTY
HWYLQKPGQSPKLLIYKVF NRFSQVPDRFSGSGSGTDF TLKISRVEAEDLGIYFCAQ
STLVPPTEFGGGTKLEIK VK-549 DVVMTQTPLSLPVSLGDQA 108 QSLVHSSE 228 KV 348
SQSTLVPTFG 468 SISCRSSQSLVHSSENTYL NTY HWYLQKPGQSPKLLIYKVS
NRFSQVPDRFSGSGSGTDF TLKISRVEAEDLGVYFCSQ STLVPTFGAGTKLELK VK-550
DVVMTQTPLSLPVSLGNQA 109 QSLVHSNG 229 KV 349 AQSTLVPPTEFG 469
SISCRSSQSLVHSNGNTYL NTY HWYLQKPGQSPRLIYKVF NRFSQVPDRFSGSGSGTDF
TLKISRVEAEDLGVYFCAQ STLVPPTEFGGGTKLEIK VK-553 DVVMTQTPLSLPVSLGDQA

110 QSLVHSNG 230 KV 350 SQSTLVSFTYF 470 SISRSSQSLVHSNTYL NTY
HWYVQKPGQSPKLLIYKVS NRFSGVPDRFSGSGSGTDF TLKISRVEAEDLG VYFCSQ
STLVPVTFGAGTKLELK VK-556 DVVMTQTPLSLPVSLGNQA 111 QSLVHSNG 231 KV 351
AQSTLVPPTFG 471 SISRSSQSLVHSNGNTYL NTY HWYLQKPGQSPRLLIYKVF
NRFPQVPDRFSGSGSGTDF TLKISRVEAEDLG VYFCAQ STLVPPTFGGGTKLEIK VK-557
DVVMTQTPLSLPVSLGDQA 112 QSLVHSSG 232 KV 352 SQSTLIPLTFG 472
SISRSSQSLVHSSGNTYL NTY HWYLQKPGQSPKLLIYKVS NRFSGVPDRFSGSGSGTHF
TLKLSRVEAEDLGIYFCSQ STLIPLTFGAGTKLELK VK-567 DIQMTQSPSSLSASLG EKV 113
QEISGY 233 AA 353 LHYANYPPTFG 473 SLTCRASQEISGYLSWLQQ
KPDGSIKRLIYAASTLDSG VPKRFSGSRSGSVYSLTIS SLESEDFADYYCLHYANYP
PTFGGGTKLEIR VK-568 DVQMTQSPSSLSASLGDTI 114 QNIYVW 234 KA 354
QQGLSYPLTFG 474 TITCHASQNIYVWLNWFQQ KPGNIPKLLISKASDLHTG
VPSRFSGSGSGTGFTLTIS SLQPEDIATYYCQQGLSYP LTFGGGTNLEIK VK-583
DVVMTQTPLSPPVSLGYQA 115 QSLVHSNG 235 KV 355 SQSTHVPYPTFG 475
SISRSSQSLVHSNGNTYL NTY NWYLQKPGQSPKLLIYKVS NRLSGVPDRFSGSGSGTDF
TLKISRVEDLGVYFCSQ STHVPYPTFGGGTKLEIK VK-584 DVVMTQTPLSLPVSLGDRA
116 QSLVHSNG 236 KV 356 SQSTHVPWTFG 476 SISRSGQSLVHSNGNTYL NTY
HWYLQRPGRSPNLLIYKVS NRFSGVPDRFSGSGSGTDF TLKISRVEAEDLG VYFCSQ
STHVPWTFGGGTKEIK VK-585 DVVMTQTPLSLPVSLGDQA 117 QSIVYSNG 237 KV 357
FQGSHPPTFG 477 SISRFSQSIVYSNGNTYL NTY QWYLQKPGQSPKLLIYKVS
NRFSGVPDRFSGSGSGTDF TLKISRVEAEDLG VYYCFQ GSHVPPTFGGGTKLEIK VK-586
DVVMTQTPLTSLVTIGQPT 118 QSLLYSNG 238 LV 358 VQNTHLPYPTFG 478
SISCKSSQSLLYSNGKTYL KTY SWLLQRPQSPKRLIYLVS KLDGVPDRFTGSGSGTDF
TLKINRVEAEDLGLYYCVQ NTHLPYPTFGGGTKLEIR VK-588 DIQMTQSTSSLSASLGDRV
119 QDISNY 239 QI 359 QQGNSLPPTFG 479 TISCRASQDISNYLNWYQQ
TPDGTVKLLIYQISRLHSG VPSRFSGSGSGTDYSLTIS NLEEEDIANYFCQQGNSLP
PTFGGGTKVEIK VK-592 DIQMNQSPSSLSASLGDTI 120 QNIDVW 240 KA 360
QQGHSYPLTFG 480 TITCHASQNIIDVWLSWYQQ KPGNIPKLLIYKASNLHTG
VPSRFSGRDSGTAFTLTIS SLQPEDIATYYCQQGHSYP LTFGSGTKLELK

(158) TABLE-US-00007 TABLE 1C Exemplary VH/VL pairings VH Name VL Name Antibody
Molecule (e.g., as listed in (e.g., as listed in Name Tables 1A, 2A, and 3A) Tables 1B, 2B, and 3B)
11 VH-11 VK-11 11v2 VH-11v2 VK-11 66 VH-66 VK-66 79 VH-79 VK-79 184 VH-184 VK-
184 184_C_Y VH-184 VK-184_C_Y 216 VH-216 VK-216_272 272 VH-272 VK-216_272 308
VH-308 VK-308 317 VH-317 VK-317 317_v2-1 VH-317v2 VK-317 317_v1-2 VH-317 VK-317v2
317_v2 VH-317v2 VK-317v2 322 VH-322 VK-322 322_v2 VH-322v2 VK-322 329 VH-329 VK-
329 330 VH-330 VK-330 332 VH-332 VK-332 335 VH-335 VK-335 336_v1 VH-336v1 VK-
336v1 336_v1-2 VH-336v1 VK-336v2 336_v2-1 VH-336v2 VK-336v1 336_v2 VH-336v2 VK-
336v2 336_v3-1 VH-336v3 VK-336v1 336_v3-2 VH-336v3 VK-336v2 338_v1 VH-338 VK-
338v1 338_v1-2 VH-338 VK-338v2 341_v1 VH-341v1 VK-341v1 341_v1-2 VH-341v1 VK-
341v2 341_v2-1 VH-341v2 VK-341v1 341_v2-2 VH-341v2 VK-341v2 343 VH-343 VK-343 399
VH-399 VK-399 402 VH-402 VK-402 416 VH-416 VK-416 429 VH-429 VK-429 430 VH-430
VK-430 440 VH-440 VK-440 453 VH-453 VK-453 454 VH-454 VK-454 465 VH-465 VK-465
475 VH-475 VK-475 481 VH-481 VK-481 497 VH-497 VK-497 502 VH-502 VK-502 503 VH-
503 VK-503 503_v2 VH-503 VK-503 504 VH-504 VK-504 507 VH-507 VK-507 508 VH-508
VK-508 510 VH-510 VK-510 511 VH-511 VK-511 511_v2 VH-511 VK-511 518 VH-511 VK-518
518_v2 VH-511 VK-518 528 VH-510 VK-528 521 VH-521 VK-521 530 VH-530 VK-530 536
VH-536 VK-530 541 VH-541 VK-541 547 VH-547 VK-547 549 VH-549 VK-549 550 VH-550
VK-550 553 VH-553 VK-553 556 VH-556 VK-556 557 VH-557 VK-557 567 VH-567 VK-567
568 VH-568 VK-568 570 VH-570 VK-567 573 VH-573 VK-567 583 VH-583 VK-583 584 VH-
584 VK-584 585 VH-585 VK-585 586 VH-586 VK-586 588 VH-588 VK-588 592 VH-592 VK-

(159) TABLE-US-00008 TABLE 2A The amino acid sequences of the heavy chain variable region (VH) and Chothia HCDRs of the exemplary anti-C5aR1 antibodies SEQ ID SEQ ID SEQ ID Name VH sequence NO: HCDR1 NO: HCDR2 NO: HCDR3 NO: VH-11 QVQLQQSGPELVKPGASVK 481 GYSFSSS 601 SPGDGD 721 RFLITSTRYVMDY 841 ISCKASGYSFSSSWMNWVK QRPKGKLEWIGRISPGDGD TRYSGKFKGKATLTADKSS STAYMQVTSLTSEDSAIYF CVRRFLITSTRYVMDYWGQ GTTVTVSS VH-11v2 QVQLQQSGPELVKPGASVK 482 GYSFSSS 602 SPGDGD 722 FLITSTRYVMDY 842 ISCKASGYSFSSSWMNWVK QRPKGKLEWIGRISPGDGD TRYSGKFKGKATLTADKSS STAYMQVTSLTSEDSAIYF CVRFLITSTRYVMDYWGQG TTVTVSS VH-66 QVQLQQSDAELVKPGASVK 483 GYTFIDH 603 SPGNGE 723 ALFYITGKYQPMY 843 ISCKASGYTFIDHAIHWVK QRPEQGLEWIGYISPGNGE IKYNEKFKAKATLTADKSS SAAYMQLNSLTSAADSAVYF CKRALFYITGKYQPMYWG QGTTVTDSS VH-79 QVQLKESGPGLVAPSQSL 484 GFSLSY 604 WTGGG 724 DGDYVYYAMAY 844 ITCTVSGFSLTSYAIWVR QPPKGKLEWLGVITGGGT KYNSALKSRLSISKDNSKS HVFLKMNSLQSDDTARYYC ARDGDYVYYAMAYWGQGT TTVTVSS VH-184 QVQLQQSGAELVKPGASVK 485 GYAFSRY 605 YPGDGD 725 SLGV 845 ISCKASGYAFSRYWMNWVK QRPKGKLEWIGQIYPGDGD TKYNGKFKGKATLTADKSS STAYMQLNSLTSEDSAVYF CTRSLGVWGTGTTTVTVSS VH-216 QIQLVQSGPELKKPGETVK 486 GYSFTTF 606 NTYSGV 726 GLGRLLAY 846 ISCKASGYSTTFGMSWVK QAPGKVLKWMGWINTYSGV PTYADDFKGRFAFSLETS A STAYLQINNKNEDTATYF CARGLGRLLAYWGQGTTLVT VSA VH-272 QVQLQQSGAELVKPGASVK 487 GYAFSSY 607 YPGDGD 727 SLGV 847 ISCKASGYAFSSYWMNWVK QRPKGKLEWIGHIYPGDGD TKYNGKFKGKATLTADKSS STAYMQVSSLTSEDSAVYF CTRSLGVWGTGTTTVTVSS VH-308 QVQLQQSGAELAKPGASVK 488 GYTFTSY 608 NPSSGY 728 GMFAMDY 848 MSCKASGYTFTSYWMHWVK QRPQGQLEWIGYINPSSGY TEYNQKFKDKATLTADKSS STAYMQLSSLTSEDSAVYY CARGMFAMDYWGQGT TTVTV SS VH-317 QVQLQQPGAELVMPGASVK 489 GYTFTSY 609 DPSDGY 729 EGF 849 LSCASGYTFTSYWLHWVR QRPQGQLEWIGEIDPSDGY SNHNQKFKGKATLTVDKSS STAYMQLSSLTSEDSAVYY CATEGFWGQGT TTVTVSS VH-317v2 QVQLQQPGAELVMPGASVK 490 GYTFTSY 610 DPSDSY 730 VAYYSNFGGFAY 850 LSCASGYTFTSYWMHWVK QRPQGQLEWIGEIDPSDSY TNYNQKFKGKATLTVDKSS STAYMQLSSLTSEDSAVYY CARVAYYSNFGGFAYWGQ TTVTVSS VH-322 QVQLQQSDAALVKPGASVK 491 GHTFTDH 611 SPGNGD 731 PLFVRGQYYITMDY 851 ISCKASGHTFTDHAIHWVK QRPEQGLEWIGYISPGNGD IKYNDKFKGKATLTADKSS STAYMQLNSLTPEDSAVYF CKGPLFVRGQYYITMDYWG QGTTTVTVSS VH-329 QVQLQQSGAELTKPGASVK 492 GYTFTNY 612 NPSSGY 732 SGGDNYGNPYYFDR 852 LSCASGYTFTNYWMHWVK QRPQGQLEWIGYLN PSSGY TKYNQKFKDKATLTADKSS STAYMQLNSLTYEDSAVYY CTRSGGDNYGNPYYFDRWG QGTTTVTVSS VH-330 QVQLKQSGPGLVQPSQSL 493 GFSLSY 613 WRGGS 733 NSQLGNAMDY 853 IACTVSGFSLTSYGVHWVR QSPKGKLEWLGVWIRGGST DYNAAFKSRLSITKDNSKS QVFFTMNRLHADDTAIYYC AKNSQLGNAMDYWGQGT TTV TVSS VH-332 QVQLQQSDAELVKPGASVK 494 GYTFTDH 614 SPGNGD 734 PLLRWRYFYPVDY 854 ISCKASGYTFTDHSIHVK QRPEQGLEWIGYISPGNGD IKYDEKFKGKATLTADTSS STAYMQLNSLTSEDSAVYF CKGPLLLRWRYFYPVDYWG QGTTTVTVSS VH-335 QVQLQQSDAALVKPGASVK 495 GYTFTDH 615 SPGNGD 735 PLLRWRYITMDY 855 ISCKASGYTFTDHAIHWVK QRPEQGLEWIGYISPGNGD IKYNEKFKGKATLTADKSS STAYMQLNSLTSEDSAVYF CKGPLLRWRYITMDYWG QGTTTVTVSS VH-336v1 EVKLEESGGGLVQPGGSMK 496 GFTFSDA 616 RNKANNH 736 GGYVVFAY 856

LSCAASGFTSFADSTDMVKNNAEATFVWAEIRKANN HATYAESVKGRFTISRDD
SKSSVYLQMNSLRAEDTGI YYCTRGGYYVFAYWGQGT VTVSS VH-336v2
QVQLKQSGPGLVQSSQSL 497 GFSLISY 617 WSGGS 737 NSQLGNAMDY 857
ITCTVSGFSLISYGVHWVR QSPGKGLEWLGVWSSGGST DYNAAFKSRLSITKDNSKS
QVFFKMNSLQADDTAIYYC AKNSQLGNAMDYWGQGT TVVSS VH-336v3
QVQLQQPGAELVKPGASVK 498 GYTFTSY 618 HPNSNS 738 SLTGTKTY 858
LSCKASGYTFTSYWMHWVK QRPQGQLEWIGMIHPNSNS TNYNEKFKSKATLTVDKSS
STAYMQLSSLTSEDSAVYY CARSLTGTKTYWGQGT TVT VSS VH-338
EVQLQQSGPELVKPGASVK 499 GYTFTDY 619 NPNNGG 739 GEGDYAY 859
MSCKASGYTFTDYNMHVVK QSHGKSLEWIGYINPNNGG TSYNQKFKGKATLTVNKSS
STAYMELRSLTSEDSAVYY CAHGEDYAYWGQGT TVTV SS VH-341v1
EFQLQQSGPELVKPGASVK 500 GYTFTKY 620 NPYNDG 740 ARATSY 860
MSCKASGYTFTKYVIHWVK QKPGQGLEWIGYINPYNDG TKYNEKFKGKARLTSDKSS
NTVYMDLSSLTSEDSAVYY CATARATSYWGQGT TVTVS S VH-341v2
QVQLQQPGAELVKPGASVK 501 GYSFTSY 621 YPGRGT 741 WGTGGRSY 861
MSCKASGYSFTSYWITWLK QRPQGQLEWIGDIYPGRGT TDYNEKLKSRATLTVDTS
TTAYMQLSSLTSEDSAVYY CARWGTGGRSYWGQGT TVT VSS VH-343
QVQLQQSGAELVKPGASVK 502 GYTFTEY 622 YPGSGS 742 HGNYDGSWFAY 862
LSCKASGYTFTEYTIHWVN QRSQGLEWIGWFYPGSGS IKYNEKFKDKATLTADKSS
HTVYMELSRSLTSEDSAVYY CARHGNYYDGSWFAYWGQ TLVTVSA VH-399
EVQLVESGGDLVKPGGSLK 503 GFTFSNY 623 TSGGTH 743 HGAYYSNPWFAY 863
LSCAASGFTFSNYGMSWVR QTPDKRLEWVATITSGGTH TYYPDSVKGRFTISRDNK
NTLYLQMSSLKSEDTAMYY CTRHGAYYSNPWFAYWGQ TLVTVS VH-402
QIQLVQSGPELKKPGETVK 504 GYTFTTF 624 NTNSGM 744 KSLFY 864
ISCKASGYTFTTFGMSWVK QAPGKGLKWMGWINTNSGM PTYTDDFRGRFAFSLETS
STAYLQISSLKNEEDTATYF CARKSLFYWGQGT TLTVSS VH-416
QVQLQQSGAELAKPGASVK 505 GYTFISY 625 NPRSDY 745 VTGTEGPYYFDY 865
LSCKASGYTFISYWMHWVK QRPQGQLEWIGYINPRSDY AKYNQKFKDKATLTNKS
STAYMQLSSLTYEDYAVYY CARVTGTEGPYYFDYWGQ TTLTVSS VH-429
QVQLQQSGAELAKPGASVK 506 GYTFSSY 626 NPRGDY 746 VTGSEGPYYFDY 866
LSCKASGYTFSSYWIHWVK QRPQGQLEWIGYINPRGDY TKYNQKFKDKATLTADKSS
STAFMQLSSLTYEDSAVYY CVRVTGSEGPYYFDYWGQ TTLTVSS VH-430
QVQLQQSGADLAKPGASVK 507 GYTFTSY 627 NPRGDY 747 VTGTEGPYYFDY 867
LSCKASGYTFTSYWIHWVK QRPQGQLEWIGYINPRGDY TKYNQKFKDKATLTADR
STAYMQLSSLTYEDYAVYY CARVTGTEGPYYFDYWGQ TTLTVSS VH-440
QIQLVQSGPELKKPGETVK 508 GYTFTAY 628 NTYSGV 748 SRYDGYFDY 868
ISCKASGYTFTAYGMSWVK QTPGKGLKWMGWINTYSGV PANADDFKGRFAFSLETS
STAYLQINNLIKNEEDTATYF CARSRYDGYFDYWGQ TTLTVSS VH-453
QVQLKQSGPGQVAPSQSL 509 GFSLINS 629 WSDGS 749 NGRLGNAMDY 869
ITCTVSGFSLINSAVHWVR QSPGKGLEWLGVWSDGST DYNATAFISRLSISRDNKS
QVFFKMNSLQVDDTAVYYC ARNGRLGNAMDYWGQGT TVVSS VH-454
QVQLKQSGPGLVAPSQSL 510 GFSLTSY 630 WGVGS 750 PYYSHYVPFAY 870
ITCTVSGFSLTSYGVDWIR QSPGKGLEWLGVWVGST NYNSALKSRLSISKDNSRS
QVFLKLNSLQTTDDTAMYYC ASPYYSHYVPFAYWGQGT LTVSA VH-465
QVQLQQSGAELAKPGASVK 511 GYIFTSY 631 NPSTTS 751 PDNSGYVGFAY 871
LSCKASGYIFTSYWMNWVK QRPQGQLEWIGYINPSTTS TKYNQKFKDKATLTADKSS
TTAYMQLTSLTYEDSAVYY CARPDNSGYVGFAYWGQGT LTVSA VH-475
QVQLKQSGPGLVAPSQSL 512 GFSLTSY 632 WGVGG 752 PYYSHYVPFAY 872
ITCTVSGFSLTSYGVDWIR QSPGKGLEWLGVWVGST NYNSALKSRLSISKDNSRS
QVFLKLNSLQTTDDTAMYYC ASPYYSHYVPFAYWGQGT LTVSA VH-481

QVQLQQSGAELAKPGASVK 513 GYSFTNY 753 NPSTDY 753 GLPFDY 873
VSCKASGYSFTRYWMHWIK QRPQGQLEWIGYINPSTDY SAYNQKFKDKATLTADKSS
STAYLQLTSLTSEDSAVYY CAGGLPHFDYWGGQTTLTV SS VH-497
QVQLKQSGPGLVAPSQSLS 514 GFSLTTY 634 WSDGS 754 NSRYGNSFAY 874
ITCTVSGFSLTTYGVHWVR QPPGKGLEWLVIWSDGST TYNSALKSRLSISKDNSKS
QVFLKMNSLQPDDTAMYYC ARNSRYGNSFAYWGQGTTLV TVSA VH-502
DVQLQESGPDLVKPSQSLS 515 GYSITSGY 635 HYSGS 755 WLPFDY 875
LTCTVTGYSITSGYSWHWI RQFPGNKLEWMGYIHYSGS TNYNPSLKSRLSITRDTSK
NQFFLQLKSVTTEDTATYY CVFWLPFDYWGGQTTLTVS S VH-503
DVQLQESGPGLVKPFQSLS 516 GYSITSDY 636 NYSGS 756 MGYRYPWFAY 876
LTCTVTGYSITSDYAWNWI RQFPGNKLEWMGYINYSGS TSYNPSLRSRLSITRDTSK
NQFFLHLNSVTTEDTATYY CARMGYRYPWFAYWGQGTTLV TVSA VH-504
EVQLKQSGGGLVKPGGSLK 517 GFTFSSY 637 SSGGSY 757 GGGNYDAMDY 877
LSCAASGFTFSSYTMWVR QTPEKRLEWVATISSGGSY TYYPDSVKGRFTISRDNK
NTLYLQMSSLKSEDTAMYY CTRGGGNYDAMDYWGQGTST VTVSS VH-507
QVQLQQSGAELAKPGASVK 518 GYTFTNY 638 NPSSAS 758 VPLPYGSSYGPYFF 878
LSCKASGYTFTNYWMHWVK DF QRPQGQLEWIGYINPSSAS SKYNQKFKDRATLTDDKSS
STAFMHLSSLTYEDSAVYY CARVPLPYGSSYGPYFFDF WGQGTTLTVSS VH-508
QVQLQQSGAELARPGASVK 519 GYTFTSY 639 NPSSGY 759 SGDYDGFAY 879
MSCKASGYTFTSYTMHWVK QRPQGQLEWIGYINPSSGY TKYNQKFKDKATLTADKSS
STAYMQLSSLTSEDSAVYY CARSGDYDGFAYWGQGTTLV TVSA VH-510
QVQLQQSGAELAKPGASVK 520 GYTFTSY 640 NPSSGY 760 VPLSYGSSYGPYFF 880
LSCKASGYTFTSYWMHWVK DY QRPQGQLEWIGYINPSSGY
TKYNQKFKDRATLTADKSS STAYMQLSSLTYEDSAVYY CARVPLSYGSSYGPYFFDY
WGQGTTLTVSS VH-511 QVQLQQSGAELAKPGASVK 521 GYTFTNY 641 NPSGDY 761
VPLSYGSGNGPYFF 881 LSCKTSGYTFTNYWMHWIK DY QRPGLGLEWIGYINPSGDY
TKHNQKFKDKATLTADRSS STAYMQLSSLTYEDSAVYY CARVPLSYGSGNGPYFFDY
WGQGTTLTVSS VH-521 QVQLQQSRAALAKPGASVK 522 GYTFTNH 642 NPINGF 762
VPLSYGGSYGPYFF 882 LSCKASGYTFTNHWMHWVK DF QRPQGQLEWIGYINPINGF
NRYNQNFKDRATLTDDKAS STAFIHLNGLTYEDFAVYY CARVPLSYGGSYGPYFFDF
WGQGTILTVSS VH-530 QVQLQQSRAALAKPGASVK 523 GYTFTNH 643 NPINGF 763
VPLSYGGSYGPYFF 883 LSCKASGYTFTNHWLHWVK DF QRPQGQLEWIGYINPINGF
SKYNQNFKNRATLTDDSSS STAFIHLNGLTYEDFAVYY CARVPLSYGGSYGPYFFDF
WGQGTILTVSS VH-536 QVQLQQSGAALAKPGASVK 524 GYSFTNY 644 NPINGY 764
VPLSYGGSYGPYFF 884 LSCKASGYSTFTNYWMHWVK DF QRPQGQLEWIGYINPINGY
GKYNQNFKDRATLTDDKSS STAFIHLNGLTYEDSAVYY CARVPLSYGGSYGPYFFDF
WGQGTILTVSS VH-541 QVQLQQSGPELVKPGESVK 525 GYTFTDY 645 YPNNGG 765
GSGPFAY 885 MSCKASGYTFTDYMDWVK QSHGKSLEWIGYFYPNNGG
VKYSQKFKDKAALTVDKSS TTAYMELHSLTFEDSAVYY CTRGSGPFAYWGQGTTLVTV SA
VH-547 QVQLKQSGPGLVAPSQSLS 526 GFSLTNY 646 WGDGI 766 ALDYSNYGFAY 886
ITCTVSGFSLTNYGVDWVR QSPGKGLEWLGVWGDGIT KYNSALKSRLSISKDNSKS
QVFLKMNSLQTDDTAMYYC ASALDYSNYGFAYWGQGTTLV TVSA VH-549
QVQLQQSGPELVKPGDSVK 527 GYTFTDY 647 YPNNGG 767 GSGPFAY 887
MSCKVSGYTFTDYIDWVK QSHGKSLEWIGYFYPNNGG AKYNQKFKSKAALTVDKSS
TTAYMELHSLTFEDSAVYY CTRGSGPFAYWGQGTTLVTV SA VH-550
QVQLKQSGPGLVAPSQSLS 528 GFSLTNC 648 WGDGL 768 ALDYSNYGFAY 888
ITCTVSGFSLTNCGVDWVR QSPGKSLEWLGVWGDGLT KYNSALKSRLSISKDNSKS
QVFLKVNSLQTDDTAVYYC ASALDYSNYGFAYWGQGTTLV TVSA VH-553
QVQLQQSGPELVKPGDSVK 529 GYTFTDY 649 YPNNGG 769 GSGPFAY 889
MSCKASGYTFTDYMDWVK QSHGKSLEWIGYFYPNNGG AKYNQKFKGKAALTVDKSS

TTAYMELHSLTSEDSAVYY CTRGSGPFAYWGQGLTV SA VH-556
 QVQLKQSGPGLVAPSQSLS 530 GFSLTNC 650 WGDGL 770 ALDYSNFGFAY 890
 ITCTVSGFSLTNCGVDWVR QSPGKSLEWLGVWGDGLT KYNSALKSRLSISKDNSKS
 QVFLKVNRLQTDDTAMYYC ASALDYSNFGFAYWGQGLT VTVSA VH-557
 QVQLQQSGPELVKPGDSVK 531 GYTFSDY 651 YPNNDG 771 GSGPFVY 891
 MSCKASGYTFSDYYMDWVK QSHGKSLEWIGYFYYPNNDG IRYNQRFKGRASLTVDKSS
 NTAYMELHSLTSEDSAVYY CARGSGPFVYWGQGLTV SA VH-567
 DVQLQESGPGLVKPSQSLS 532 GYSITSDS 652 SYSGR 772 ASIGFDY 892
 LTCTVTGYSITSDSAWNWI RQFPGNKLEWMGYISYSGR ISYNPSLKSRLSITRDTSK
 NQFFLQFNSVTTEDTAKYY CARASIGFDYWGGTTLTV SS VH-568
 EVQLQQSGADLVKPGASVK 533 GFNIKDS 653 DPTNVN 773 RLRQTYAMDY 893
 LSCTVSGFNKDSYIHWLK QRPQGGLWIGRIDPTNVN TKYDPKFQGGKASITTDTS
 NTAYLQLSSLTSENTAVYY CARRLRQTYAMDYWGQGT VTVSS VH-570
 DVQLQESGPGLVKPSQSLS 534 GYSITSDS 654 SYSGR 774 ASIGFDY 894
 LTCTVTGYSITSDSAWNWI RQFPGNKLEWMGYISYSGR ISYNPSLKSRLSITRDTSK
 NQIFLQFNSVTTEDTAKYY CARASIGFDYWGGTTLTV SS VH-573
 DVQLQESGPGLVKPSQSLS 535 GYSITSDS 655 SYSGR 775 ASIGFDY 895
 LTCTVTGYSITSDSAWNWI RQFPGNKLEWMGYISYSGR ISYNPSLKSRLSITRDTSK
 NQFFLQFYSVTTEDTARYY CARASIGFDYWGGTTLTV SS VH-583
 EVQLVESGGGLVKPGGSLK 536 GFTFNAY 656 STGGN 776 GYQRFSGFAY 896
 LSCAASGFTFNAYAMSWVR QTPEKRLEWVASISTGGNT YCPDSVKDRFTVSRDNVRN
 ILYLQMSSLRSEDAMYYC TRGYQRFSGFAYWGQGLTV TVSA VH-584
 DVQLQESGPDLVKPSQSLS 537 GYSITSGY 657 HNSGS 777 SIGDY 897
 LTCTVTGYSITSGYSWHWI RQSPGNKLEWLAYIHNSGS TNYNPSLKSRLSITRDTSK
 NQFFLKLNSVTTEDTATYY CARSIGDYWGQGTTLAVSS VH-585
 DVQLQESGPGLVKPSQSLS 538 GYSITSDY 658 SYSGS 778 YGGNYPTYAMDY 898
 LTCTVTGYSITSDYAWNWI RQFPGNKLEWMGYISYSGS TSYNPSLKSRLSITRDTSK
 NQFFLQLSSVTTEDTATYY CARYGGNYPTYAMDYWGQG TSVTVSS VH-586
 DVQLQESGPDLVKPSQSLS 539 GFSLTDSY 659 HYSGR 779 YDFAY 899
 LTCIVAGFSLTDSYSWHWI RQFPGNKLEWMGYIHYSGR TNYNPSLKTQFSITRNTSK
 NQFFLQLISVPTEDTATYY CARYDFAYWGRGTSVTVSS VH-588
 DVQLQESGPGLVKPSQSLS 540 GYSITSDY 660 SYSGS 780 TTGGYFDY 900
 LTCTVTGYSITSDYAWNWI RQFPGNKLEWMGYISYSGS IRYNPSLKSRLSITRDTSK
 NQFFLQLNSVTTEDTATYY CAITTGGYFDYWGGTTLT VSS VH-592
 QVQLKQSGAELVRPGSSVG 541 GYAFTNF 661 YPGDDD 781 TEVKRRRSFAY 901
 ISCKASGYAFTNFWMNWVR QRPQGGLWIGQLYPGDDD THYNGKFKGKVTTLADRSS
 GTAYMQLSRLTSEDSAVYF CAVTEVKRRRSFAYWGQGT LVTVSA

(160) TABLE-US-00009 TABLE 2B The amino acid sequences of the light chain variable region (VL) and Chothia LCDRs of the exemplary anti-C5aR1 antibodies
 SEQ ID SEQ ID SEQ ID SEQ ID Name VL sequence NO: LCDR1 NO: LCDR2 NO: LCDR3 NO: VK-11 DVVMTQTPLSLPVSLGDQA 542 RSSQSLVHSNGNTY 662 KVS NRFS 782 SQSTLVPPT 902 SISRSSQSLVHSNGNTYL LH HWYLQKPGQSPKLLIYKVS NRFS GVPDRFSGSGSGTDF TLKISRVEAEDLG VYFCSQ STLVPPTFGGGTKLEIK VK-66 DIVMTQSQKFMSTTAGDRV 543 KASQNVGSAVV 663 SSSIRYT 783 QQYNSFPLT 903 SITCKASQNVGSAVVWYQQ KPGRSPKLLIYSSSIRYTG VPDRFTGSGSGTDFTLTIN SVQSEDLADYFCQQYNSFP LTFGAGTKLEIK VK-79 DVVMTQTPLSLPVSLGDQA 544 RSSQSLVHSNGNTY 664 KVS NRFS 784 SQSTHVPPT 904 SISRSSQSLVHSNGNTYL LH HWYLQKPGQSPKLLIYKVS NRFS GVPDRFSGSGSGTDF KLKISRVEAEDLG VYFCSQ STHVPPTFGGGTKLEIK VK-184 DVVMTQTPLTSLVITIGQPA 545 KSSQSLLHSNGKTY 665 LVSKLDS 785 VQGTHVPYT 905 SISCKSSQSLLHSNGKTYL LN

NWLLQRPQGQSPKRLIYLVSKLDSGVDPDRFTGSGSGTDF TLKISRVEAEDLG VYCCVQ
GTHVPYTFGGGKLEIK VK-184_C_Y DVVMTQTPLTSLVTIGQPA 546
KSSQSLLHSNGKTY 666 LVSKLDS 786 VQGTHVPYT 906 SISCKSSQSLLHSNGKTYL LN
NWLLQRPQGQSPKRLIYLVSKLDSGVDPDRFTGSGSGTDF TLKISRVEAEDLG VYYCVQ
GTHVPYTFGGGKLEIK VK-216_272 DVVMTQTPLTSLVTIGQPA 547 KSSQSLLDSDGKTY
667 LVSKLDS 787 LQATHFPWT 907 SISCKSSQSLLDSDGKTYL LN
NWLLQRPQGQSPKRLIYLVSKLDSGVDPDRFTGSGSGTDF TLKISRVEAEDLG VYYCLQ
ATHFPWTFGGGKLEIK VK-308 DIVMTQSQNF MSTSVGDRV 548 KASQYVGTYVA 668
SASYRHT 788 QQYSSSPYT 908 SVTCKASQYVGTYVAWYQQ KPGQSPKALIYSASYRHTG
VPDRFTGSGSGTDFTLTIS NVQSEDLADYFCQQYSSSP YTFGGGKLEIK VK-317
DIVMTQSPSSLSVSAGEKV 549 KSSQSLLKSGNQKN 669 GASTRES 789 QNDHSHPYT 909
TMSCKSSQSLLKSGNQKNY YLA LAWHQQKPGQPPKLLIYGA
STRESGVPDRFTGSGSGTD FTLTISSVQAEDLAVYYCQ NDHSHPYTFGGGKLEIK VK-
317v2 DIVMSQSPSSLA VSAGEKV 550 KSSQSLLNSRTRKN 670 WASTRES 790 KQSYNLYT
910 TMSCKSSQSLLNSRTRKNY YLA LAWYQQKPGQSPKLLIYWA
STRESGVPDRFTGSGSGTD FTLTISSVQAEDLAVYYCK QSYNLYTFGGGKLEIK VK-322
DIVMTQSQKFMSTTVGDRV 551 KASQNVGA AVV 671 SASYRYS 791 QQYNSFPLT 911
SITCKASQNVGA AVVWYQQ KPGQSPKLLIYSASYRYS VPDRFTGSGSGTDFTLTIS
NMQSEDLADYFCQQYNSFP LTFGGGKLEIK VK-329 DVVMTQTPLSLTVSLGDQA 552
RSSQSLVHSNGNTY 672 KVS NRFS 792 SQSTLVPLT 912 SISRSSQSLVHSNGNTYL LH
HWYLQKPGQSPKFLIYKVS NRFSGVDPDRFSGSGSGTDF RLKISRVEAEDLG VYFCSQ
STLVPLTFGAGTKLEIK VK-330 DIVMSQSPSSLA VSAGEKV 553 KSSQSLFNSRTRKN 673
WASTRES 793 NQSYDLLT 913 TMSCKSSQSLFNSRTRKNY YLA
LAWYQQKPGQSPKLLIYWA STRESGVPDRFTGSGSGTD FTLTINSVQAEDLALYYCN
QSYDLLTFGAGTKLEIK VK-332 DIVMTQSQKFMSSAVGDRV 554 KASQNVGA AVA 674
SASIRYT 794 QQYNSFPLT 914 TITCKASQNVGA AVAWYQQ KPGQSPKLLLYSASIRYTG
VPDRFTGSGSGTDFTLTIS NIQSEDLAHFFCQQYNSFP LTFGGGKLEIK VK-335
DIVMTQSQKFMSTTVGDRV 555 KASQNVGA AVV 675 SASIRYT 795 QQYNSFPLT 915
SITCKASQNVGA AVVWYQQ KPGQSPKLLIYSASYRYTG VPDRFTGSGSGTDFTLTIS
NMQSEDVADYFCQQYNSFP LTFGGGKLEIK VK-336v1 DIVMSQSPSSLA VSAGEKV 556
KSSQSLLSSRTRKN 676 WASTRES 796 NQSYDLLT 916 TMSCKSSQSLLSSRTRKNY YLA
LAWYQQKPGQSPKLLIYWA STRESGVPDRFTGSGSGTD FTLTISSVQAEDLALYFCN
QSYDLLTFGAGTKLEIK VK-336v2 DIVMTQSPATLSVTPGDRV 557 RASQSIDYLH 677
YASQIS 797 QNGHSFPPT 917 SLSCRASQSIDYLHWYQQ KSHESPRLLIKYASQSIG
IPSRFSGSGSGSDFTLSIN SVEPEDVG VYYCQNGHSFP PTFGAGTKLEIK VK-338v1
DIVMTQSQKFMSTTLGDRV 558 KASQSVGA AVA 678 SASIRYA 798 QQYNSFPLT 918
SIPCKASQSVGA AVAWYQQ KPGQSPKLLIYSASIRYAG VPDRFTGSGSGTDFTLTVS
NMRSEDLADYFCQQYNSFP LTFGGGKLEIK VK-338v2 DIQMTQSSSYLSVSLGGRV 559
KASDHINN WLA 679 GATSLET 799 QQYWSTPYT 919 TITCKASDHINN WLA WYQQ
KPGNAPRLLISGATSLETG VPSRFSGSGSGKDYTLSIT SLQTEDVATYYCQQYWSTP
YTFGGGKLEIK VK-341v1 DVVMTQTPLTSLVTIGQPV 560 KSSQSLLDSDGKTY 680
LVSKLDS 800 LQATHFPHT 920 SISCKSSQSLLDSDGKTYL LN NWLLQRPGESPKLLIYLVSKLDSGVDPDRFTGSGSGTDF TLKISRVEAEDLG VFYCLQ ATHFPHTFGGGGKLEIK VK-341v2 DIQMTQSPSSLSASLGERV 561 RASQEIKTYLS 681 AATTLES 801 LQYASYPWT 921 SLTCRASQEIKTYLSWLQQ KPDGTIKRLIYAATTLESV VPKRFSGSWSGSEYSLTIS SLESEDFADYYCLQYASYP WTFGGGKLEIK VK-343 DVQITQSPSYLAASPGETI 562 RASKSISKYLA 682 SGSTLQS 802 QQHDEYPWT 922 TINCRASKSISKYLA WYQE KPGKTNKLLIYSGSTLQSG IPSRFSGSGSGTDFTLTIS SLEPEDFAMYYCQQHDEYP WTFGGGKLEIK VK-399 DIVMTQSQKFMSTSVGDRV 563 KASQNVGENVS 683 SASYRYS 803 QQYNSSPWT 923 SVTCKASQNVGENVSWYQQ

KPGQFPAALYASRYRSYRSPDFTGSDGTDLAEYFCQQYNSSP
WTFGGGTKLEIK VK-402 DVVMTQSPLTSLVTIGQPA 564 RSSRSLDSDGKTK 684
LVSKLDS 804 WQGTHFPWT 924 SISRSSRSLDSDGKTKL LH
HWLLQRPGQSPKSLIYLVSKLDSGVPNRFTGGGSGTDF TLKINRVEAEDLGVYYCWQ
GTHFPWTFGVGKLEIK VK-416 DVVMTQTPLSLPVSLGGQA 565 RSSQSLVHSNGNTY
685 KVSNRIS 805 SQSTLVPLT 925 SISRSSQSLVHSNGNTYL LH
HWYLQKPGQSPKFLIYKVS NRISGVPDRFSGSGSGTDF TLKISRVEAEDLGVYFCSQ
STLVPLTFGAGTKLELR VK-429 DVVMTQTPLSLPVSLGDQA 566 RSSQSLVHSNGNTY 686
KVSNRFS 806 SQSTLVPLT 926 SISRSSQSLVHSNGNTYL LQ
QWYLQKPGQSPKLLIYKVS NRFSGVPDRFSGSGSGTEF TLKIRRVEAEDLGVFLCSQ
STLVPLTFGAGTKLELK VK-430 DVVMTQTPLSLPVSLGDQV 567 RSSQSLVHSNGNTY 687
KVSNRFS 807 SQSTLVPLT 927 SISRSSQSLVHSNGNTYL LQ
QWYLQKPGQSPKLLIYKVS NRFSGVPDRFSGSGSGTDF TLKISRVEAEDLGVYFCSQ
STLVPLTFGAGTKLELR VK-440 ENVLTQSPAIMSASPGEKV 568 SASSSVSYMH 688
STSNLAS 808 QQRSSYPPT 928 TITCSASSSVSYMHWFQQK PGTSPKLWIYSTSNLASGV
PTRFSGSGSGTSYSLTISR MEAEDAATYYCQQRSSYPP TFGGGALEIK VK-453
DIVMTQSPSSLAVSVGEKV 569 KSSQSLFSSRTRKN 689 WASTRES 809 KQSYNLLT 929
TMSCKSSQSLFSSRTRKNY YLA LAWYQQKPGQSPKLLIYWA STRESGVPYRFTGSGSGTD
FTLTISVQTEDLAVYYCK QSYNLLTFGAGTKLEL VK-454 DVVMTQTPLSLPVSLGDQA
570 RSSQSLVHSNGNTY 690 RVSNRFS 810 SQSTHVPPT 930 SISRSSQSLVHSNGNTYL
LH HWYLQKPGQSPKLLIYRVS NRFSGVPDRFSGSGSGTDF TLKISRVEAEDLGVYFCSQ
STHVPPTFGGGTKLEIK VK-465 DVVMTQTPLSLPVSLGDQA 571 RSSQSLVHSTGNTF 691
KVSNRFS 811 SQSTLVPLT 931 SVSCRSSQSLVHSTGNTFL LH
HWYLQKPGQSPKLLIYKVS NRFSGVPDRFSGSGSGTDF TLKISRVEAEDLGVYFCSQ
STLVPLTFGAGTKLELK VK-475 DVVMTQTPLSLPVSLGDQA 572 RSSQSLVHSNGNTY 692
KVSNRFS 812 SQSTLVPT 932 SISRSSQSLVHSNGNTYL LH
HWYLQKPGQSPKLLIYKVS NRFSGLPDRFSGSGSGTDF TLKISRVEAEDLGVYFCSQ
STLVPTFGGGTKLEIK VK-481 DVVMTQTPLSLPVSLGDQA 573 RSSQSLVHNNGVTY 693
KVSNRFS 813 SQSTHVPIT 933 SISRSSQSLVHNNGVTYL LH
HWYLQKPGQSPKLLIYKVS NRFSGVPDRFSGSGSGTDF TLKISRVEAEDLGVYFCSQ
STHVPITFGAGTKLELK VK-497 DVVMTQTPLSLPVSLGDQA 574 RSSQSLVHSNGNTY 694
KVSNRFS 814 SQSTHVPPT 934 SISRSSQSLVHSNGNTYL LH
HWYLQKPGQSPKLLIYKVS NRFSGVPDRFSGSGSGTDF TLKISRVEAEDLGVYFCSQ
STHVPPTFGSGTKLEIK VK-502 DVVMTQTPLSLPVSLGDQA 575 RSSQSLVHSNGNTY 695
KVSNRFS 815 SQSTHVPYT 935 SISRSSQSLVHSNGNTYL LH
HWYLQKSGQSPKLLIYKVS NRFSGVPDRFSGSGSGTDF TLKISRVEAEDLGVYFCSQ
STHVPYTFGGGKLEIK VK-503 DIVLTQSPASLAVSLGQRA 576 RASKSVSTSGYSYL 696
LASNLES 816 QHNRELPT 936 TISCRASKSVSTSGYSYLH H WYQQKPGQPPKLLIYLASN
LESGVPARFSGSGSGTDFT LNIHPVEEEDAATYYCQHN RELPTFGGGTKLEIK VK-504
DVVMTQSPLTSLVTIGQPA 577 KSSQSLLYSNGKTY 697 LVSKLDS 817 VQGTHFPHT 937
SISCKSSQSLLYSNGKTYL LN NWLLQRPGQSPKRLIYLVSKLDSGVPDRFTGSGSGTDF
TLKISRVEAEDLGVYYCVQ GTHFPHTFGGGTKLEIK VK-507 DVVMTQTPLSLPVSLGDHA
578 RSSQSLIHSNGNNY 698 TVSNRFS 818 SQSTLVPLT 938 SISRSSQSLIHSNGNNYL LH
HWYLQKPGQSPKLLIYTVS NRFSGVPDRFSGSGSGTDF TLKISRVEAEDLGVYFCSQ
STLVPLTFGSGTKLEIK VK-508 DIVMTQSPSSLAMSVGQKV 579 KSSQSLLNSSSQKN 699
FASTRES 819 QQHYSTPPT 939 TMSCKSSQSLLNSSSQKNY YLA
LAWYQQKPGQSPKLLIYFA STRESGVPDRFIGSGSGTD FTLTISNVQAEDLADYFCQ
QHYSTPPTFGGGTKLEIK VK-510 DVVMTQTPLSLPVSLGDHA 580 RSSQSLIHSNGNNY
700 KVSNRFS 820 SQSTLVPLT 940 SISRSSQSLIHSNGNNYL LH
HWYLQKPGQSPQLLIYKVS NRFSGVPDRFSGSGSGTDF TLKISRVEAEDLGVYFCSQ

STLVPLTFGSGTKLEL VK-511 DVVMTQTPLSLPVSLGDHA 581 RSSQSLVHSNGNTY 701
KVS NRFS 821 SQSTLVPLT 941 SISCRSSQSLVHSNGNTYL LH HWYLQRPQGQSPKLLIHKVS
NRFS GVPDRFSGSGSGTDF TLKISRVEAEDVGIFYCSQ STLVPLTFGSGTKLEL VK-518
DVVMTQTPLSLPVSLGDHA 582 RSSQSLVHSNGNTY 702 KVS NRFS 822 SQSTLVPLT 942
SISCRSSQSLVHSNGNTYL LH HWYLQRPQGQSPKLLIHKVS NRFS GVPDRFSGSGSGTDF
TLKISRVEAEDVGIFYCSQ STLVPLTFGSGTKLEL VK-528 DVVMTQTPLSLPVSLGDHA
583 RSSQSLIHSNGNNY 703 KVS NRFS 823 SQSTLVPLT 943 SISCRSSQSLIHSNGNNYL LH
HWYLQKPGQSPKLLIYKVS NRFS GVPDRFSGSGSGTDF TLKISRVEAEDLGVYFCSQ
STLVPLTFGSGTKLEL VK-521 DVVMTQSPSLPVSLGDHA 584 RSSQSLIHSNGNNY 704
TVS NRFS 824 SQSTLVPLT 944 SISCRSSQSLIHSNGNNYL LH HWYLQKPGQSPKLLIYTVS
NRFS GVPDRFSGSGSGTDF TLKISRVEAEDLGVYFCSQ STLVPLTFGSGTKLEV VK-530
DVVMTQTPLSLPVSLGDHA 585 RSSQSLIHSNGNNY 705 TVS NRFS 825 SQSTLVPLT 945
SISCRSSQSLIHSNGNNYL LH HWYLQKPGQSPKLLIYTVS NRFS GVPDRFSGSGSGTDF
TLKISRVEAEDLGVYFCSQ STLVPLTFGSGTKLEV VK-541 DIVMTQTPLSLPVSLGDQA
586 RSSQSLVHSSGNTY 706 KVS NRFS 826 SQSTLVPVT 946 SISCRSSQSLVHSSGNTYL LH
HWYLQKPGQSPKLLIYKVS NRFS GVPDRFSGSGSGTDF TLRISRVAEDLGVYFCSQ
STLVPVTFGAGTELEL VK-547 DVVMTQTPLSLPVSLGNQA 587 RSSQRLVHSNGNTY
707 KVF NRFS 827 AQSTLVPPT 947 SISCRSSQRLVHSNGNTYL LH
HWYLQKPGQSPKLLIYKVF NRFS GVPDRFSGSGSGTDF TLKISRVEAEDLGIYFCAQ
STLVPPTFGGGTKLEIK VK-549 DVVMTQTPLSLPVSLGDQA 588 RSSQSLVHSSNTY 708
KVS NRFS 828 SQSTLVPVT 948 SISCRSSQSLVHSSNTYL LH HWYLQKPGQSPKLLIYKVS
NRFS GVPDRFSGSGSGTDF TLKISRVEAEDLGVYFCSQ STLVPVTFGAGTKLEL VK-550
DVVMTQTPLSLPVSLGNQA 589 RSSQSLVHSNGNTY 709 KVF NRFS 829 AQSTLVPPT 949
SISCRSSQSLVHSNGNTYL LH HWYLQKPGQSPRLLIYKVF NRFS GVPDRFSGSGSGTDF
TLKISRVEAEDLGVYFCAQ STLVPPTFGGGTKLEIK VK-553 DVVMTQTPLSLPVSLGDQA
590 RSSQSLVHSSNTY 710 KVS NRFS 830 SQSTLVPVT 950 SISCRSSQSLVHSSNTYL LH
HWYVQKPGQSPKLLIYKVS NRFS GVPDRFSGSGSGTDF TLKISRVEAEDLGVYFCSQ
STLVPVTFGAGTKLEL VK-556 DVVMTQTPLSLPVSLGNQA 591 RSSQSLVHSNGNTY
711 KVF NRFP 831 AQSTLVPPT 951 SISCRSSQSLVHSNGNTYL LH
HWYLQKPGQSPRLLIYKVF NRFP GVPDRFSGSGSGTDF TLKISRVEAEDLGVYFCAQ
STLVPPTFGGGTKLEIK VK-557 DVVMTQTPLSLPVSLGDQA 592 RSSQSLVHSSGNTY 712
KVS NRFS 832 SQSTLIPLT 952 SISCRSSQSLVHSSGNTYL LH HWYLQKPGQSPKLLIYKVS
NRFS GVPDRFSGSGSGTHF TLKLSRVEAEDLGIYFCSQ STLIPLTFGAGTKLEL VK-567
DIQMTQSPSSLSASLGEKV 593 RASQEISGYLS 713 AASTLDS 833 LHYANYPPT 953
SLTCRASQEISGYLSWLQQ KPDGSIKRLIYAASLDSG VPKRFSGSRSGSVYSLTIS
SLESEDFADYYCLHYANY PTFGGGTKLEIR VK-568 DVQMTQSPSSLSASLGTI 594
HASQNIYVWLN 714 KASDLHT 834 QQGLSYPLT 954 TITCHASQNIYVWLNWFQQ
KPGNIPKLLISKASDLHTG VPSRFSGSGSGTGFTLTIS SLQPEDIATYYCQQGLSYP
LTFGGGTNLEIK VK-583 DVVMTQTPLSPVSLGYQA 595 RSSQSLVHSNGNTY 715
KVS NRLS 835 SQSTHVPYT 955 SISCRSSQSLVHSNGNTYL LN
NWYLQKPGQSPKLLIYKVS NRLSGVPDRFSGSGSGTDF TLKISRVEDLGVYFCSQ
STHVPYTFGGGTKLEIK VK-584 DVVMTQTPLSLPVSLGDRA 596 RSGQSLVHSNGNTY
716 KVS NRFS 836 SQSTHVPWT 956 SISCRSGQSLVHSNGNTYL LH
HWYLQRPGRSPNLLIYKVS NRFS GVPDRFSGSGSGTDF TLKISRVEAEDLGVYFCSQ
STHVPWTFGGGTKLEIK VK-585 DVVMTQTPLSLPVSLGDQA 597 RFSQSIVYSNGNTY
717 KVS NRFS 837 FQGSHVPPT 957 SISCRFSQSIVYSNGNTYL LQ
QWYLQKPGQSPKLLIYKVS NRFS GVPDRFSGSGSGTDF TLKISRVEAEDLGVYYCFQ
GSHVPPTFGGGTKLEIK VK-586 DVVMTQTPLTSLVTIGQPT 598 KSSQSLLYSNGKTY 718
LVSKLDS 838 VQNTHLPYT 958 SISCKSSQSLLYSNGKTYL LS SWLLQRPQGQSPKRLLIYLV
KLDSGVPDRFTGSGSGTDF TLKINRVEAEDLGLYYCVQ NTHLPYTFGGGTKLEIR VK-588

DIQMTQSTSSLSASLGDRV 599 RASQDISNYLN 719 QISRHS 839 QQGNSLPPT 959
TISCRAQDISNYLNWYQQ TPDGTVKLLIYQISRLHSG VPSRFSGSGSGTDYSLTIS
NLEEEDIANYFCQQGNSLP PTFGGGTKVEIK VK-592 DIQMNQSPSSLSASLGDTI 600
HASQNIDVWLS 720 KASNLHT 840 QQGHSYPLT 960 TITCHASQNIDVWLSWYQQ
KPGNIPKLLIYKASNLHTG VPSRFSGRDSGTAFTLTIS SLQPEDIATYYCQQGHSYP
LTFGSGTKLELK
(161) TABLE-US-00010 TABLE 3A The amino acid sequences of the heavy
chain variable region (VH) and Kabat HCDRs of the exemplary anti-C5aR1
antibodies SEQ ID SEQ ID SEQ ID Name VH sequence NO: HCDR1 NO:
HCDR2 NO: HCDR3 NO: VH-11 QVQLQQSGPELVKPGASVK 961 SSWMN 1081
RISPGDGDTRYSGKFKG 1201 RFLITSTRYVMDY 1321 ISCKASGYSFSSSWMNWVK
QRPKGKLEWIGRISPGDGD TRYSGKFKGKATLTADKSS STAYMQVTSLTSEDSAIYF
CVRRLITSTRYVMDYWGQ GTTVTVSS VH-11v2 QVQLQQSGPELVKPGASVK 962
SSWMN 1082 RISPGDGDTRYSGKFKG 1202 FLITSTRYVMDY 1322
ISCKASGYSFSSSWMNWVK QRPKGKLEWIGRISPGDGD TRYSGKFKGKATLTADKSS
STAYMQVTSLTSEDSAIYF CVRFLITSTRYVMDYWGQG TTVTVSS VH-66
QVQLQQSDAELVKPGASVK 963 DHAIH 1083 YISPGNGEIKYNEKFKA 1203
ALFYITGKYQPM 1323 ISCKASGYTFIDHAIHWVK Y QRPEQGLEWIGYISPGNGE
IKYNEKFKAKATLTADKSS SAAYMQLNSLTADSADSAVYF CKRALFYITGKYQPM 1323
QGTITVDSS VH-79 QVQLKESGPGLVAPSQSL 964 SYAIS 1084 VIWTGGGTKYNSALKS
1204 DGDYVYYAMAY 1324 ITCTVSGFSLTSYAISWVR QPPKGKLEWLGVIWTGGGT
KYNSALKSRLSISKDNSKS HVFLKMNSLQSDDTARYYC ARDGDYVYYAMAYWGQGT
TVTVSS VH-184 QVQLQQSGAELVKPGASVK 965 RYWMN 1085 QIYPGDGDTKYNGKFKG
1205 SLGV 1325 ISCKASGYAFSRYWMNWVK QRPKGKLEWIGQIYPGDGD
TKYNGKFKGKATLTADKSS STAYMQLNSLTSEDSADSAVYF CTRSLGVWGTGTTTVTVSS VH-
216 QIQLVQSGPELVKPGASVK 966 TFGMS 1086 WINTYSGVPTYADDFKG 1206
GLGRLAY 1326 ISCKASGYSTFTFGMSWVK QAPGKVLKWMGWINTYSGV
PTYADDFKGRFAFSLETS STAYLQINNLTNEDTATYF CARGLGRLLAYWGQGTITV VSA
VH-272 QVQLQQSGAELVKPGASVK 967 SYWMN 1087 HIYPGDGDTKYNGKFKG 1207
SLGV 1327 ISCKASGYAFSSYWMNWVK QRPKGKLEWIGHIYPGDGD
TKYNGKFKGKATLTADKSS STAYMQVSSLTSEDSADSAVYF CTRSLGVWGTGTTTVTVSS VH-
308 QVQLQQSGAELAKPGASVK 968 SYWMH 1088 YINPSSGYTEYNQKFKD 1208
GMFAMDY 1328 MSCKASGYTFTSYWMHWVK QRPQGQLEWIGYINPSSGY
TEYNQKFKDKATLTADKSS STAYMQLSSLTSEDSADSAVY CARGMFAMDYWGQGTITV SS
VH-317 QVQLQQPGAELVMPGASVK 969 SYWLH 1089 EIDPSDGYSNHNQKFKG 1209
EGF 1329 LSCASGYTFTSYWLHWVR QRPQGQLEWIGEIDPSDGY
SNHNQKFKGKATLTVDKSS STAYMQLSSLTSEDSADSAVY CATEGFWGQGTITVTVSS VH-
317v2 QVQLQQPGAELVMPGASVK 970 SYWMH 1090 EIDPSDSYTNYNQKFKG 1210
VAYYSNFGGFAY 1330 LSCASGYTFTSYWMHWVK QRPQGQLEWIGEIDPSDSY
TNYNQKFKGKATLTVDKSS STAYMQLSSLTSEDSADSAVY CARVAYYSNFGGFAYWGQG
TTTVTVSS VH-322 QVQLQQSDAELVKPGASVK 971 DHAIH 1091 YISPGNGDIKYNDKFKG
1211 PLFVRGQYYITMD 1331 ISCKASGHTFTDHAIHWVK Y QRPEQGLEWIGYISPGNGD
IKYNDKFKGKATLTADKSS STAYMQLNSLTPEDSAVYF CKGPLFVRGQYYITMDYWG
QGTITVTVSS VH-329 QVQLQQSGAELTKPGASVK 972 NYWMH 1092
YLNPSGKYTKYNQKFKD 1212 SGGDNYGNPYYFD 1332 LSCASGYTFTNYWMHWVK R
QRPQGQLEWIGYLNPSGKYTKYNQKFKDKATLTADKSS STAYMQLNSLTYEDSAVY
CTRSGGDNYGNPYYFDRWG QGTITVTVSS VH-330 QVQLKQSGPGLVQPSQSL 973
SYGVH 1093 VIWRGGSTDYNAAFKS 1213 NSQLGNAMDY 1333
IACVSGFSLTSYGVHWVR QSPKGKLEWLGVIWRGGST DYNAAFKSRLSITKDNSKS
QVFTMNRLHADDTAIYC AKNSQLGNAMDYWGQGTITV TVSS VH-332

QVQLQQSDAELVKPGASVK 974 DHSIH 1094 YISPGNGDIKYDEKFKG 1214
PLLLRWRYFYPVD 1334 ISCKASGYTFTDHSIHWVK Y QRPEQGLEWIGYISPGNGD
IKYDEKFKGKATLTADTSS STAYMQLNSLTSEDSAVYF CKGPLLLRWRYFYPVDYWG
QGTTVTVSS VH-335 QVQLQQSDAALVKPGASVK 975 DHAIH 1095
YISPGNGDIKYNEKFKG 1215 PLLVRWRYYITMD 1335 ISCKASGYTFTDHAIHWVK Y
QRPEQGLEWIGYISPGNGD IKYNEKFKGKATLTADKSS STAYMQLNSLTSEDSAVYF
CKGPLLVRWRYYITMDYWG QGTTVTVSS VH-336v1 EVKLEESGGGLVQPGGSMK 976
DAWMD 1096 EIRNKANNHATYYAESV 1216 GGYVVFAY 1336
LSCAASGFTFSDAWMDWVR KG QSPEKGLEWVVAEIRNKANN
HATYYAESVKGRFTISRDD SKSSVYLQMNSLRAEDTGI YYCTRGGYYVFAYWGQGTT
VTVSS VH-336v2 QVQLKQSGPGLVQSSQSLS 977 SYGVH 1097 VIWSGGSTDYNAAFKS
1217 NSQLGNAMDY 1337 ITCTVSGFSLISYGVHWVR QSPGKGLEWLGVWSGGST
DYNAAFKSRLSITKDNSKS QVFFKMNSLQADDTAIYYC AKNSQLGNAMDYWGQGTTV
TVSS VH-336v3 QVQLQQPGAELVKPGASVK 978 SYWMH 1098 MIHPNSNSTNYNEKFKS
1218 SLTGTKTY 1338 LSCKASGYTFTSYWMHWVK QRPQGQLEWIGMIHPNSNS
TNYNEKFKSKATLTVDKSS STAYMQLSSLTSEDSAVYY CARSLTGTKTYWGQGTTVT VSS
VH-338 EVQLQQSGPELVKPGASVK 979 DYNMH 1099 YINPNNGGTSYNQKFKG 1219
GEGDYAY 1339 MSCKASGYTFTDYNMHWVK QSHGKSLEWIGYINPNNGG
TSYNQKFKGKATLTVNKSS STAYMELRSLTSEDSAVYY CAHGEGDYAYWGQGTTVTV SS
VH-341v1 EFQLQQSGPELVKPGASVK 980 KYVIH 1100 YINPYNDGTKYNEKFKG 1220
ARATSY 1340 MSCKASGYTFTKYVIHWVK QKPGQGLEWIGYINPYNDG
TKYNEKFKGKARLTSDKSS NTVYMDLSSLTSEDSAVYY CATARATSYWGQGTTVTVS S
VH-341v2 QVQLQQPGAELVKPGASVK 981 SYWIT 1101 DIYPGRGTTDYNEKLKS 1221
WGTTGRSY 1341 MSCKASGYSFTSYWITWLK QRPQGQLEWIGDIYPGRGT
TDYNEKLKSRATLTVDTSSTAYMQLSSLTSEDSAVYY CARWGTTGRSYWGQGTTVT
VSS VH-343 QVQLQQSGAELVKPGASVK 982 EYTIH 1102 WFYPGSGSIKYNEKFKD 1222
HGNYYDGSWFAY 1342 LSCKASGYTFTEYTIHWVN QRSGQGLEWIGWFYPGSGS
IKYNEKFKDKATLTADKSS HTVYMELSRSLTSEDSAVYF CARHGNYYDGSWFAYWGQG
TLVTVSA VH-399 EVQLVESGGDLVKPGGSLK 983 NYGMS 1103 TITSGGTHTYYPDSVKG
1223 HGAYYSNPWFAY 1343 LSCAASGFTFSNYGMSWVR QTPDKRLEWVATITSGGTH
TYYPDSVKGRFTISRDNALNTLYLQMSSLKSEDTAMYY CTRHGAYYSNPWFAYWGQG
TLVTVS VH-402 QIQLVQSGPELKKPGETVK 984 TFGMS 1104 WINTNSGMPTYTDDFRG
1224 KSLFY 1344 ISCKASGYTFTTFGMSWVK QAPGKGLKWMGWINTNSGM
PTYTDDFRGRFAFSLETSASTAYLQISSLKNEEDTATYF CARKSLFYWGQGTTTLTVSS VH-
416 QVQLQQSGAELAKPGASVK 985 SYWMH 1105 YINPRSDYAKYNQKFKD 1225
VTGTEGPYYFDY 1345 LSCKASGYTFISYWMHWVK QRPQGQLEWIGYINPRSDY
AKYNQKFKDKATLTITNKSS STAYMQLSSLTYEDYAVYY CARVTGTEGPYYFDYWGQG
TTLTVSS VH-429 QVQLQQSGAELAKPGASVK 986 SYWIH 1106 YINPRGDYTKYNQKFKD
1226 VTGSEGPYYFDY 1346 LSCKASGYTFSSYWIHWVK QRPQGQLEWIGYINPRGDY
TKYNQKFKDKATLTADKSS STAFMQLSSLTYEDSAVYY CVRVTGSEGPYYFDYWGQG
TTLTVSS VH-430 QVQLQQSGADLAKPGASVK 987 SYWIH 1107
YINPRGDYTKYNQKFKD 1227 VTGTEGPYYFDY 1347 LSCKASGYTFTSYWIHWVK
QRPQGQLEWIGYINPRGDY TKYNQKFKDKATLTADRSS STAYMQLSSLTYEDYAVYY
CARVTGTEGPYYFDYWGQG TTLTVSS VH-440 QIQLVQSGPELKKPGETVK 988 AYGMS
1108 WINTYSGVPANADDFKG 1228 SRYDGYFDY 1348 ISCKASGYTFTAYGMSWVK
QTPGKGLKWMGWINTYSGVPANADDFKGRAFSLETSASTAYLQINNLIKNEEDTATYF
CARSRYDGYFDYWGQGTTL TVSS VH-453 QVQLKQSGPGQVAPSQSLS 989 NSAVH 1109
VIWSDGSTDYNTAFIS 1229 NGRLGNAMDY 1349 ITCTVSGFSLINSAVHWVR
QSPGKGLEWLGVWSDGST DYNTAFISRLSISRDNSKS QVFFKMRSLLQVDDTAVYYC
ARNGRLGNAMDYWGQGTSV TVSS VH-454 QVQLKQSGPGLVAPSQSLS 990 SYGVD 1110

VIWGSTNYNSALX 1230 PYYSHYVPFAY 1350 ITCTVSGFSLTSYGVDWIR
QSPGKGLEWLGVWGVGST NYNSALKSRLSISKDNSRS QVFLKLNSLQTDDTAMYCY
ASPYSHYVPFAYWGQGT VTVSA VH-465 QVQLQQSGAELAKPGASVK 991 SYWMN
1111 YINPSTTSTKYNQKFKD 1231 PDNSGYVGFAY 1351 LSCKASGYFTSYWMNWVK
QRPQGQLEWIGYINPSTTS TKYNQKFKDKATLTADKSS TTAYMQLTSLTYEDSAVYY
CARPDNSGYVGFAYWGQGT LTVSA VH-475 QVQLKQSGPGLVAPSQSLS 992 SYGVD
1112 VIWGVGGTNYNSALKS 1232 PYYSHYVPFAY 1352 ITCTVSGFSLTSYGVDWIR
QSPGKGLEWLGVWGVGGT NYNSALKSRLSISKDNSRS QVFLKLNSLQTDDTAMYCY
ASPYSHYVPFAYWGQGT VTVSA VH-481 QVQLQQSGAELAKPGASVQ 993 RYWMH
1113 YINPSTDYSAYNQKFKD 1233 GLPHFDY 1353 VSCKASGYSFTRYWMHWIK
QRPQGQLEWIGYINPSTDY SAYNQKFKDKATLTADKSS STAYLQLTSLTSEDSAVYY
CAGGLPHFDYWGQGTTLTV SS VH-497 QVQLKQSGPGLVAPSQSLS 994 TYGVH 1114
VIWSDGSTTYNSALKS 1234 NSRYGNSFAY 1354 ITCTVSGFSLTTYGVHWVR
QPPGKGLEWLVIWSDGST TYNSALKSRLSISKDNSKS QVFLKMNSLQPDDTAMYCY
ARNSRYGNSFAYWGQGT VTVSA VH-502 DVQLQESGPDLVKPSQSLS 995 SGYSW 1115
YIHYSGSTNYNPSLKS 1235 WLPFDY 1355 LTCTVTGYSITSGYSWHWI H
RQFPGNKLEWMGYIHYSGS TNYNPSLKSRLSITRDTSK NQFFLQLKSVTTEDTATYY
CVFWLPFDYWGQGTTLTVS S VH-503 DVQLQESGPGLVKPFQSLS 996 SDYAW 1116
YINYSGSTSYNPSLRS 1236 MGYRYPWFAY 1356 LTCTVTGYSITSDYAWNWI N
RQFPGNKLEWMGYINYSGS TSYNPSLRSRLSITRDTSK NQFFLHLNSVTTEDTATYY
CARMGYRYPWFAYWGQGT VTVSA VH-504 EVQLKQSGGGLVKPGGSLK 997 SYTMS
1117 TISSGGSYTYYPDSVK 1237 GGGNYDAMDY 1357 LSCAASGFTFSSYTMSWVR
QTPEKRLEWVATISSGGSY TYYPDSVKGRFTISRDNAL NTLYLQMSSLKSEDTAMYY
CTRGGGNYDAMDYWGQGT VTVSS VH-507 QVQLQQSGAELAKPGASVK 998 NYWMH
1118 YINPSSASSKYNQKFKD 1238 VPLPYGSSYGPYF 1358 LSCKASGYTFTNYWMHWVK
FDF QRPQGQLEWIGYINPSSAS SKYNQKFKDRATLTADKSS STAFMHLSSLTYEDSAVYY
CARVPLPYGSSYGPYFFDF WGQGTTLTVSS VH-508 QVQLQQSGAELARPGASVK 999
SYTMH 1119 YINPSSGYTKYNQKFKD 1239 SGDYDGFAY 1359
MSCKASGYTFTSYTMHWVK QRPQGQLEWIGYINPSSGY TKYNQKFKDKATLTADKSS
STAYMQLSSLTSEDSAVYY CARSGDYDGFAYWGQGT VTVSA VH-510
QVQLQQSGAELAKPGASVK 1000 SYWMH 1120 YINPSSGYTKYNQKFKD 1240
VPLSYGSSYGPYF 1360 LSCKASGYTFTSYWMHWVK FDY QRPQGQLEWIGYINPSSGY
TKYNQKFKDRATLTADKSS STAYMQLSSLTYEDSAVYY CARVPLSYGSSYGPYFFDY
WGQGTTLTVSS VH-511 QVQLQQSGAELAKPGASVK 1001 NYWMH 1121
YINPSGDYTKHNQKFKD 1241 VPLSYGSGNGPYF 1361 LSCKTSGYTFTNYWMHWIK
FDY QRPGLGLEWIGYINPSGDY TKHNQKFKDKATLTADRSS STAYMQLSSLTYEDSAVYY
CARVPLSYGSGNGPYFFDY WGQGTTLTVSS VH-521 QVQLQQSRAALAKPGASVK 1002
NHWMH 1122 YINPINGFNRYNQNFKD 1242 VPLSYGSSYGPYF 1362
LSCKASGYTFTNHWMHWVK FDF QRPQGQLEWIGYINPINGF
NRYNQNFKDRATLTADKSS STAFIHLNGLTYEDFAVYY CARVPLSYGSSYGPYFFDF
WGQGTILTVSS VH-530 QVQLQQSRAALAKPGASVK 1003 NHWLH 1123
YINPINGFSKYNQNFKN 1243 VPLSYGSSYGPYF 1363 LSCKASGYTFTNHWLHWVK FDF
QRPQGQLEWIGYINPINGF SKYNQNFKNRATLTADSSS STAFIHLNGLTYEDFAVYY
CARVPLSYGSSYGPYFFDF WGQGTILTVSS VH-536 QVQLQQSGAALAKPGASVK 1004
NYWMH 1124 YINPINGYGKYNQNFKD 1244 VPLSYGSSYGPYF 1364
LSCKASGYSFTNYWMHWVK FDF QRPQGQLEWIGYINPINGY
GKYNQNFKDRATLTADKSS STAFIHLNGLTYEDSAVYY CARVPLSYGSSYGPYFFDF
WGQGTILTVSS VH-541 QVQLQQSGPELVKPGESVK 1005 DYYMD 1125
YFYPNNGGVKYSQKFKD 1245 GSGPFAY 1365 MSCKASGYTFTDYYMDWVK
QSHGKSLEWIGYFYPNNGG VKYSQKFKDKAALTVDKSS TTAYMELHSLTFEDSAVYY

CTRGSGPFAYWGQGLTV SA VH-447 QVQLKQSGPGLVAPSSLS 1006 NYGVD 1126
VIWGDGITKYNSALKS 1246 ALDYSNYGFAY 1366 ITCTVSGFSLTNYGVDWVR
QSPGKGLEWLGVWGDGIT KYNSALKSRLSISKDNSKS QVFLKMNSLQTDDTAMYYC
ASALDYSNYGFAYWGQGLT VTVSA VH-549 QVQLKQSGPELVKPGDSVK 1007 DYYID
1127 YFYNNNGGAKYNQKFKS 1247 GSGPFAY 1367 MSCKVSGYTFTDYYIDWVK
QSHGKSLEWIGYFYFNNNGG AKYNQKFKSKAALTVDKSS TTAYMELHSLTFEDSAVYY
CTRGSGPFAYWGQGLTV SA VH-550 QVQLKQSGPGLVAPSSLS 1008 NCGVD 1128
VIWGDGLTKYNSALKS 1248 ALDYSNYGFAY 1368 ITCTVSGFSLTNCGVDWVR
QSPGKSLEWLGVWGDGLT KYNSALKSRLSISKDNSKS QVFLKVNSLQTDDTAVYYC
ASALDYSNYGFAYWGQGLT VTVSA VH-553 QVQLKQSGPELVKPGDSVK 1009 DYYMD
1129 YFYNNNGGAKYNQKFKG 1249 GSGPFAY 1369 MSCKASGYTFTDYYMDWVK
QSHGKSLEWIGYFYFNNNGG AKYNQKFKGKAALTVDKSS TTAYMELHSLTFEDSAVYY
CTRGSGPFAYWGQGLTV SA VH-556 QVQLKQSGPGLVAPSSLS 1010 NCGVD 1130
VIWGDGLTKYNSALKS 1250 ALDYSNFGFAY 1370 ITCTVSGFSLTNCGVDWVR
QSPGKSLEWLGVWGDGLT KYNSALKSRLSISKDNSKS QVFLKVNRLQTDDTAMYYC
ASALDYSNFGFAYWGQGLT VTVSA VH-557 QVQLKQSGPELVKPGDSVK 1011 DYYMD
1131 YFYNNNDGIRYNQRFKG 1251 GSGPFVY 1371 MSCKASGYTFSDYYMDWVK
QSHGKSLEWIGYFYFNNNDG IRYNQRFKGRASLTVDKSS NTAYMELHSLTSEDSAVYY
CARGSGPFVYWGQGLTV SA VH-567 DVQLQESGPGLVKPSQSL 1012 SDSAW 1132
YISYSGRISYNPSLKS 1252 ASIGFDY 1372 LTCTVTGYSITSDSAWNWI N
RQFPGNKLEWMGYISYSGR ISYNPSLKSRLSITRDTSK NQFFLQFNSVTTEDTAKYY
CARASIGFDYWGQGTTLTV SS VH-568 EVQLKQSGADLVKPGASVK 1013 DSYIH 1133
RIDPTNVNTKYDPKFQ 1253 RLRQTYAMDY 1373 LSCTVSGFNIKDSYIHWLK
QRPQGQLEWIGRIDPTNVN TKYDPKFQKGASITTDTS NTAYLQLSSLTSENTAVYY
CARRLRQTYAMDYWGQGTSS VTVSS VH-570 DVQLQESGPGLVKPSQSL 1014 SDSAW
1134 YISYSGRISYNPSLKS 1254 ASIGFDY 1374 LTCTVTGYSITSDSAWNWI N
RQFPGNKLEWMGYISYSGR ISYNPSLKSRLSITRDTSK NQIFLQFNSVTTEDTAKYY
CARASIGFDYWGQGTTLTV SS VH-573 DVQLQESGPGLVKPSQSL 1015 SDSAW 1135
YISYSGRISYNPSLKS 1255 ASIGFDY 1375 LTCTVTGYSITSDSAWNWI N
RQFPGNKLEWMGYISYSGR ISYNPSLKSRLSITRDTSK NQFFLQFYSVTTEDTARYY
CARASIGFDYWGQGTTLTV SS VH-583 EVQLVESGGGLVKPGGSLK 1016 AYAMS 1136
SISTGGNTYCPDSVKD 1256 GYQRFSGFAY 1376 LSCAASGFTFNAYAMSWVR
QTPEKRLEWVASISTGGNT YCPDSVKDRFTVSRDNVRN ILYLQMSSLRSEDAMYYC
TRGYQRFSGFAYWGQGLTV TVS VH-584 DVQLQESGPDLVKPSQSL 1017 SGYSW 1137
YIHNSGSTNYPNPSLKS 1257 SIGDY 1377 LTCTVTGYSITSGYSWHWI H
RQSPGNKLEWLAYIHNSGS TNYNPSLKSRLSITRDTSK NQFFLKLNSVTTEDTATYY
CARSIGDYWGQGTTLAVSS VH-585 DVQLQESGPGLVKPSQSL 1018 SDYAW 1138
YISYSGSTSYNPSLKS 1258 YGGNYPTYAMDY 1378 LTCTVTGYSITSDYAWNWI N
RQFPGNKLEWMGYISYSGS TSYNPSLKSRLSITRDTSK NQFFLQLSSVTTEDTATYY
CARYGGNYPTYAMDYWGQG TSVTVSS VH-586 DVQLQESGPDLVKPSQSL 1019
DSYSW 1139 YIHYSGRTNYPNPSLKT 1259 YDFAY 1379 LTCIVAGFSLTDSYSWHWI H
RQFPGNKLEWMGYIHYSGR TNYNPSLKTQFSITRNTSK NQFFLQLISVPTEDTATYY
CARYDFAYWGRGTSVTVSS VH-588 DVQLQESGPGLVKPSQSL 1020 SDYAW 1140
YISYSGSIRYNPSLKS 1260 TTGGYFDY 1380 LTCTVTGYSITSDYAWNWI N
RQFPGNKLEWMGYISYSGS IRYNPSLKSRLSITRDTSK NQFFLQLNSVTTEDTATYY
CAITTGGYFDYWGQGTTLT VSS VH-592 QVQLKQSGAELVRPGSSVG 1021 NFWMN 1141
QLYPGDDDDTHYNGKFKG 1261 TEVKRRRSFAY 1381 ISCKASGYAFTNFWMNWVR
QRPQGQLEWIGQLYPGDDDD THYNGKFKGKVTLTADRSS GTAYMQLSRLTSEDSAVYF
CAVTEVKRRRSFAYWGQGT LTVSA

variable region (VL) and LCDR3 of the exemplary anti-C5aR1 antibodies are provided as follows. SEQ SEQ SEQ SEQ ID ID ID ID Name VL sequence NO: LCDR1 NO: LCDR2 NO: LCDR3 NO: VK-11 DVVMTQTPLSLPVSLGDQASI 1022 RSSQSLVHSNGNTYLH 1142 KVSNRFS 1262 SQSTLVPPT 1382 SCRSSQSLVHSNGNTYLHWYL QKPGQSPKLLIYKVSNRFSGV PDRFSGSGSGTDFTLKISRVE AEDLGVYFCSQSTLVPPTFGG GTKLEIK VK-66 DIVMTQSQKFMSTTAGDRVSI 1023 KASQNVGSAVV 1143 SSSIRYT 1263 QQYNSFPLT 1383 TCKASQNVGSAVVWYQQKPGR SPKLLIYSSSIRYTGVPDRFT GSGSGTDFTLTINSVQSEDLA DYFCQQYNSFPLTFGAGTKLE IK VK-79 DVVMTQTPLSLPVSLGDQASI 1024 RSSQSLVHSNGNTYLH 1144 KVSNRFS 1264 SQSTHVPPT 1384 SCRSSQSLVHSNGNTYLHWYL QKPGQSPKLLIYKVSNRFSGV PDRFSGSGSGTDGFKLKISRVE AEDLGVYFCSQSTHVPPTFGG GTKLEIK VK-184 DVVMTQTPLTSLVTIGQPASI 1025 KSSQSLLHSNGKTYLN 1145 LVSKLDS 1265 VQGTHVPYT 1385 SCKSSQSLLHSNGKTYLNWLL QRPQGSPKRLIYLVSKLDSGV PDRFTGSGSGTDFTLKISRVE AEDLGVYCCVQGTHVPYTFGG GTKLEIK VK-184_C_Y DVVMTQTPLTSLVTIGQPASI 1026 KSSQSLLHSNGKTYLN 1146 LVSKLDS 1266 VQGTHVPYT 1386 SCKSSQSLLHSNGKTYLNWLL QRPQGSPKRLIYLVSKLDSGV PDRFTGSGSGTDFTLKISRVE AEDLGVYYCVQGTHVPYTFGG GTKLEIK VK-216_272 DVVMTQTPLTSLVTIGQPASI 1027 KSSQSLLDSDGKTYLN 1147 LVSKLDS 1267 LQATHFPWT 1387 SCKSSQSLLDSDGKTYLNWLL QRPQGSPKRLIYLVSKLDSGV PDRFTGSGSGTDFTLKISRVE AEDLGVYYCLQATHFPWTFGG GTKLEIK VK-308 DIVMTQSQNFMSTSVGDRVSV 1028 KASQYVGTYVA 1148 SASYRHT 1268 QQYSSSPYT 1388 TCKASQYVGTYVAWYQQKPGQ SPKALIYSASYRHTGVPDRFT GSGSGTDFTLTISNVQSEDLA DYFCQQYSSSPYTFGGGTKLE IK VK-317 DIVMTQSPSSLSVSAGEKVTM 1029 KSSQSLLKSGNQKNYL 1149 GASTRES 1269 QNDHSHPYT 1389 SCKSSQSLLKSGNQKNYLAWH A QKPGQPPKLLIYGASTRESG VPDRFTGSGSGTDFTLTISV QAEDLAVYYCQNDHSHPYTFG GGTKLEIK VK-317v2 DIVMSQSPSSLAVSAGEKVTM 1030 KSSQSLLNSRTRKKNYL 1150 WASTRES 1270 KQSYNLYT 1390 SCKSSQSLLNSRTRKKNYLAWY A QKPGQSPKLLIYWASTRESG VPDRFTGSGSGTDFTLTISV QAEDLAVYYCKQSYNLYTFGG GTKLEIK VK-322 DIVMTQSQKFMSTTVGDRVSI 1031 KASQNVGAADV 1151 SASYRYS 1271 QQYNSFPLT 1391 TCKASQNVGAADVWYQQKPGQ SPKLLIYSASYRYSGVPDRFT GSGSGTDFTLTISNMQSEDLA DYFCQQYNSFPLTFGGGTKLE IK VK-329 DVVMTQTPLSLTVSLGDQASI 1032 RSSQSLVHSNGNTYLH 1152 KVSNRFS 1272 SQSTLVPLT 1392 SCRSSQSLVHSNGNTYLHWYL QKPGQSPKFLLIYKVSNRFSGV PDRFSGSGSGTDGFKLKISRVE AEDLGVYFCSQSTLVPLTFGA GTKLEIK VK-330 DIVMSQSPSSLAVSVGEKVTM 1033 KSSQSFLNSRTRKKNYL 1153 WASTRES 1273 NQSYDLLT 1393 SCKSSQSFLNSRTRKKNYLAWY A QKPGQSPKLLIYWASTRESG VPDRFTGSGSGTDFTLTINSV QAEDLALYYCNQSYDLLTFGA GTKLEIK VK-332 DIVMTQSQKFMSSAVGDRVTI 1034 KASQNVGAAVA 1154 SASIRYT 1274 QQYNSFPLT 1394 TCKASQNVGAAVAWYQQKPGQ SPKLLLYSASIRYTGVPDRFT GSGSGTDFTLTISNIQSEDLA HFFCQQYNSFPLTFGGGTKLE IK VK-335 DIVMTQSQKFMSTTVGDRVSI 1035 KASQNVGAADV 1155 SASYRYT 1275 QQYNSFPLT 1395 TCKASQNVGAADVWYQQKPGQ SPKLLIYSASYRYTGVPDRFT GSGSGTDFTLTISNMQSEDLA DYFCQQYNSFPLTFGGGTKLE IK VK-336v1 DIVMSQSPSSLAVSAGEKVTM 1036 KSSQSLLSSRTRKKNYL 1156 WASTRES 1276 NQSYDLLT 1396 SCKSSQSLLSSRTRKKNYLAWY A QKPGQSPKLLIYWASTRESG VPDRFTGSGSGTDFTLTISV QAEDLALYFCNQSYDLLTFGA GTKLEIK VK-336v2 DIVMTQSPATLSVTPGDRVSL 1037 RASQSIDYLH 1157 YASQSID 1277 QNGHSFPPT 1397 SCRASQSIDYLHWYQQKSHE SPRLLIKYASQSIDGIPSRFS GSGSGSDFTLSINSVEPEDVG

VYYCQNGHFGPPTFKMSTTTLGDRVSI 1038
KASQSVGAAVA 1158 SASIRYA 1278 QQYNSFPLT 1398 PCKASQSVGA AVAWYQQKPGQ
SPKLLIYASIRYAGVPDRFT GSGSGTDFTLTVSNMRSEDLA
DYFCQQYNSFPLTFGGGKLE IK VK-338v2 DIQMTQSSSYLSVSLGGRVTI 1039
KASDHINNWL 1159 GATSLET 1279 QQYWSTPYT 1399 TCKASDHINNWLAWYQQKPGN
APRLISGATSLETGVPSRFS GSGSGKDYTLSITSLQTEDVA
TYYCQQYWSTPYTFGGGKLE IK VK-341v1 DVVMTQTPLTSLVTIGQPVS I 1040
KSSQSLLESDGKTYLN 1160 LVSKLDS 1280 LQATHFPHT 1400
SCKSSQSLLESDGKTYLNWLL QRPGESPKLLIYLVSKLDSGV
PDRFTGSGSGTDFTLKISRVE AEDLGVFYCLQATHFPHTFGG GTKLEIK VK-341v2
DIQMTQSPSSLSASLGERVSL 1041 RASQEIKTYLS 1161 AATTLES 1281 LQYASYPWT 1401
TCRASQEIKTYLSWLQQKPDG TIKRLIYAATTLESVVPKRFS GSWSGSEYSLTISSLESEDA
DYYCLQYASYPWTFGGGKLE IK VK-343 DVQITQSPSYLAASPGETITI 1042
RASKSISKYLA 1162 SGSTLQS 1282 QQHDEYPWT 1402 NCRASKSISKYLA WYQEKPGK
TNKLLIYSGSTLQSGIPSRFS GSGSGTDFTLTISSLEPEDFA
MYYCQQHDEYPWTFGGGKLE IK VK-399 DIVMTQSQKFMSTSVGDRVSV 1043
KASQNVGENVS 1163 SASYRYS 1283 QQYNSSPWT 1403
TCKASQNVGENVSWYQQKPGQ FPKALIYSASYRYS GVPDRFT
GSGSGTDFSLTISNVQSEDLA EYFCQQYNSSPWTFGGGKLE IK VK-402
DVVMTQSPLTSLVTIGQPASI 1044 RSSRSLDSDGKTKLH 1164 LVSKLDS 1284
WQGTHFPWT 1404 SCRSSRSLDSDGKTKLHWLL QRPQGSPKSLIYLVSKLDSGV
PNRFTGGGSGTDFTLKINRVE AEDLGVYYCWQGTHFPWTFGV GTKLEIK VK-416
DVVMTQTPLSLPVSLGGQASI 1045 RSSQSLVHSNGNTYLH 1165 KVS NRIS 1285
SQSTLVPLT 1405 SCRSSQSLVHSNGNTYLHWYL QKPGQSPKFLIYKVS NRISGV
PDRFSGSGSGTDFTLKISRVE AEDLGVYFCSQSTLVPLTFGA GTKLELR VK-429
DVVMTQTPLSLPVSLGDQASI 1046 RSSQSLVHSNGNTYLQ 1166 KVS NRFS 1286
SQSTLVPLT 1406 SCRSSQSLVHSNGNTYLQWYL QKPGQSPKLLIYKVS NRFSGV
PDRFSGSGSGTEFTLKIRRVE AEDLGVFLCSQSTLVPLTFGA GTKLELK VK-430
DVVMTQTPLSLPVSLGDQVSI 1047 RSSQSLVHSNGNTYLQ 1167 KVS NRFS 1287
SQSTLVPLT 1407 SCRSSQSLVHSNGNTYLQWYL QKPGQSPKLLIYKVS NRFSGV
PDRFSGSGSGTDFTLKISRVE AEDLGVYFCSQSTLVPLTFGA GTKLELR VK-440
ENVLTQSPAIMSASPGEKVTI 1048 SASSSVSYMH 1168 STSNLAS 1288 QQRSSYPPT 1408
TCSASSSVSYMHWFQQKPGTS PKLWIYSTSNLASGVPTRFSG
SGSGTSYSLTISRMEAEDAAT YYCQQRSSYPPTFGGGAKLEI K VK-453
DIVMTQSPSSLAVSVGEKVTM 1049 KSSQSLFSSRTRKNYL 1169 WASTRES 1289
KQSYNLLT 1409 SCKSSQSLFSSRTRKNYLAWY A QKPGQSPKLLIYWASTRESG
VPYRFTGSGSGTDFTLTISSV QTEDLAVYYCKQSYNLLTFGA GTKLEL VK-454
DVVMTQTPLSLPVSLGDQASI 1050 RSSQSLVHSNGNTYLH 1170 RVSNRFS 1290
SQSTHVPPT 1410 SCRSSQSLVHSNGNTYLHWYL QKPGQSPKLLIYRVSNRFSGV
PDRFSGSGSGTDFTLKISRVE AEDLGVYFCSQSTHVPPTFGG GTKLEIK VK-465
DVVMTQTPLSLPVSLGDQASV 1051 RSSQSLVHSTGNTFLH 1171 KVS NRFS 1291
SQSTLVPLT 1411 SCRSSQSLVHSTGNTFLHWYL QKPGQSPKLLIYKVS NRFSGV
PDRFSGSGSGTDFTLKISRVE AEDLGVYFCSQSTLVPLTFGA GTKLELK VK-475
DVVMTQTPLSLPVSLGDQASI 1052 RSSQSLVHSNGNTYLH 1172 KVS NRFS 1292
SQSTLVPT 1412 SCRSSQSLVHSNGNTYLHWYL QKPGQSPKLLIYKVS NRFSGL
PDRFSGSGSGTDFTLKISRVE AEDLGVYFCSQSTLVPTFGG GTKLEIK VK-481
DVVMTQTPLSLPVSLGDQASI 1053 RSSQSLVHNNGVTYLH 1173 KVS NRFS 1293
SQSTHVPIT 1413 SCRSSQSLVHNNGVTYLHWYL QKPGQSPKLLIYKVS NRFSGV
PDRFSGSGSGTDFTLKISRVE AEDLGVYFCSQSTHVPITFGA GTKLELK VK-497
DVVMTQTPLSLPVSLGDQASI 1054 RSSQSLVHSNGNTYLH 1174 KVS NRFS 1294

SQSTHVPPT 1414 SCRSSQSLVHSNGNTYLHWYL QKPGQSPKLLIYKVS NRFSGV
PDRFSGSGSGTDFTLKISRVE AEDLGVYFCSQSTHVPPTFGS GTKLEIK VK-502
DVVMTQTPLSLPVSLGDQASI 1055 RSSQSLVHSNGNTYLH 1175 KVS NRFS 1295
SQSTHVPYT 1415 SCRSSQSLVHSNGNTYLHWYL QKSGQSPKLLIYKVS NRFSGV
PDRFSGSGSGTDFTLKISRVE AEDLGVYFCSQSTHVPYTFGG GTKLEIK VK-503
DIVLTQSPASLAVSLGQRATI 1056 RASKSVSTSGYSYLH 1176 LASNLES 1296
QHNREL PPT 1416 SCRASKSVSTSGYSYLHWYQQ KPGQPPKLLIYLASNLES GP
ARFSGSGSGTDFTLNIHPVEE EDAATYYCQHNREL PPTFGGG TKLEIK VK-504
DVVMTQSPLTSLVTIGQPASI 1057 KSSQSLLYSNGKTYLN 1177 LVSKLDS 1297
VQGTHFPHT 1417 SCKSSQSLLYSNGKTYLNWLL QRPQGQSPKRLIYLVSKLDSGV
PDRFTGSGSGTDFTLKISRVE AEDLGVYYCVQGTHFPHTFGG GTKLEIK VK-507
DVVMTQTPLSLPVSLGDHASI 1058 RSSQSLIHSNGN NYLH 1178 TVSNRFS 1298
SQSTLVPLT 1418 SCRSSQSLIHSNGN NYLHWYL QKPGQSPKLLIYTVSNRFSGV
PDRFSGSGSGTDFTLKISRVE AEDLGVYFCSQSTLVPLTFGS GTKLEIK VK-508
DIVMTQSPSSLAMSVGQKVTM 1059 KSSQSLN SSSQKNYL 1179 FASTRES 1299
QQHYSTPPT 1419 SCKSSQSLN SSSQKNYLAWY A QKPGQSPKLLIYFASTRESG
VPDRFIGSGSGTDFTLTISNV QAEDLADYFCQQHYSTPPTFG GGTKLEIK VK-510
DVVMTQTPLSLPVSLGDHASI 1060 RSSQSLIHSNGN NYLH 1180 KVS NRFS 1300
SQSTLVPLT 1420 SCRSSQSLIHSNGN NYLHWYL QKPGQSPQLLIYKVS NRFSGV
PDRFSGSGSGTDFTLKISRVE AEDLGVYFCSQSTLVPLTFGS GTKLELR VK-511
DVVMTQTPLSLPVSLGDHASI 1061 RSSQSLVHSNGNIYLH 1181 KVS NRFS 1301
SQSTLVPLT 1421 SCRSSQSLVHSNGNIYLHWYL QRPQGQSPKLLIHKVS NRFSGV
PDRFSGSGSGTDFTLKISRVE AEDVGIYFCSQSTLVPLTFGS GTKLELK VK-518
DVVMTQTPLSLPVSLGDHASI 1062 RSSQSLVHSNGNTYLH 1182 KVS NRFS 1302
SQSTLVPLT 1422 SCRSSQSLVHSNGNTYLHWYL QRPQGQSPKLLIHKVS NRFSGV
PDRFSGSGSGTDFTLKISRVE AEDVGIYFCSQSTLVPLTFGS GTKLELK VK-528
DVVMTQTPLSLPVSLGDHASI 1063 RSSQSLIHSNGN NYLH 1183 KVS NRFS 1303
SQSTLVPLT 1423 SCRSSQSLIHSNGN NYLHWYL QKPGQSPKLLIYKVS NRFSGV
PDRFSGSGSGTDFTLKISRVE AEDLGVYFCSQSTLVPLTFGS GTKLELK VK-521
DVVMTQSPLSLPVSLGDHASI 1064 RSSQSLIHSNGN NYLH 1184 TVSNRFS 1304
SQSTLVPLT 1424 SCRSSQSLIHSNGN NYLHWYL QKPGQSPKLLIYTVSNRFSGV
PDRFSGSGSGTDFTLKISRVE AEDLGVYFCSQSTLVPLTFGS GTKLEVK VK-530
DVVMTQTPLSLPVSLGDHASI 1065 RSSQSLIHSNGN NYLH 1185 TVSNRFS 1305
SQSTLVPLT 1425 SCRSSQSLIHSNGN NYLHWYL QKPGQSPKLLIYTVSNRFSGV
PDRFSGSGSGTDFTLKISRVE AEDLGVYFCSQSTLVPLTFGS GTKLEVK VK-541
DIVMTQTPLSLPVSLGDQASI 1066 RSSQSLVHSSGNTYLH 1186 KVS NRFS 1306
SQSTLVPVT 1426 SCRSSQSLVHSSGNTYLHWYL QKPGQSPKLLIYKVS NRFSGV
PDRFSGSGSGTDFTLRISVA AEDLGVYFCSQSTLVPVTFGA GTELELK VK-547
DVVMTQTPLSLPVSLGNQASI 1067 RSSQRLVHSNGNTYLH 1187 KVFNRF 1307
AQSTLVPPT 1427 SCRSSQRLVHSNGNTYLHWYL QKPGQSPKLLIYKVFNRFSGV
PDRFSGSGSGTDFTLKISRVE AEDLGIYFCAQSTLVPPTFGG GTKLEIK VK-549
DVVMTQTPLSLPVSLGDQASI 1068 RSSQSLVHSSNTYLH 1188 KVS NRFS 1308
SQSTLVPVT 1428 SCRSSQSLVHSSNTYLHWYL QKPGQSPKLLIYKVS NRFSGV
PDRFSGSGSGTDFTLKISRVE AEDLGVYFCSQSTLVPVTFGA GTKLELK VK-550
DVVMTQTPLSLPVSLGNQASI 1069 RSSQSLVHSNGNTYLH 1189 KVFNRF 1309
AQSTLVPPT 1429 SCRSSQSLVHSNGNTYLHWYL QKPGQSPRLIYKVFNRFSGV
PDRFSGSGSGTDFTLKISRVE AEDLGVYFCAQSTLVPPTFGG GTKLEIK VK-553
DVVMTQTPLSLPVSLGDQASI 1070 RSSQSLVHSSNTYLH 1190 KVS NRFS 1310
SQSTLVPVT 1430 SCRSSQSLVHSSNTYLHWYV QKPGQSPKLLIYKVS NRFSGV
PDRFSGSGSGTDFTLKISRVE AEDLGVYFCSQSTLVPVTFGA GTKLELK VK-556

DVVMTQTPLSLPVSLGQASI 1071 RSSQSLVHSSGNTYLH 1191 KVSNNRFP 1311
 AQSTLVPPT 1431 SCRSSQSLVHSSGNTYLHWYL QKPGQSPRLLIYKVFNRFPGV
 PDRFSGSGSGTDFTLKISRVE AEDLGVYFCAQSTLVPPTFGG GTKLEIK VK-557
 DVVMTQTPLSLPVSLGQASI 1072 RSSQSLVHSSGNTYLH 1192 KVSNNRFS 1312
 SQSTLIPLT 1432 SCRSSQSLVHSSGNTYLHWYL QKPGQSPKLLIYKVSNNRFSGV
 PDRFSGSGSGTHFTLKISRVE AEDLGIYFCSQSTLIPLTFGA GTKLEL VK-567
 DIQMTQSPSSLSASLGEKVSL 1073 RASQEISGYLS 1193 AASTLDS 1313 LHYANYPPT
 1433 TCRASQEISGYLSWLQKPDG SIKRLIYAAASTLDSGVPKRFS
 GSRSGSVYSLTISSELEDEFA DYYCLHYANYPPTFGGGGTKLE IR VK-568
 DVQMTQSPSSLSASLGDTITI 1074 HASQNIYVWLN 1194 KASDLHT 1314 QQGLSYPLT
 1434 TCHASQNIYVWLNWFQQKPGN IPKLLISKASDLHTGVPSRFS
 GSGSGTGFTLTISLQPEDIA TYCYCQQGLSYPLTFGGGTNLE IK VK-583
 DVVMTQTPLSPPVSLGYQASI 1075 RSSQSLVHSSGNTYLN 1195 KVSNNRLS 1315
 SQSTHVPYT 1435 SCRSSQSLVHSSGNTYLNWYL QKPGQSPKLLIYKVSNNRLSGV
 PDRFSGSGSGTDFTLKISRVE TEDLGVYFCSQSTHVPYTFGG GTKLEIK VK-584
 DVVMTQTPLSLPVSLGDRASI 1076 RSGQSLVHSSGNTYLH 1196 KVSNNRFS 1316
 SQSTHVPWT 1436 SCRSGQSLVHSSGNTYLHWYL QRPGRSPNLLIYKVSNNRFSGV
 PDRFSGSGSGTDFTLKISRVE AEDLGVYFCSQSTHVPWTFGG GTKLEIK VK-585
 DVVMTQTPLSLPVSLGQASI 1077 RFSQSIVYSNGNTYLQ 1197 KVSNNRFS 1317
 FQGSHVPPT 1437 SCRFSQSIVYSNGNTYLQWYL QKPGQSPKLLIYKVSNNRFSGV
 PDRFSGSGSGTDFTLKISRVE AEDLGVYYCFQGSHVPPTFGG GTKLEIK VK-586
 DVVMTQTPLTSLVTIGQPTSI 1078 KSSQSLLYSNGKTYLS 1198 LVSKLDS 1318
 VQNTHLPYT 1438 SCKSSQSLLYSNGKTYLSWLL QRPQSPKRLIYLVSKLDSGV
 PDRFTGSGSGTDFTLKINRVE AEDLGLYYCVQNTHLPYTFGG GTKLEIR VK-588
 DIQMTQSTSSLSASLGDRVTI 1079 RASQDISNYLN 1199 QISRLHS 1319 QQGNLSPPT 1439
 SCRASQDISNYLNWYQQTPDG TVKLLIYQISRLHSGVPSRFS
 GSGSGTDYSLTISNLEEEDIA NYFCQQGNLSPPTFGGGGTKVE IK VK-592
 DIQMNQSPSSLSASLGDTITI 1080 HASQNIDVWLS 1200 KASNLHT 1320 QQGHSYPLT
 1440 TCHASQNIDVWLSWYQQKPGN IPKLLIYKASNLHTGVPSRFS
 GRDSGTAFTLTISLQPEDIA TYCYCQQGHSYPLTFGSGGTKLE LK

(163) In an embodiment, the antibody molecule comprises one, two, or three CDRs of the VH described herein, e.g., in Table 1A, 2A, or 3A (e.g., any of VH-11, VH-11v2, VH-66, VH-79, VH-184, VH-216, VH-272, VH-308, VH-317, VH-317v2, VH-322, VH-329, VH-330, VH-332, VH-335, VH-336v1, VH-336v2, VH-336v3, VH-338, VH-341v1, VH-341v2, VH-343, VH-399, VH-402, VH-416, VH-429, VH-430, VH-440, VH-453, VH-454, VH-465, VH-475, VH-481, VH-497, VH-502, VH-503, VH-504, VH-507, VH-508, VH-510, VH-511, VH-521, VH-530, VH-536, VH-541, VH-547, VH-549, VH-550, VH-553, VH-556, VH-557, VH-567, VH-568, VH-570, VH-573, VH-583, VH-584, VH-585, VH-586, VH-588, or VH-592), using the IMGT, Kabat, or Chothia definitions of CDRs.

(164) In an embodiment, the antibody molecule comprises one, two, or three CDRs of the VL described herein, e.g., in Table 1B, 2B, or 3B (e.g., any of VK-11, VK-66, VK-79, VK-184, VK-184_C_Y, VK-216_272, VK-308, VK-317, VK-317v2, VK-322, VK-329, VK-330, VK-332, VK-335, VK-336v1, VK-336v2, VK-338v1, VK-338v2, VK-341v1, VK-341v2, VK-343, VK-399, VK-402, VK-416, VK-429, VK-430, VK-440, VK-453, VK-454, VK-465, VK-475, VK-481, VK-497, VK-502, VK-503, VK-504, VK-507, VK-508, VK-510, VK-511, VK-518, VK-528, VK-521, VK-530, VK-541, VK-547, VK-549, VK-550, VK-553, VK-556, VK-557, VK-567, VK-568, VK-583, VK-584, VK-585, VK-586, VK-588, VK-592), using the IMGT, Kabat, or Chothia definitions of CDRs.

(165) In an embodiment, the antibody molecule comprises one, two, or three CDRs of the VH of an antibody molecule described herein, e.g., in Table 1A, 2A, or 3A (e.g., any of antibodies 11v2, 322,

329, 330, 336v2-1, 402, 429, 430, 440, 453, 454, 465, 475, 507, 510, 511, 518, 528, 541, 547, 549, 550, 553, 556, 567, 568, 570, 573, 583, 584, 585, 586, 588, or 592, e.g., as listed in Table 1C), using the IMGT, Kabat, or Chothia definitions of CDRs. In an embodiment, the antibody molecule comprises one, two, or three CDRs of the VH of an antibody molecule described herein, e.g., in Table 1A, 2A, or 3A (e.g., any of antibodies 11v2, 322, 329, 330, 336v2-1, 402, 429, 430, 440, 453, 454, 465, 475, 507, 510, 511, 518, 528, 541, 547, 549, 550, 553, 556, 567, 568, 570, 573, 583, 584, 585, 586, 588, or 592, e.g., as listed in Table 1C), using the IMGT, Kabat, or Chothia definitions of CDRs.

(166) In an embodiment, the antibody molecule comprises one, two, or three CDRs of the VL region of an antibody molecule described herein, e.g., in Table 1B, 2B, or 3B (e.g., any of antibodies 11v2, 322, 329, 330, 336v2-1, 402, 429, 430, 440, 453, 454, 465, 475, 507, 510, 511, 518, 528, 541, 547, 549, 550, 553, 556, 567, 568, 570, 573, 583, 584, 585, 586, 588, or 592 (e.g., as listed in Table 1C))), using the IMGT, Kabat, or Chothia definitions of CDRs.

(167) In an embodiment, the antibody molecule comprises one, two, or three CDRs of the VH described herein, e.g., in Table 1A, 2A, or 3A (e.g., any of VH-11, VH-11v2, VH-66, VH-79, VH-184, VH-216, VH-272, VH-308, VH-317, VH-317v2, VH-322, VH-329, VH-330, VH-332, VH-335, VH-336v1, VH-336v2, VH-336v3, VH-338, VH-341v1, VH-341v2, VH-343, VH-399, VH-402, VH-416, VH-429, VH-430, VH-440, VH-453, VH-454, VH-465, VH-475, VH-481, VH-497, VH-502, VH-503, VH-504, VH-507, VH-508, VH-510, VH-511, VH-521, VH-530, VH-536, VH-541, VH-547, VH-549, VH-550, VH-553, VH-556, VH-557, VH-567, VH-568, VH-570, VH-573, VH-583, VH-584, VH-585, VH-586, VH-588, or VH-592), using the IMGT, Kabat, or Chothia definitions of CDRs; and one, two, or three CDRs of the VL described herein, e.g., in Table 1B, 2B, or 3B (e.g., any of VK-11, VK-66, VK-79, VK-184, VK-184_C_Y, VK-216_272, VK-308, VK-317, VK-317v2, VK-322, VK-329, VK-330, VK-332, VK-335, VK-336v1, VK-336v2, VK-338v1, VK-338v2, VK-341v1, VK-341v2, VK-343, VK-399, VK-402, VK-416, VK-429, VK-430, VK-440, VK-453, VK-454, VK-465, VK-475, VK-481, VK-497, VK-502, VK-503, VK-504, VK-507, VK-508, VK-510, VK-511, VK-518, VK-528, VK-521, VK-530, VK-541, VK-547, VK-549, VK-550, VK-553, VK-556, VK-557, VK-567, VK-568, VK-583, VK-584, VK-585, VK-586, VK-588, VK-592), using the IMGT, Kabat, or Chothia definitions of CDRs.

(168) In an embodiment, the antibody molecule comprises one, two, or three CDRs of the VH of an antibody molecule described herein, e.g., in Table 1A, 2A, or 3A (e.g., any of antibodies 11v2, 322, 329, 330, 336v2-1, 402, 429, 430, 440, 453, 454, 465, 475, 507, 510, 511, 518, 528, 541, 547, 549, 550, 553, 556, 567, 568, 570, 573, 583, 584, 585, 586, 588, or 592, e.g., as listed in Table 1C), using the IMGT, Kabat, or Chothia definitions of CDRs; and one, two, or three CDRs of the VL region of an antibody molecule described herein, e.g., in Table 1B, 2B, or 3B (e.g., any of antibodies 11v2, 322, 329, 330, 336v2-1, 402, 429, 430, 440, 453, 454, 465, 475, 507, 510, 511, 518, 528, 541, 547, 549, 550, 553, 556, 567, 568, 570, 573, 583, 584, 585, 586, 588, or 592 (e.g., as listed in Table 1C))), using the IMGT, Kabat, or Chothia definitions of CDRs.

(169) In an embodiment, the antibody molecule comprises one, two, or three HCDRs described in Table 1A, 2A, or 3A. In an embodiment, the antibody molecule comprises one, two, or three LCDRs described in Table 1B, 2B, or 3B. In an embodiment, the antibody molecule comprises one, two, or three HCDRs described in Table 1A, 2A, or 3A; and one, two, or three LCDRs described in Table 1B, 2B, or 3B.

(170) In an embodiment, the antibody molecule comprises one, two, or three CDRs of the VL region of an antibody molecule described herein, e.g., in Table 1B, 2B, or 3B (e.g., any of antibodies 11v2, 322, 329, 330, 336v2-1, 402, 429, 430, 440, 453, 454, 465, 475, 507, 510, 511, 518, 528, 541, 547, 549, 550, 553, 556, 567, 568, 570, 573, 583, 584, 585, 586, 588, or 592 (e.g., as listed in Table 1C))), using the IMGT, Kabat, or Chothia definitions of CDRs.

(171) In an embodiment, the antibody molecule comprises one, two, or three CDRs of the VH described herein, e.g., in Table 1A, 2A, or 3A (e.g., any of VH-11, VH-11v2, VH-66, VH-79, VH-

184, VH-216, VH-272, VH-308, VH-317, VH-317v2, VH-322, VH-329, VH-330, VH-332, VH-335, VH-336v1, VH-336v2, VH-336v3, VH-338, VH-341v1, VH-341v2, VH-343, VH-399, VH-402, VH-416, VH-429, VH-430, VH-440, VH-453, VH-454, VH-465, VH-475, VH-481, VH-497, VH-502, VH-503, VH-504, VH-507, VH-508, VH-510, VH-511, VH-521, VH-530, VH-536, VH-541, VH-547, VH-549, VH-550, VH-553, VH-556, VH-557, VH-567, VH-568, VH-570, VH-573, VH-583, VH-584, VH-585, VH-586, VH-588, or VH-592), using the IMGT, Kabat, or Chothia definitions of CDRs; and one, two, or three CDRs of the VL described herein, e.g., in Table 1B, 2B, or 3B (e.g., any of VK-11, VK-66, VK-79, VK-184, VK-184_C_Y, VK-216_272, VK-308, VK-317, VK-317v2, VK-322, VK-329, VK-330, VK-332, VK-335, VK-336v1, VK-336v2, VK-338v1, VK-338v2, VK-341v1, VK-341v2, VK-343, VK-399, VK-402, VK-416, VK-429, VK-430, VK-440, VK-453, VK-454, VK-465, VK-475, VK-481, VK-497, VK-502, VK-503, VK-504, VK-507, VK-508, VK-510, VK-511, VK-518, VK-528, VK-521, VK-530, VK-541, VK-547, VK-549, VK-550, VK-553, VK-556, VK-557, VK-567, VK-568, VK-583, VK-584, VK-585, VK-586, VK-588, VK-592), using the IMGT, Kabat, or Chothia definitions of CDRs.

(172) In an embodiment, the antibody molecule comprises one, two, or three CDRs of the VH of an antibody molecule described herein, e.g., in Table 1A, 2A, or 3A (e.g., any of antibodies 11v2, 322, 329, 330, 336v2-1, 402, 429, 430, 440, 453, 454, 465, 475, 507, 510, 511, 518, 528, 541, 547, 549, 550, 553, 556, 567, 568, 570, 573, 583, 584, 585, 586, 588, or 592, e.g., as listed in Table 1C), using the IMGT, Kabat, or Chothia definitions of CDRs; and one, two, or three CDRs of the VL region of an antibody molecule described herein, e.g., in Table 1B, 2B, or 3B (e.g., any of antibodies 11v2, 322, 329, 330, 336v2-1, 402, 429, 430, 440, 453, 454, 465, 475, 507, 510, 511, 518, 528, 541, 547, 549, 550, 553, 556, 567, 568, 570, 573, 583, 584, 585, 586, 588, or 592 (e.g., as listed in Table 1C)), using the IMGT, Kabat, or Chothia definitions of CDRs.

(173) In an embodiment, the antibody molecule comprises one, two, or three HCDRs described in Table 1A, 2A, or 3A. In an embodiment, the antibody molecule comprises one, two, or three LCDRs described in Table 1B, 2B, or 3B. In an embodiment, the antibody molecule comprises one, two, or three HCDRs described in Table 1A, 2A, or 3A; and one, two, or three LCDRs described in Table 1B, 2B, or 3B.

(174) In an embodiment, the antibody molecule comprises HCDR1, HCDR2, and HCDR3 described in the same row of Table 1A, 2A, or 3A. In an embodiment, the antibody molecule comprises LCDR1, LCDR2, and LCDR3 described in the same row described in the same row of Table 1B, 2B, or 3B. In an embodiment, the antibody molecule comprises HCDR1, HCDR2, and HCDR3 described in the same row of Table 1A, 2A, or 3A; and LCDR1, LCDR2, and LCDR3 described in the same row of Table 1B, 2B, or 3B.

(175) In an embodiment, the antibody molecule comprises one, two, three, or four frameworks of the VH described herein, e.g., in Table 1A, 2A, or 3A (e.g., any of VH-11, VH-11v2, VH-66, VH-79, VH-184, VH-216, VH-272, VH-308, VH-317, VH-317v2, VH-322, VH-329, VH-330, VH-332, VH-335, VH-336v1, VH-336v2, VH-336v3, VH-338, VH-341v1, VH-341v2, VH-343, VH-399, VH-402, VH-416, VH-429, VH-430, VH-440, VH-453, VH-454, VH-465, VH-475, VH-481, VH-497, VH-502, VH-503, VH-504, VH-507, VH-508, VH-510, VH-511, VH-521, VH-530, VH-536, VH-541, VH-547, VH-549, VH-550, VH-553, VH-556, VH-557, VH-567, VH-568, VH-570, VH-573, VH-583, VH-584, VH-585, VH-586, VH-588, or VH-592), using the IMGT, Kabat, or Chothia definitions of CDRs.

(176) In an embodiment, the antibody molecule comprises one, two, three, or four frameworks of the VH of an antibody molecule described herein, e.g., in Table 1A, 2A, or 3A (e.g., any of antibodies 11v2, 322, 329, 330, 336v2-1, 402, 429, 430, 440, 453, 454, 465, 475, 507, 510, 511, 518, 528, 541, 547, 549, 550, 553, 556, 567, 568, 570, 573, 583, 584, 585, 586, 588, or 592, e.g., as listed in Table 1C), using the IMGT, Kabat, or Chothia definitions of CDRs. In an embodiment, the antibody molecule comprises one, two, three, or four frameworks of the VH of an antibody molecule described herein, e.g., in Table 1A, 2A, or 3A (e.g., any of antibodies 11v2, 322, 329,

330, 336v2-1, 402, 429, 430, 440, 453, 454, 465, 475, 507, 510, 511, 518, 528, 541, 547, 549, 550, 553, 556, 567, 568, 570, 573, 583, 584, 585, 586, 588, or 592, e.g., as listed in Table 1C), using the IMGT, Kabat, or Chothia definitions of CDRs. In an embodiment, the antibody molecule comprises one, two, three, or four frameworks of the VL described herein, e.g., in Table 1B, 2B, or 3B (e.g., any of VK-11, VK-66, VK-79, VK-184, VK-184_C_Y, VK-216_272, VK-308, VK-317, VK-317v2, VK-322, VK-329, VK-330, VK-332, VK-335, VK-336v1, VK-336v2, VK-338v1, VK-338v2, VK-341v1, VK-341v2, VK-343, VK-399, VK-402, VK-416, VK-429, VK-430, VK-440, VK-453, VK-454, VK-465, VK-475, VK-481, VK-497, VK-502, VK-503, VK-504, VK-507, VK-508, VK-510, VK-511, VK-518, VK-528, VK-521, VK-530, VK-541, VK-547, VK-549, VK-550, VK-553, VK-556, VK-557, VK-567, VK-568, VK-583, VK-584, VK-585, VK-586, VK-588, VK-592), using the IMGT, Kabat, or Chothia definitions of CDRs.

(177) In an embodiment, the antibody molecule comprises one, two, three, or four frameworks of the VL region of an antibody molecule described herein, e.g., in Table 1B, 2B, or 3B (e.g., any of antibodies 11v2, 322, 329, 330, 336v2-1, 402, 429, 430, 440, 453, 454, 465, 475, 507, 510, 511, 518, 528, 541, 547, 549, 550, 553, 556, 567, 568, 570, 573, 583, 584, 585, 586, 588, or 592 (e.g., as listed in Table 1C))), using the IMGT, Kabat, or Chothia definitions of CDRs.

(178) In an embodiment, the antibody molecule comprises one, two, three, or four frameworks of the VH described herein, e.g., in Table 1A, 2A, or 3A (e.g., any of VH-11, VH-11v2, VH-66, VH-79, VH-184, VH-216, VH-272, VH-308, VH-317, VH-317v2, VH-322, VH-329, VH-330, VH-332, VH-335, VH-336v1, VH-336v2, VH-336v3, VH-338, VH-341v1, VH-341v2, VH-343, VH-399, VH-402, VH-416, VH-429, VH-430, VH-440, VH-453, VH-454, VH-465, VH-475, VH-481, VH-497, VH-502, VH-503, VH-504, VH-507, VH-508, VH-510, VH-511, VH-521, VH-530, VH-536, VH-541, VH-547, VH-549, VH-550, VH-553, VH-556, VH-557, VH-567, VH-568, VH-570, VH-573, VH-583, VH-584, VH-585, VH-586, VH-588, or VH-592), using the IMGT, Kabat, or Chothia definitions of CDRs; and one, two, three, or four frameworks of the VL described herein, e.g., in Table 1B, 2B, or 3B (e.g., any of VK-11, VK-66, VK-79, VK-184, VK-184_C_Y, VK-216_272, VK-308, VK-317, VK-317v2, VK-322, VK-329, VK-330, VK-332, VK-335, VK-336v1, VK-336v2, VK-338v1, VK-338v2, VK-341v1, VK-341v2, VK-343, VK-399, VK-402, VK-416, VK-429, VK-430, VK-440, VK-453, VK-454, VK-465, VK-475, VK-481, VK-497, VK-502, VK-503, VK-504, VK-507, VK-508, VK-510, VK-511, VK-518, VK-528, VK-521, VK-530, VK-541, VK-547, VK-549, VK-550, VK-553, VK-556, VK-557, VK-567, VK-568, VK-583, VK-584, VK-585, VK-586, VK-588, VK-592), using the IMGT, Kabat, or Chothia definitions of CDRs.

(179) In an embodiment, the antibody molecule comprises one, two, three, or four frameworks of the VH of an antibody molecule described herein, e.g., in Table 1A, 2A, or 3A (e.g., any of antibodies 11v2, 322, 329, 330, 336v2-1, 402, 429, 430, 440, 453, 454, 465, 475, 507, 510, 511, 518, 528, 541, 547, 549, 550, 553, 556, 567, 568, 570, 573, 583, 584, 585, 586, 588, or 592, e.g., as listed in Table 1C), using the IMGT, Kabat, or Chothia definitions of CDRs; and one, two, three, or four frameworks of the VL region of an antibody molecule described herein, e.g., in Table 1B, 2B, or 3B (e.g., any of antibodies 11v2, 322, 329, 330, 336v2-1, 402, 429, 430, 440, 453, 454, 465, 475, 507, 510, 511, 518, 528, 541, 547, 549, 550, 553, 556, 567, 568, 570, 573, 583, 584, 585, 586, 588, or 592 (e.g., as listed in Table 1C))), using the IMGT, Kabat, or Chothia definitions of CDRs. In an embodiment, the antibody molecule comprises one, two, three, or four frameworks of the VL region of an antibody molecule described herein, e.g., in Table 1B, 2B, or 3B (e.g., any of antibodies 11v2, 322, 329, 330, 336v2-1, 402, 429, 430, 440, 453, 454, 465, 475, 507, 510, 511, 518, 528, 541, 547, 549, 550, 553, 556, 567, 568, 570, 573, 583, 584, 585, 586, 588, or 592 (e.g., as listed in Table 1C))), using the IMGT, Kabat, or Chothia definitions of CDRs.

(180) In an embodiment, the antibody molecule comprises one, two, three, or four frameworks of the VH described herein, e.g., in Table 1A, 2A, or 3A (e.g., any of VH-11, VH-11v2, VH-66, VH-79, VH-184, VH-216, VH-272, VH-308, VH-317, VH-317v2, VH-322, VH-329, VH-330, VH-332, VH-335, VH-336v1, VH-336v2, VH-336v3, VH-338, VH-341v1, VH-341v2, VH-343, VH-399,

VH-416, VH-429, VH-430, VH-440, VH-453, VH-454, VH-465, VH-475, VH-481, VH-497, VH-502, VH-503, VH-504, VH-507, VH-508, VH-510, VH-511, VH-521, VH-530, VH-536, VH-541, VH-547, VH-549, VH-550, VH-553, VH-556, VH-557, VH-567, VH-568, VH-570, VH-573, VH-583, VH-584, VH-585, VH-586, VH-588, or VH-592), using the IMGT, Kabat, or Chothia definitions of CDRs; and one, two, three, or four frameworks of the VL described herein, e.g., in Table 1B, 2B, or 3B (e.g., any of VK-11, VK-66, VK-79, VK-184, VK-184_C_Y, VK-216_272, VK-308, VK-317, VK-317v2, VK-322, VK-329, VK-330, VK-332, VK-335, VK-336v1, VK-336v2, VK-338v1, VK-338v2, VK-341v1, VK-341v2, VK-343, VK-399, VK-402, VK-416, VK-429, VK-430, VK-440, VK-453, VK-454, VK-465, VK-475, VK-481, VK-497, VK-502, VK-503, VK-504, VK-507, VK-508, VK-510, VK-511, VK-518, VK-528, VK-521, VK-530, VK-541, VK-547, VK-549, VK-550, VK-553, VK-556, VK-557, VK-567, VK-568, VK-583, VK-584, VK-585, VK-586, VK-588, VK-592), using the IMGT, Kabat, or Chothia definitions of CDRs.

(181) In an embodiment, the antibody molecule comprises one, two, three, or four frameworks of the VH of an antibody molecule described herein, e.g., in Table 1A, 2A, or 3A (e.g., any of antibodies 11v2, 322, 329, 330, 336v2-1, 402, 429, 430, 440, 453, 454, 465, 475, 507, 510, 511, 518, 528, 541, 547, 549, 550, 553, 556, 567, 568, 570, 573, 583, 584, 585, 586, 588, or 592, e.g., as listed in Table 1C), using the IMGT, Kabat, or Chothia definitions of CDRs; and one, two, three, or four frameworks of the VL region of an antibody molecule described herein, e.g., in Table 1B, 2B, or 3B (e.g., any of antibodies 11v2, 322, 329, 330, 336v2-1, 402, 429, 430, 440, 453, 454, 465, 475, 507, 510, 511, 518, 528, 541, 547, 549, 550, 553, 556, 567, 568, 570, 573, 583, 584, 585, 586, 588, or 592 (e.g., as listed in Table 1C))), using the IMGT, Kabat, or Chothia definitions of CDRs. In an embodiment, the antibody molecule comprises all heavy chain frameworks described in the same row of Table 1A, 2A, or 3A. In an embodiment, the antibody molecule comprises all light chain frameworks described in the same row described in the same row of Table 1B, 2B, or 3B. In an embodiment, the antibody molecule comprises all heavy chain frameworks described in the same row of Table 1A, 2A, or 3A; and all light chain frameworks described in the same row of Table 1B, 2B, or 3B.

(182) In an embodiment, the antibody molecule comprises a VH comprising an amino acid sequence described in Table 1A, 2A, or 3A (e.g., any of VH-11, VH-11v2, VH-66, VH-79, VH-184, VH-216, VH-272, VH-308, VH-317, VH-317v2, VH-322, VH-329, VH-330, VH-332, VH-335, VH-336v1, VH-336v2, VH-336v3, VH-338, VH-341v1, VH-341v2, VH-343, VH-399, VH-402, VH-416, VH-429, VH-430, VH-440, VH-453, VH-454, VH-465, VH-475, VH-481, VH-497, VH-502, VH-503, VH-504, VH-507, VH-508, VH-510, VH-511, VH-521, VH-530, VH-536, VH-541, VH-547, VH-549, VH-550, VH-553, VH-556, VH-557, VH-567, VH-568, VH-570, VH-573, VH-583, VH-584, VH-585, VH-586, VH-588, or VH-592), or an amino acid sequence substantially identical thereto. In an embodiment, the antibody molecule comprises a VL comprising an amino acid sequence described in Table 1B, 2B, or 3B (e.g., any of VK-11, VK-66, VK-79, VK-184, VK-184_C_Y, VK-216_272, VK-308, VK-317, VK-317v2, VK-322, VK-329, VK-330, VK-332, VK-335, VK-336v1, VK-336v2, VK-338v1, VK-338v2, VK-341v1, VK-341v2, VK-343, VK-399, VK-402, VK-416, VK-429, VK-430, VK-440, VK-453, VK-454, VK-465, VK-475, VK-481, VK-497, VK-502, VK-503, VK-504, VK-507, VK-508, VK-510, VK-511, VK-518, VK-528, VK-521, VK-530, VK-541, VK-547, VK-549, VK-550, VK-553, VK-556, VK-557, VK-567, VK-568, VK-583, VK-584, VK-585, VK-586, VK-588, VK-592), or an amino acid sequence substantially identical thereto.

(183) In an embodiment, the antibody molecule comprises a VH comprising an amino acid sequence described in Table 1A, 2A, or 3A (e.g., any of VH-11, VH-11v2, VH-66, VH-79, VH-184, VH-216, VH-272, VH-308, VH-317, VH-317v2, VH-322, VH-329, VH-330, VH-332, VH-335, VH-336v1, VH-336v2, VH-336v3, VH-338, VH-341v1, VH-341v2, VH-343, VH-399, VH-402, VH-416, VH-429, VH-430, VH-440, VH-453, VH-454, VH-465, VH-475, VH-481, VH-497, VH-502, VH-503, VH-504, VH-507, VH-508, VH-510, VH-511, VH-521, VH-530, VH-536, VH-541,

VH-547, VH-549, VH-550, VH-553, VH-556, VH-557, VH-567, VH-568, VH-570, VH-573, VH-583, VH-584, VH-585, VH-586, VH-588, or VH-592), or an amino acid sequence substantially identical thereto; and a VL comprising an amino acid sequence described in Table 1B, 2B, or 3B (e.g., any of VK-11, VK-66, VK-79, VK-184, VK-184_C_Y, VK-216_272, VK-308, VK-317, VK-317v2, VK-322, VK-329, VK-330, VK-332, VK-335, VK-336v1, VK-336v2, VK-338v1, VK-338v2, VK-341v1, VK-341v2, VK-343, VK-399, VK-402, VK-416, VK-429, VK-430, VK-440, VK-453, VK-454, VK-465, VK-475, VK-481, VK-497, VK-502, VK-503, VK-504, VK-507, VK-508, VK-510, VK-511, VK-518, VK-528, VK-521, VK-530, VK-541, VK-547, VK-549, VK-550, VK-553, VK-556, VK-557, VK-567, VK-568, VK-583, VK-584, VK-585, VK-586, VK-588, VK-592), or an amino acid sequence substantially identical thereto. In an embodiment, the antibody molecule comprises a VH comprising an amino acid sequence described in Table 1A, 2A, or 3A (e.g., any of VH-11, VH-11v2, VH-66, VH-79, VH-184, VH-216, VH-272, VH-308, VH-317, VH-317v2, VH-322, VH-329, VH-330, VH-332, VH-335, VH-336v1, VH-336v2, VH-336v3, VH-338, VH-341v1, VH-341v2, VH-343, VH-399, VH-402, VH-416, VH-429, VH-430, VH-440, VH-453, VH-454, VH-465, VH-475, VH-481, VH-497, VH-502, VH-503, VH-504, VH-507, VH-508, VH-510, VH-511, VH-521, VH-530, VH-536, VH-541, VH-547, VH-549, VH-550, VH-553, VH-556, VH-557, VH-567, VH-568, VH-570, VH-573, VH-583, VH-584, VH-585, VH-586, VH-588, or VH-592), or an amino acid sequence substantially identical thereto; and a VL comprising an amino acid sequence described in Table 1B, 2B, or 3B (e.g., any of VK-11, VK-66, VK-79, VK-184, VK-184_C_Y, VK-216_272, VK-308, VK-317, VK-317v2, VK-322, VK-329, VK-330, VK-332, VK-335, VK-336v1, VK-336v2, VK-338v1, VK-338v2, VK-341v1, VK-341v2, VK-343, VK-399, VK-402, VK-416, VK-429, VK-430, VK-440, VK-453, VK-454, VK-465, VK-475, VK-481, VK-497, VK-502, VK-503, VK-504, VK-507, VK-508, VK-510, VK-511, VK-518, VK-528, VK-521, VK-530, VK-541, VK-547, VK-549, VK-550, VK-553, VK-556, VK-557, VK-567, VK-568, VK-583, VK-584, VK-585, VK-586, VK-588, VK-592), or an amino acid sequence substantially identical thereto. In an embodiment, the antibody molecule comprises a VH of an antibody molecule described herein, e.g., in Table 1A, 2A, or 3A (e.g., any of antibodies 11v2, 322, 329, 330, 336v2-1, 402, 429, 430, 440, 453, 454, 465, 475, 507, 510, 511, 518, 528, 541, 547, 549, 550, 553, 556, 567, 568, 570, 573, 583, 584, 585, 586, 588, or 592, e.g., as listed in Table 1C). In an embodiment, the antibody molecule comprises a VL of an antibody molecule described herein, e.g., in Table 1B, 2B, or 3B (e.g., any of antibodies 11v2, 322, 329, 330, 336v2-1, 402, 429, 430, 440, 453, 454, 465, 475, 507, 510, 511, 518, 528, 541, 547, 549, 550, 553, 556, 567, 568, 570, 573, 583, 584, 585, 586, 588, or 592, e.g., as listed in Table 1C). In an embodiment, the antibody molecule comprises a VH of an antibody molecule described herein, e.g., in Table 1A, 2A, or 3A (e.g., any of antibodies 11v2, 322, 329, 330, 336v2-1, 402, 429, 430, 440, 453, 454, 465, 475, 507, 510, 511, 518, 528, 541, 547, 549, 550, 553, 556, 567, 568, 570, 573, 583, 584, 585, 586, 588, or 592, e.g., as listed in Table 1C); and a VL of an antibody molecule described herein, e.g., in Table 1B, 2B, or 3B (e.g., any of antibodies 11v2, 322, 329, 330, 336v2-1, 402, 429, 430, 440, 453, 454, 465, 475, 507, 510, 511, 518, 528, 541, 547, 549, 550, 553, 556, 567, 568, 570, 573, 583, 584, 585, 586, 588, or 592, e.g., as listed in Table 1C).

(184) In an embodiment, the antibody molecule further comprises a heavy chain constant region. In an embodiment, the heavy chain constant region is an IgG1 constant region, e.g., any of SEQ ID NOS: 1441-1443, or a functional portion thereof. In another embodiment, the heavy chain constant region is an IgG2 constant region, e.g., any of SEQ ID NOS: 1444-1447, or a functional portion thereof. In an embodiment, the antibody molecule further comprises a light chain constant region. In an embodiment, the antibody molecule further comprises a heavy chain constant region and a light chain constant region. In certain embodiments, the antibody molecule comprises a heavy chain constant region, a light chain constant region, and variable regions that comprise one, two, three, four, five, or six CDRs of a VH described in Table 1A, 2A, or 3A, and/or a VL described in Table 1B, 2B, or 3B. In certain embodiments, the antibody molecule comprises a heavy chain

constant region, a light chain constant region, and variable regions that comprise one, two, three, four, five, or six CDRs of an antibody molecule described in Table 1C. In an embodiment, the antibody molecule comprises a heavy chain constant region, a light chain constant region, a VH described in Table 1A, 2A, or 3A, and a VL described in Table 1B, 2B, or 3B.

(185) Exemplary heavy chain constant regions are described below.

(186) TABLE-US-00012 Exemplary IgG1 constant regions >IGHG1*01 (SEQ ID NO: 1441) STKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGV

HTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDKKVE
PKSCDKTHTCPPCPAPELLGGPSVFLFPPKPKDTLMISRTPEVTCVVDV
VSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWL
NGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQV
SLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPVLDSDGSFFLYSKLTV

DKSRWQQGNVFSCSVMHEALHNHYTQKSLSLSP >IGHG1*03 (SEQ ID NO: 1442)

ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSG
VHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDKRV
EPKSCDKTHTCPPCPAPELLGGPSVFLFPPKPKDTLMISRTPEVTCVVDV
DVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDW
LNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQ
VSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPVLDSDGSFFLYSKLT

VDKSRWQQGNVFSCSVMHEALHNHYTQKSLSLSP >IGHG1*04 (SEQ ID NO: 1443)

ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSG
VHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDKKV
EPKSCDKTHTCPPCPAPELLGGPSVFLFPPKPKDTLMISRTPEVTCVVDV
DVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDW
LNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQ
VSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPVLDSDGSFFLYSKLT

VDKSRWQQGNVFSCSVMHEALHNHYTQKSLSLSP Exemplary IgG2 constant regions
>IGHG2*01 (SEQ ID NO: 1444)

STKGPSVFPLAPCSRSTSESTAALGCLVKDYFPEPVTVSWNSGALTSGV
HTFPAVLQSSGLYSLSSVVTVPSSNFGTQTYTCNVDHKPSNTKVDKTVE
RKCCVECPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVDSHE
DPEVQFNWYVDGVEVHNAKTKPREEQFNSTFRVVSFLTVVHQDWLNGKE
YKCKVSNKGLPAPIEKTISKTKGQPREPQVYTLPPSREEMTKNQVSLTC
LVKGFYPSDIAVEWESNGQPENNYKTTTPMLDSDGSFFLYSKLTVDKSR

WQQGNVFSCSVMHEALHNHYTQKSLSLSP >IGHG2*02 (SEQ ID NO: 1445)

STKGPSVFPLAPCSRSTSESTAALGCLVKDYFPEPVTVSWNSGALTSGV
HTFPAVLQSSGLYSLSSVVTVTSSNFGTQTYTCNVDHKPSNTKVDKTVE
RKCCVECPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVDSHE
DPEVQFNWYVDGMEVHNAKTKPREEQFNSTFRVVSFLTVVHQDWLNGKE
YKCKVSNKGLPAPIEKTISKTKGQPREPQVYTLPPSREEMTKNQVSLTC
LVKGFYPSDIAVEWESNGQPENNYKTTTPMLDSDGSFFLYSKLTVDKSR

WQQGNVFSCSVMHEALHNHYTQKSLSLSP >IGHG2*04 (SEQ ID NO: 1446)

STKGPSVFPLAPCSRSTSESTAALGCLVKDYFPEPVTVSWNSGALTSGV
HTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYTCNVDHKPSNTKVDKTVE
RKCCVECPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVDSHE
DPEVQFNWYVDGVEVHNAKTKPREEQFNSTFRVVSFLTVVHQDWLNGKE
YKCKVSNKGLPAPIEKTISKTKGQPREPQVYTLPPSREEMTKNQVSLTC
LVKGFYPSDIAVEWESNGQPENNYKTTTPMLDSDGSFFLYSKLTVDKSR

WQQGNVFSCSVMHEALHNHYTQKSLSLSP >IGHG2*06 (SEQ ID NO: 1447)

STKGPSVFPLAPCSRSTSESTAALGCLVKDYFPEPVTVSWNSGALTSGV

HTFPAVLQSSVLTSSVTVTPSSNFTQTYTCTCNVDHKPSNTKVDKTVE
RKCCVECPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVSHE
DPEVQFNWYVDGVEVHNAKTKPREEQFNSTFRVSVLTVVHQDNLNGKE
YKCKVSNKGLPAPIEKTISKTKGQPREPQVYTLPPSREEMTKNQVSLTCL
LVKGFYPSDISVEWESNGQPENNYKTTPPMLDSDGSFFLYSKLTVDKSR
WQQGNVFSCSVMHEALHNHYTQKSLSLSP

(187) In an embodiment, the antibody molecule is capable of binding, or substantially binding, to human C5aR1. In an embodiment, the antibody molecule binds to C5aR1 with high affinity, e.g., with a dissociation constant ($K_{sub.D}$) of less than about 100 nM, less than about 80 nM, less than about 60 nM, less than about 50 nM, less than about 40 nM, less than about 25 nM, less than about 10 nM, and more typically, about $10^{-0.001}$ nM, about $10^{-0.01}$ nM, about $10^{-0.01}$ nM, about 5-0.01 nM, about 3-0.05 nM, about 1-0.1 nM, or stronger, e.g., less than about 80, 70, 60, 50, 40, 30, 20, 10, 8, 6, 4, 3, 2, 1, 0.5, 0.2, 0.1, 0.05, 0.01, 0.005, or 0.001 nM. In an embodiment, the antibody molecule binds to C5aR1 with a $K_{sub.off}$ slower than $1 \times 10^{sup.4}$, $5 \times 10^{sup.-5}$, or $1 \times 10^{sup.-5}$ s.^{sup.-1}. In an embodiment, the antibody molecule binds to C5aR1 with a $K_{sub.on}$ faster than $1 \times 10^{sup.4}$, $5 \times 10^{sup.4}$, $1 \times 10^{sup.5}$, or $5 \times 10^{sup.5}$ M.^{sup.-1}s.^{sup.-1}.

(188) In an embodiment, the antibody molecule inhibits binding of human C5aR1 to human C5a by 50% or more, e.g., 60% or more, 70% or more, 80% or more, 85% or more, 90% or more, 95% or more, 96% or more, 97% or more, 98% or more, 99% or more, or 100%, as determined by a method described herein (e.g., normalized to the no antibody control).

(189) In an embodiment, the antibody molecule binds to a linear or conformational epitope on C5aR1. In an embodiment, the antibody molecule binds to an epitope conserved between human C5aR1 and mouse C5aR1. In an embodiment, the antibody molecule binds to an epitope described herein. In an embodiment, the antibody molecule binds, or substantially binds, to the same, similar, or overlapping epitope on C5aR1, as a second antibody molecule (e.g., an antibody molecule described in Table 1C). In an embodiment, the antibody molecule competes with a second antibody molecule (e.g., an antibody molecule described in Table 1C) for binding to C5aR1.

(190) In an embodiment, the epitope is a linear epitope. In an embodiment, the epitope is a conformational epitope.

(191) In an embodiment, the antibody molecule binds to the epitope of an antibody molecule as listed in Table 1C. In an embodiment, the epitope of the antibody molecule of Table 1C comprises a cyclic ECL2 region (e.g., comprised in Site II), and/or comprising a sulfated N-terminal (e.g., comprised in Site I) and/or a non-sulfated N-terminal (e.g., comprised in Site I).

(192) In an embodiment, the antibody molecule comprises a VH, wherein the VH comprises three HCDRs (HCDR1, HCDR2, and HCDR3), wherein the VH comprises one, two, or all of the following: (i) an HCDR1 comprising an amino acid sequence that differs by no more than 1, 2, or 3 amino acid residues from, or has at least 85, 90, 95, 99 or 100% homology with, the amino acid sequence of the HCDR1 of antibody 11v2; (ii) an HCDR2 comprising an amino acid sequence that differs by no more than 1, 2, or 3 amino acid residues from, or has at least 85, 90, 95, 99 or 100% homology with, the amino acid sequence of the HCDR2 of antibody 11v2; or (iii) an HCDR3 comprising an amino acid sequence that differs by no more than 1, 2, or 3 amino acid residues from, or has at least 85, 90, 95, 99 or 100% homology with, the amino acid sequence of the HCDR3 of antibody 11v2.

(193) In an embodiment, the antibody molecule comprises a VH, wherein the VH comprises three HCDRs (HCDR1, HCDR2, and HCDR3), wherein the VH comprises one, two, or all of the following: (i) an HCDR1 comprising an amino acid sequence that differs by no more than 1, 2, or 3 amino acid residues from, or has at least 85, 90, 95, 99 or 100% homology with, the amino acid sequence of the HCDR1 of antibody 322; (ii) an HCDR2 comprising an amino acid sequence that differs by no more than 1, 2, or 3 amino acid residues from, or has at least 85, 90, 95, 99 or 100% homology with, the amino acid sequence of the HCDR2 of antibody 322; or (iii) an HCDR3

comprising an amino acid sequence that differs by no more than 1, 2, or 3 amino acid residues from, or has at least 85, 90, 95, 99 or 100% homology with, the amino acid sequence of the HCDR3 of antibody 586.

(224) In an embodiment, the antibody molecule comprises a VH, wherein the VH comprises three HCDRs (HCDR1, HCDR2, and HCDR3), wherein the VH comprises one, two, or all of the following: (i) an HCDR1 comprising an amino acid sequence that differs by no more than 1, 2, or 3 amino acid residues from, or has at least 85, 90, 95, 99 or 100% homology with, the amino acid sequence of the HCDR1 of antibody 588; (ii) an HCDR2 comprising an amino acid sequence that differs by no more than 1, 2, or 3 amino acid residues from, or has at least 85, 90, 95, 99 or 100% homology with, the amino acid sequence of the HCDR2 of antibody 588; or (iii) an HCDR3 comprising an amino acid sequence that differs by no more than 1, 2, or 3 amino acid residues from, or has at least 85, 90, 95, 99 or 100% homology with, the amino acid sequence of the HCDR3 of antibody 588.

(225) In an embodiment, the antibody molecule comprises a VH, wherein the VH comprises three HCDRs (HCDR1, HCDR2, and HCDR3), wherein the VH comprises one, two, or all of the following: (i) an HCDR1 comprising an amino acid sequence that differs by no more than 1, 2, or 3 amino acid residues from, or has at least 85, 90, 95, 99 or 100% homology with, the amino acid sequence of the HCDR1 of antibody 592; (ii) an HCDR2 comprising an amino acid sequence that differs by no more than 1, 2, or 3 amino acid residues from, or has at least 85, 90, 95, 99 or 100% homology with, the amino acid sequence of the HCDR2 of antibody 592; or (iii) an HCDR3 comprising an amino acid sequence that differs by no more than 1, 2, or 3 amino acid residues from, or has at least 85, 90, 95, 99 or 100% homology with, the amino acid sequence of the HCDR3 of antibody 592.

(226) In an embodiment, the antibody molecule comprises a VL, wherein the VL comprises three LCDRs (LCDR1, LCDR2, and LCDR3), wherein the VL comprises one, two, or all of the following: (i) an LCDR1 comprising an amino acid sequence that differs by no more than 1, 2, or 3 amino acid residues from, or has at least 85, 90, 95, 99 or 100% homology with, the amino acid sequence of the LCDR1 of antibody 11v2; (ii) an LCDR2 comprising an amino acid sequence that differs by no more than 1, 2, or 3 amino acid residues from, or has at least 85, 90, 95, 99 or 100% homology with, the amino acid sequence of the LCDR2 of antibody 11v2; or (iii) an LCDR3 comprising an amino acid sequence that differs by no more than 1, 2, or 3 amino acid residues from, or has at least 85, 90, 95, 99 or 100% homology with, the amino acid sequence of the LCDR3 of antibody 11v2.

(227) In an embodiment, the antibody molecule comprises a VL, wherein the VL comprises three LCDRs (LCDR1, LCDR2, and LCDR3), wherein the VL comprises one, two, or all of the following: (i) an LCDR1 comprising an amino acid sequence that differs by no more than 1, 2, or 3 amino acid residues from, or has at least 85, 90, 95, 99 or 100% homology with, the amino acid sequence of the LCDR1 of antibody 322; (ii) an LCDR2 comprising an amino acid sequence that differs by no more than 1, 2, or 3 amino acid residues from, or has at least 85, 90, 95, 99 or 100% homology with, the amino acid sequence of the LCDR2 of antibody 322; or (iii) an LCDR3 comprising an amino acid sequence that differs by no more than 1, 2, or 3 amino acid residues from, or has at least 85, 90, 95, 99 or 100% homology with, the amino acid sequence of the LCDR3 of antibody 322.

(228) In an embodiment, the antibody molecule comprises a VL, wherein the VL comprises three LCDRs (LCDR1, LCDR2, and LCDR3), wherein the VL comprises one, two, or all of the following: (i) an LCDR1 comprising an amino acid sequence that differs by no more than 1, 2, or 3 amino acid residues from, or has at least 85, 90, 95, 99 or 100% homology with, the amino acid sequence of the LCDR1 of antibody 329; (ii) an LCDR2 comprising an amino acid sequence that differs by no more than 1, 2, or 3 amino acid residues from, or has at least 85, 90, 95, 99 or 100% homology with, the amino acid sequence of the LCDR2 of antibody 329; or (iii) an LCDR3

from, or has at least 85, 90, 95, 99 or 100% homology with, the amino acid sequence of the LCDR3 of antibody 588.

(259) In an embodiment, the antibody molecule comprises a VL, wherein the VL comprises three LCDRs (LCDR1, LCDR2, and LCDR3), wherein the VL comprises one, two, or all of the following: (i) an LCDR1 comprising an amino acid sequence that differs by no more than 1, 2, or 3 amino acid residues from, or has at least 85, 90, 95, 99 or 100% homology with, the amino acid sequence of the LCDR1 of antibody 592; (ii) an LCDR2 comprising an amino acid sequence that differs by no more than 1, 2, or 3 amino acid residues from, or has at least 85, 90, 95, 99 or 100% homology with, the amino acid sequence of the LCDR2 of antibody 592; or (iii) an LCDR3 comprising an amino acid sequence that differs by no more than 1, 2, or 3 amino acid residues from, or has at least 85, 90, 95, 99 or 100% homology with, the amino acid sequence of the LCDR3 of antibody 592.

(260) In an embodiment, the antibody molecule comprises: (i) a VH comprising one, two, or all of the following: an HCDR1 comprising an amino acid sequence that differs by no more than 1, 2, or 3 amino acid residues from, or has at least 85, 90, 95, 99 or 100% homology with, the amino acid sequence of the HCDR1 of antibody 11v2; an HCDR2 comprising an amino acid sequence that differs by no more than 1, 2, or 3 amino acid residues from, or has at least 85, 90, 95, 99 or 100% homology with, the amino acid sequence of the HCDR2 of antibody 11v2; or an HCDR3 comprising an amino acid sequence that differs by no more than 1, 2, or 3 amino acid residues from, or has at least 85, 90, 95, 99 or 100% homology with, the amino acid sequence of the HCDR3 of antibody 11v2, and (ii) a VL comprising one, two, or all of the following: an LCDR1 comprising an amino acid sequence that differs by no more than 1, 2, or 3 amino acid residues from, or has at least 85, 90, 95, 99 or 100% homology with, the amino acid sequence of the LCDR1 of antibody 11v2; an LCDR2 comprising an amino acid sequence that differs by no more than 1, 2, or 3 amino acid residues from, or has at least 85, 90, 95, 99 or 100% homology with, the amino acid sequence of the LCDR2 of antibody 11v2; or an LCDR3 comprising an amino acid sequence that differs by no more than 1, 2, or 3 amino acid residues from, or has at least 85, 90, 95, 99 or 100% homology with, the amino acid sequence of the LCDR3 of antibody 11v2.

(261) In an embodiment, the antibody molecule comprises: (i) a VH comprising one, two, or all of the following: an HCDR1 comprising an amino acid sequence that differs by no more than 1, 2, or 3 amino acid residues from, or has at least 85, 90, 95, 99 or 100% homology with, the amino acid sequence of the HCDR1 of antibody 322; an HCDR2 comprising an amino acid sequence that differs by no more than 1, 2, or 3 amino acid residues from, or has at least 85, 90, 95, 99 or 100% homology with, the amino acid sequence of the HCDR2 of antibody 322; or an HCDR3 comprising an amino acid sequence that differs by no more than 1, 2, or 3 amino acid residues from, or has at least 85, 90, 95, 99 or 100% homology with, the amino acid sequence of the HCDR3 of antibody 322, and (ii) a VL comprising one, two, or all of the following: an LCDR1 comprising an amino acid sequence that differs by no more than 1, 2, or 3 amino acid residues from, or has at least 85, 90, 95, 99 or 100% homology with, the amino acid sequence of the LCDR1 of antibody 322; an LCDR2 comprising an amino acid sequence that differs by no more than 1, 2, or 3 amino acid residues from, or has at least 85, 90, 95, 99 or 100% homology with, the amino acid sequence of the LCDR2 of antibody 322; or an LCDR3 comprising an amino acid sequence that differs by no more than 1, 2, or 3 amino acid residues from, or has at least 85, 90, 95, 99 or 100% homology with, the amino acid sequence of the LCDR3 of antibody 322.

(262) In an embodiment, the antibody molecule comprises: (i) a VH comprising one, two, or all of the following: an HCDR1 comprising an amino acid sequence that differs by no more than 1, 2, or 3 amino acid residues from, or has at least 85, 90, 95, 99 or 100% homology with, the amino acid sequence of the HCDR1 of antibody 329; an HCDR2 comprising an amino acid sequence that differs by no more than 1, 2, or 3 amino acid residues from, or has at least 85, 90, 95, 99 or 100% homology with, the amino acid sequence of the HCDR2 of antibody 329; or an HCDR3 comprising

or 100% homology with, the amino acid sequence of the VH of antibody 588; and (ii) a VL comprising an amino acid sequence that differs by no more than 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, or 15 amino acid residues from, or has at least 85, 90, 95, 96, 97, 98, 99, or 100% homology with, the amino acid sequence of the VL of antibody 588. In an embodiment the antibody molecule is antibody 588.

(326) In an embodiment, the antibody molecule comprises a VH comprising an amino acid sequence that differs by no more than 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, or 15 amino acid residues from, or has at least 85, 90, 95, 96, 97, 98, 99, or 100% homology with, the amino acid sequence of the VH of antibody 592. In an embodiment, the antibody molecule comprises a VL comprising an amino acid sequence that differs by no more than 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, or 15 amino acid residues from, or has at least 85, 90, 95, 96, 97, 98, 99, or 100% homology with, the amino acid sequence of the VL of antibody 592. In an embodiment, the antibody molecule comprises: (i) a VH comprising an amino acid sequence that differs by no more than 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, or 15 amino acid residues from, or has at least 85, 90, 95, 96, 97, 98, 99, or 100% homology with, the amino acid sequence of the VH of antibody 592; and (ii) a VL comprising an amino acid sequence that differs by no more than 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, or 15 amino acid residues from, or has at least 85, 90, 95, 96, 97, 98, 99, or 100% homology with, the amino acid sequence of the VL of antibody 592. In an embodiment the antibody molecule is antibody 592.

(327) In an aspect, the disclosure features an anti-C5aR1 antibody molecule, which: a) competes for binding to C5aR1 (e.g., human C5aR1) with an antibody molecule comprising the heavy chain complementary determining regions (HCDR1, HCDR2 and HCDR3) and the light chain complementary determining regions (LCDR1, LCDR2 and LCDR3) of any of antibodies 11v2, 322, 329, 330, 336v2-1, 402, 429, 430, 440, 453, 454, 465, 475, 507, 510, 511, 518, 528, 541, 547, 549, 550, 553, 556, 567, 568, 570, 573, 583, 584, 585, 586, 588, or 592 (e.g., as listed in Table 1C); or b) binds, or substantially binds, to an epitope on C5aR1 (e.g., human C5aR1) that completely or partially overlaps with the epitope of an antibody molecule comprising the heavy chain complementary determining regions (HCDR1, HCDR2 and HCDR3) and the light chain complementary determining regions (LCDR1, LCDR2 and LCDR3) of any of antibodies 11v2, 322, 329, 330, 336v2-1, 402, 429, 430, 440, 453, 454, 465, 475, 507, 510, 511, 518, 528, 541, 547, 549, 550, 553, 556, 567, 568, 570, 573, 583, 584, 585, 586, 588, or 592 (e.g., as listed in Table 1C).

(328) In an embodiment, the disclosure features an anti-C5aR1 antibody molecule, which: a) competes for binding to C5aR1 (e.g., human C5aR1) with an antibody molecule comprising the heavy chain complementary determining regions (HCDR1, HCDR2 and HCDR3) and the light chain complementary determining regions (LCDR1, LCDR2 and LCDR3) of any of antibodies 11v2, 322, 329, 583, 66 and 336v2; or b) binds, or substantially binds, to an epitope on C5aR1 (e.g., human C5aR1) that completely or partially overlaps with the epitope of an antibody molecule comprising the heavy chain complementary determining regions (HCDR1, HCDR2 and HCDR3) and the light chain complementary determining regions (LCDR1, LCDR2 and LCDR3) of any of antibodies 11v2, 322, 329, 583, 66 and 336v2.

(329) In an embodiment, the anti-C5aR1 antibody molecule is a synthetic antibody molecule. In an embodiment, the anti-C5aR1 antibody molecule is an isolated antibody molecule. In an embodiment, the anti-C5aR1 antibody molecule is a human or humanized antibody molecule, e.g., comprising one or more framework regions derived from human framework germline sequence.

(330) In an embodiment, the anti-C5aR1 antibody molecule competes for binding with two, three, four, five, six, seven, eight, nine, ten, or more of the antibody molecules that comprise the HCDR1, HCDR2, HCDR3, LCDR1, LCDR2, and LCDR3 of any of antibodies 11v2, 322, 329, 330, 336v2-1, 402, 429, 430, 440, 453, 454, 465, 475, 507, 510, 511, 518, 528, 541, 547, 549, 550, 553, 556, 567, 568, 570, 573, 583, 584, 585, 586, 588, or 592 (e.g., as listed in Table 1C).

(331) In an embodiment, the anti-C5aR1 antibody molecule binds, or substantially binds, to an

epitope that completely or partially overlaps with the epitopes of two, three, four, five, six, seven, eight, nine, ten, or more of the antibody molecules that comprise the HCDR1, HCDR2, HCDR3, LCDR1, LCDR2, and LCDR3 of any of antibodies 11v2, 322, 329, 330, 336v2-1, 402, 429, 430, 440, 453, 454, 465, 475, 507, 510, 511, 518, 528, 541, 547, 549, 550, 553, 556, 567, 568, 570, 573, 583, 584, 585, 586, 588, or 592 (e.g., as listed in Table 1C).

(332) In an embodiment, the anti-C5aR1 antibody molecule is a monospecific or monoparatopic antibody molecule. In an embodiment, the anti-C5aR1 antibody molecule is a multispecific or multiparatopic (e.g., a bispecific or biparatopic) antibody molecule.

(333) In an embodiment, the antibody molecule that comprises the HCDR1, HCDR2, HCDR3, LCDR1, LCDR2, and LCDR3 of any of antibodies 11v2, 322, 329, 330, 336v2-1, 402, 429, 430, 440, 453, 454, 465, 475, 507, 510, 511, 518, 528, 541, 547, 549, 550, 553, 556, 567, 568, 570, 573, 583, 584, 585, 586, 588, or 592 (e.g., as listed in Table 1C) molecule is a monospecific or monoparatopic antibody molecule. In an embodiment, the antibody molecule that comprises the HCDR1, HCDR2, HCDR3, LCDR1, LCDR2, and LCDR3 of any of antibodies 11v2, 322, 329, 330, 336v2-1, 402, 429, 430, 440, 453, 454, 465, 475, 507, 510, 511, 518, 528, 541, 547, 549, 550, 553, 556, 567, 568, 570, 573, 583, 584, 585, 586, 588, or 592 (e.g., as listed in Table 1C) is a multispecific or multiparatopic (e.g., a bispecific or biparatopic) antibody molecule.

(334) In an embodiment, the antibody molecule that comprises the HCDR1, HCDR2, HCDR3, LCDR1, LCDR2, and LCDR3 of any of antibodies 11v2, 322, 329, 330, 336v2-1, 402, 429, 430, 440, 453, 454, 465, 475, 507, 510, 511, 518, 528, 541, 547, 549, 550, 553, 556, 567, 568, 570, 573, 583, 584, 585, 586, 588, or 592 (e.g., as listed in Table 1C) comprises a heavy chain variable region and a light chain variable region of any of antibodies 11v2, 322, 329, 330, 336v2-1, 402, 429, 430, 440, 453, 454, 465, 475, 507, 510, 511, 518, 528, 541, 547, 549, 550, 553, 556, 567, 568, 570, 573, 583, 584, 585, 586, 588, or 592 (e.g., as listed in Table 1C).

(335) In an embodiment, the antibody molecule that comprises the HCDR1, HCDR2, HCDR3, LCDR1, LCDR2, and LCDR3 of any of antibodies 11v2, 322, 329, 330, 336v2-1, 402, 429, 430, 440, 453, 454, 465, 475, 507, 510, 511, 518, 528, 541, 547, 549, 550, 553, 556, 567, 568, 570, 573, 583, 584, 585, 586, 588, or 592 (e.g., as listed in Table 1C) is antibody 11v2, 322, 329, 330, 336v2-1, 402, 429, 430, 440, 453, 454, 465, 475, 507, 510, 511, 518, 528, 541, 547, 549, 550, 553, 556, 567, 568, 570, 573, 583, 584, 585, 586, 588, or 592 (e.g., as listed in Table 1C).

(336) In an embodiment, the antibody molecule comprises a heavy chain variable region (VH) having an amino acid sequence described any of Tables 1A, 2A, or 3A. In an embodiment, the antibody molecule comprises a light chain variable region (VL) having an amino acid sequence described in any of Tables 1B, 2B, or 3B. In an embodiment, the antibody molecule comprises a heavy chain variable region (VH) having an amino acid sequence described in any of Tables 1A, 2A, or 3A and a light chain variable region (VL) having an amino acid sequence described in any of Tables 1B, 2B, or 3B.

(337) In an embodiment, the anti-C5aR1 antibody molecule binds, or substantially binds, to C5aR1 (e.g., human C5aR1), at an EC₅₀ of 100 nM or less, e.g., 80 nM or less, 60 nM or less, 40 nM or less, 20 nM or less, 10 nM or less, 9 nM or less, 8 nM or less, 7 nM or less, 6 nM or less, 5 nM or less, 4 nM or less, 3 nM or less, 2 nM or less, 1 nM or less, 0.8 nM or less, 0.6 nM or less, 0.4 nM or less, 0.2 nM or less, 0.1 nM or less, 0.05 nM or less, 0.02 nM or less, 0.01 nM or less, 0.005 nM or less, 0.002 nM or less, or 0.001 nM or less, e.g., between 0.001 nM and 100 nM, e.g., between 0.001 nM and 50 nM, between 0.01 nM and 20 nM, between 0.1 nM and 20 nM, e.g., between 0.1 nM and 10 nM, between 0.5 nM and 5 nM, between 1 nM and 5 nM, between 0.001 nM and 0.1 nM, between 0.001 nM and 0.01 nM, between 0.001 nM and 0.005 nM, between 0.01 nM and 0.05 nM, or between 0.01 nM and 0.1 nM, e.g., as determined by a method described herein.

(338) In an embodiment, the antibody molecule inhibits, or substantially inhibits, binding of C5aR1 (e.g., human C5aR1) to C5a (e.g., human C5a). In an embodiment, the antibody molecule inhibits,

or substantially inhibits, binding of C5aR1 (e.g., human C5aR1) to C5a (e.g., human C5a), at an EC₅₀ of 100 nM or less, e.g., 80 nM or less, 60 nM or less, 40 nM or less, 20 nM or less, 10 nM or less, 9 nM or less, 8 nM or less, 7 nM or less, 6 nM or less, 5 nM or less, 4 nM or less, 3 nM or less, 2 nM or less, 1 nM or less, 0.8 nM or less, 0.6 nM or less, 0.4 nM or less, 0.2 nM or less, 0.1 nM or less, 0.05 nM or less, 0.02 nM or less, 0.01 nM or less, 0.005 nM or less, 0.002 nM or less, or 0.001 nM or less, e.g., between 0.001 nM and 100 nM, e.g., between 0.001 nM and 50 nM, between 0.01 nM and 20 nM, between 0.1 nM and 20 nM, e.g., between 0.1 nM and 10 nM, between 0.5 nM and 5 nM, between 1 nM and 5 nM, between 0.001 nM and 0.1 nM, between 0.001 nM and 0.01 nM, between 0.001 nM and 0.005 nM, between 0.01 nM and 0.05 nM, or between 0.01 nM and 0.1 nM, e.g., as determined by a method described herein.

(339) In an embodiment, the anti-C5aR1 antibody molecule binds to one or more epitopes described herein. In an embodiment, the anti-C5aR1 antibody molecule binds, or substantially binds, to one or more, e.g., 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, or all, residues (e.g., consecutive residues) within a region of human C5aR1 Site I sequence, e.g., comprising SEQ ID NO: 1449 or 1452. In an embodiment, the anti-C5aR1 antibody molecule binds, or substantially binds, to one or more, e.g., 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, or all, residues (e.g., consecutive residues) within a region of human C5aR1 Site II sequence, e.g., comprising any of SEQ ID NOs: 1450, 1453, 1454, or 1455. In an embodiment, the anti-C5aR1 antibody molecule binds, or substantially binds, to one or more, e.g., 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, or all, residues (e.g., consecutive residues) within a core epitope region of human C5aR1 Site II sequence, e.g., comprising amino acids 1-20 of SEQ ID NO: 1450. In an embodiment, the anti-C5aR1 antibody molecule binds, or substantially binds, to one or more, e.g., 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, or all, residues (e.g., consecutive residues) within a region of human C5aR1 Site I sequence, e.g., comprising SEQ ID NO: 1449 or 1452; and one or more, e.g., 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, or all, residues (e.g., consecutive residues) within a region of human C5aR1 Site II sequence, e.g., comprising any of SEQ ID NOs: 1450, 1453, 1454, or 1455. In an embodiment, the anti-C5aR1 antibody molecule binds, or substantially binds, to one or more, e.g., 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, or all, residues (e.g., consecutive residues) within a region of human C5aR1 Site I sequence, e.g., comprising SEQ ID NO: 1449 or 1452; and one or more, e.g., 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, or all, residues (e.g., consecutive residues) within a core epitope region of human C5aR1 Site II sequence, e.g., comprising amino acids 1-20 of SEQ ID NO: 1450.

(340) In an embodiment, the anti-C5aR1 antibody molecule binds, or substantially binds, to a linear epitope. In an embodiment, the anti-C5aR1 antibody molecule binds, or substantially binds, to a conformational epitope. In an embodiment, the anti-C5aR1 antibody molecule binds, or substantially binds, to two or more epitopes on C5aR1. In an embodiment, the anti-C5aR1 antibody molecule binds to two or more conformational epitopes. In an embodiment, the anti-C5aR1 antibody molecule binds, or substantially binds, to at least one linear epitope and at least one conformational epitope.

(341) In an embodiment, the anti-C5aR1 antibody molecule is an IgG antibody molecule, e.g., comprising a heavy chain constant region of IgG, e.g., chosen from IgG1, IgG2 (e.g., IgG2a), IgG3, or IgG4, e.g., IgG2 or IgG4. In an embodiment, the anti-C5aR1 antibody molecule is an IgG1 antibody molecule. In another embodiment, the anti-C5aR1 antibody molecule is an IgG2 antibody molecule. In an embodiment, the anti-C5aR1 antibody molecule comprises a light chain constant region of kappa or lambda light chain.

(342) In an embodiment, the anti-C5aR1 antibody molecule comprises two heavy chain variable regions and two light chain variable regions. In an embodiment, the anti-C5aR1 antibody molecule

comprises a Fab, F(ab')₂, Fv, Fd, or a single chain Fv fragment (scFv).

(343) In an embodiment, the anti-C5aR1 antibody molecule comprises an Fc region. In an embodiment, the Fc region comprises one or more mutations described in International Publication No. WO 2018/052556, U.S. Application Publication No. US 2018/0037634, Booth et al., MAbs. 2018; 10 (7): 1098-1110, the contents of the aforesaid publications are incorporated by reference in their entirety.

(344) In some embodiments, the monospecific antibody capable of binding to C5aR1, wherein the antibody inhibits G α signaling.

(345) In some embodiments, the monospecific antibody capable of binding to C5aR1, wherein the antibody inhibits neutrophil chemotaxis.

(346) In some embodiments, the monospecific antibody capable of binding to C5aR1, inhibits calcium release.

(347) In an embodiment, the monospecific antibody capable of binding to C5aR1, but not to C5aR2 or other GPCRs.

(348) In an embodiment, the monospecific antibody capable of binding one or more amino acid residues of C5aR1 at Site I (SEQ ID NO: 1449 or SEQ ID NO: 1452) or one or more amino acid residues of C5aR1 at Site II (SEQ ID NO: 1450). In some embodiments, the monospecific antibody is capable of binding C5aR1 at Site II (SEQ ID NO: 1450).

(349) In an embodiment, the monospecific antibody binding C5aR1, can compete with an antibody binding to C5aR1 at Site I (SEQ ID NO: 1449 or SEQ ID NO: 1452) or at Site II (SEQ ID NO: 1450).

(350) In some embodiments the monospecific antibody binding C5aR1 binds one or more of amino acid residues T8 (threonine 8), D10 (aspartate 10), Y11 (tyrosine 11), Y14 (tyrosine 14) and/or D15 (aspartate 15).

(351) In some embodiments the monospecific antibody binding C5aR1 binds to amino acid residues R175-G189 in SEQ ID NO: 1448.

(352) In some embodiments the monospecific antibody binding C5aR1 binds to amino acid residues E180-P183 in SEQ ID NO: 1448.

(353) In some embodiments the monospecific antibody binding C5aR1 binds to amino acid residues E180-P184 in SEQ ID NO: 1448.

(354) In some embodiments the monospecific antibody binding C5aR1 binds to amino acid residues E178-P183 in SEQ ID NO: 1448.

(355) Multispecific and Multiparatopic Antibody Molecules

(356) Without wishing to be bound by theory, it is believed that in some embodiments, C5a binds to Site I and Site II of C5aR1. Antibody molecules binding to these sites should therefore cause orthosteric inhibition dependent on antibody affinity and, possibly, not on the antibody valency. In an embodiment, a cocktail of site I and II antibodies or a biparatopic antibody induces C5aR1 clustering. Such clustering can enhance potency of biparatopic antibodies (or a combination of site I+site II antibodies). In an embodiment, use of site I/site II combination or biparatopic antibodies can result in one or more of the following: receptor clustering (analogous to multimeric antigen), which could result in a slower off rate that translates to longer residence time and more potent and potentially complete or total inhibition of C5a binding; antibody mediated C5aR1 clustering, which could lead to internalization of receptors and making them unavailable for signaling, and/or C5aR1 clustering induced signal attenuation.

(357) Without wishing to be bound by theory, it is believed that in some embodiments, a combination of site I and site II antibodies or a biparatopic antibody that bind to both Site I and Site II can theoretically form a lattice or large cluster of C5aR1 (FIG. 1). In an embodiment, an antibody mediated GPCR cluster can comprise GPCR molecules at the most 120 Å apart (e.g., the maximum distance between the two Fab domains of an IgG1). In an embodiment, effector proteins such as adenyl cyclase and ion channels are excluded from this GPCR cluster/raft. In an

embodiment, effector proteins are anchored to the membrane through their lipid moiety.

(358) Without wishing to be bound by theory, it is believed that in some embodiments, antibody-bound C5aR1 molecules that participate in clustering are inactive; however, if a Site II is exposed due to dissociation of the bound antibody molecule, it can, in some instances, facilitate the binding of agonist and get activated. In such an event, migration of a complex from outside the raft to the activated GPCR may be considerably slowed, thus reducing the GPCR signaling rate. On the other hand, a molecule that is present inside the cluster as an inclusion may be able to efficiently bind to the activated GPCR molecule. Upon activation, the molecules may migrate to their respective effector molecules outside the raft to cause signal cascade; this slow migration is contemplated to reduce the GPCR signaling rate. Thus, the antibody molecule-mediated GPCR clustering/raft may hinder efficient migration of the G-proteins between the activated GPCR and the effector molecules, resulting in signal attenuation. This signal attenuation can, in some instances, be independent of the antibody molecule's ability to inhibit the receptor molecule.

(359) The antibody molecules described herein can, in some instances, be multispecific or multiparatopic antibody molecules.

(360) In an embodiment, the antibody molecule is a multispecific or multiparatopic antibody molecule, e.g., it comprises a plurality of immunoglobulin variable region sequences, wherein a first immunoglobulin variable region sequence of the plurality has binding specificity for a first epitope and a second immunoglobulin variable region sequence of the plurality has binding specificity for a second epitope. In an embodiment, the first and second epitopes are on the same antigen, e.g., the same protein (or subunit of a multimeric protein). In an embodiment, the first and second epitopes overlap. In an embodiment, the first and second epitopes do not overlap. In an embodiment, the first and second epitopes are on different antigens, e.g., different proteins (or different subunits of a multimeric protein). In an embodiment, the first and second epitopes are on different molecules, e.g., different molecules of the same antigen. In an embodiment, a multispecific or multiparatopic antibody molecule comprises a third, fourth or fifth immunoglobulin variable region. In an embodiment, a multispecific or multiparatopic antibody molecule is a bispecific or biparatopic antibody molecule, a trispecific or triparatopic antibody molecule, or tetraspecific or tetraparatopic antibody molecule.

(361) In an embodiment, the multispecific or multiparatopic antibody molecule is a bispecific or biparatopic antibody molecule. A bispecific or biparatopic antibody has specificity for no more than two antigens or epitopes. A bispecific or biparatopic antibody molecule is typically characterized by a first immunoglobulin variable region sequence which has binding specificity for a first epitope and a second immunoglobulin variable region sequence that has binding specificity for a second epitope. In an embodiment, the first and second epitopes are on the same antigen, e.g., the same protein (or subunit of a multimeric protein). In an embodiment, the first and second epitopes overlap. In an embodiment, the first and second epitopes do not overlap. In an embodiment, the first and second epitopes are on different antigens, e.g., the different proteins (or different subunits of a multimeric protein). In an embodiment, the first and second epitopes are on different molecules, e.g., different molecules of the same antigen. In an embodiment, a bispecific or biparatopic antibody molecule comprises a heavy chain variable region sequence and a light chain variable region sequence which have binding specificity for a first epitope and a heavy chain variable region sequence and a light chain variable region sequence which have binding specificity for a second epitope. In an embodiment, a bispecific or biparatopic antibody molecule comprises a half antibody having binding specificity for a first epitope and a half antibody having binding specificity for a second epitope. In an embodiment, a bispecific or biparatopic antibody molecule comprises a half antibody, or fragment thereof, having binding specificity for a first epitope and a half antibody, or fragment thereof, having binding specificity for a second epitope. In an embodiment, a bispecific or biparatopic antibody molecule comprises a scFv, or fragment thereof, have binding specificity for a first epitope and a scFv, or fragment thereof, have binding specificity

for a second epitope. In an embodiment the first epitope is located on C5aR1 (e.g., Site I, e.g., comprising an N-terminal region as described herein) and the second epitope is located on C5aR1 (e.g., Site II, e.g., comprising an ECL1, ECL2, and/or ECL3, as described herein).

(362) In an embodiment, the antibody molecule is multispecific (e.g., bispecific or trispecific). In an embodiment, the antibody molecule binds to C5aR1 (e.g., human C5aR1) and to one or more additional molecules. In an embodiment, the antibody molecule is multiparatopic (e.g., biparatopic). In an embodiment, the antibody molecule binds to two or more distinct epitopes on C5aR1 (e.g., human C5aR1). In an embodiment, the antibody is a biparatopic antibody that binds to Site I and Site II of C5aR1 (e.g., human C5aR1).

(363) In an embodiment, the antibody molecule is a biparatopic antibody molecule capable of simultaneously binding to Site I and Site II of C5aR1 (e.g., human C5aR1). Such a biparatopic antibody molecule can, in an embodiment, exhibit higher affinity (e.g., 2, 3, 4, 5, 6, 7, 8, 9, 10, 15, 20, 25, 30, 40, 50, 60, 70, 80, 90, 100, 500, or 1000-fold higher affinity) and/or higher avidity (e.g., 2, 3, 4, 5, 6, 7, 8, 9, 10, 15, 20, 25, 30, 40, 50, 60, 70, 80, 90, 100, 500, or 1000-fold higher avidity) for C5aR1 (e.g., human C5aR1) compared to a monoparatopic antibody molecule.

(364) In an embodiment, the biparatopic antibodies are orthosteric inhibitors, e.g., they compete with C5a for binding to C5aR1, and the binding of these antibodies to C5aR1 inhibit C5a binding and C5a mediated signaling. Biparatopic antibodies against Site I and Site II (and combination of Site I and Site II antibodies) may, in an embodiment, be able to engage with both sites on C5aR1 involved in C5a binding and therefore compete more effectively with C5a even in the presence of higher concentrations of C5a.

(365) The antibody molecules described herein can be a multispecific or multiparatopic antibody molecule. In an embodiment, the antibody molecule binds, or substantially binds, to one or more, e.g., 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, or all, residues (e.g., consecutive residues) within a region of human C5aR1 Site I sequence, e.g., comprising SEQ ID NO: 1449 or 1452; and one or more, e.g., 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, or all, residues (e.g., consecutive residues) within a region of human C5aR1 Site II sequence, e.g., comprising any of SEQ ID NOs: 1450, 1453, 1454, or 1455. In an embodiment, the antibody molecule binds, or substantially binds, to one or more, e.g., 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, or all, residues (e.g., consecutive residues) within a region of human C5aR1 Site I sequence, e.g., comprising SEQ ID NO: 1449 or 1452; and one or more, e.g., 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, or more, residues (e.g., consecutive residues) within a core epitope region of human C5aR1 Site II sequence, e.g., comprising amino acids 1-20 of SEQ ID NO: 1450.

(366) The multispecific or multiparatopic antibody molecules described herein (e.g., bispecific or biparatopic antibody molecules described herein) can be derived from monospecific or monoparatopic antibody molecules described herein. For example, biparatopic scFv-Fc molecules can be produced with knob-hole technology (e.g., including hole mutations: Y349C, T366S, L368A, Y407V; knob mutations: S354C, T366W).

(367) In some embodiments, the multispecific or biparatopic antibody capable of binding to C5aR1, wherein the antibody inhibits G α signaling.

(368) In some embodiments, the multispecific or biparatopic antibody capable of binding to C5aR1, wherein the antibody inhibits neutrophil chemotaxis.

(369) In some embodiments, the multispecific or biparatopic antibody capable of binding to C5aR1, inhibits calcium release.

(370) In an embodiment, the multispecific or biparatopic antibody capable of binding to C5aR1, but not to C5aR2 or other GPCRs.

(371) Animal Models

(372) The antibody molecules described herein can be evaluated in vivo, e.g., using various animal

models. For example, an animal model can be used to test the efficacy of an antibody molecule described herein in inhibiting C5aR1 and/or in treating or preventing a disorder described herein, e.g., a C5aR1-associated disorder, e.g., ANCA-vasculitis. Animal models can also be used, e.g., to investigate for side effects, measure concentrations of antibody molecules in situ, demonstrate correlations between an C5aR1 function and a disorder described herein (e.g., ANCA-vasculitis). (373) Exemplary animal models for the disorders described herein, e.g., a C5aR1-associated disorder, e.g., ANCA-vasculitis, are known in the art, e.g., as described in Salama and Little, *Curr Opin Rheumatol.* 2012; 24 (1): 1-7; Coughlan et al., *Clin Exp Immunol.* 2012; 169 (3): 229-237; and Hutton et al., *Semin Nephrol.* 2017; 37 (5): 418-435. Exemplary types of animals that can be used to evaluate the antibody molecules described herein include, but are not limited to, mice, rats, rabbits, guinea pigs, and monkeys.

(374) Pharmaceutical Compositions and Kits

(375) In some aspects, this disclosure provides compositions, e.g., pharmaceutically acceptable compositions, which include an antibody molecule described herein, formulated together with a pharmaceutically acceptable carrier.

(376) As used herein, “pharmaceutically acceptable carrier” includes any and all solvents, dispersion media, isotonic and absorption delaying agents, and the like that are physiologically compatible. The carrier can be suitable for intravenous, intramuscular, subcutaneous, parenteral, rectal, spinal or epidermal administration (e.g., by injection or infusion). In certain embodiments, less than about 5%, e.g., less than about 4%, 3%, 2%, or 1% of the antibody molecules in the pharmaceutical composition are present as aggregates. In other embodiments, at least about 95%, e.g., at least about 96%, 97%, 98%, 98.5%, 99%, 99.5%, 99.8%, or more of the antibody molecules in the pharmaceutical composition are present as monomers. In an embodiment, the level of aggregates or monomers is determined by chromatography, e.g., high performance size exclusion chromatography (HP-SEC).

(377) The compositions set out herein may be in a variety of forms. These include, for example, liquid, semi-solid and solid dosage forms, such as liquid solutions (e.g., injectable and infusible solutions), dispersions or suspensions, liposomes, and suppositories. A suitable form depends on the intended mode of administration and therapeutic application. Typical suitable compositions are in the form of injectable or infusible solutions. One suitable mode of administration is parenteral (e.g., intravenous, subcutaneous, intraperitoneal, intramuscular). In an embodiment, the antibody molecule is administered by intravenous infusion or injection. In certain embodiments, the antibody is administered by intramuscular or subcutaneous injection.

(378) The phrases “parenteral administration” and “administered parenterally” as used herein means modes of administration other than enteral and topical administration, usually by injection, and includes, without limitation, intravenous, intramuscular, intraarterial, intrathecal, intracapsular, intraorbital, intracardiac, intradermal, intraperitoneal, transtracheal, subcutaneous, subcuticular, intraarticular, subcapsular, subarachnoid, intraspinal, epidural and intrasternal injection and infusion.

(379) Therapeutic compositions typically should be sterile and stable under the conditions of manufacture and storage. The composition can be formulated as a solution, microemulsion, dispersion, liposome, or other ordered structure suitable to high antibody concentration. Sterile injectable solutions can be prepared by incorporating the active compound (i.e., antibody or antibody portion) in the required amount in an appropriate solvent with one or a combination of ingredients enumerated above, as required, followed by filtered sterilization. Generally, dispersions are prepared by incorporating the active compound into a sterile vehicle that contains a basic dispersion medium and the required other ingredients from those enumerated above. In the case of sterile powders for the preparation of sterile injectable solutions, the preferred methods of preparation are vacuum drying and freeze-drying that yields a powder of the active ingredient plus any additional desired ingredient from a previously sterile-filtered solution thereof. The proper

fluidity of a solution can be maintained, for example, by the use of a coating such as lecithin, by the maintenance of the required particle size in the case of dispersion and by the use of surfactants. Prolonged absorption of injectable compositions can be brought about by including in the composition an agent that delays absorption, for example, monostearate salts and gelatin.

(380) The antibody molecules described herein can be administered by a variety of methods. Several are known in the art, and for many therapeutic, prophylactic, or diagnostic applications, an appropriate route/mode of administration is intravenous injection or infusion. For example, the antibody molecules can be administered by intravenous infusion at a rate of less than 10 mg/min; preferably less than or equal to 5 mg/min to reach a dose of about 1 to 100 mg/m², preferably about 5 to 50 mg/m², about 7 to 25 mg/m² and more preferably, about 10 mg/m². As will be appreciated by the skilled artisan, the route and/or mode of administration will vary depending upon the desired results. In certain embodiments, the active compound may be prepared with a carrier that will protect the compound against rapid release, such as a controlled release formulation, including implants, transdermal patches, and microencapsulated delivery systems. Biodegradable, biocompatible polymers can be used, such as ethylene vinyl acetate, polyanhydrides, polyglycolic acid, collagen, polyorthoesters, and polylactic acid. Many methods for the preparation of such formulations are patented or generally known to those skilled in the art. See, e.g., *Sustained and Controlled Release Drug Delivery Systems*, J. R. Robinson, ed., Marcel Dekker, Inc., New York, 1978.

(381) In certain embodiments, an antibody molecule can be orally administered, for example, with an inert diluent or an assimilable edible carrier. The antibody molecule (and other ingredients, if desired) may also be enclosed in a hard or soft shell gelatin capsule, compressed into tablets, or incorporated directly into the subject's diet. For oral therapeutic administration, the antibody molecule may be incorporated with excipients and used in the form of ingestible tablets, buccal tablets, troches, capsules, elixirs, suspensions, syrups, wafers, and the like. To administer an antibody molecule by other than parenteral administration, it may be necessary to coat the compound with, or co-administer the compound with, a material to prevent its inactivation. Therapeutic, prophylactic, or diagnostic compositions can also be administered with medical devices, and several are known in the art.

(382) Dosage regimens are adjusted to provide the desired response (e.g., a therapeutic, prophylactic, or diagnostic response). For example, a single bolus may be administered, several divided doses may be administered over time or the dose may be proportionally reduced or increased as indicated by the exigencies of the therapeutic situation. It is especially advantageous to formulate parenteral compositions in dosage unit form for ease of administration and uniformity of dosage. Dosage unit form as used herein refers to physically discrete units suited as unitary dosages for the subjects to be treated; each unit contains a predetermined quantity of active compound calculated to produce the desired therapeutic effect in association with the required pharmaceutical carrier. The specification for the dosage unit forms are dictated by and directly dependent on (a) the unique characteristics of the antibody molecule and the particular therapeutic, prophylactic, or diagnostic effect to be achieved, and (b) the limitations inherent in the art of compounding such an antibody molecule for the treatment of sensitivity in individuals.

(383) An exemplary, non-limiting range for a therapeutically, prophylactically, or diagnostically effective amount of an antibody molecule is about 0.1-50 mg/kg body weight of a subject, e.g., about 0.1-30 mg/kg, e.g., about 1-30, 1-15, 1-10, 1-5, 5-10, or 1-3 mg/kg, e.g., about 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 15, 20, 30, 40, or 50 mg/kg. The antibody molecule can be administered by intravenous infusion at a rate of less than 10 mg/min, e.g., less than or equal to 5 mg/min to reach a dose of about 1 to 100 mg/m², e.g., about 5 to 50 mg/m², about 7 to 25 mg/m², e.g., about 10 mg/m². It is to be noted that dosage values may vary with the type and severity of the condition to be alleviated. It is to be further understood that for any particular subject, specific dosage regimens should be adjusted over time according to the individual need and the professional judgment of the person

administering or supervising the administration of the compositions, and that dosage ranges set forth herein are exemplary only and are not intended to limit the scope or practice of the claimed compositions.

(384) The pharmaceutical compositions herein may include a “therapeutically effective amount,” “prophylactically effective amount,” or “diagnostically effective amount” of an antibody molecule described herein.

(385) A “therapeutically effective amount” refers to an amount effective, at dosages and for periods of time necessary, to achieve the desired therapeutic result. A therapeutically effective amount of the antibody molecule may vary according to factors such as the disease state, age, sex, and weight of the individual, and the ability of the antibody or antibody portion to elicit a desired response in the individual. A therapeutically effective amount is also one in which any toxic or detrimental effect of the antibody molecule is outweighed by the therapeutically beneficial effects. A “therapeutically effective dosage” typically inhibits a measurable parameter by at least about 20%, e.g., by at least about 40%, by at least about 60%, or by at least about 80% relative to untreated subjects. The measurable parameter may be, e.g., hematuria, colored urine, foamy urine, pain, swelling (edema) in the hands and feet, or high blood pressure. The ability of an antibody molecule to inhibit a measurable parameter can be evaluated in an animal model system predictive of efficacy in treating or preventing a C5aR1-associated disorder, e.g., ANCA-vasculitis.

Alternatively, this property of a composition can be evaluated by examining the ability of the antibody molecule to inhibit C5aR1, e.g., by an in vitro assay.

(386) A “prophylactically effective amount” refers to an amount effective, at dosages and for periods of time necessary, to achieve the desired prophylactic result. Typically, since a prophylactic dose is used in subjects prior to or at an earlier stage of disease, the prophylactically effective amount will be less than the therapeutically effective amount.

(387) A “diagnostically effective amount” refers to an amount effective, at dosages and for periods of time necessary, to achieve the desired diagnostic result. Typically, a diagnostically effective amount is one in which a disorder, e.g., a disorder described herein, e.g., a C5aR1-associated disorder, e.g., ANCA-vasculitis, can be diagnosed in vitro, ex vivo, or in vivo.

(388) Also within this disclosure is a kit that comprises an antibody molecule, described herein. The kit can include one or more other elements including: instructions for use; other reagents, e.g., a label, a therapeutic agent, or an agent useful for chelating, or otherwise coupling, an antibody molecule to a label or therapeutic agent, or a radioprotective composition; devices or other materials for preparing the antibody molecule for administration; pharmaceutically acceptable carriers; and devices or other materials for administration to a subject.

(389) Nucleic Acids

(390) The present disclosure also features nucleic acids comprising nucleotide sequences that encode the antibody molecules (e.g., heavy and light chain variable regions and CDRs of the antibody molecules), as described herein.

(391) For example, the present disclosure features a first and second nucleic acid encoding heavy and light chain variable regions, respectively, of an antibody molecule chosen from one or more of the antibody molecules disclosed herein, e.g., an antibody molecule of Table 1C, or a portion of an antibody molecule, e.g., the variable regions of any of Tables 1A, 2A, 3A, 1B, 2B, or 3B. The nucleic acid can comprise a nucleotide sequence encoding any one of the amino acid sequences in the tables herein, or a sequence substantially identical thereto (e.g., a sequence at least about 85%, 90%, 95%, 99% or more identical thereto, or which differs by no more than 3, 6, 15, 30, or 45 nucleotides from the sequences shown in the tables herein).

(392) In certain embodiments, the nucleic acid comprises a nucleotide sequence encoding at least one, two, or three CDRs from a heavy chain variable region having an amino acid sequence as set forth in the tables herein, or a sequence substantially homologous thereto (e.g., a sequence at least about 85%, 90%, 95%, 99% or more identical thereto, and/or having one or more substitutions,

e.g., conserved substitutions). In an embodiment, the nucleic acid comprises a nucleotide sequence encoding at least one, two, or three CDRs from a light chain variable region having an amino acid sequence as set forth in the tables herein, or a sequence substantially homologous thereto (e.g., a sequence at least about 85%, 90%, 95%, 99% or more identical thereto, and/or having one or more substitutions, e.g., conserved substitutions). In an embodiment, the nucleic acid comprises a nucleotide sequence encoding at least one, two, three, four, five, or six CDRs from heavy and light chain variable regions having an amino acid sequence as set forth in the tables herein, or a sequence substantially homologous thereto (e.g., a sequence at least about 85%, 90%, 95%, 99% or more identical thereto, and/or having one or more substitutions, e.g., conserved substitutions).

(393) The nucleic acids disclosed herein include deoxyribonucleotides or ribonucleotides, or analogs thereof. The polynucleotide may be either single-stranded or double-stranded, and if single-stranded may be the coding strand or non-coding (antisense) strand. A polynucleotide may comprise modified nucleotides, such as methylated nucleotides and nucleotide analogs. The sequence of nucleotides may be interrupted by non-nucleotide components. A polynucleotide may be further modified after polymerization, such as by conjugation with a labeling component. The nucleic acid may be a recombinant polynucleotide, or a polynucleotide of genomic, cDNA, semisynthetic, or synthetic origin which either does not occur in nature or is linked to another polynucleotide in a non-natural arrangement.

(394) In some aspects, the application features host cells and vectors containing the nucleic acids described herein. The nucleic acids may be present in a single vector or separate vectors present in the same host cell or separate host cell, as described in more detail below.

(395) Vectors

(396) Further provided herein are vectors that comprise nucleotide sequences encoding the antibody molecules (e.g., heavy and light chain variable regions and CDRs of the antibody molecules), as described herein.

(397) In an embodiment, the vector comprises a nucleic acid described herein. For example, the vector can comprises a first and second nucleic acid encoding heavy and light chain variable regions, respectively, of an antibody molecule chosen from one or more of the antibody molecules disclosed herein, e.g., an antibody molecule of Table 1C, or a portion of an antibody molecule, e.g., the variable regions of any of Tables 1A, 2A, 3A, 1B, 2B, or 3B.

(398) In certain embodiments, the vector comprises a nucleotide sequence encoding at least one, two, or three CDRs from a heavy chain variable region having an amino acid sequence as set forth in the tables herein, or a sequence substantially homologous thereto (e.g., a sequence at least about 85%, 90%, 95%, 99% or more identical thereto, and/or having one or more substitutions, e.g., conserved substitutions). In an embodiment, the vector comprises a nucleotide sequence encoding at least one, two, or three CDRs from a light chain variable region having an amino acid sequence as set forth in the tables herein, or a sequence substantially homologous thereto (e.g., a sequence at least about 85%, 90%, 95%, 99% or more identical thereto, and/or having one or more substitutions, e.g., conserved substitutions). In an embodiment, the vector comprises a nucleotide sequence encoding at least one, two, three, four, five, or six CDRs from heavy and light chain variable regions having an amino acid sequence as set forth in the tables herein, or a sequence substantially homologous thereto (e.g., a sequence at least about 85%, 90%, 95%, 99% or more identical thereto, and/or having one or more substitutions, e.g., conserved substitutions).

(399) The vectors include, but are not limited to, a virus, plasmid, cosmid, lambda phage or a yeast artificial chromosome (YAC). Numerous vector systems can be employed. For example, one class of vectors utilizes DNA elements which are derived from animal viruses such as, for example, bovine papilloma virus, polyoma virus, adenovirus, vaccinia virus, baculovirus, retroviruses (Rous Sarcoma Virus, MMTV or MOMLV) or SV40 virus. Another class of vectors utilizes RNA elements derived from RNA viruses such as Semliki Forest virus, Eastern Equine Encephalitis virus and Flaviviruses.

(400) Additionally, cells which have stably integrated the DNA into their chromosomes may be selected by introducing one or more markers which allow for the selection of transfected host cells. The marker may provide, for example, prototrophy to an auxotrophic host, biocide resistance (e.g., antibiotics), or resistance to heavy metals such as copper, or the like. The selectable marker gene can be either directly linked to the DNA sequences to be expressed or introduced into the same cell by cotransformation. Additional elements may also be needed for optimal synthesis of mRNA. These elements may include splice signals, as well as transcriptional promoters, enhancers, and termination signals.

(401) Once the expression vector or DNA sequence containing the constructs has been prepared for expression, the expression vectors may be transfected or introduced into an appropriate host cell. Various techniques may be employed to achieve this, such as, for example, protoplast fusion, calcium phosphate precipitation, electroporation, retroviral transduction, viral transfection, gene gun, lipid-based transfection or other conventional techniques. In the case of protoplast fusion, the cells are grown in media and screened for the appropriate activity.

(402) Methods and conditions for culturing the resulting transfected cells and for recovering the antibody molecule produced are known to those skilled in the art and may be varied or optimized depending upon the specific expression vector and mammalian host cell employed, based upon the present description.

(403) Cells

(404) The present disclosure also provides cells (e.g., host cells) comprising a nucleic acid encoding an antibody molecule as described herein. For example, the host cells may comprise a nucleic acid molecule having a nucleotide sequence encoding an amino acid sequence described in any of Tables 1A, 1B, 2A, 2B, 3A, or 3B, a sequence substantially homologous thereto (e.g., a sequence at least about 85%, 90%, 95%, 99% or more identical thereto, and/or capable of hybridizing under the stringency conditions described herein), or a portion of one of said nucleic acids.

(405) In an embodiment, the host cells are genetically engineered to comprise nucleic acids encoding the antibody molecule described herein.

(406) In certain embodiments, the host cells are genetically engineered by using an expression cassette. The phrase “expression cassette,” refers to nucleotide sequences, which are capable of affecting expression of a gene in hosts compatible with such sequences. Such cassettes may include a promoter, an open reading frame with or without introns, and a termination signal. Additional factors necessary or helpful in effecting expression may also be used, such as, for example, an inducible promoter.

(407) The disclosure also provides host cells comprising the vectors described herein.

(408) The cell can be, but is not limited to, a eukaryotic cell, a bacterial cell, an insect cell, or a human cell. Suitable eukaryotic cells include, but are not limited to, Vero cells, HeLa cells, COS cells, CHO cells, HEK293 cells, BHK cells and MDCKII cells. Suitable insect cells include, but are not limited to, Sf9 cells. In an embodiment, the cell (e.g., host cell) is an isolated cell.

(409) Uses of Antibody Molecules

(410) The antibody molecules disclosed herein, as well as the pharmaceutical compositions disclosed herein, have in vitro, ex vivo, and in vivo therapeutic, prophylactic, and/or diagnostic utilities.

(411) In an embodiment, the antibody molecule reduces (e.g., inhibits, blocks, or neutralizes) one or more biological activities of C5aR1. For example, these antibodies molecules can be administered to cells in culture, in vitro or ex vivo, or to a subject, e.g., a human subject, e.g., in vivo, to reduce (e.g., inhibits, blocks, or neutralizes) one or more biological activities of C5aR1. In an embodiment, the antibody molecule inhibits, or substantially inhibit, binding of C5aR1, e.g., human C5aR1, to C5a. In an embodiment, the antibody molecule reduces (e.g., inhibits, blocks, or neutralizes) chemotaxis priming. In an embodiment, the antibody molecule reduces (e.g., inhibits,

blocks, or neutralizes) neutrophil activation. In an embodiment, the antibody molecule reduces (e.g., inhibits, blocks, or neutralizes) endothelium injury associated with C5aR1 signaling. For example, the antibody molecules described herein can be used to reduce elevated complement activation or to treat or prevent a disorder associated with elevated complement activation.

(412) In an embodiment, an antibody molecule targeting Site I or the N-terminus of C5aR1 (e.g., an antibody molecule described herein) is used. In an embodiment, an antibody molecule targeting Site II of C5aR1 (e.g., an antibody molecule described herein) is used. In an embodiment, a combination of an antibody molecule targeting Site I of C5aR1 (e.g., an antibody molecule described herein) and an antibody molecule targeting Site II (e.g., comprising ECL2) of C5aR1 (e.g., an antibody molecule described herein) is used. In an embodiment, a combination of an antibody molecule targeting Site I of C5aR1 (e.g., an antibody molecule described herein) and an antibody molecule targeting Site II (e.g., comprising ECL2) of C5aR1 (e.g., an antibody molecule described herein) achieves an additive or synergistic effect, e.g., on C5aR1 signaling. In an embodiment, an antibody molecule targeting Site I, and Site II (e.g., comprising ECL2) of C5aR1 (e.g., an antibody molecule described herein) is used. In an embodiment, an antibody molecule targeting Site I, and Site II (e.g., comprising ECL2) C5aR1 (e.g., an antibody molecule described herein) achieves an additive or synergistic effect, e.g., on C5aR1 signaling, e.g., compared to monovalent controls. Without wishing to be bound by theory, it is believed that in some embodiments, such a combination use, or use of a multispecific or multiparatopic (e.g., bispecific or biparatopic) antibody molecules, can have certain gain of function due to a combination of valency and/or specificity, e.g., receptor clustering that promotes internalization or prevents G-protein binding and/or efficient migration of the G-proteins between the activated GPCR and the effector molecules.

(413) Accordingly, in an aspect, the disclosure provides a method of treating, preventing, or diagnosing a disorder, e.g., a disorder described herein (e.g., a C5aR1-associated disorder, e.g., ANCA-vasculitis), in a subject, comprising administering to the subject an antibody molecule described herein, such that the disorder is treated, prevented, or diagnosed. For example, the disclosure provides a method comprising contacting the antibody molecule described herein with cells in culture, e.g. in vitro or ex vivo, or administering the antibody molecule described herein to a subject, e.g., in vivo, to treat, prevent, or diagnose a disorder (e.g., a C5aR1-associated disorder, e.g., ANCA-vasculitis).

(414) As used herein, the term “subject” is intended to include human and non-human animals. In an embodiment, the subject is a human subject, e.g., a human patient having a disorder described herein (e.g., a C5aR1-associated disorder, e.g., ANCA-vasculitis), or at risk of having a disorder described herein (e.g., a C5aR1-associated disorder, e.g., ANCA-vasculitis). The term “non-human animals” includes mammals and non-mammals, such as non-human primates. In an embodiment, the subject is a human. The methods and compositions described herein are suitable for treating human patients a disorder described herein (e.g., a C5aR1-associated disorder, e.g., ANCA-vasculitis). Patients having a disorder described herein (e.g., a C5aR1-associated disorder, e.g., ANCA-vasculitis) include those who have developed a disorder described herein (e.g., a C5aR1-associated disorder, e.g., ANCA-vasculitis) but are (at least temporarily) asymptomatic, patients who have exhibited a symptom of a disorder described herein (e.g., a C5aR1-associated disorder, e.g., ANCA-vasculitis), or patients having a disorder related to or associated with a disorder described herein (e.g., a C5aR1-associated disorder, e.g., ANCA-vasculitis).

(415) Methods of Treating or Preventing Disorders

(416) The antibody molecules described herein can be used to treat or prevent disorders associated with C5aR1 and/or C5 or symptoms thereof. Without wishing to be bound by theory, it is believed that in some embodiments, targeting the C5a/C5aR1 axis (e.g., by targeting the agonist precursor C5, the agonist C5a, or the receptor C5aR1) as described herein provides a useful approach to treat or prevent various disorders, e.g., to limit disease progression.

(417) Exemplary disorders or conditions that can be treated or prevented by the antibody molecules described herein include, but are not limited to, a C5aR1-associated disorder or a C5-associated disorder. In an embodiment, the disorder is associated with neutrophil recruitment, activation, and/or NETosis. In an embodiment, the disorder is associated with complement system activation and/or coagulation system activation. In an embodiment, the disorder is associated with a C5aR-mediated inflammatory response. In an embodiment, the disorder is associated with monocyte chemoattractant protein-1 (MCP-1) and/or renal inflammation. In an embodiment, the disorder is associated with chemotaxis (e.g., chemotaxis priming). In an embodiment, the disorder is associated with endothelium injury.

(418) In an embodiment, the exemplary disorders, for example, a C5aR1-associated disorder, can be treated using an antibody capable of binding one or more amino acid residues of C5aR1 at Site I (SEQ ID NO: 1449 or SEQ ID NO: 1452) or one or more amino acid residues of C5aR1 at Site II (SEQ ID NO: 1450). In some embodiments, the exemplary disorders, for example, a C5aR1-associated disorder, can be treated using an antibody capable of binding C5aR1 at Site II (SEQ ID NO: 1450).

(419) In an embodiment, the exemplary disorders, for example, a C5aR1-associated disorder, can be treated using an antibody capable of competing with an antibody binding to C5aR1 at Site I (SEQ ID NO: 1449 or SEQ ID NO: 1452) or at Site II (SEQ ID NO: 1450).

(420) In an embodiment, the exemplary disorders, for example, a C5aR1-associated disorder, can be treated using an antibody capable of binding to C5aR1, but not to C5aR2 or other GPCRs.

(421) In an embodiment, the exemplary disorders, for example, a C5aR1-associated disorder, can be treated using an antibody capable of binding to C5aR1, comprising a heavy chain, light chain, HCDRs and LCDRs presented in Tables 1A, 1B, 2A, 2B, 3A and 3B.

(422) In an embodiment, the exemplary disorders, for example, a C5aR1-associated disorder, can be treated using an antibody capable of binding to C5aR1, the antibody comprising the amino acid sequences AYAMS (SEQ ID NO: 1456), SISTGGNTY (SEQ ID NO: 1457), and GYQRFSGFAY (SEQ ID NO: 1458) or a variant thereof. In some embodiments, the antibody variant for the treatment of a C5aR1-associated disorder, comprises an amino acid sequence that differs by no more than 5, 4, 3, 2 or 1 amino acids from the amino acids of SEQ ID NO: 1456, SEQ ID NO: 1457 or SEQ ID NO: 1458. In some embodiments, the exemplary disorders, for example, a C5aR1-associated disorder, can be treated using an antibody capable of binding to C5aR1, comprises HCDRs, wherein the HCDRs comprise SEQ ID NOs: 1456, 1457 and/or 1458 are part of HCDR.

(423) In some embodiments, the exemplary disorders, for example, a C5aR1-associated disorder, can be treated using an antibody capable of binding to C5aR1, the antibody comprising the amino acid sequences RSSQSLVHSNGNTYLN (SEQ ID NO: 1459), KVSNRLS (SEQ ID NO: 1460), and SQSTHVPYT (SEQ ID NO: 1461). In some embodiments, the antibody variant for the treatment of a C5aR1-associated disorder, comprises an amino acid sequence that differs by no more than 5, 4, 3, 2 or 1 amino acids from the amino acids of SEQ ID Nos: 1459, 1460 and/or 1461. In some embodiments, the antibody comprising SEQ ID NOs: 1459, 1460 and/or 1461, bind Site II defined by SEQ ID NO: 1450. In some embodiments, the exemplary disorders, for example, a C5aR1-associated disorder, can be treated using an antibody capable of binding to C5aR1, comprises HCDRs, wherein the HCDRs comprise SEQ ID NOs: 1459, 1460 and/or 1461 are part of HCDR.

(424) In some embodiments, the exemplary disorders, for example, a C5aR1-associated disorder, can be treated using an antibody capable of binding to C5aR1, wherein the antibody binds C5aR1 with an affinity of 10 pM to 50 nM.

(425) In some embodiments, the exemplary disorders, for example, a C5aR1-associated disorder, can be treated using an antibody capable of binding to C5aR1, wherein the antibody inhibits G α signaling.

(426) In some embodiments, the exemplary disorders, for example, a C5aR1-associated disorder,

can be treated using an antibody capable of binding to C5aR1, wherein the antibody inhibits neutrophil chemotaxis.

(427) In some embodiments, the exemplary disorders, for example, a C5aR1-associated disorder, can be treated using an antibody capable of binding to C5aR1, wherein the antibody inhibits calcium release.

(428) In an embodiment, the exemplary disorders, for example, a C5aR1-associated disorder, can be treated using a combination of an antibody capable of binding to C5aR1, and an additional therapeutic or a second therapeutic. In some embodiments, the additional therapeutic for the treatment of a C5aR1 associated disease is a small molecule, such as, but not limited to avacopan.

(429) In an embodiment, the exemplary disorders, for example, a C5aR1-associated disorder, can be treated using a multispecific or biparatopic antibody capable of binding one or more amino acid residues of C5aR1 at Site I (SEQ ID NO: 1449 or SEQ ID NO: 1452) and one or more amino acid residues of C5aR1 at Site II (SEQ ID NO: 1450).

(430) In an embodiment, the exemplary disorders, for example, a C5aR1-associated disorder, can be treated using a multispecific or biparatopic antibody capable of competing with an antibody binding to C5aR1 at Site I (SEQ ID NO: 1449 or SEQ ID NO: 1452) and at Site II (SEQ ID NO: 1450).

(431) In an embodiment, the exemplary disorders, for example, a C5aR1-associated disorder, can be treated using a multispecific or biparatopic antibody capable of binding to C5aR1, the antibody comprising the amino acid sequences AYAMS (SEQ ID NO: 1456), SISTGGNTY (SEQ ID NO: 1457), and GYQRFSGFAY (SEQ ID NO: 1458) or a variant thereof; and an antibody comprising the amino acid sequences RSSQSLVHSNGNTYLN (SEQ ID NO: 1459), KVSNRLS (SEQ ID NO: 1460), and SQSTHVPYT (SEQ ID NO: 1461) or a variant thereof. In some embodiments, the multispecific or biparatopic antibody variant for the treatment of a C5aR1-associated disorder, comprises an amino acid sequence that differs by no more than 5, 4, 3, 2 or 1 amino acids from the amino acids of SEQ ID Nos: 1459, 1460 and/or 1461. In some embodiments, the multispecific or biparatopic antibody variant for the treatment of a C5aR1-associated disorder, comprises an amino acid sequence that differs by no more than 5, 4, 3, 2 or 1 amino acids from the amino acids of SEQ ID NO: 1456, SEQ ID NO: 1457 or SEQ ID NO: 1458.

(432) In an embodiment, the exemplary disorders, for example, a C5aR1-associated disorder, can be treated using a multispecific or biparatopic antibody capable of competing with an antibody binding to C5aR1 at Site I (SEQ ID NO: 1449 or SEQ ID NO: 1452) and at Site II (SEQ ID NO: 1450).

(433) In some embodiments, the exemplary disorders, for example, a C5aR1-associated disorder, can be treated using a multispecific or biparatopic antibody capable of binding to C5aR1, wherein the antibody binds C5aR1 with an affinity of 10 pM to 50 nM.

(434) In some embodiments, the exemplary disorders, for example, a C5aR1-associated disorder, can be treated using a multispecific or biparatopic antibody capable of binding to C5aR1, wherein the antibody inhibits G α signaling.

(435) In some embodiments, the exemplary disorders, for example, a C5aR1-associated disorder, can be treated using a multispecific or biparatopic antibody capable of binding to C5aR1, wherein the antibody inhibits neutrophil chemotaxis.

(436) In an embodiment, the exemplary disorders, for example, a C5aR1-associated disorder, can be treated using a multispecific or biparatopic antibody capable of binding to C5aR1, but not to C5aR2 or other GPCRs.

(437) In an embodiment, the exemplary disorders, for example, a C5aR1-associated disorder, can be treated using a combination of a multispecific or a biparatopic antibody capable of binding to C5aR1, and an additional therapeutic or a second therapeutic. In some embodiments, the additional therapeutic for the treatment of a C5aR1 associated disease is a small molecule, such as, but not limited to avacopan.

(438) Exemplary disorders or conditions that can be treated or prevented by the antibody molecules described herein also include, but are not limited to, vasculitis (e.g., ANCA-vasculitis), an autoimmune disorder (e.g., rheumatoid arthritis or lupus), or cancer. In an embodiment, the disorder is associated with aberrant expression of C5aR1 or C5a. In an embodiment, the antibody molecule is used to treat a subject having a disorder described herein or is at risk of developing a disorder described herein. Exemplary disorders that can be treated with an antibody molecule described herein are listed, e.g., in Table 7. Table 7 also lists exemplary agents that can be combined with an antibody molecule described herein for treating such disorders.

(439) TABLE-US-00013 TABLE 7 Exemplary disorders treatable with anti-C5aR1 antibody molecules

Indication	Combination Agent(s)
ANCA-Vasculitis	Avacopan
C3 glomerulopathy/aHUS	Avacopan
IgA nephropathy	Avacopan
Hidradenitis Suppurativa	Avacopan, IFX-1
Pyoderma gangrenosum	IFX-1
Sepsis	IFX-1
Rheumatoid arthritis	IPH5401, PMX53
Osteoarthritis	PMX53
Psoriasis	PMX53
Age-related macular degeneration (AMD)	PMX53
Graft versus host disease (GvHD)	ALXN-1007
Anti-phospholipid syndrome	ALXN-1007
Cancer	MOR210 (TJ210) + CKIs
Cancer	IPH5401 + CKIs
Amyotrophic lateral sclerosis (ALS)	Huntington's disease
Alzheimer's disease	Neuropathic pain

(440) Further examples of diseases and disorders that can be treated with an antibody molecule described herein are listed, for example, in Table 2 of Sadik et al. (*Semin. Immunol.* 37:21-29 (2018)); incorporated herein by reference in its entirety.

(441) The antibody molecules described herein are typically administered at a frequency that keeps a therapeutically effective level of antibody molecules in the patient's system until the patient recovers. For example, the antibody molecules may be administered at a frequency that achieves a serum concentration sufficient for at least about 1, 2, 5, 10, 20, 30, or 40 antibody molecules to bind each C5aR1 molecule. In an embodiment, the antibody molecules are administered every 1, 2, 3, 4, 5, 6, or 7 days, every 1, 2, 3, 4, 5, or 6 weeks, or every 1, 2, 3, 4, 5, or 6 months.

(442) Methods of administering various antibody molecules are known in the art and are described below. Suitable dosages of the antibody molecules used will depend on the age and weight of the subject and the particular drug used.

(443) In an embodiment, the antibody molecule is administered to the subject (e.g., a human subject) intravenously. In an embodiment, the antibody molecule is administered to the subject at a dose between 0.1 mg/kg and 50 mg/kg, e.g., between 0.2 mg/kg and 25 mg/kg, between 0.5 mg/kg and 10 mg/kg, between 0.5 mg/kg and 5 mg/kg, between 0.5 mg/kg and 3 mg/kg, between 0.5 mg/kg and 2.5 mg/kg, between 0.5 mg/kg and 2 mg/kg, between 0.5 mg/kg and 1.5 mg/kg, between 0.5 mg/kg and 1 mg/kg, between 1 mg/kg and 1.5 mg/kg, between 1 mg/kg and 2 mg/kg, between 1 mg/kg and 2.5 mg/kg, between 1 mg/kg and 3 mg/kg, between 1 mg/kg and 2.5 mg/kg, or between 1 mg/kg and 5 mg/kg. In an embodiment, the antibody molecule is administered to the subject at a fixed dose between 10 mg and 1000 mg, e.g., between 10 mg and 500 mg, between 10 mg and 250 mg, between 10 mg and 150 mg, between 10 mg and 100 mg, between 10 mg and 50 mg, between 250 mg and 500 mg, between 150 mg and 500 mg, between 100 mg and 500 mg, between 50 mg and 500 mg, between 25 mg and 250 mg, between 50 mg and 150 mg, between 50 mg and 100 mg, between 100 mg and 150 mg, between 100 mg and 200 mg, or between 150 mg and 250 mg. In an embodiment, the antibody molecule is administered once a week, twice a week, once every two weeks, once every three weeks, once every four weeks, once every eight weeks, once a month, once every two months, or once every three months. In an embodiment, the antibody molecule is administered between 0.5 mg/kg and 3 mg/kg or between 50 mg and 150 mg, once a week, twice a week, once every two weeks, or once every four weeks.

(444) The antibody molecules can be used by themselves or conjugated to a second agent, e.g., a bacterial agent, toxin, or protein, e.g., a second anti-C5aR1 antibody molecule. This method includes: administering the antibody molecule, alone or conjugated to a second agent, to a subject requiring such treatment. The antibody molecules can be used to deliver a variety of therapeutic

agents, e.g., a toxin, or mixtures thereof.

(445) Vasculitis

(446) The antibody molecule described herein can be used to treat or prevent vasculitis (e.g., ANCA-vasculitis). Vasculitis is a group of disorders that destroy blood vessels by inflammation. Vasculitis is primarily caused by leukocyte migration and resultant damage. Exemplary types of vasculitis include, but are not limited to, ANCA-vasculitis, Henoch-Schönlein purpura, acute proliferative glomerulonephritis (e.g., post-streptococcal glomerulonephritis), microscopic polyarteritis (poly-angiitis), Wegener's granulomatosis, polyarteritis nodosa,

(447) ANCA-Vasculitis

(448) In an embodiment, the antibody molecule is used to treat or prevent anti-neutrophil cytoplasmic autoantibody (ANCA)-vasculitis (also known as antineutrophil cytoplasmic antibody-associated vasculitis or ANCA-associated vasculitis (AAV)).

(449) ANCA-vasculitis is an autoimmune disease characterized by an immune response to either myeloperoxidase (MPO-ANCA) or proteinase 3 (PR3-ANCA), proteins which are expressed primarily in neutrophils. The result generally includes inflammation and destruction of small and medium-sized blood vessels, rapidly progressive glomerulonephritis, and the presence of circulating antibodies against MPO-ANCA and PR3-ANCA. The blood vessels of the kidney or lung are most commonly affected by this disease. ANCAs lead to small vessel vasculitis and crescentic glomerulonephritis. In the United States, 40,000-75,000 new cases are reported each year. Patients are prone to relapse with 11-16% of patients progressing to end-stage renal disease and death within one year of diagnosis. The current treatment regimen consists of high-dose steroids, typically prednisone, in combination with immunosuppressants, such as cyclophosphamide and rituximab. The steroids eventually need to be tapered off due to its side-effect. It is believed that the treatments contribute to ~60% of mortality rate. Without wishing to be bound by theory, it is believed that the antibody molecules described herein can have certain clinical benefits of C5aR1 targeting (e.g., BVAS score, reduced uACR, or reduced urinary MCP-1), which can allow for reduced use or replacement of high dose glucocorticoids in induction therapy for improved safety profile and quality of life.

(450) ANCA-vasculitis progression generally involves two key facets: the initial production and presence of ANCAs and C5a-C5aR1 signaling in neutrophils. The potent anaphylatoxin, C5a, promotes chemotaxis and activation of neutrophils, a key driver in inflammatory diseases driven by type III hypersensitivities such as ANCA-vasculitis. The binding of C5a to C5aR1 is one of the terminal events in the complement pathway that has pleiotropic effects, with one major consequence being the migration, trafficking, and activation of leukocytes, including neutrophils. ANCA-vasculitis symptoms can be induced in mice by passive transfer of anti-MPO serum, indicating that the presence of ANCAs alone can initiate the disease in wild type mice. Additionally, passive transfer of anti-MPO serum to C5aR1^{-/-} mice does not induce ANCA-vasculitis symptoms, indicating that C5aR1 is critical for disease progression in mice. Additionally, administration of C5aR1 antagonists in an MPO-ANCA model can limit the humoral response to MPO.

(451) MPO and PR3 are stored in neutrophil granules and are released upon activation. Resting neutrophils do not have cell surface associated MPO and have low levels of cell surface associated PR3. Neutrophil priming is required for further ANCA-induced activation, which can be accomplished by various cytokines including TNF-alpha, IL-1, IL-6, IL-18, fMLF, and C5a. ANCA-vasculitis disease onset is often preceded by an infection, providing a possible mechanism for cytokine release, neutrophil priming, and subsequent ANCA activation. C5a production and C5aR1 signaling promote further recruitment and activation of neutrophils.

(452) Standards for measuring ANCA-vasculitis include, e.g., Birmingham Vasculitis Activity Score as described, e.g., in Luqmani et al (1994) QJM 87 (11): 671-678; Luqmani et al. (1997) Baillieres Clin Rheumatol 11 (2): 423-446; Mukhtyar et al., (2009) *Ann Rheum Dis*. 68 (12): 1827-

1832.

(453) Other treatments that can be used in combination with the antibody molecule described herein to treat ANCA include, e.g., a glucocorticoid (e.g., prednisolone), cyclophosphamide, an anti-CD20 antibody (e.g., rituximab), a small molecule selective inhibitor of C5a receptor (e.g., avacopan (CCX168)), plasma exchange (PEX), or a maintenance therapy (e.g., an immunosuppressant, e.g., azathioprine or methotrexate).

(454) Henoch-Schönlein Purpura

(455) In an embodiment, the antibody molecule is used to treat or prevent Henoch-Schönlein purpura.

(456) Henoch-Schönlein purpura (HSP, also known as anaphylactoid purpura, purpura rheumatica, or Schonlein-Henoch purpura) is a disease of the skin and other organs that most commonly affects children. HSP is a systemic vasculitis (inflammation of blood vessels) and is characterized by deposition of immune complexes of IgA and complement component 3 (C3) on arterioles, capillaries, and venules. In the skin, the disease causes palpable purpura (small hemorrhages); often with joint and abdominal pain. With kidney involvement, there may be a loss of small amounts of blood and protein in the urine; in a small proportion of cases, the kidney involvement proceeds to chronic kidney disease even irreversible kidney damage. HSP is often preceded by an infection, such as a throat infection.

(457) Symptoms of Henoch-Schönlein purpura include, e.g., rash (purpura), swollen or sore joints (arthritis), gastrointestinal symptoms (e.g., abdominal pain, nausea, vomiting or bloody stools), and kidney involvement (e.g., protein or blood in the urine). Serum levels of IgA are high in HSP patients.

(458) Standards for defining Henoch-Schönlein purpura include, e.g., the 1990 American College of Rheumatology (ACR) classification (Mills et al. (1990). Arthritis and Rheumatism 33 (8): 1114-21), the 1994 Chapel Hill Consensus Conference (CHCC) (Jennette et al. (1994) Arthritis and Rheumatism 37 (2): 187-92), and the 2006 European League Against Rheumatism (EULAR) and Pediatric Rheumatology Society (PReS) classification, which includes palpable purpura as a mandatory criterion, together with at least one of the following findings: diffuse abdominal pain, predominant IgA deposition (confirmed on skin biopsy), acute arthritis in any joint, and renal involvement (as evidenced by the presence of blood and/or protein in the urine) (Ozen et al. (2006) Annals of Rheumatic Diseases 65 (7): 936-41).

(459) Other treatments that can be used in combination with the antibody molecule described herein to treat Henoch-Schonlein purpura include, e.g., analgesics for the abdominal and joint pains, steroids (e.g., oral steroids or a combination of intravenous methylprednisolone (steroid), cyclophosphamide and dipyridamole followed by prednisone). Other regimens also include, e.g., steroids/azathioprine, and steroids/cyclophosphamide (with or without heparin and warfarin), or intravenous immunoglobulin (IVIG).

(460) Acute Proliferative Glomerulonephritis

(461) In another embodiment, the antibody molecule is used to treat acute proliferative glomerulonephritis, e.g., post-streptococcal glomerulonephritis.

(462) Acute proliferative glomerulonephritis is a disorder of the glomeruli (glomerulonephritis), or small blood vessels in the kidneys. It is a common complication of bacterial infections, typically skin infection by *Streptococcus* bacteria types 12, 4 and 1 (impetigo) but also after streptococcal pharyngitis, for which it is also known as postinfectious or poststreptococcal glomerulonephritis. The infection causes blood vessels in the kidneys to develop inflammation, which hampers the renal organs ability to filter urine.

(463) The pathophysiology of this disorder is consistent with an immune complex mediated mechanism. This disorder produces proteins that have different antigenic determinants, which in turn have an affinity for sites in the glomerulus. As soon as binding occurs to the glomerulus, via interaction with properdin, complement is activated. Complement fixation causes the generation of

additional inflammatory mediators.

(464) Symptoms of acute proliferative glomerulonephritis include, e.g., hematuria, oliguria, edema, hypertension, fever, headache, malaise, anorexia, and nausea.

(465) Other treatments that can be used in combination with the antibody molecule described herein to treat acute proliferative glomerulonephritis includes, e.g., blood pressure (BP) control and control of the amount of potassium in individuals with oliguric acute kidney injury.

(466) Autoimmune Disorders

(467) The antibody molecules described herein can be used to treat or prevent an autoimmune disorder.

(468) Exemplary autoimmune disorders that can be treated or prevented by the antibody molecule described herein include, but are not limited to, lupus, acute Disseminated Encephalomyelitis (ADEM), acute necrotizing hemorrhagic leukoencephalitis, Addison's disease, agammaglobulinemia, alopecia areata, amyloidosis, ankylosing spondylitis, anti-GBM/anti-TBM nephritis, antiphospholipid syndrome (APS), autoimmune angioedema, autoimmune aplastic anemia, autoimmune dysautonomia, autoimmune hepatitis, autoimmune hyperlipidemia, autoimmune immunodeficiency, autoimmune inner ear disease (AIED), autoimmune myocarditis, autoimmune oophoritis, autoimmune pancreatitis, autoimmune retinopathy, autoimmune thrombocytopenia purpura (ATP), autoimmune thyroid disease, autoimmune urticaria, axonal & neuronal neuropathies, Baló disease, Behçet's disease, bullous pemphigoid, cardiomyopathy, Castleman disease, celiac disease, Chagas disease, chronic fatigue syndrome, chronic inflammatory demyelinating polyneuropathy (CIDP), chronic recurrent multifocal osteomyelitis (CRMO), Churg-Strauss syndrome, cicatricial pemphigoid/benign mucosal pemphigoid, Crohn's disease, Cogan's syndrome, cold agglutinin disease, congenital heart block, coxsackie myocarditis, CREST disease, essential mixed cryoglobulinemia, demyelinating neuropathies, dermatitis herpetiformis, dermatomyositis, Devic's disease (neuromyelitis optica), discoid lupus, Dressler's syndrome, endometriosis, eosinophilic esophagitis, eosinophilic fasciitis, Erythema nodosum, experimental allergic encephalomyelitis, Evans syndrome, fibromyalgia, fibrosing alveolitis, giant cell arteritis (temporal arteritis), giant cell myocarditis, glomerulonephritis, Goodpasture's syndrome, granulomatosis with polyangiitis (GPA) (formerly called Wegener's Granulomatosis), Graves' disease, Guillain-Barre syndrome, Hashimoto's encephalitis, Hashimoto's thyroiditis, hemolytic anemia, Henoch-Schönlein purpura, Herpes gestationis, hypogammaglobulinemia, idiopathic thrombocytopenia purpura (ITP), IgA nephropathy, IgG4-related sclerosing disease, immunoregulatory lipoproteins, inclusion body myositis, interstitial cystitis, juvenile arthritis, juvenile diabetes (type 1 diabetes), juvenile myositis, Kawasaki syndrome, Lambert-Eaton syndrome, Leukocytoclastic vasculitis, Lichen planus, Lichen sclerosus, Ligneous conjunctivitis, linear IgA disease (LAD), lupus (SLE), Lyme disease, chronic, Ménière's disease, microscopic polyangiitis, mixed connective tissue disease (MCTD), Mooren's ulcer, Mucha-Habermann disease, multiple sclerosis, myasthenia gravis, myositis, narcolepsy, neuromyelitis optica (Devic's), neutropenia, ocular cicatricial pemphigoid, optic neuritis, palindromic rheumatism, PANDAS (Pediatric Autoimmune Neuropsychiatric Disorders Associated with *Streptococcus*), paraneoplastic cerebellar degeneration, paroxysmal nocturnal hemoglobinuria (PNH), Parry Romberg syndrome, Parsonage-Turner syndrome, Pars planitis (peripheral uveitis), pemphigus, peripheral neuropathy, perivenous encephalomyelitis, pernicious anemia, POEMS syndrome, polyarteritis nodosa, Type I, II, & III autoimmune polyglandular syndromes, polymyalgia rheumatica, polymyositis, postmyocardial infarction syndrome, postpericardiotomy syndrome, progesterone dermatitis, primary biliary cirrhosis, primary sclerosing cholangitis, psoriasis, psoriatic arthritis, idiopathic pulmonary fibrosis, pyoderma gangrenosum, pure red cell aplasia, Raynaud's phenomenon, reactive arthritis, reflex sympathetic dystrophy, Reiter's syndrome, relapsing polychondritis, restless legs syndrome, retroperitoneal fibrosis, rheumatic fever, rheumatoid arthritis, sarcoidosis, Schmidt syndrome, scleritis, scleroderma, Sjögren's syndrome, sperm & testicular autoimmunity, stiff

person syndrome, subacute bacterial endocarditis (SBE), Susac's syndrome, sympathetic ophthalmia, Takayasu's arteritis, temporal arteritis/Giant cell arteritis, thrombocytopenia purpura (TTP), Tolosa-Hunt syndrome, transverse myelitis, Type 1 diabetes, ulcerative colitis, undifferentiated connective tissue disease (UCTD), uveitis, vasculitis, vesiculobullous dermatosis, vitiligo, Wegener's granulomatosis (also known as Granulomatosis with Polyangiitis (GPA). (469) In an embodiment, the autoimmune disorder is rheumatoid arthritis, systemic lupus erythematosus, a linear IgA bullous disease (e.g., linear immunoglobulin A (IgA) dermatosis), or IgA-mediated epidermolysis bullosa acquisita.

(470) In an embodiment, the antibody molecule is used to treat rheumatoid arthritis. Other treatments that can be used in combination with the antibody molecule described herein to treat rheumatoid arthritis includes, e.g., an NSAID, a steroid (e.g., corticosteroid), a disease-modifying antirheumatic drug (DMARD) (e.g., methotrexate (TREXALL®), leflunomide (ARAVA®), hydroxychloroquine (PLAQUENIL®), or sulfasalazine (AZULFIDINE®), a biologic response modifier (e.g., abatacept (ORENCIA®), adalimumab (HUMIRA®), anakinra (KINERET®), certolizumab (CIMZIA®), etanercept (ENBREL®), golimumab (SIMPONI®), infliximab (REMICADE®), rituximab (RITUXAN®) and tocilizumab (ACTEMRA®), or Tofacitinib (XELJANZ®)), or surgery.

(471) In an embodiment, the antibody molecule is used to treat systemic lupus erythematosus. Other treatments that can be used in combination with the antibody molecule described herein to treat rheumatoid arthritis includes, e.g., an NSAID, an antimalarial drug (e.g., hydroxychloroquine (PLAQUENIL®), corticosteroid (e.g., prednisone), an immunosuppressant (e.g., azathioprine (IMURAN®, AZASAN®), mycophenolate (CELLCEPT®), leflunomide (ARAVA®), or methotrexate (TREXALL®)), or a BAFF inhibitor (e.g., belimumab (BENLYSTA®).

(472) In an embodiment, the antibody molecule is used to treat a linear IgA bullous disease (e.g., linear immunoglobulin A (IgA) dermatosis). Other treatments that can be used in combination with the antibody molecule described herein to treat a linear IgA bullous disease (e.g., linear immunoglobulin A (IgA) dermatosis) include, e.g., corticosteroids (e.g., prednisone or prednisolone), an antibiotic (e.g., tetracycline, erythromycin, sulfapyridine), colchicine, or mycophenolate mofetil.

(473) In an embodiment, the antibody molecule is used to treat IgA-mediated epidermolysis bullosa acquisita. Other treatments that can be used in combination with the antibody molecule described herein to treat IgA-mediated epidermolysis bullosa acquisita includes, e.g., an antibiotic, an anti-inflammatory drug (e.g., corticosteroid), or surgery.

(474) Cancer

(475) The antibody molecules described herein can be used to treat or prevent a cancer or a metastatic lesion thereof.

(476) In an embodiment, the cancer is a solid tumor. In an embodiment, the cancer is a hematological cancer (e.g., a leukemia, a lymphoma, and a multiple myeloma). In an embodiment, the cancer is a soft tissue sarcoma. In an embodiment, the cancer is associated with C5aR1 activation.

(477) Exemplary cancers that can be treated or prevented by the polypeptides described herein include, but are not limited to, acute lymphoblastic leukemia (ALL), acute myeloid leukemia (AML), adrenocortical carcinoma, Kaposi sarcoma, an AIDS-related lymphoma, primary central nervous system (CNS) lymphoma, anal cancer, appendix cancer, astrocytoma, atypical teratoid/rhabdoid tumor, basal cell carcinoma, bile duct cancer, bladder cancer, bone cancer (e.g., Ewing sarcoma or osteosarcoma and malignant fibrous histiocytoma), brain tumor (e.g., astrocytomas, brain stem glioma, central nervous system atypical teratoid/rhabdoid tumor, central nervous system embryonal tumor, central nervous system germ cell tumor, craniopharyngioma, or ependymoma), breast cancer, bronchial tumor, Burkitt lymphoma, carcinoid tumor (e.g., gastrointestinal carcinoid tumor), cardiac (heart) tumor, embryonal tumor, germ cell tumor,

lymphoma, cervical cancer, cholangiocarcinoma, chordoma, chronic lymphocytic leukemia (CLL), chronic myelogenous leukemia (CML), chronic myeloproliferative neoplasm, colon cancer, colorectal cancer, craniopharyngioma, cutaneous T-cell lymphoma, ductal carcinoma in situ (DCIS), endometrial cancer, ependymoma, esophageal cancer, esthesioneuroblastoma, Ewing sarcoma, extracranial germ cell tumor, extragonadal germ cell tumor, eye cancer (e.g., intraocular melanoma or retinoblastoma), fallopian tube cancer, fibrous histiocytoma of bone, osteosarcoma, gallbladder cancer, gastric (stomach) cancer, gastrointestinal carcinoid tumor, gastrointestinal stromal tumors (GIST), germ cell tumor (e.g., central nervous system tumor, extracranial tumor, extragonadal tumor, ovarian cancer, or testicular cancer), gestational trophoblastic disease, glioma, hairy cell leukemia, head and neck cancer, hepatocellular (liver) cancer, Hodgkin lymphoma, hypopharyngeal cancer, intraocular melanoma, islet cell tumor, pancreatic neuroendocrine tumor, Kaposi sarcoma, kidney cancer (e.g., renal cell cancer or Wilms tumor), Langerhans cell histiocytosis (LCH), laryngeal cancer, leukemia (e.g., acute lymphoblastic leukemia (ALL), acute myeloid leukemia (AML), chronic lymphocytic leukemia (CLL), chronic myelogenous leukemia (CML), or hairy cell leukemia), lip and oral cavity cancer, liver cancer, lung cancer (e.g., non-small cell lung cancer (NSCLC) or small cell lung cancer), lymphoma (e.g., AIDS-related, Burkitt lymphoma, cutaneous T-cell lymphoma, Hodgkin lymphoma, non-Hodgkin lymphoma, or primary central nervous system (CNS) lymphoma), Waldenström macroglobulinemia, male breast cancer, malignant fibrous histiocytoma of bone and osteosarcoma, melanoma (e.g., intraocular (eye) melanoma), Merkel cell carcinoma, mesothelioma, metastatic squamous neck cancer, midline tract carcinoma, mouth cancer, multiple endocrine neoplasia syndrome, multiple myeloma/plasma cell neoplasm, mycosis fungoides, myelodysplastic syndrome, myelodysplastic/myeloproliferative neoplasm, chronic myeloproliferative neoplasm, nasal cavity and paranasal sinus cancer, nasopharyngeal cancer, neuroblastoma, oral cancer, lip and oral cavity cancer, oropharyngeal cancer, osteosarcoma and malignant fibrous histiocytoma of bone, ovarian cancer (e.g., epithelial ovarian cancer or germ cell ovarian tumor), pancreatic cancer, pancreatic neuroendocrine tumors (islet cell tumors), papillomatosis, paraganglioma, paranasal sinus and nasal cavity cancer, parathyroid cancer, penile cancer, pharyngeal cancer, pheochromocytoma, pituitary tumor, pleuropulmonary blastoma, peritoneal cancer, prostate cancer, rectal cancer, retinoblastoma, rhabdomyosarcoma, salivary gland cancer, sarcoma (e.g., Ewing sarcoma, Kaposi sarcoma, osteosarcoma, rhabdomyosarcoma, soft tissue sarcoma, or uterine sarcoma), Sézary syndrome, skin cancer (e.g., melanoma, Merkel cell carcinoma, or nonmelanoma skin cancer), small intestine cancer, squamous cell carcinoma, testicular cancer, throat cancer, thymoma and thymic carcinoma, thyroid cancer, transitional cell cancer of the renal pelvis and ureter, urethral cancer, endometrial uterine cancer, vaginal cancer, vulvar cancer, or a metastatic lesion thereof.

(478) Combination Therapies

(479) The antibody molecules can be used in combination with other therapies. For example, the combination therapy can include an antibody molecule co-formulated with, and/or co-administered with, one or more additional therapeutic agents, e.g., one or more additional therapeutic agents described herein. In other embodiments, the antibody molecules are administered in combination with other therapeutic treatment modalities, e.g., other therapeutic treatment modalities described herein. Such combination therapies may advantageously utilize lower dosages of the administered therapeutic agents, thus avoiding possible toxicities or complications associated with the various monotherapies.

(480) Administered “in combination”, as used herein, means that two (or more) different treatments are delivered to the subject before, or during the course of the subject's affliction with a disorder. In an embodiment, two or more treatments are delivered prophylactically, e.g., before the subject has the disorder or is diagnosed with the disorder. In another embodiment, the two or more treatments are delivered after the subject has developed or diagnosed with the disorder. In an embodiment, the delivery of one treatment is still occurring when the delivery of the second begins, so that there is

overlap. This is sometimes referred to herein as “simultaneous” or “concurrent delivery.” In other embodiments, the delivery of one treatment ends before the delivery of the other treatment begins. In an embodiment of either case, the treatment is more effective because of combined administration. For example, the second treatment is more effective, e.g., an equivalent effect is seen with less of the second treatment, or the second treatment reduces symptoms to a greater extent, than would be seen if the second treatment were administered in the absence of the first treatment, or the analogous situation is seen with the first treatment. In an embodiment, delivery is such that the reduction in a symptom, or other parameter related to the disorder is greater than what would be observed with one treatment delivered in the absence of the other. The effect of the two treatments can be partially additive, wholly additive, or greater than additive. The delivery can be such that an effect of the first treatment delivered is still detectable when the second is delivered. (481) In certain embodiments, the additional agent is a second antibody molecule, e.g., an antibody molecule different from a first antibody molecule. Exemplary antibody molecules that can be used in combination include, but are not limited to, an antibody molecule as described herein, e.g., in Table 1C.

(482) In an embodiment, the antibody molecule is administered in combination with a second therapy to treat or prevent ANCA-vasculitis.

(483) In an embodiment, the antibody molecule is administered in combination with a small molecule C5aR1 antagonist, such as avacopan or NDT9513727. In an embodiment, the antibody molecule is administered in combination with a high-dose steroid, e.g., prednisone. In an embodiment, the antibody molecule is administered in combination with an immunosuppressant, e.g., cyclophosphamide or rituximab.

(484) In an embodiment, the antibody molecule is administered in combination with an agent described in Table 7, e.g., to treat a disorder described in Table 7.

(485) In an embodiment, the antibody molecule is administered in combination with prednisone and/or cyclophosphamide. In an embodiment, prednisone is administered at a dose between 0.2 mg/kg and 2 mg/kg, e.g., between 0.5 mg/kg and 1 mg/kg, e.g., once a day. In an embodiment, cyclophosphamide is administered at a dose between 0.2 g and 2 g, e.g., between 0.5 g and 1 g, e.g., once a day.

(486) In an embodiment, the antibody molecule is administered in combination with rituximab (RITUXAN®). Rituximab is a chimeric anti-CD20 monoclonal antibody. In an embodiment, rituximab is administered at a dose between 100 mg/m² and 500 mg/m², e.g., between 200 mg/m² and 450 mg/m² or between 300 mg/m² and 400 mg/m², intravenously, e.g., once weekly, once every two weeks, once every four weeks, or once every eight weeks.

(487) Exemplary therapies that can be used in combination with an antibody molecule or composition described herein to treat or prevent other disorders are also described in the section of “Methods of Treating or Preventing Disorders” herein.

(488) Methods of Diagnosis

(489) In some aspects, the present disclosure provides a diagnostic method for detecting the presence of C5aR1 in vitro (e.g., in a biological sample, such as a biopsy or blood sample) or in vivo (e.g., in vivo imaging in a subject). The method includes: (i) contacting the sample with an antibody molecule described herein, or administering to the subject, the antibody molecule; (optionally) (ii) contacting a reference sample, e.g., a control sample (e.g., a control biological sample, such as a biopsy or blood sample) or a control subject with an antibody molecule described herein; and (iii) detecting formation of a complex between the antibody molecule and C5aR1 in the sample or subject, or the control sample or subject, wherein a change, e.g., a statistically significant change, in the formation of the complex in the sample or subject relative to the control sample or subject is indicative of the presence of C5aR1 in the sample. The antibody molecule can be directly or indirectly labeled with a detectable substance to facilitate detection of the bound or unbound antibody. Suitable detectable substances include various enzymes, prosthetic groups, fluorescent

materials, luminescent materials and radioactive materials, as described above and described in more detail below.

(490) The term “sample,” as it refers to samples used for detecting a polypeptide (e.g., C5aR1) or a nucleic acid encoding the polypeptide includes, but is not limited to, cells, cell lysates, proteins or membrane extracts of cells, body fluids such as blood, or tissue samples such as biopsies.

(491) Complex formation between the antibody molecule, and C5aR1, can be detected by measuring or visualizing either the antibody molecule bound to C5aR1 or unbound antibody molecule. Any suitable detection assays can be used, and conventional detection assays include an enzyme-linked immunosorbent assays (ELISA), a radioimmunoassay (RIA) or tissue immunohistochemistry. Alternative to labeling the antibody molecule, the presence of C5aR1 can be assayed in a sample by a competition immunoassay utilizing standards labeled with a detectable substance and an unlabeled antibody molecule. In this assay, the biological sample, the labeled standards and the antibody molecule are combined and the amount of labeled standard bound to the unlabeled binding molecule is determined. The amount of C5aR1 in the sample is inversely proportional to the amount of labeled standard bound to the antibody molecule.

(492) The antibody molecules described herein can be used to diagnose disorders that can be treated or prevented by the antibody molecules described herein. The detection or diagnostic methods described herein can be used in combination with other methods described herein to treat or prevent a disorder described herein.

EXAMPLES

Example 1: Development of Anti-C5aR1 Antibody Molecules

(493) This Example describes the development of anti-C5aR1 antagonistic antibody.

(494) Antagonistic antibodies that inhibit (e.g., prevent) the binding and/or signaling of the soluble agonist C5a to C5aR1 were generated. Briefly, mice were immunized with relevant antigens, followed by creation of hybridomas and interrogation of the hybridoma repertoire for C5aR1-specific antibodies that showed antagonistic activity against C5a-mediated C5aR1 signaling, e.g., resulting in inhibited calcium mobilization and chemotaxis of neutrophils in the presence of C5a.

(495) Immunization

(496) Mice were immunized with different immunogens, including C5aR1 DNA, C5aR1 virus-like particles (VLPs) and/or C5aR1 peptides representing N-terminal region and extracellular loop 2 (ECL2) of human C5aR1. Immunizations were performed in both wild type CDI mice and in C5aR1 knock-out mice. DNA immunization was performed by hydrodynamic tail vein injection or intradermal injection followed by electroporation. VLP immunization was performed by subcutaneous injection with the use of adjuvant at an adjacent injection site. A final boost was performed with VLPs and/or C5aR1 peptides by tail vein injection 72 and 48 hours prior to splenic harvest.

(497) Subsequent splenic fusion followed by hybridoma screening successfully identified functional antibodies that blocked C5aR1 signaling. Hybridoma cell lines were created by polyethylene glycol fusion of spleen cells with a myeloma fusion partner. Cells were plated into 384 well plates and grown for 2-3 weeks. Upon outgrowth, the supernatant was screened for reactivity to C5aR1 and C5aR2 in a VLP ELISA. Clones with specificity for C5aR1 were expanded for cryopreservation and RNA extraction. RT-PCR was performed to amplify antibody heavy and light chain transcripts followed by Sanger sequencing. Upon sequence identification, recombinant antibodies were expressed, purified, and tested for their ability to prevent C5aR1 signaling in a panel of orthogonal assays.

(498) The following approaches were utilized to increase the number and diversity of anti-C5aR1 antibodies: 1. DNA immunization in combination with heterologous C5aR1-VLP protein boosts were used to increase the human C5aR1 immune response. DNA immunization typically elicits a weaker immune response compared to protein immunization, so it was expected that C5aR1-VLPs should augment the response. 2. Next generation sequencing (NGS) of mouse splenocytes were

used to identify clonal siblings of antibodies discovered by hybridoma screening. Due to the limited depth of hybridoma screening, it was expected that clonal variants, some of which may have improved functionality or biophysical properties, would be uncovered by NGS. Clonal variants were recombinantly expressed and their ability to block C5aR1 signaling were compared to the parental molecule. 3. C5aR1 knock-out mice were also immunized to reduce the tolerance imposed by the homologous mouse C5aR1. Despite sequence differences, human C5a can bind to both human and mouse C5aR1, indicating that structural homology is strong between the two species. 4. The use of genetic adjuvants, including DNA encoding IL-12a and CD40L, were used during DNA immunization to augment the humoral response to C5aR1. 5. All hybridoma plating and screening were performed in 384 well plates to increase the throughput.

(499) Antibodies against Site I and antibodies against Site II were both identified and selected for their ability to inhibit C5a binding to C5aR1.

(500) In Vitro Hybridoma Screening Strategy

(501) Hybridoma supernatant was screened for reactivity to C5aR1 using a high-throughput 384-well VLP-based ELISA. Briefly, C5aR1 or C5aR2 VLPs were immobilized to a MaxiSorp ELISA plate at a concentration of 30 µg/mL and incubated overnight at 4° C. The following morning, plates were washed 3 times with 1×PBS and plates were blocked for 30 minutes with 100 µL of PBSA (1×PBS with 3% BSA). A serial titration of anti-C5aR1 antibody was performed in the presence of PBSA and incubated for 1 hour at room temperature. Plates were washed 6 times with PBSA. Anti-human-HRP was diluted in PBSA, added to all wells, and incubated for 45 minutes at room temperature. Plates were washed 6 times with PBS. TMB substrate was added to all wells and incubated for 10 minutes before the addition of stop solution (0.1 M sulfuric acid). Absorbance at 450 nm was measured on a standard plate reader. A four-parameter curve fit was used to generate the EC50 value of the antibody titration in nM. The results of the VLP ELISA are shown in Table 5 below.

(502) Table 5. Reactivity of antibodies to C5aR1 and C5aR2 by VLP ELISA (EC.sub.50 values reported in nM of antibody)

(503) TABLE-US-00014 TABLE 5 Reactivity of antibodies to C5aR1 and C5aR2 by VLP ELISA (EC.sub.50 values reported in nM of antibody)

Antibody	C5aR1	C5aR2	Anti-C5aR1
Clone_11	>3.16E-8	<1E-12	Clone_11
Clone_11_V2	1.20E-11	>3.16E-8	Clone_66
Clone_66	7.00E-12	>3.16E-8	Clone_79
Clone_79	6.07E-12	>3.16E-8	Clone_184
Clone_184	1.42E-11	>3.16E-8	Clone_216
Clone_216	1.11E-11	>3.16E-8	Clone_272
Clone_272	7.12E-11	>3.16E-8	Clone_317_V1-2
Clone_317_V1-2	1.12E-10	>3.16E-8	Clone_317_V2
Clone_317_V2	5.39E-10	1.51E-09	Clone_317_V2-1
Clone_317_V2-1	7.66E-12	>3.16E-8	Clone_322
Clone_322	1.4E-12	>3.16E-8	Clone_322_V2
Clone_322_V2	1.05E-12	>3.16E-8	Clone_330
Clone_330	2.42E-10	>3.16E-8	Clone_336_V1
Clone_336_V1	1.87E-09	9.38E-09	Clone_336_V1-2
Clone_336_V1-2	3.05E-10	4.29E-10	Clone_336_V2
Clone_336_V2	2.14E-08	>3.16E-8	Clone_336_V2-1
Clone_336_V2-1	1.92E-10	1.71E-08	Clone_336_V3-1
Clone_336_V3-1	6.63E-11	>3.16E-8	Clone_336_V3-2
Clone_336_V3-2	1.27E-11	.sup. 3E-08	Clone_338_V1
Clone_338_V1	1.28E-09	>3.16E-8	Clone_338_V1-2
Clone_338_V1-2	8.5E-09	>3.16E-8	Clone_341_V1
Clone_341_V1	1.64E-11	>3.16E-8	Clone_341_V1-2
Clone_341_V1-2	2.65E-08	>3.16E-8	Clone_341_V2-1
Clone_341_V2-1	9.57E-09	>3.16E-8	Clone_341_V2-2
Clone_341_V2-2	1.51E-08	>3.16E-8	Clone_399
Clone_399	5.72E-12	>3.16E-8	Clone_402
Clone_402	.sup. <1E-12	>3.16E-8	Clone_429
Clone_429	6.33E-09	>3.16E-8	Clone_430
Clone_430	8.14E-10	>3.16E-8	Clone_440
Clone_440	6.5E-12	>3.16E-8	Clone_453
Clone_453	7.65E-12	>3.16E-8	Clone_454
Clone_454	>3.16E-8	>3.16E-8	Clone_465
Clone_465	6.7E-11	>3.16E-8	Clone_475
Clone_475	1.65E-10	>3.16E-8	Clone_481
Clone_481	8.45E-12	>3.16E-8	Clone_497
Clone_497	5.82E-10	>3.16E-8	Clone_503
Clone_503	3.82E-11	>3.16E-8	Clone_503_V2
Clone_503_V2	1.91E-08	>3.16E-8	Clone_507
Clone_507	3.81E-10	>3.16E-8	Clone_508
Clone_508	>3.16E-8	>3.16E-8	Clone_510
Clone_510	1.23E-09	>3.16E-8	Clone_511
Clone_511	2.01E-09	>3.16E-8	Clone_511_V2
Clone_511_V2	1.46E-09	>3.16E-8	Clone_518
Clone_518	1.2E-09	>3.16E-8	Clone_518_V2
Clone_518_V2	1.39E-09	>3.16E-8	Clone_528
Clone_528	7.34E-10	>3.16E-8	Clone_541
Clone_541	7.93E-10	>3.16E-8	Clone_547
Clone_547	8.08E-10	>3.16E-8	Clone_549
Clone_549	1.11E-10	>3.16E-8	Clone_550
Clone_550	3.45E-10	>3.16E-8	Clone_553
Clone_553	7.61E-11	>3.16E-8	Clone_556
Clone_556	5.03E-11	>3.16E-8	Clone_567
Clone_567	2.06E-11	>3.16E-8	Clone_568
Clone_568	1.11E-12	>3.16E-8	Clone_570
Clone_570	1.71E-11	>3.16E-8	Clone_573
Clone_573	1.64E-11	>3.16E-8	

>3.16E-8 Clone_583 .sup. <1E-12 >3.16E-8 Clone_584 2.82E-12 2.71E-10 Clone_585
 7.31E-12 >3.16E-8 Clone_586 .sup. <1E-12 >3.16E-8 Clone_588 1.61E-11 >3.16E-8
 Clone_592 .sup. <1E-12 >3.16E-8

(504) Follow up screening to ensure specificity was performed by measuring cell surface binding to cells expressing, for example, human C5aR1, human C5aR2, mouse C5aR1, and control cells that were mock transfected. Briefly, cell surface binding was performed on various C5aR1 expressing cell lines (transiently expressed Expi293 cells, stable C5aR1-U937 cells, and neutrophils isolated from whole blood). Briefly, a titration of antibody diluted in flow buffer (1×PBS with 2% fetal bovine serum) was incubated with cells for 1 hour at 4° C. Cells were washed twice with flow buffer. Anti-human-APC secondary antibody was diluted 1:400 in flow buffer and incubated with cells for 30 minutes. Cells were washed twice with flow buffer and resuspended in 100 µL of flow buffer. Fluorescence was measured using the iQue Screen Plus flow cytometer. The results of the cell binding assay are shown in Table 6 below.

(505) Table 6. Binding of exemplary anti-C5aR1 antibodies to human C5aR1 transiently expressed on the surface of Expi293 cells. Values represent the geometric mean fluorescent intensity for staining with 100 nM of antibody.

(506) TABLE-US-00015 TABLE 6 Binding of exemplary anti-C5aR1 antibodies to human C5aR1 transiently expressed on the surface of Expi293 cells. Values represent the geometric mean fluorescent intensity for staining with 100 nM of antibody. Cell Surface binding to human C5aR1 transiently expressed on the surface of Expi293 Cells Human Mock transfection Human Mock transfection Human Mock Antibody C5aR1 Antibody C5aR1 Antibody C5aR1 transfection
 Clone_11v2 310,187 4,976 Clone_454 416,030 6,232 Clone_556 550,678 7,127 Clone_66 280,444
 5,471 Clone_465 202,626 4,683 Clone_567 516,557 7,270 Clone_79 663,741 6,820 Clone_475
 597,201 7,673 Clone_568 604,145 6,676 Clone_184 456,014 8,512 Clone_503 586,998 7,927
 Clone_570 497,164 7,289 Clone_317v2-1 8,106 4,535 Clone_507 224,419 5,573 Clone_573
 398,667 7,208 Clone_322_v2 420,835 7,347 Clone_510 176,751 5,389 Clone_583 259,523 5,291
 Clone_336_v2-1 223,156 5,007 Clone_511v2 222,224 5,790 Clone_584 317,914 6,223 Clone_329
 327,683 4,776 Clone_518v2 202,739 5,576 Clone_585 58,312 5,837 Clone_330 265,443 4,938
 Clone_528 235,709 5,011 Clone_586 560,702 8,826 Clone_402 315,064 5,974 Clone_541 207,547
 4,862 Clone_588 621,217 6,392 Clone_429 191,235 5,217 Clone_547 586,606 6,575 Clone_592
 564,574 9,800 Clone_430 247,781 5,015 Clone_549 241,071 4,741 2nd only 5,346 3,052
 Clone_440 289,803 5,301 Clone_550 609,348 7,458 Isotype control 6,553 3,722 Clone_453
 258,623 5,398 Clone_553 590,125 6,871

(507) Reactive clones were purified by protein A/G chromatography. Antibodies that showed specific binding to human C5aR1 were recombinantly expressed and the purified antibodies tested for functionality in different functional assays.

(508) In Vitro Functional Analysis

(509) One of the primary mechanisms by which neutrophils migrate to the site of inflammation is by chemotaxis induced by a C5a/C5adesArg gradient is detected by C5aR1 signaling. Anti-C5aR1 antibodies were thus assessed in a chemotaxis assay for their ability to block C5a induced migration of neutrophils. The assay involved cell migration in a double chamber well separated by filters. The cells and inhibitors were applied into the upper chamber and the lower chamber was filled with the chemoattractant, causing the cells to migrate to the lower chamber. The inhibition activity was evaluated by counting the migrated cell numbers. Specifically, a titration of C5a was added to the bottom chamber a 96 transwell plate. Cells expressing human C5aR1 (C5aR1-U937 cells or human neutrophils) were incubated with or without antagonist and then seeded into the top chamber of the transwell plate. The cells were incubated for 2 hours. Cells that migrated to the bottom chamber were quantified using the CyQUANT cell proliferation kit following the manufacturer's protocol. The assay was performed using engineered cells stably expressing C5aR1 as well as using human neutrophils.

(510) Intracellular calcium is released into the cytoplasm by stimulation of GPCRs and acts as a second messenger of GPCR signaling, commonly used to quantify GPCR agonism and antagonism. As such, calcium flux assays using calcium sensitive dye were utilized to detect cytosolic changes in calcium concentration. Cells were incubated with esterified (inactive) calcium dye. The dye penetrated the cell membrane and became active once inside the cell. The active form of the dye became fluorescent after binding to intracellular calcium and the fluorescence was used to determine C5aR1 signaling in response to C5a addition. Specifically, stable C5aR1-U937 cells or neutrophils isolated from whole blood were stained with Fluo-4 Direct Calcium Assay kit from Thermo Fisher for 1 hour at 37° C. Next, antibody or antagonist was incubated with the cells for 30 minutes. The basal fluorescence with an excitation at 494 nm and emission at 516 nm was measured for 15 seconds. C5a was added to the cells and the fluorescence was measured over a span of four minutes. The basal read from before C5a stimulation and the max signal after C5a stimulation were used to calculate the response ratio. The potency of the antibodies was assessed for their ability to inhibit intracellular calcium release in C5aR1 expressing cells in the presence of C5a. Similar to the chemotaxis assay, the calcium flux assays were performed using engineered cells stably expressing C5aR1 as well as using human neutrophils.

(511) In addition, a GeneBLAzer reporter assay was used, which mimics the calcium flux assay. The GeneBLAzer assay kit and C5aR1 cell line were commercially available from Thermo Fisher (Catalog #K1544). The assay was performed as recommend by the manufacturer. Briefly, antibodies or antagonists were incubated for 30 minutes at 37° C. C5a was then added to the cells and incubated at 37° C. for an additional 4-5 hours. Beta-lactamase substrate was then added and incubated for 2 hours at room temperature. Fluorescent measurements for each well with an excitation/emission of 409/460 (blue) and 409/530 (green) were measured. An increase in the blue to green ratio was proportional to C5aR1 activation and was used to calculate the percent of activation for the cells in each well.

(512) Of the antibodies generated and screened, several were assessed using the GeneBLAzer assay. Antibody clones 66 and 583, which target Site I, were observed to be highly potent and have similar activity in the GeneBLAzer assay compared to Site II antibody clones 11v2 and 329 (FIGS. 2A-2D). In addition, a combination of antibody 66 (which targets Site I) and antibody 79 (which targeted site II) outperformed the activity of each antibody on its own in the GeneBLAzer assay (FIGS. 3A-3C, show that the combination of Site I and Site II targeting antibodies can have an additive or synergistic effect on C5aR1 antagonism.

(513) In Vivo Screening Strategy

(514) The antibodies that most potently inhibit C5aR1 mediated signaling in in vitro assays, will be evaluated in pre-clinical animal models. The antibodies will be evaluated in pharmacodynamic models as well as in an anti-MPO induced necrotizing and crescentic glomerulonephritis (NCGN) disease model.

(515) Administration of C5a intravenously causes a rapid reduction in blood neutrophil counts, which is observable within one minute after administration of C5a. Antagonism of C5aR1 can prevent the rapid reduction of blood neutrophil levels. Administration of anti-C5aR1 antibodies prior C5a challenge is expected to reduce neutropenia in these animals. Given the limited identity between human and rodent C5aR1, it is challenging to use wild type mice for evaluation of therapeutics targeting human C5aR1. Indeed, none of the antibodies reported in the literature or small molecules in clinical development cross-react with mouse C5aR1. To test our lead antibodies in rodents, we plan to employ two distinct strategies. The first is through the use of humanized mice with multi-lineage human immune cells. NGS or NSG-SGM3 mice are injected with human CD34+ stem cells and engraftment is confirmed within 12-16 weeks. The engraftment is stable for 12 months before the mice succumb to graft-versus-host disease. NGS-SGM3 mice express human IL3, GM-CSF, and SCF that help to support stable engraftment of myeloid lineages and regulatory T-cells. The second strategy is performing experiments in transgenic mice which has human C5aR1

transgene in place of native mouse C5aR1. Toward this Visterra is working with Jackson Laboratories to create human C5aR1 knock-in mice. Non-human primates present another model to study the pharmacodynamics of the lead antibodies.

(516) ANCA-vasculitis like disease can be induced in mice by passive transfer of anti-MPO IgG. Administration of anti-MPO IgG in mice leads them to develop urinary abnormalities (hematuria, albuminuria and leukocyturia), crescentic glomerulonephritis, and vasculitis. The severity of the disease severity can be enhanced by injection of lipopolysaccharide. Xiao et al. (2014, J Am Soc Nephrol 25 (2): p. 225-31) have previously demonstrated that oral administration of avacopan reduced the severity of anti-MPO NCGN in human C5aR1 transgenic mice. The antibodies identified herein will be evaluated in a similar model.

Example 2: Epitope Mapping

(517) This Example describes the mapping of epitopes of each antibody isolated from immunization.

(518) The epitope mapping was undertaken, using biolayer interferometry, on a series of exemplary anti-C5aR1 antibodies generated as described in Example 1. Briefly, for each antibody, biotinylated peptides representing ECL2, the N-terminus with tyrosine sulfation, and the N-terminus without tyrosine sulfation, were immobilized in a streptavidin biosensor. Subsequently, the biosensor was exposed to 10 µg/mL of antibody to detect antibody binding to the immobilized peptide. The results of the epitope mapping study are shown in Table 4 (“+” indicates binding to the listed peptide; “-” indicates lack of binding to the listed peptide).

(519) TABLE-US-00016 TABLE 4 Epitopes on C5aR1 for exemplary anti-C5aR1 antibody molecules

Antibody	Cyclic	ECL2	Sulfated	N-terminal	Non-sulfated	N-terminal
66	-	+	-	79	+	-
- 184	-	+	+	317	+	-
- 322_v2	-	+	-	329	+	-
- 332	-	+	-	335	-	+
- 336v2-1	+	-	-	343	+	-
- 402	-	+	+	429	+	-
- 453	+	-	-	511	+	-
- 518	+	-	-	583	-	+
- 584	+	-	-	588	+	-
- 11v2	+	-	-			

(520) A superposition of bound ECL2 peptide-mAb329 co-crystal with C5aR1 yielded a model of 329 Fab-C5aR1 complex (FIG. 4A), which revealed that of the 15 residues, inner ten are C5aR1 ECL2 residues and nine are making contacts with the Fab domain. As shown in FIG. 4B, the central 5 residues made most interactions with Fab E180 & Y181 are core epitopes whose Ala mutation abolished mAb binding while mutation of Y182, P183 & P184 reduced binding. E180 makes most ES and HB interactions with the LCDR1 residues. E179 made a salt bridge with HCDR2: K59, however epitope mapping showed that E179A mutation did not reduce antibody affinity.

(521) The model of the mab 329 with the ECL2 peptide revealed that residues R35, H101, D191, S193, H194, E266, P267, S268, F272, L273, K276 are the most essential epitopes for antibody binding. Further observation of the antibody-peptide model are noted in Table 8. Fab-C5aR1 model the antibody-receptor interface increases by 80% (580 to 1045 Å²).

(522) Based on the crystal structure, the underlined residues on epitope (as shown below) are the residues that are involved in critical contacts for antibody 329 and 583, respectively.

(523) mAb329

(524) Paratopes Residues:

(525) a) VH: W33, H35, W47, Y50, K58, G97, D98, P100c, Y100d b) VL: R24, S25, S26, Q27, S28, V30, H30a, S30b, N30c, G30d, Y32, G66, S67, G68, T69, D70, S91, T92, L93, V94, L96 Epitope Residues on C5aR1: c) R35, H101, V176, V177, R178, E179, E180, Y181, F182, P183 P184, K185, L187, D191, S193, H194, E266, P267, S268, F272, L273, K276

mAb583

Paratopes residues: d) VH: A31, Y32, A33, S50,151, S52, T53, G54, G55,N56, T57, Y58, Q97, R98, F99, S100 e) VL: H27d, S28e, N28, G29,N30, T31, Y32, K50, V51, S52,N53, S91, T92, H93,94, V95, P96, Y97 Epitope residues on C5aR1:

T7, T8, P9, D10, Y11, G12, H13, Y14, DIS, D16, K17, D18

(526) TABLE-US-00017 TABLE 8 The structural description of the interface between ECL2 and

mAb 329 C5aR1 binding (Model) VH VL Total VH VL Interface (Å.sup.2) 214 366 580 271 774
Hydrogen bonds 2 9 11 2 11 Salt bridges 1 1 2 1 1 paratopes W33, H35, W47, Q27, S28, V30,
W33, H35, W47, R24, S25, S26, Q27, Y50, K59, G101, H30a, S30b, N30c, Y50, K58, G100a,
S28, V30, H30a, S30b, D102, P107, Y108 Y37, S96, T97, D102, P107, Y108 N30c, G30d, Y32,
L98, V99, L101 G66, S67, G68, T69, D70, S91, T92, L93, V94, L97 (Chothia) Epitopes
  E179, Y18, F182, P183, P184 R35, H101, V176, V177,
R178, E179, E180, Y181, F182, P184, K185, L187, D191, S193, H194, E266, P267, S268, F272,
L273, K276

Site 1 and Site II mAb 583, 11, 336, 329 and 66 Epitope Map

(527) Ala substitutions of T8 & D10 moderately reduces its affinity while Y11 & D15 abolishes mAb binding. A Phe substitution is tolerated at Y11 but acidic and acidic AA at G12 abolished binding. N-glycosylation of at N17 or D17P also abolishes mAb binding. N5 glycan is tolerated but a tandem glycosylation (N5 & N17) is not tolerated. mAb core epitopes are T8-G12 but nature of AA present in D15-D18 influence its binding. The results of the Ala substitutions of mAb583 are shown in FIG. 5A. Epitope mapping of 583 shows a noncontiguous core epitopes spanning 9aa. Clone 583 was previously characterized as an N-terminal binding antibody that does not required sulfation at positions Y11 and Y14. The data reveals that mutations T8A and D10A moderately reduce binding. Mutation Y11A abolishes binding. Additionally, in agreement with previous data, Y11F does not abolish binding, confirming that tyrosine sulfation is not required. Mutation G12A/T are tolerated, while G12E/D abolish binding, in agreement with the lack of cross-reactivity to cyno and rhesus monkey. Mutations are tolerated at positions 13 and 14, while mutations at D15 abolish binding and the insertion of a glycan at K17N abolishes binding. The core epitope appears to be located near T8-G12, with additional contributions from D15-D18.

(528) The results of ala substitution mutations of Ab66 are shown in FIG. 5B. Clone 66 was previously characterized as a site I binder and its binding is dependent on tyrosine sulfation at positions 11Y and/or 14Y. The mutagenesis data confirmed that Y11 and Y14 are critical for binding; however, sulfation is only required at position 14, as the mutation Y11F did not impact binding, while Y14F abolished binding. Alanine scanning clearly indicates that the core epitope spans 5 amino acids from D10 to Y14. It should be noted that the only the only non-human species to retain clone 66 binding was baboon, which is the only NHP that retains the DYGHY motif. Cyno and rhesus contain G12E, which abolishes binding.

(529) Peptide epitope mapping data indicated that clones 11, 329, and 336 all bound to a cyclic ECL2 peptide spanning from residues R175 to C188. However, it was unknown if additional contacts were made with ECL1 or ECL3. The mutagenesis data did not reveal any contacts with ECL1 or ECL3. The mutation W102A on ECL1 did abolish binding to all site II antibodies; however, this residue contributes to the ECL2 conformation and the loss of binding is likely due to a supportive role at the base of ECL2 and not a direct binding contact with the paratope. The data is highlighted in FIG. 5C-5E.

(530) The core epitope of clone 11 is from E180-P183; the core epitope of clone 329 is from E180-P184; and the core epitope of clone 336 is from R178-P183. C188A abolishes binding for all clones, as it forms a canonical disulfide bond with ECL1 that is important for the ECL2 conformation. Additionally, the introduction of N-glycans near the C5a binding vestibule did not impact site II binding. The protrusion of the core epitope from the surface and the tolerance of glycans near the vestibule suggest that clones 11, 329, and 336 may not directly block the C5a binding vestibule. The data for clones 11, 329, and 336 is highlighted in FIG. 5F.

Example 3: Antibody Binding to Human Neutrophils

(531) This Example describes the binding of C5aR1 on human neutrophils by the C5aR1 antibodies.

(532) Exemplary anti-C5aR1 antibodies generated as described in Example 1 were tested for binding to human neutrophils. Briefly, cryopreserved human neutrophils were obtained from Asarte

Biologics. Cell surface binding and flow cytometry were performed on half-log titrations of anti-C5aR1 antibodies ranging from at 10.sup.-7.5 to 10.sup.-12 M. As shown in FIG. 6, the anti-C5aR1 antibodies generated herein showed dose-dependent binding to human neutrophils compared to isotype controls. Clones 66 and 329 showed the highest level of neutrophil binding. Clones 332 and 335 showed intermediate levels of neutrophil binding, while clone 335 showed the comparatively lowest levels of neutrophil binding.

Example 4: Inhibition of G α Signaling by Anti-C5aR1 Antibodies

(533) This Example describes the inhibition of G α signaling by C5aR1 antibodies.

(534) Exemplary anti-C5aR1 antibodies generated as described in Example 1 were screened for antagonist activity against human C5aR1 using the GeneBLAzer assay, as described in Example 1. Briefly, cells were pre-incubated with 10 nM of antibody followed by addition of 10 nM C5a. The engineered induction of beta-lactamase expression by C5a-C5aR1 signaling was measured using the BLA substrate and calculating the ratio of blue and green fluorescence. As shown in FIG. 7, a number of the anti-C5aR1 antibodies generated herein showed comparable (e.g., clones 322, 402, 365, 603, 210, 511, 518, 549, 553, 556, and 584) or improved (e.g., clones 11v2, 329, 330, 336, 429, 430, 453, 465, 507, 541, and 583) antagonism of C5aR1 compared to reference antibodies or to Avacopan.

Example 5: Inhibition of Chemotaxis by Anti-C5aR1 Antibodies

(535) This Example describes the inhibition of C5a induced chemotaxis by C5aR1 antagonistic antibodies.

(536) In this example, the functional activity of the exemplary anti-C5aR1 antibodies was assessed by measuring their impact on cell chemotaxis, which is known to be induced by C5aR1 activity.

(537) Anti-C5aR1 antibodies were assessed in a chemotaxis assay for their ability to block C5a induced migration of neutrophils. The assay involved cell migration in a double chamber well separated by filters. The cells and inhibitors were applied into the upper chamber and the lower chamber was filled with the chemoattractant, causing the cells to migrate to the lower chamber. The inhibition activity was evaluated by counting the migrated cell numbers. Specifically, a titration of C5a was added to the bottom chamber a 96 transwell plate. Cells expressing human C5aR1 (C5aR1-U937 cells or human neutrophils) were incubated with or without antagonist and then seeded into the top chamber of the transwell plate. The cells were incubated for 2 hours. Cells that migrated to the bottom chamber were quantified using the CyQUANT cell proliferation kit following the manufacturer's protocol. The assay was performed using engineered cells stably expressing C5aR1 as well as using human neutrophils.

(538) In a first experiment, C5aR1-U937 cells were seeded into the top chamber of a 96 Transwell plate in the presence of no antibody (antagonist), or of 1, 10, or 100 nM antibody (antagonist). The bottom chamber contained a half-log titration of C5a ranging from 10⁶ to 10.sup.-9.5 M. As shown in FIGS. 8A-8F, the exemplary anti-C5aR1 antibodies generally inhibited cell chemotaxis in a dose-dependent manner, with the strongest effects detected for clones 11v2, 329, and 583.

(539) In a second experiment, human neutrophils were seeded into the top chamber of a 96 Transwell plate in the presence of no antagonist, 1, 10, or 100 nM antagonist. The bottom chamber contained a half-log titration of C5a ranging from 10.sup.-6 to 10.sup.-9.5 M. As shown in FIGS. 9A-9D, the exemplary anti-C5aR1 antibodies generally inhibited human neutrophil chemotaxis in a dose-dependent manner, with the strongest effects detected for clone 329.

(540) In a third setting, a combination of clones 66, 11, 329, 336v2, 583 and 322 were probed to detect inhibition of C5a induced chemotaxis. FIGS. 10A-10I show inhibition of chemotaxis with different combination of clones of anti-C5aR1 antibodies. It was observed that combination of Site I and Site II antibodies resulted in synergistic inhibition of chemotaxis.

Example 6: Functional Characterization of Anti-C5aR1 Antibody

(541) This Example describes the calcium flux by C5aR1 antibodies in the presence of C5a.

(542) Exemplary anti-C5aR1 antibody clone 329, generated as described in Example 1, was

assessed for its ability to inhibit C5aR1 signaling. Clones 11v2, 329, and 583 were tested for their ability to inhibit C5a induced calcium efflux. FIGS. **11A-11C** show inhibition of the C5a induced calcium efflux by C5aR1 antibodies in U937 cells. FIGS. **12A-12C** show inhibition of the C5a induced calcium efflux by C5aR1 antibodies in human neutrophils.

(543) In one experiment, antibody 329 was examined in a GeneBLAzer assay as described in Example 1. As shown in FIG. **13A**, C5aR1 signaling was inhibited in a dose-dependent fashion by antibody 329. In a second experiment, the ability of antibody 329 to inhibit C5a-induced calcium flux was measured in C5aR1-U937 cells. As shown in the bottom panel of FIG. **13B**, C5a-induced calcium flux was inhibited by antibody 329 in a dose-dependent manner.

Example 7: Combination of Anti-C5aR1 Antibody with Small Molecule Inhibitor of C5aR1

(544) An exemplary anti-C5aR1 antibody targeting Site II, generated as described in Example 1 was tested in combination with Avacopan, an allosteric small molecule C5aR1 inhibitor. C5aR1 signaling was assessed using the GeneBLAzer assay, as described in Example 1. As shown in FIGS. **14A-14C**, the combination of anti-Site II antibody 329 and Avacopan resulted in greater potency in blocking C5aR1 activity than Avacopan alone.

Example 8: Inhibition of G α Signaling by Combination of Anti-C5aR1 Antibodies

(545) This example describes inhibition of G α signaling by combination of Site I and Site II antibodies and Site I-Site II biparatopic antibody in inhibition of C5a mediated G α signaling.

(546) Anti-C5aR1 antibodies targeting Site I and Site II (clone 583, clone 11, clone 329 and clone 336) were tested for inhibition of C5aR1 mediated G α signaling using Gene BLAzer assay, as described in Example 4.

(547) In a first setting, antibodies bind to Site I and Site II, respectively, were tested in the GeneBLAzer assay, as described in Example 1. As shown in FIGS. **15A-15C**, the combination of anti-Site I antibody 66 and the anti-Site II antibody 329 resulted in greater potency in blocking C5aR1 activity than either antibody alone.

(548) In a second experiment, Site II antibody 329 was tested in combination with each of the Site I antibodies 66 and 583 in a cell binding assay. Antibodies were assessed either as a single agent or in a combination of Site I and Site II antibodies in C5aR1-U937 cells. As shown in FIGS. **16A-16B**, each combination of a Site I antibody with a Site II antibody resulted in an increase in the saturation level compared to the antibodies individually or an isotype control (HYHEL10), indicating that the combination of the Site I and Site II antibodies resulted in co-engagement of Site I and Site II of C5aR1.

(549) In a third experiment, the combination of Site I and Site II antibodies was assessed for their capacity to inhibit calcium flux in C5aR1-U937 cells. As shown in FIGS. **17A-17C**, the Site I antibody 66 and the Site II antibody 329 each inhibited calcium flux as single agents in a dose-dependent manner. The combination of antibodies 66 and 329 appeared to induce calcium flux inhibition to a greater degree than the single agents.

(550) In a fourth experiment, the combination of Site I and Site II antibodies was assessed for their capacity to inhibit chemotaxis. As shown in FIGS. **18A-18C**, the Site I antibody 583 and the Site II antibody 329 each inhibited chemotaxis as single agents in a dose-dependent manner.

(551) In a fifth setting, an IgG4 scFv antibody comprising two different variable regions, capable of binding two different epitopes on C5aR1—first to Site I, the second to Site II were used to analyze for inhibition of C5aR1 mediated G α signaling using GeneBLAzer assay. The scFv was linked to the heavy chain of the antibody. FIG. **19A-FIG. 19B** show inhibition of C5aR1 mediated G α signaling as a biparatopic antibody.

(552) In a sixth setting, an IgG4 scFv antibody comprising two different variable regions, capable of binding two different epitopes on C5aR1—one to Site I, the second to Site II were used to analyze for inhibition of C5aR1 mediated G α signaling using GeneBLAzer assay in U937 cells. The scFv was linked to the light chain of the antibody. FIG. **19C-FIG. 19E** show inhibition of C5aR1 mediated G α signaling as a biparatopic antibody.

(553) The combination of antibodies and the biparatopic antibody inhibited C5aR1 mediated G α signaling, chemotaxis and calcium flux to a greater degree than the single agents.

Example 9: Inhibition of Chemotaxis by Anti-C5aR1 biparatopic antibodies

(554) This example describes a method of inhibition of C5a chemotaxis using combination of Site I and Site II antibodies and Site I-Site II biparatopic antibody.

(555) Anti-C5aR1 antibodies targeting Site I and Site II (clone 583, clone 11, clone 329 and clone 336) were tested for inhibition of C5aR1 mediated G α signaling using Gene blaze assay, as described in Example 5, in U937 cells, in the presence of increasing concentrations of C5-alpha. In some embodiments, the biparatopic antibody comprised a scFv was linked to the light chain of the antibody (for example, 329 (L)-583-scFv). In some embodiments, the biparatopic antibody comprised a scFv was linked to the heavy chain of the antibody (for example, 329 (H)-583-scFv). FIG. 20A-FIG. 20E show inhibition of C5aR1 mediated G α signaling as a biparatopic antibody.

(556) In a second setting, combination of clone 583 and clone 329 were analyzed for inhibition of C5aR1 mediated chemotaxis, in the presence of increasing concentrations of C5-alpha. FIG. 20E-FIG. 20G show inhibition of C5aR1 mediated chemotaxis as an antibody cocktail.

(557) It was observed that biparatopic antibody and antibody cocktail targeted to site I and site II provided robust inhibition of C5aR1 mediated chemotaxis.

Example 10: In Vivo Inhibition of Neutropenia in Mice

(558) This example describes method of inhibition of neutropenia in mice by anti-C5aR1 antibodies, in comparison with avacopan.

(559) Human C5aR1 transgenic mice were obtained from Jackson Laboratories. The hC5aR1 mice were challenged with 100 μ g/kg hC5a. The mice were treated with the antibodies at the indicated doses listed in Table 9.

(560) TABLE-US-00018 TABLE 9 Dosages and antibodies for analyzing the effect of anti-C5aR1 antagonistic antibodies on Neutropenia in Mice. "bp" implies biparatopic. Dose Group N Mice Strain Test Article (mg/kg) Route Frequency Blood Collection Time-points 1 5 C5a knock-in Avacopan 30 mg/kg PO SD0 at -5 hr SD0 at -5 min, 1 min, 5 min (hom $\times$ hom) and T 2 hrs C5a 0.1 mg/kg IV SD0 at 0 hr 2 5 C5a knock-in 329 IgG4 20 mg/kg IV SD0 at -5 hr SD0 at -5 min, 1 min, 5 min (hom $\times$ hom) and T 2 hrs C5a 0.1 mg/kg SD0 at 0 hr 3 5 C5a knock-in 583 IgG4 20 mg/kg IV SD0 at -5 hr SD0 at -5 min, 1 min, 5 min (hom $\times$ hom) and T 2 hrs C5a 0.1 mg/kg SD0 at 0 hr 4 5 C5a knock-in 329-583 bp 20 mg/kg IV SD0 at -5 hr SD0 at -5 min, 1 min, 5 min (hom $\times$ hom) and T 2hrs C5a 0.1 mg/kg SD0 at 0 hr 5 6 C5a knock-in PBS 10 mL/kg IV SD0 at -5 hr SD0 at -5 min, 1 min, 5 min (hom $\times$ hom) and T 2 hrs C5a 0.1 mg/Kg SD0 at 0 hr 6 6 C5a knock-in PEG 400/ 10 mL/Kg PO SD0 at -5 hr SD0 at -5 min, 1 min, 5 min (hom $\times$ hom) Solutol (70:30) and T 2 hrs C5a 0.1 mg/kg IV SD0 at 0 hr

(561) The blood was drawn before the injection of the antibodies and 1 min, 5 min and 2 hours after intravenous injection of the antibodies. FIG. 21A-21C shows the percent CD11b expression after injection of different antibodies. It was observed that neutropenia was decreased after injection of antibodies in mice.

Example 11: Biparatopic Anti-C5aR1 Antibodies

(562) A combination of Site I antibodies and Site II antibodies, as well as a biparatopic antibody that engages Site I and Site II, were tested for functionality in comparison with standard monoclonal antibodies targeting C5aR1. The data from these experiments indicates that two full length IgG molecules can engage with C5aR1 at the same time. Further, a cocktail of Site I and Site II antibodies as well as a biparatopic antibody targeting site I and site II antibodies were more potent at blocking C5aR1 signaling than the parental mAbs (FIGS. 22A-22C). A biparatopic antibody targeting site I and site II antibodies were more potent at blocking C5aR1 signaling than the parental mAbs (FIGS. 22A-22B and 22D). This improvement in potency was not observed when two site I or two site II antibodies were used. Therefore, the data indicates that a Site I-Site II directed biparatopic antibody would be a more potent inhibitor of C5aR1 signaling than a

monospecific antibody targeting one site.

INCORPORATION BY REFERENCE

(563) All publications, patents, and Accession numbers mentioned herein are hereby incorporated by reference in their entirety as if each individual publication or patent was specifically and individually indicated to be incorporated by reference.

EQUIVALENTS

(564) While specific embodiments of the subject invention have been discussed, the above specification is illustrative and not restrictive. Many variations of the invention will become apparent to those skilled in the art upon review of this specification and the claims below. The full scope of the invention should be determined by reference to the claims, along with their full scope of equivalents, and the specification, along with such variations.

Claims

1. A nucleic acid molecule comprising a nucleotide sequence encoding an antibody molecule that binds to complement component 5a receptor 1 (C5aR1), wherein the antibody molecule comprises a heavy chain variable region (VH) and a light chain variable region (VL), wherein the VH comprises an HCDR1 comprising an amino acid sequence of SEQ ID NO: 612, an HCDR2 comprising an amino acid sequence of SEQ ID NO: 732, and an HCDR3 comprising an amino acid sequence of SEQ ID NO: 852, and wherein the VL comprises an LCDR1 comprising an amino acid sequence that is identical to or differs by no more than 2 amino acid residues from SEQ ID NO: 672, an LCDR2 comprising an amino acid sequence of SEQ ID NO: 792, and an LCDR3 comprising an amino acid sequence that is identical to or differs by no more than 2 amino acid residues from SEQ ID NO: 912.
 2. A host cell comprising the nucleic acid molecule of claim 1.
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